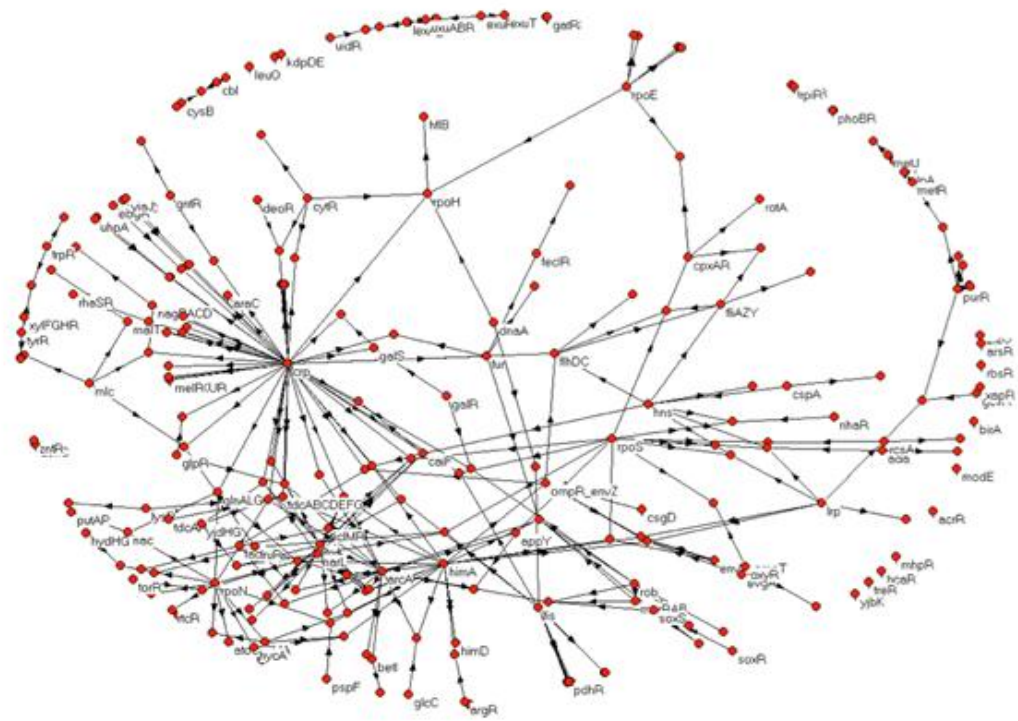
# Computational Biology

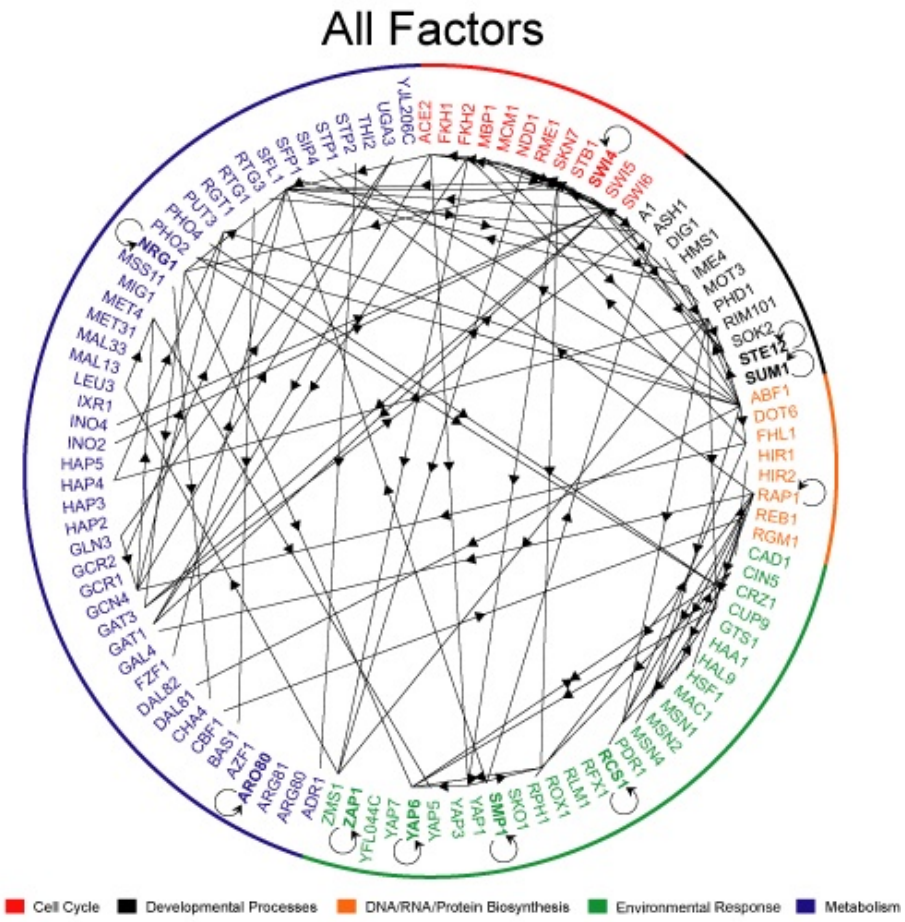
## Lecture 3: *Gene Regulation Networks*




*Gene regulation, chemical equations, motifs*

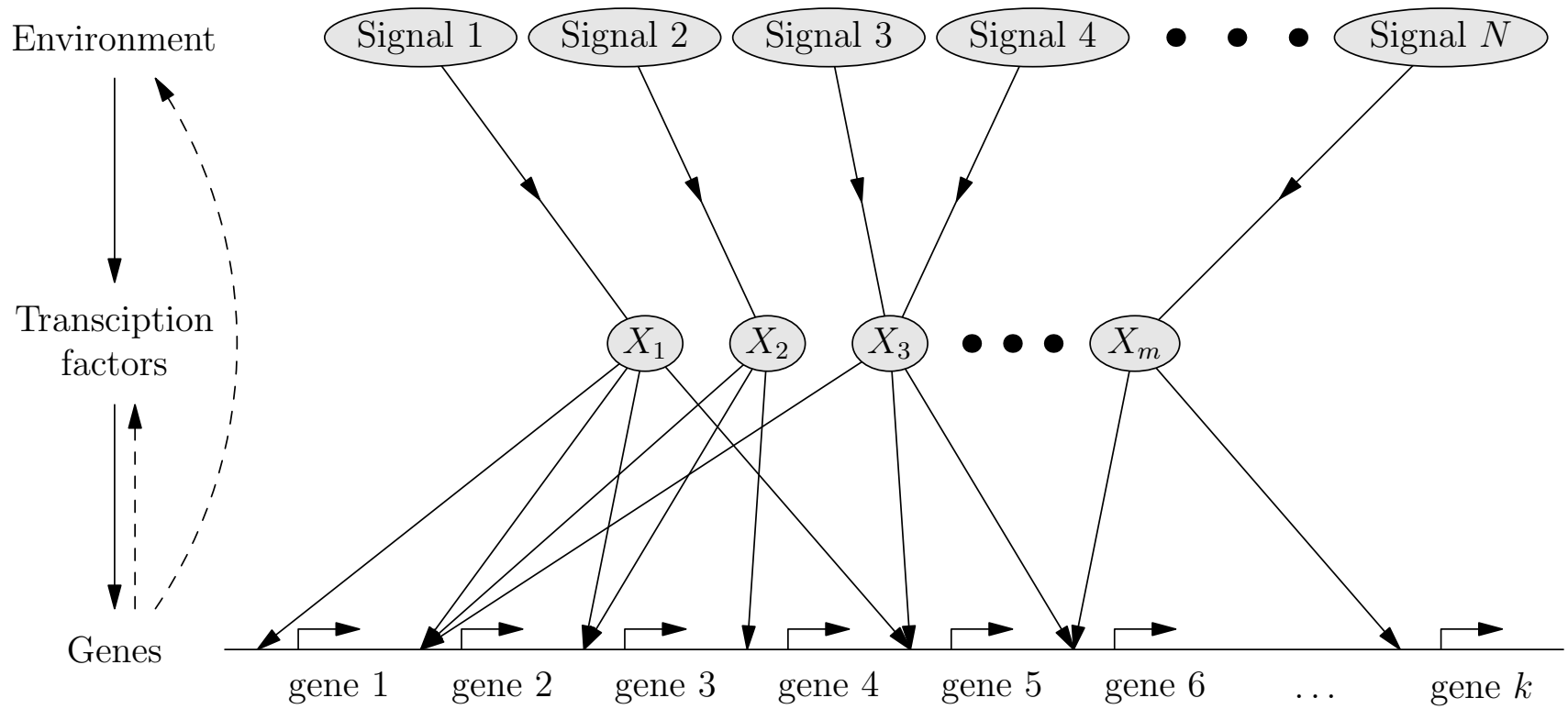
# Outline

1. **Gene Regulation**
2. Chemical Equations
3. Motifs



# Transciption Networks

- Signalling molecules combine with transcription factors (proteins) which excite or inhibit the transcription of other proteins



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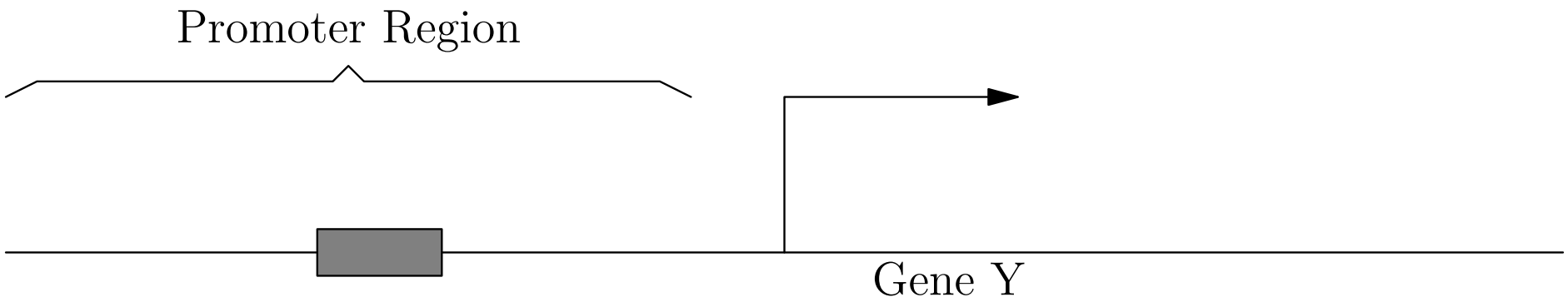

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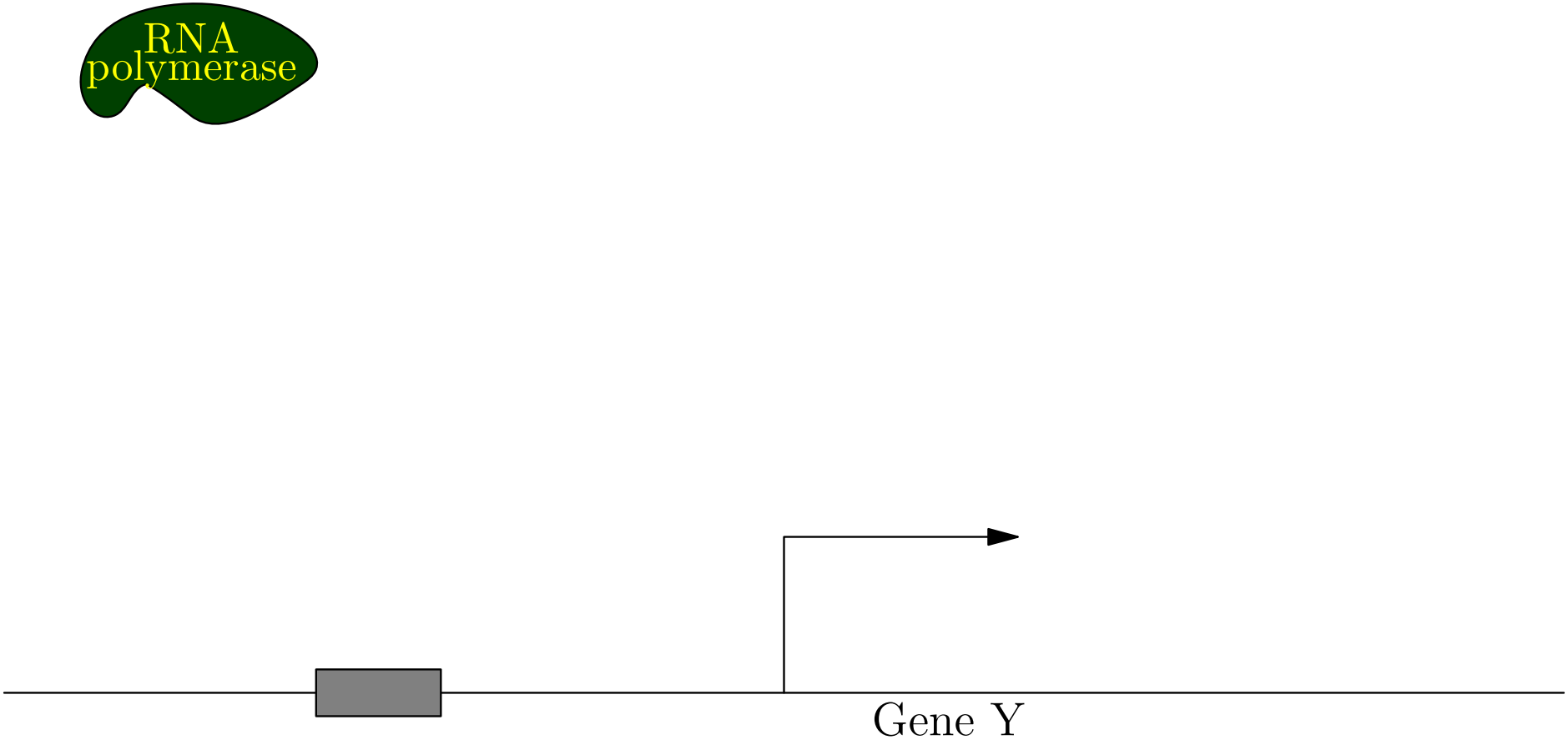

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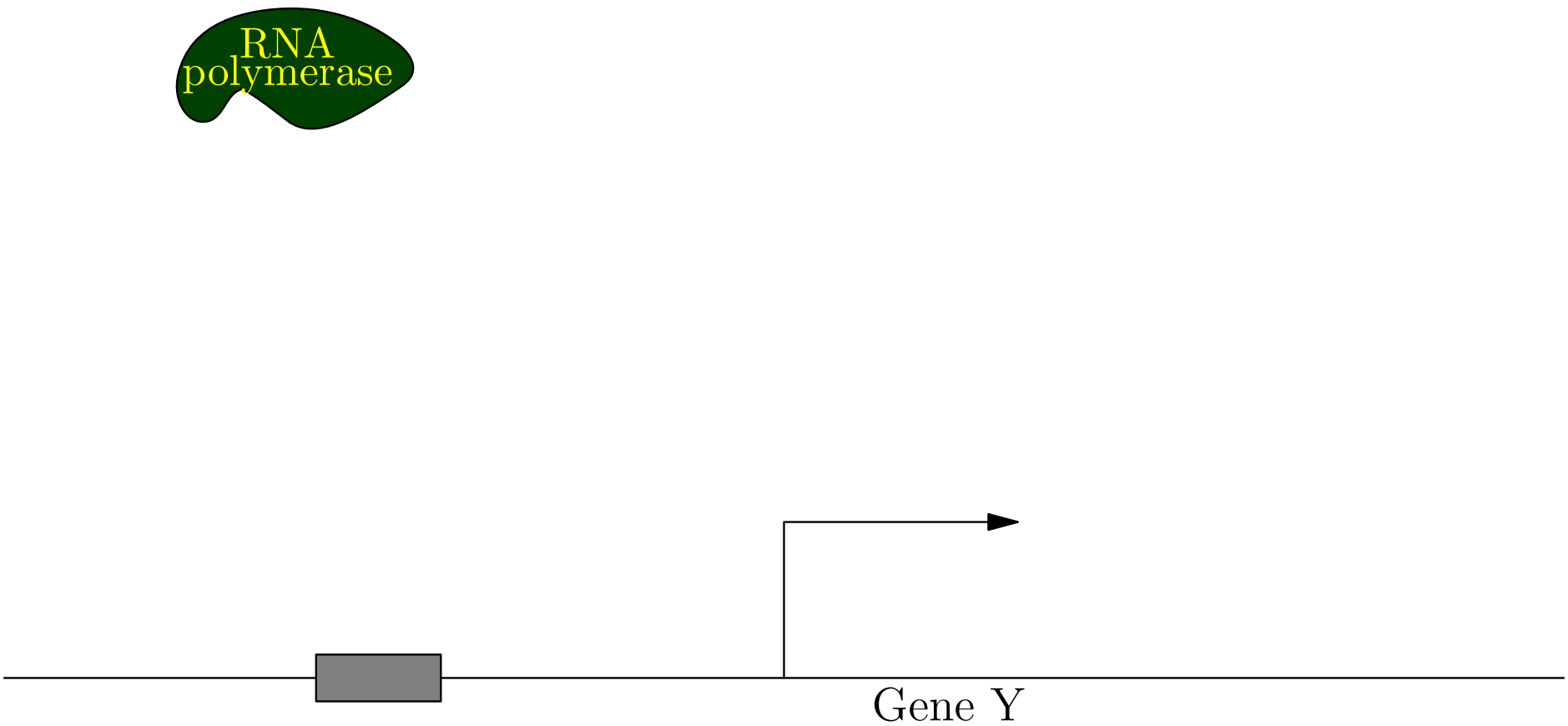

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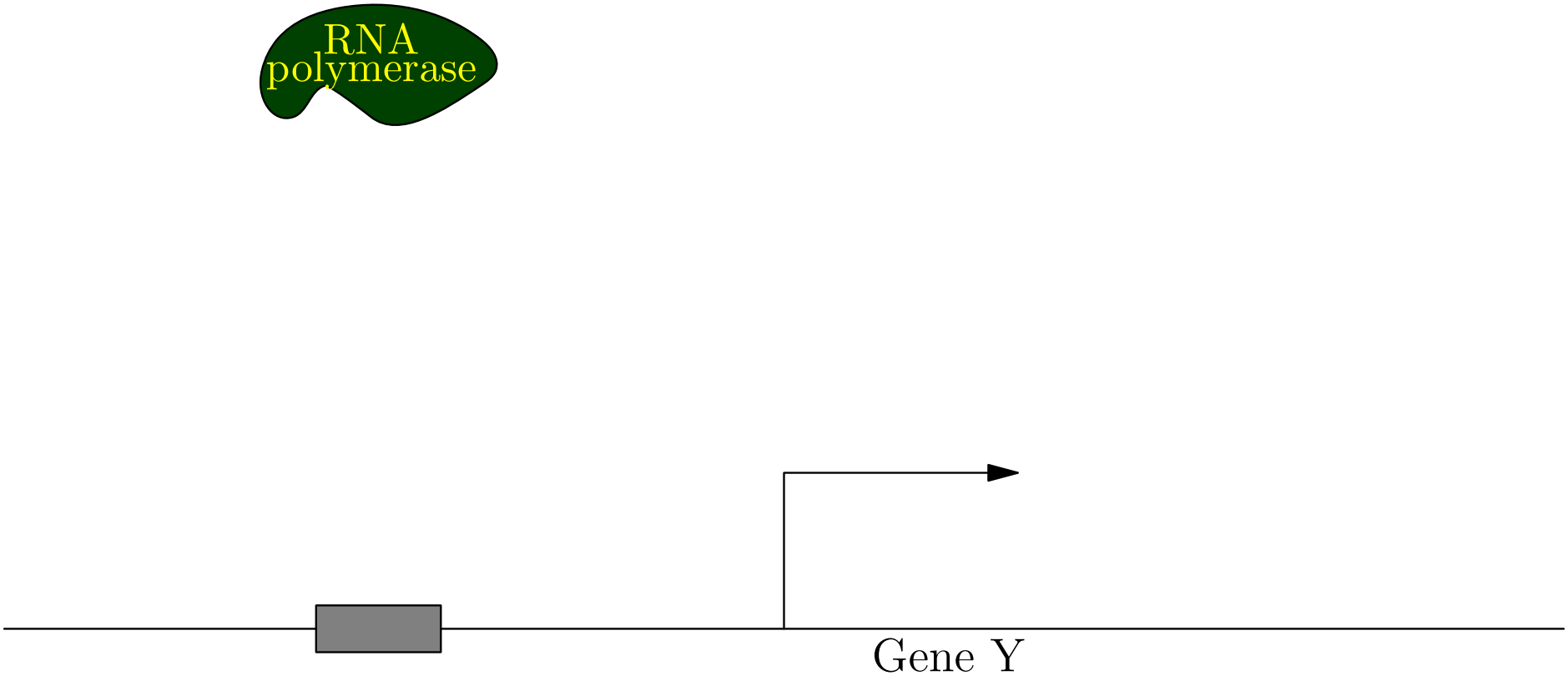

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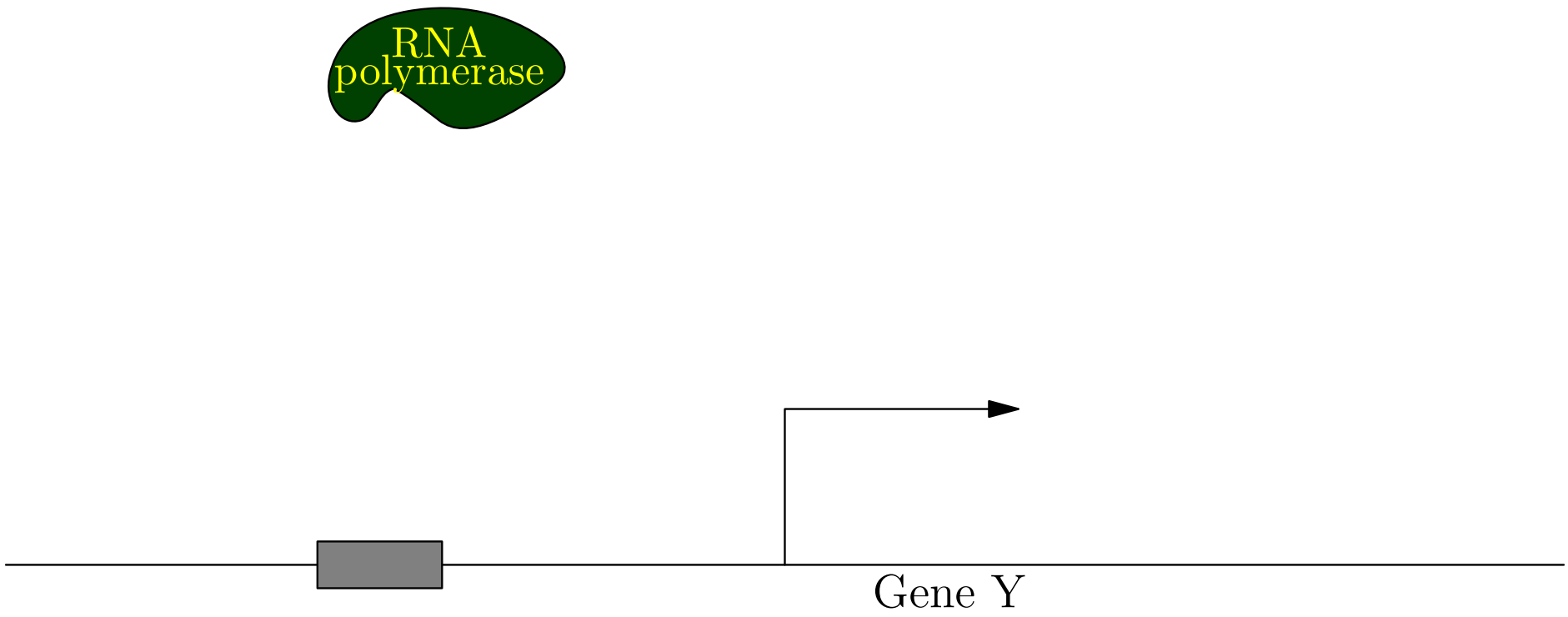

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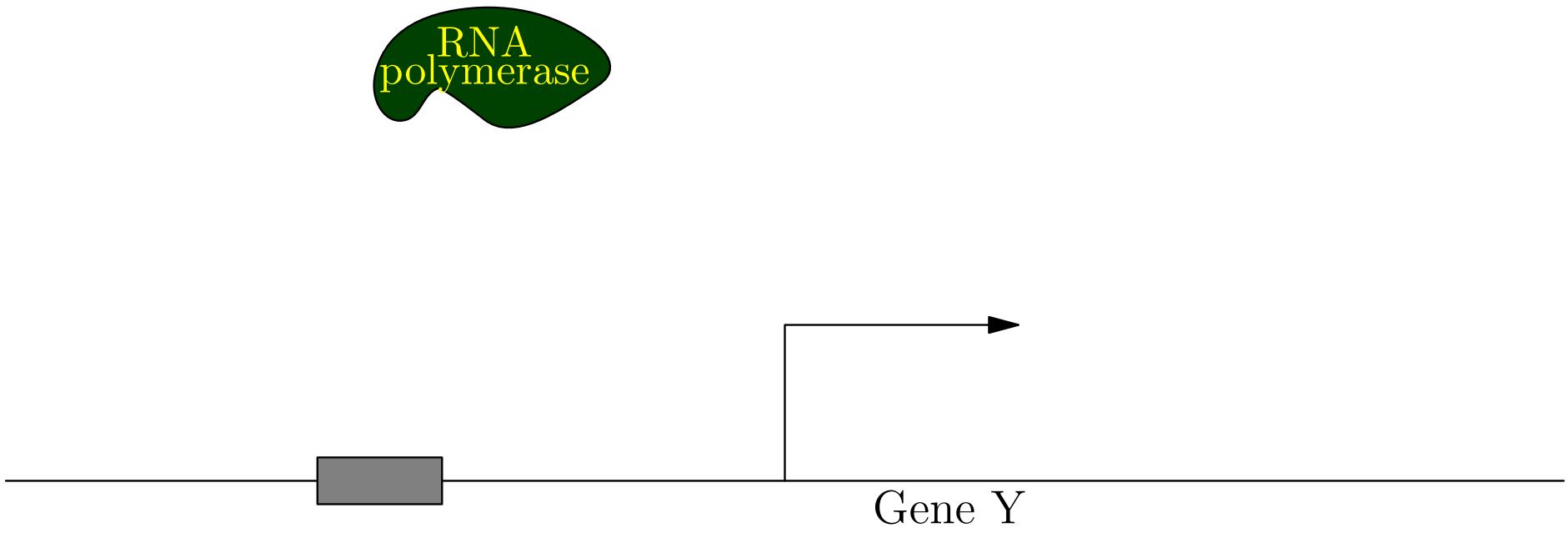

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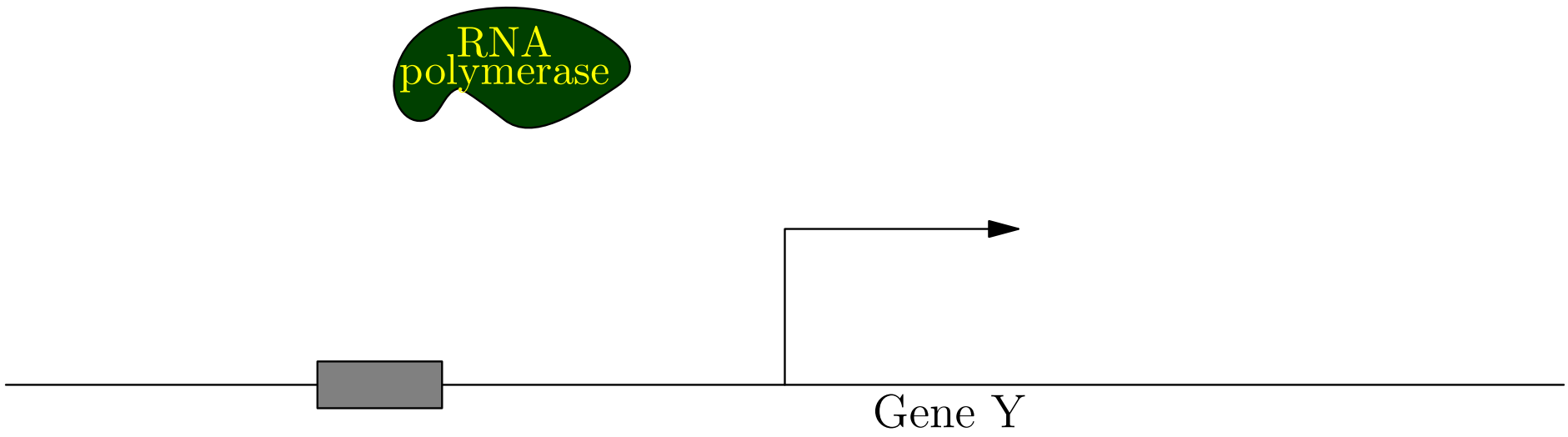

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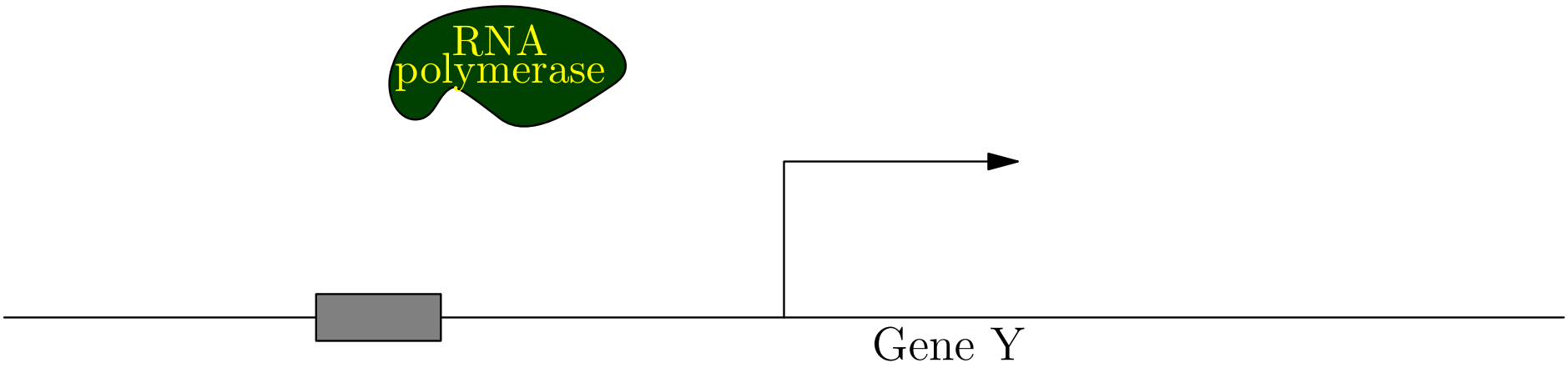

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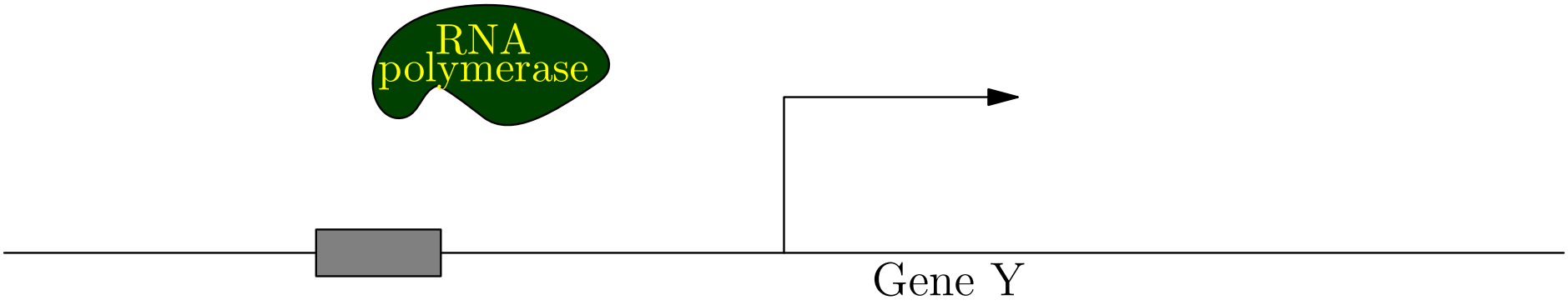

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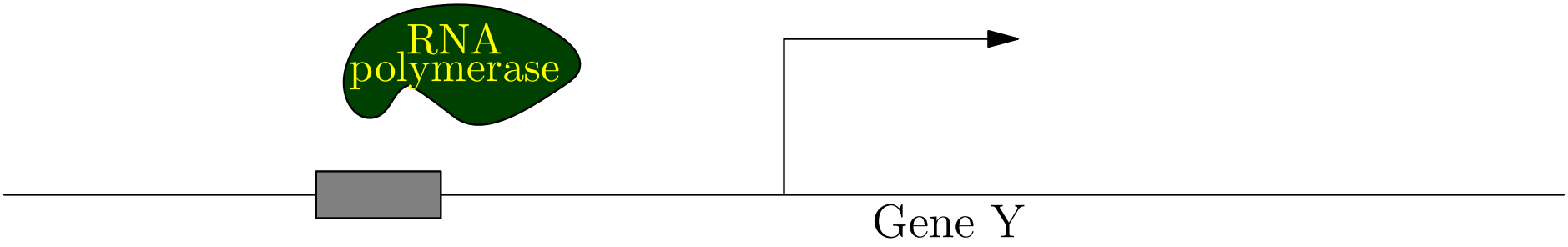

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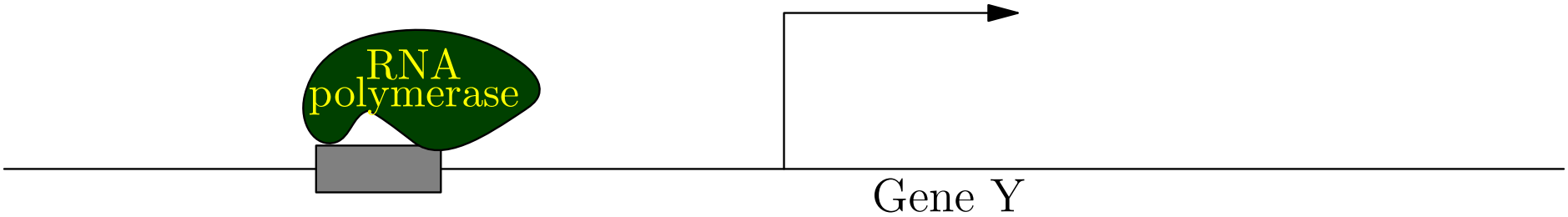

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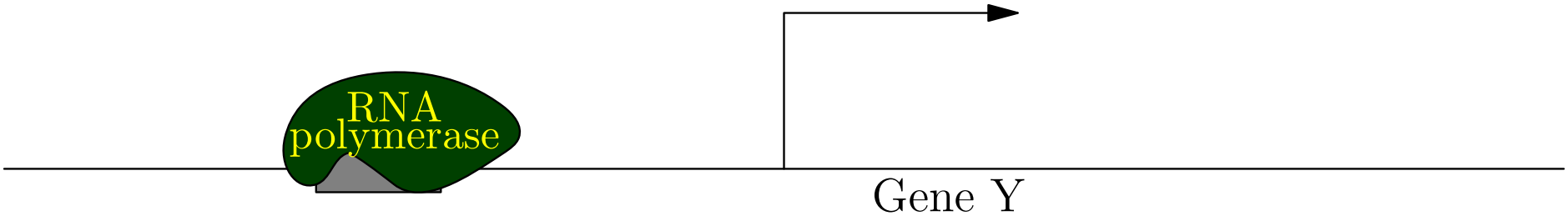

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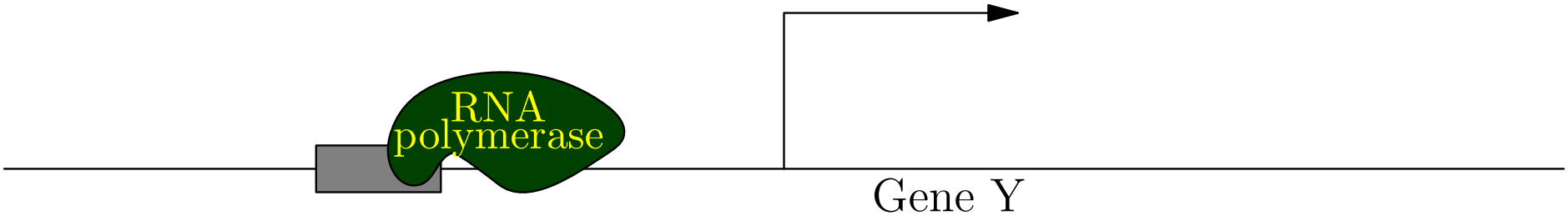

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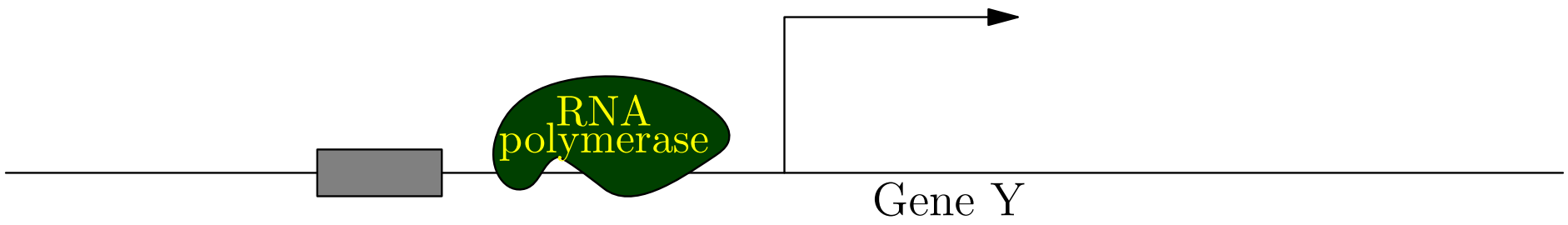

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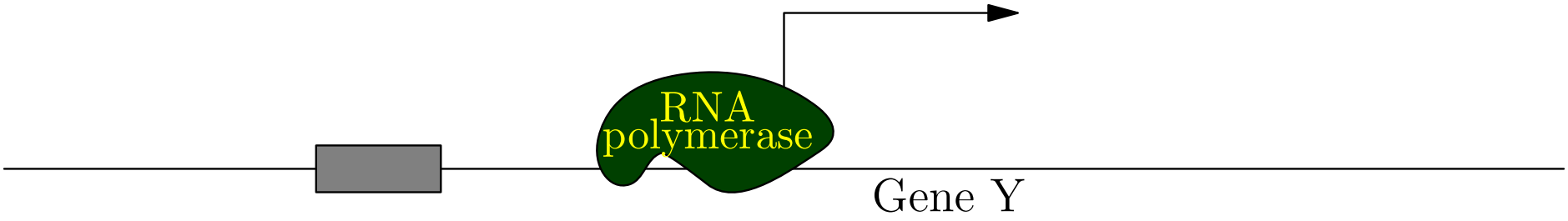

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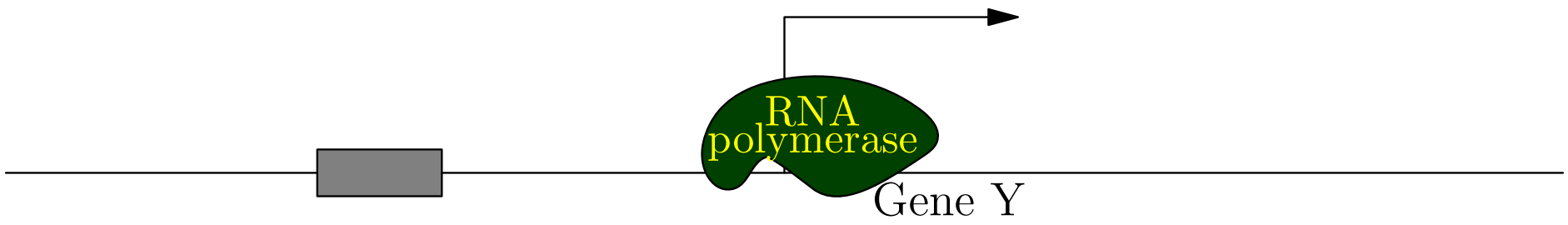

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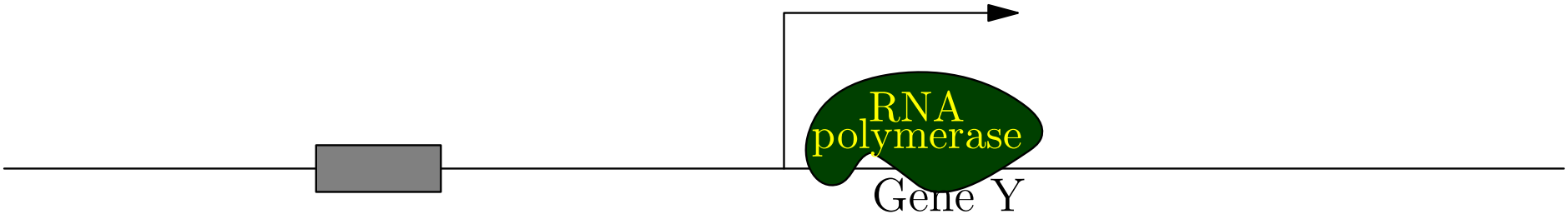

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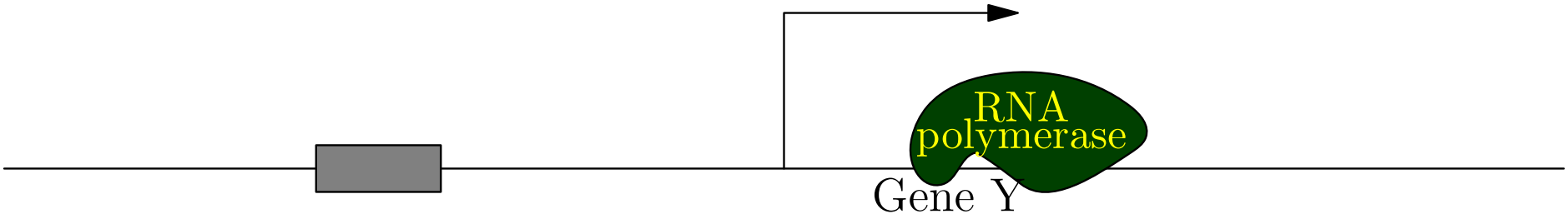

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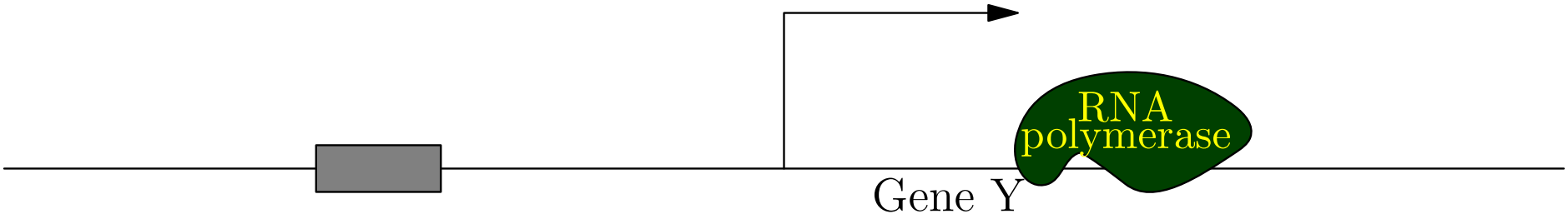

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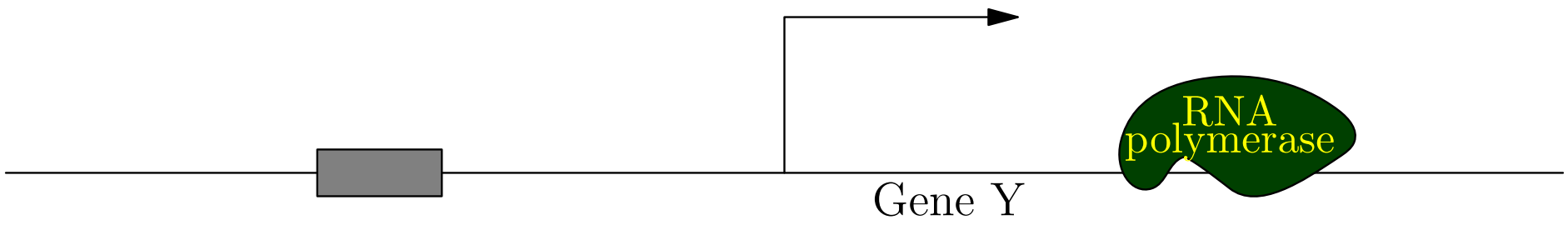

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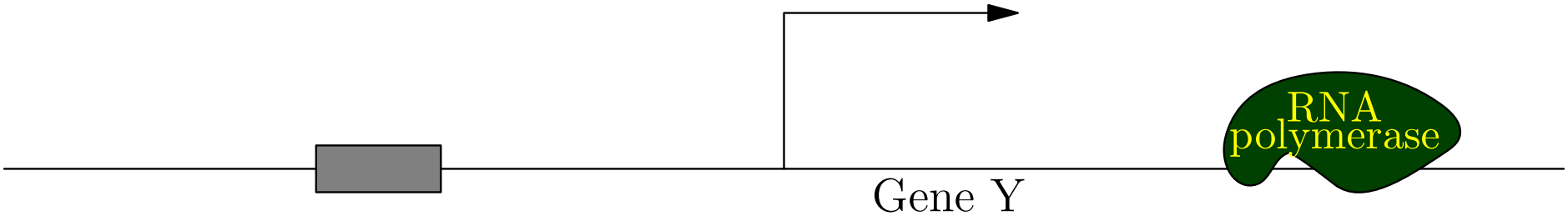

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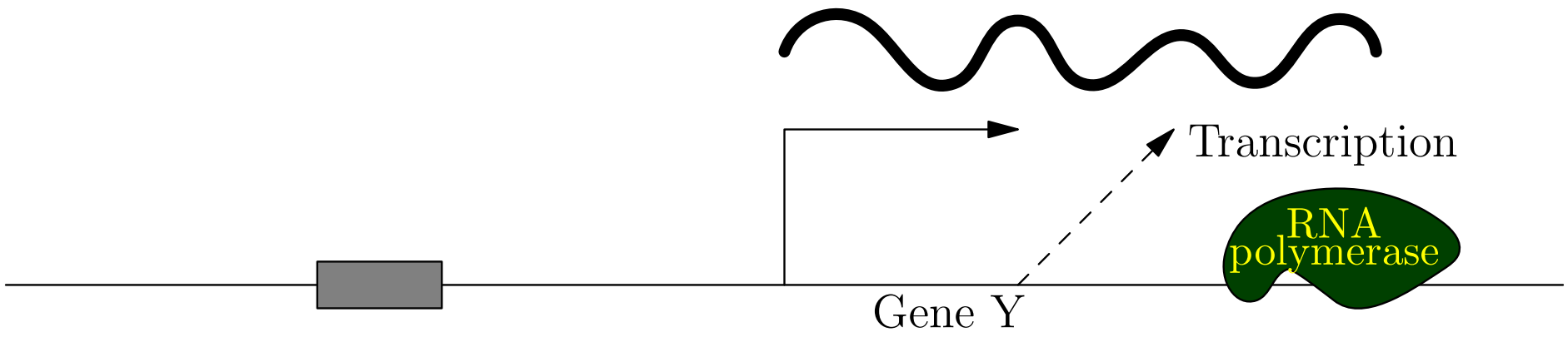

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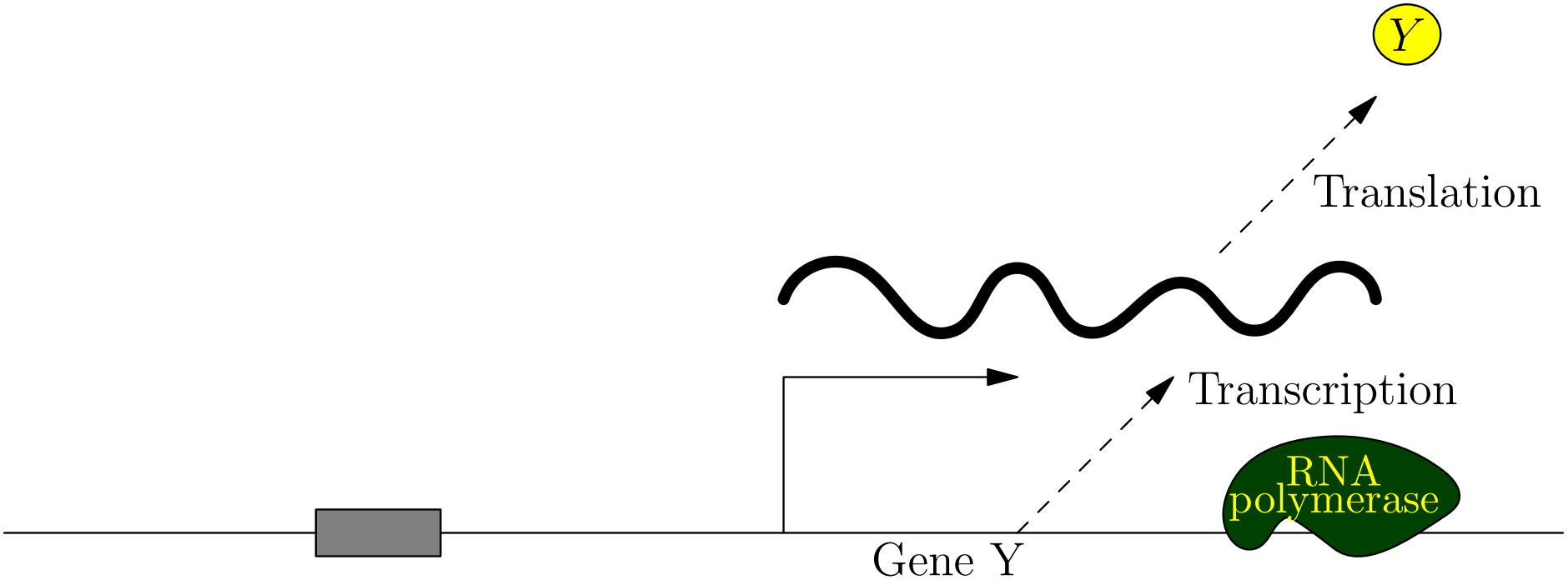

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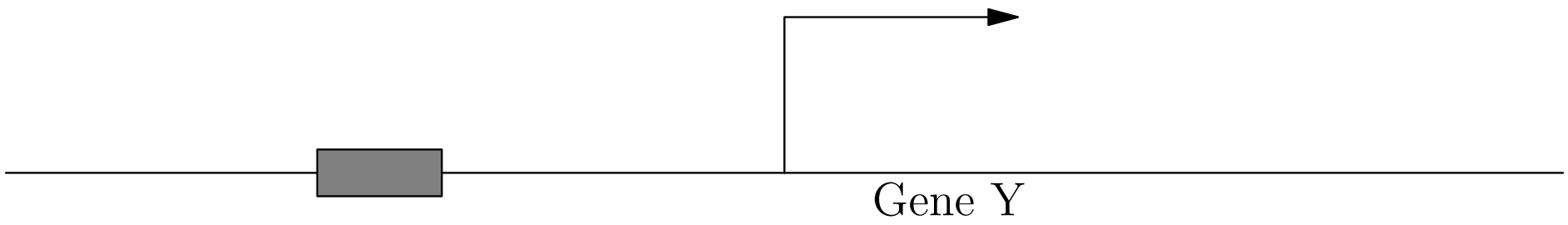

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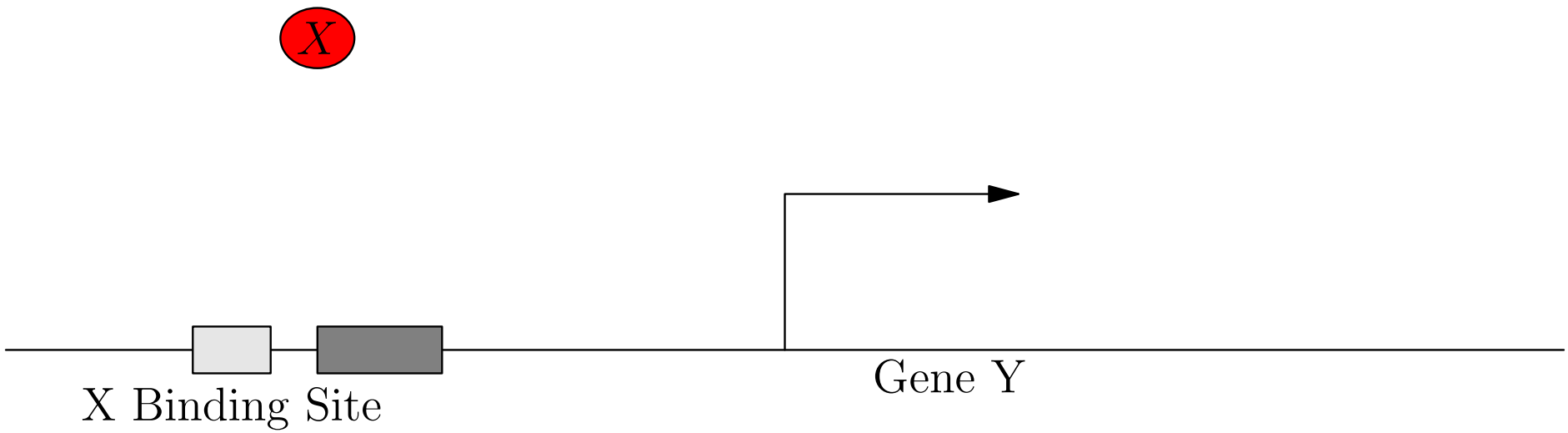

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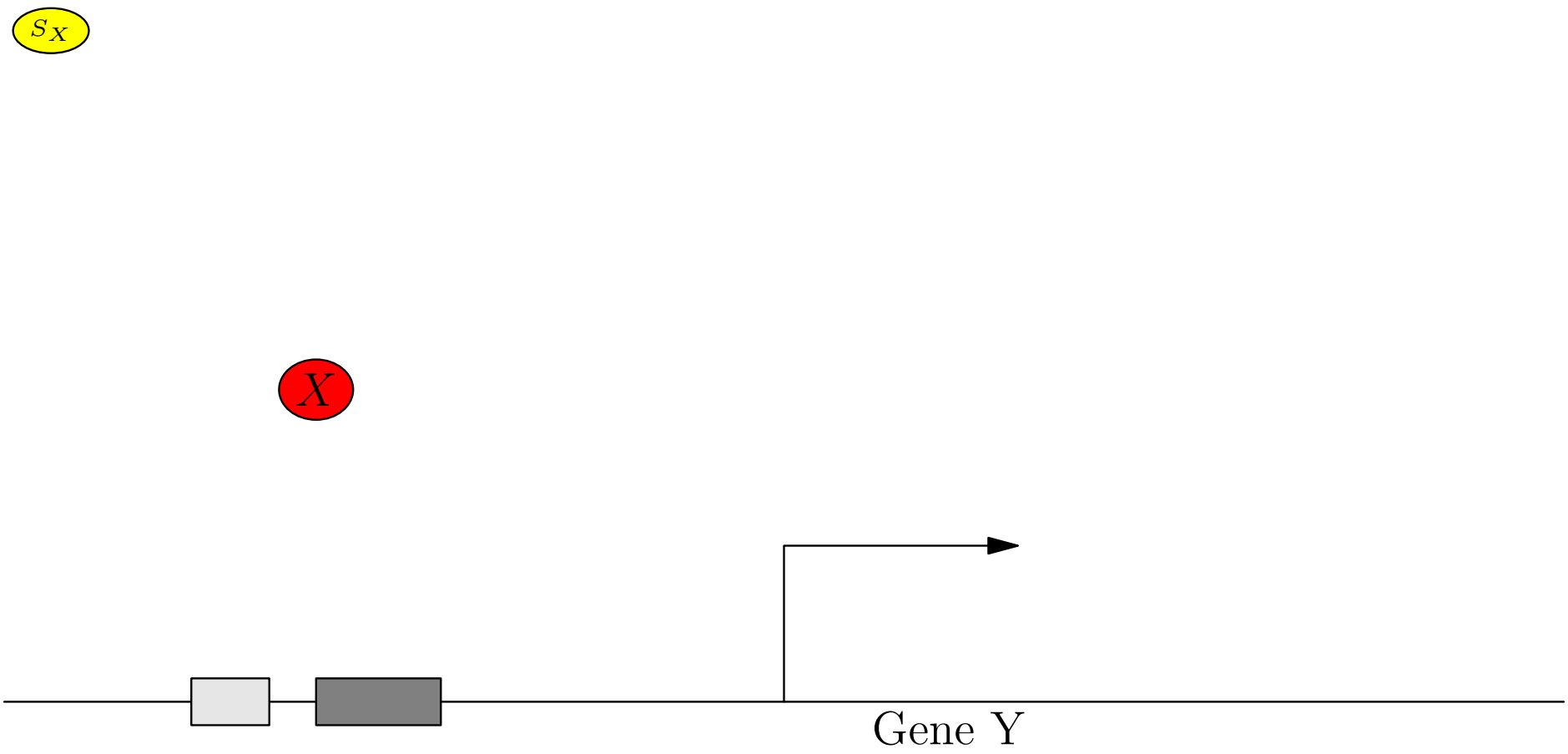

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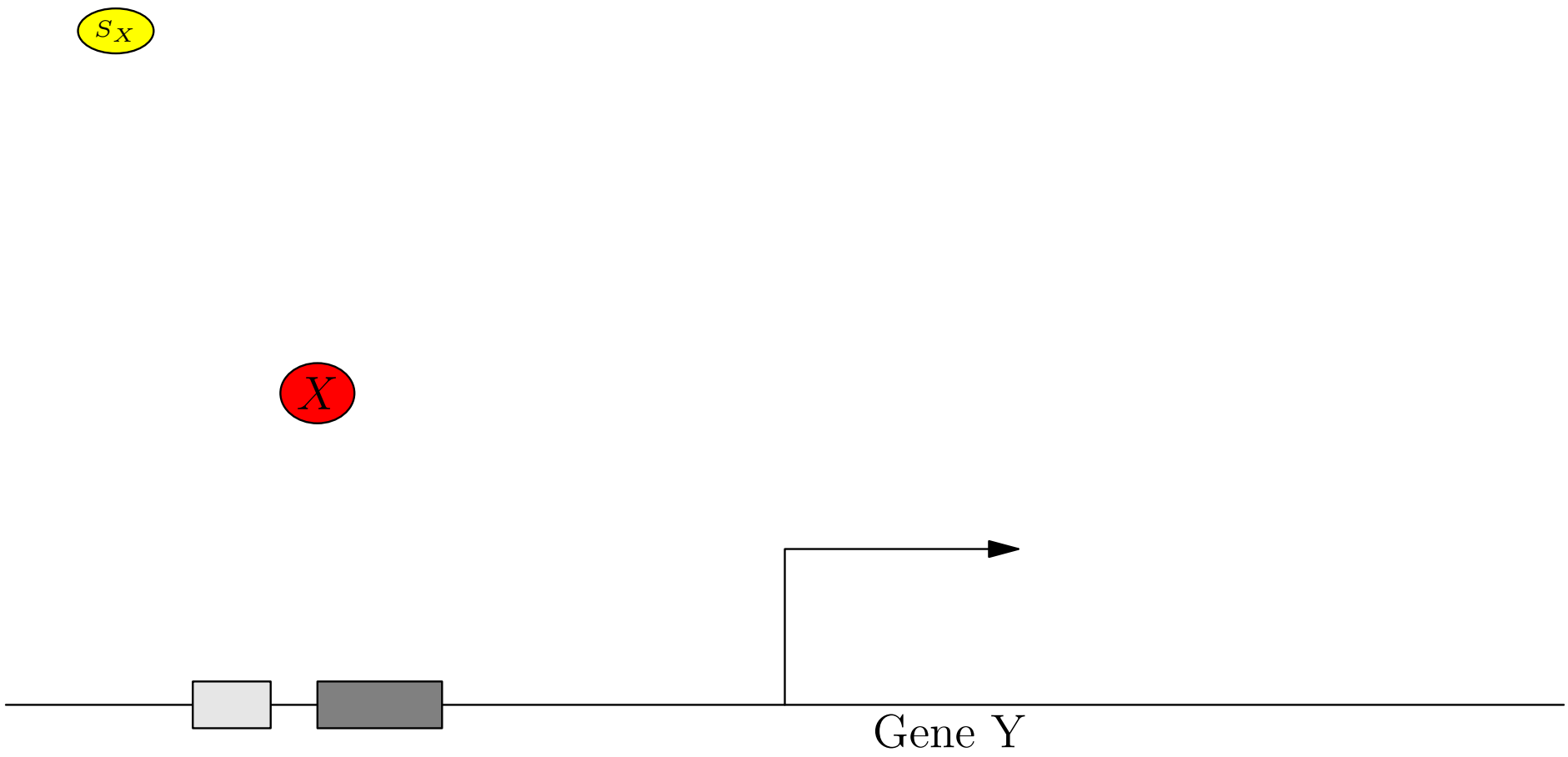

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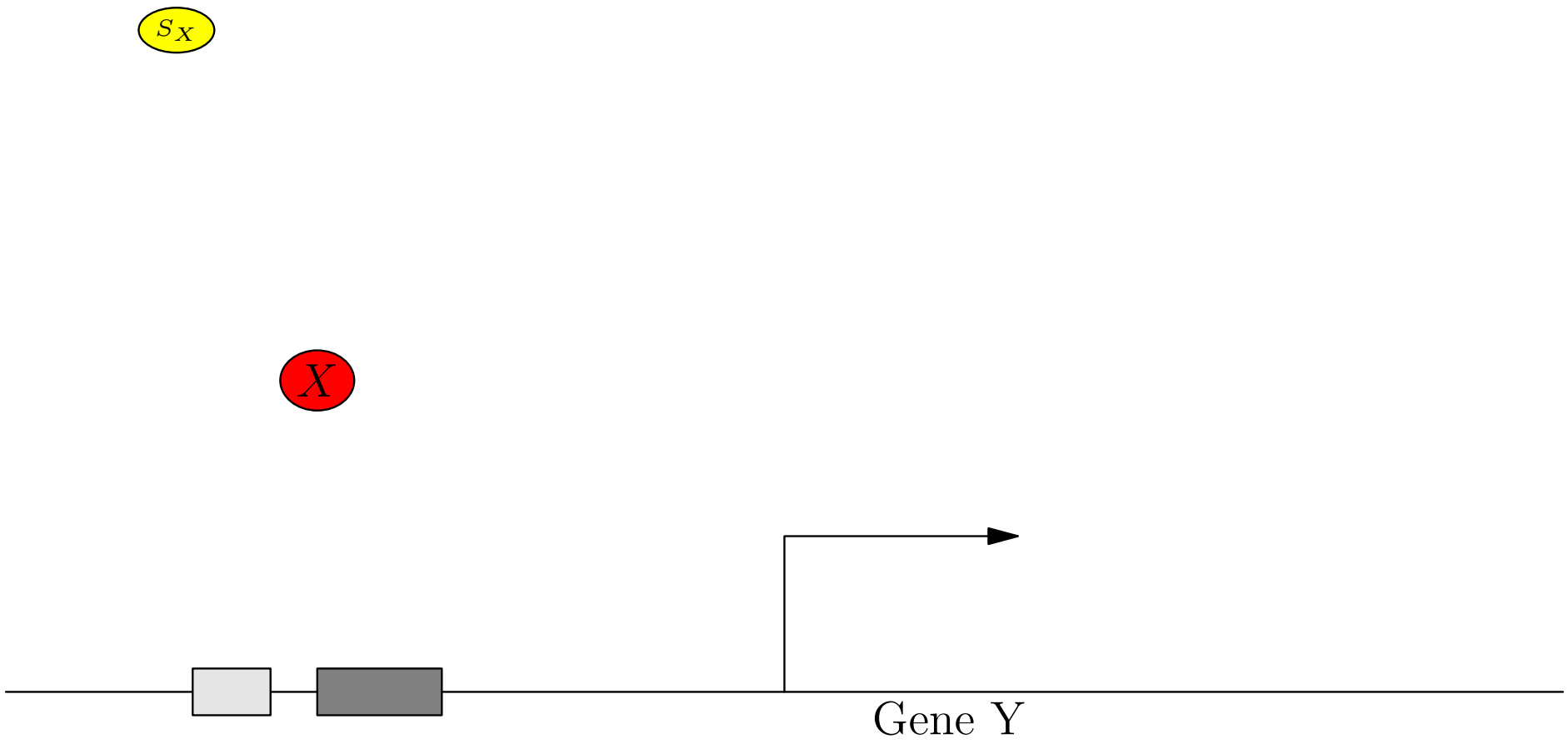

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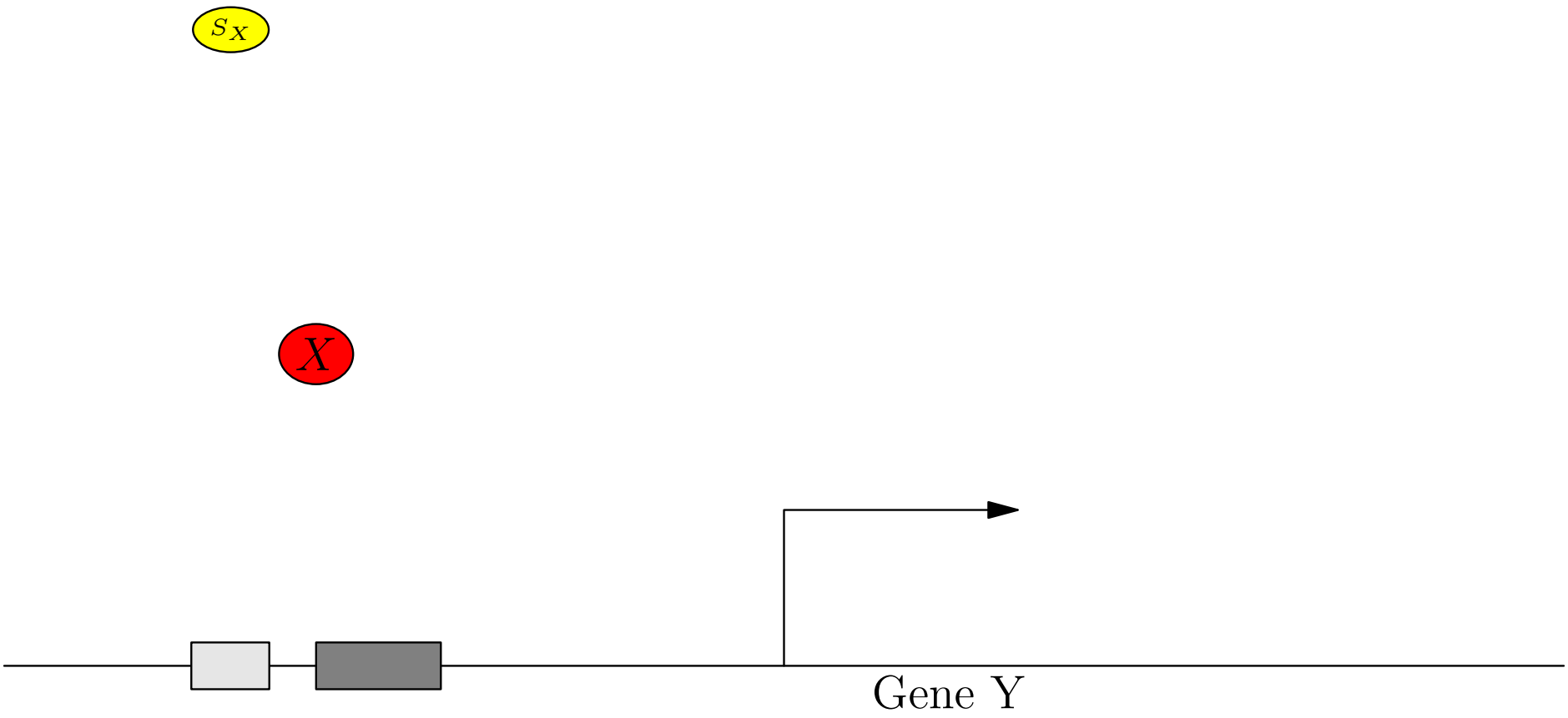

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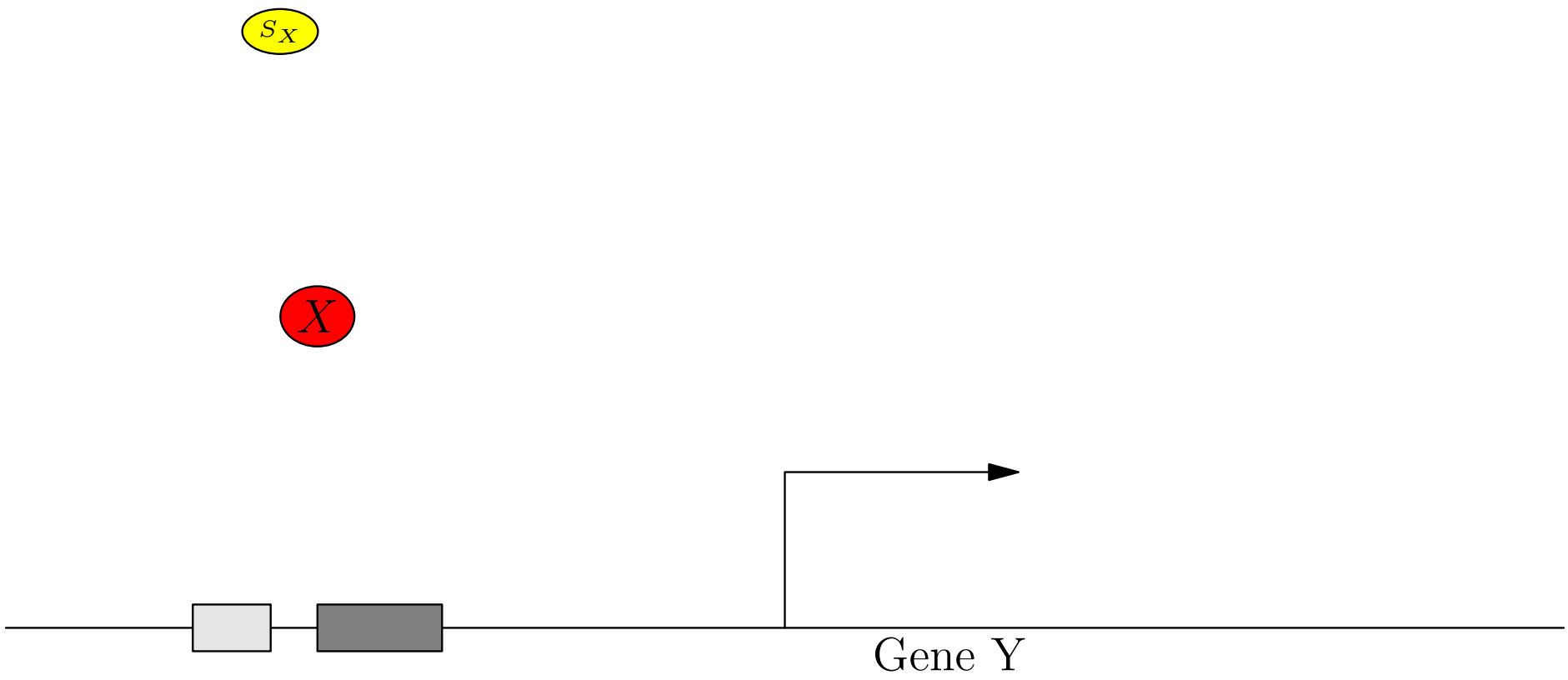

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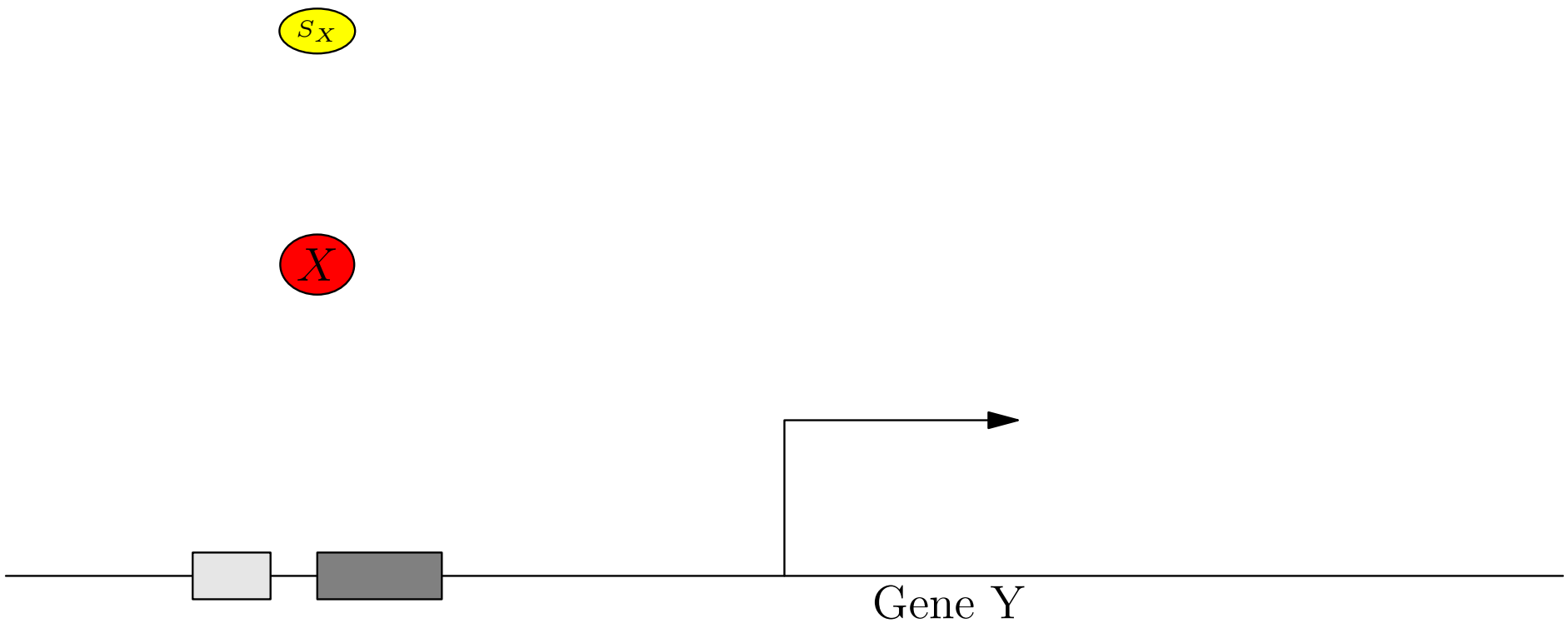

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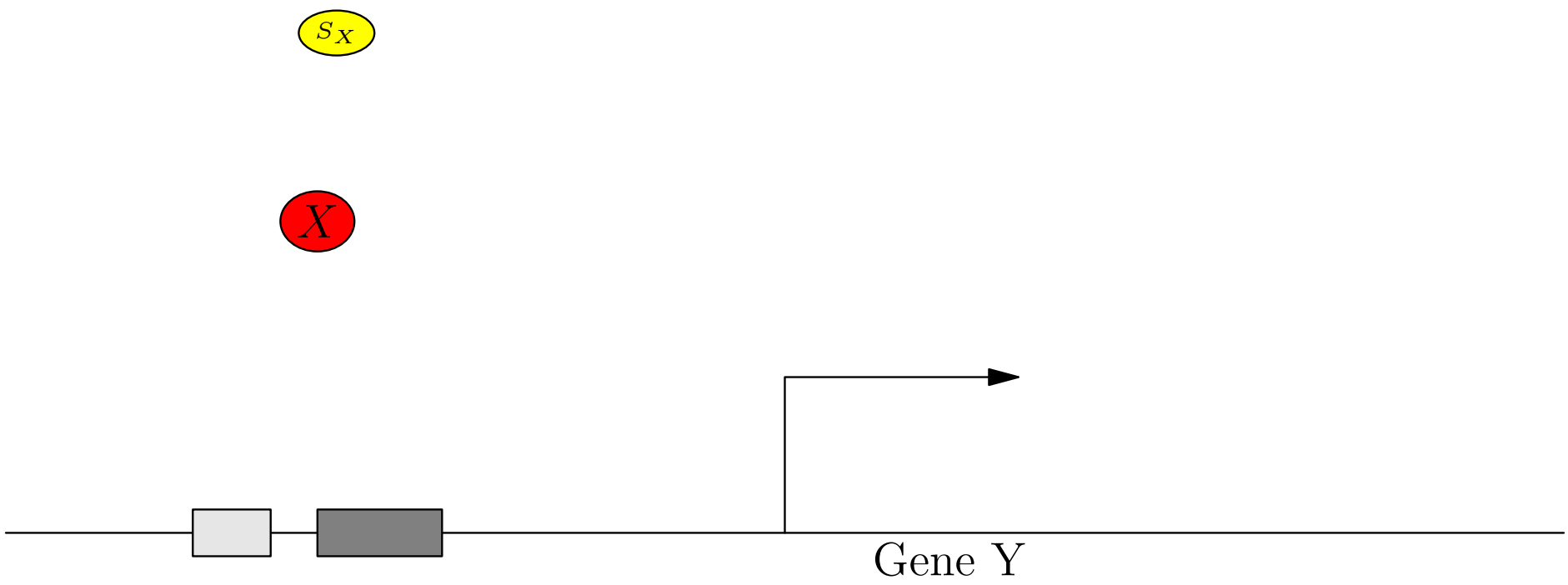

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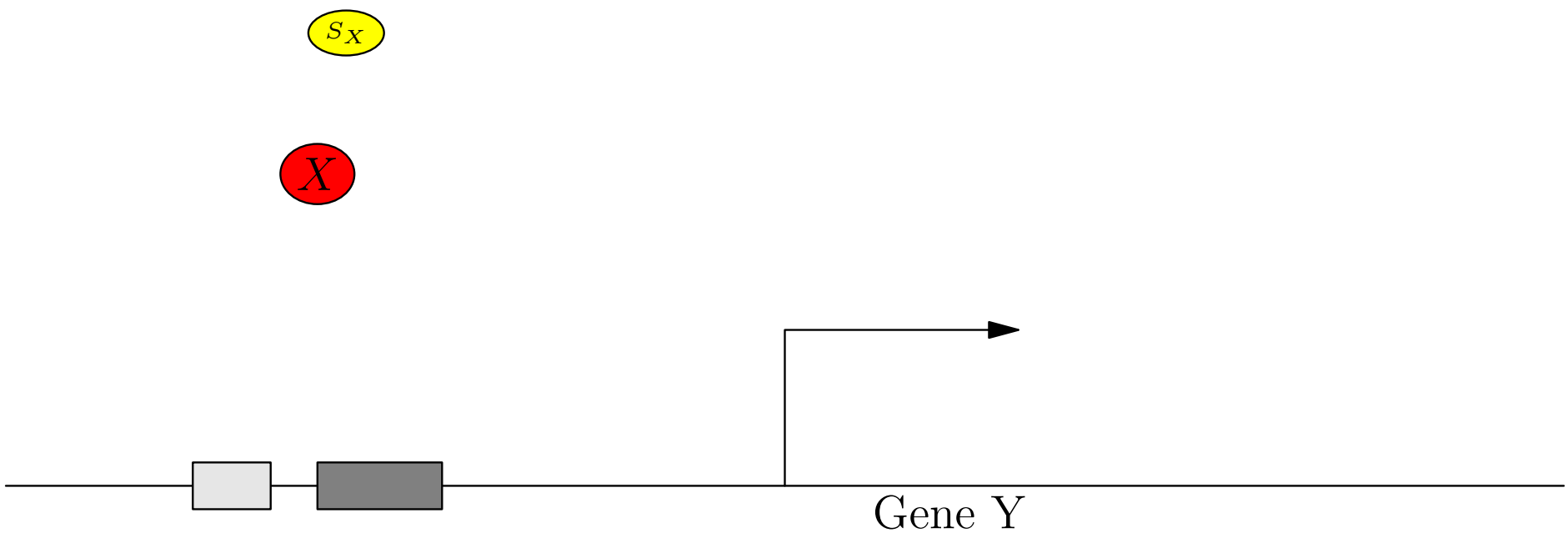

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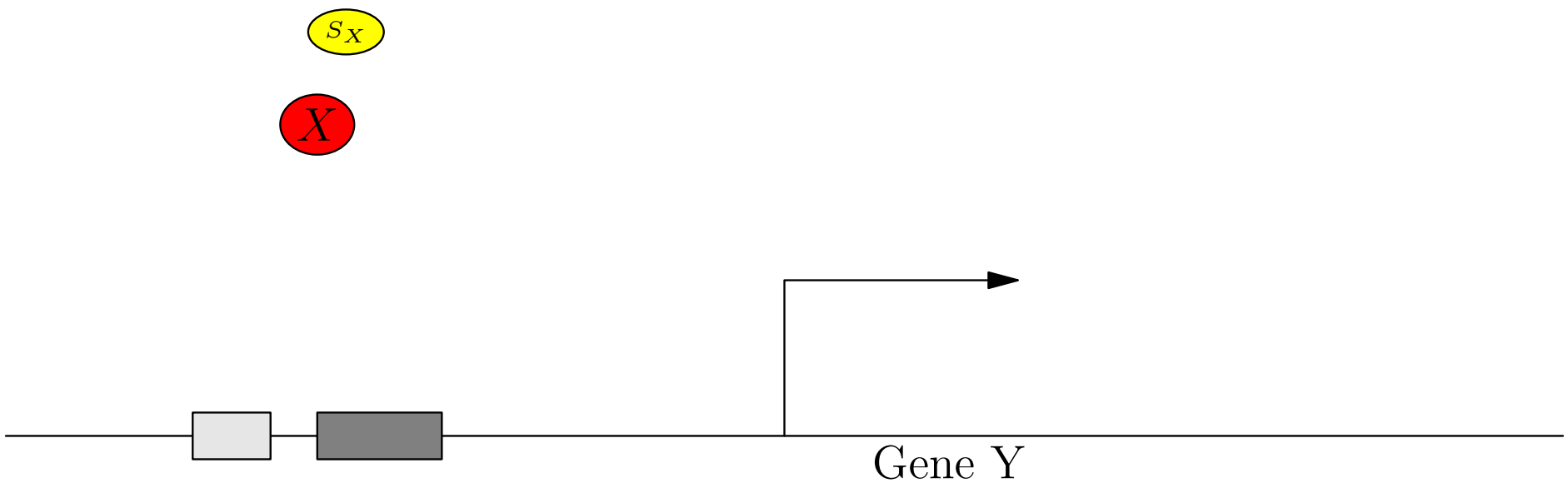

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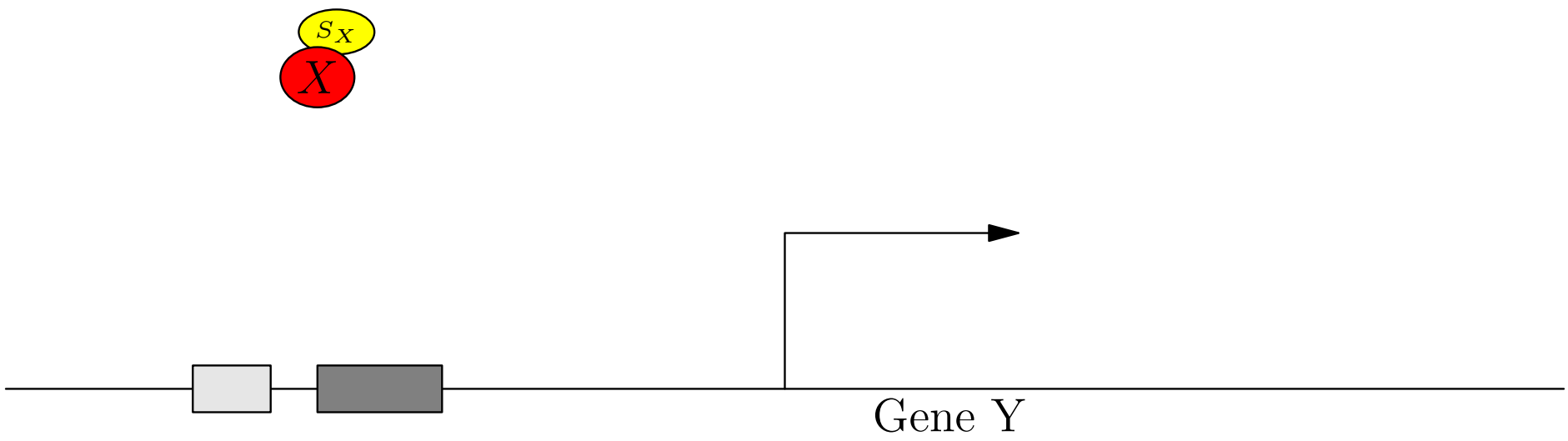

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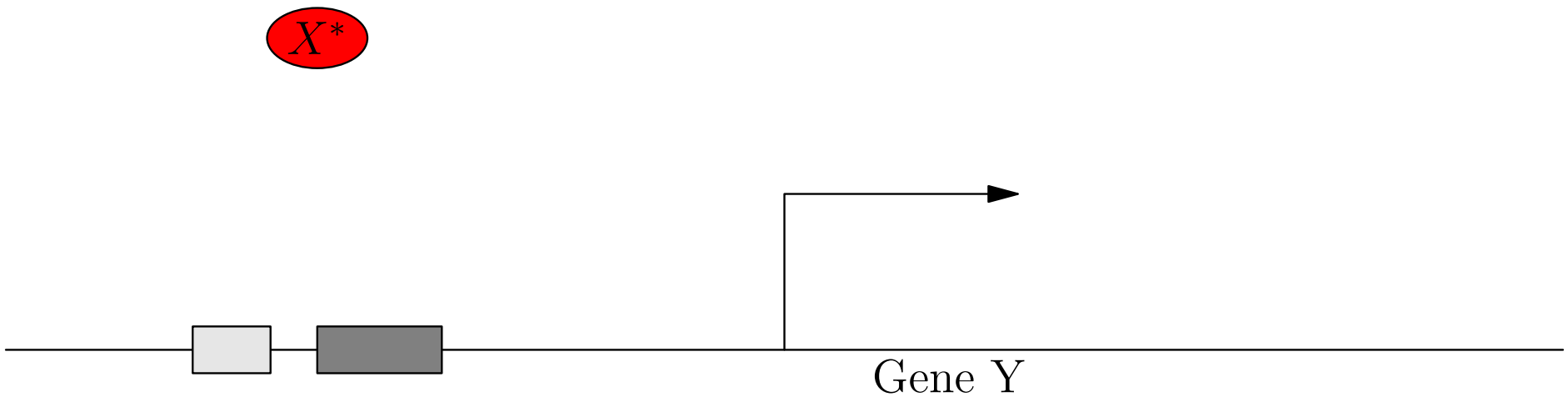

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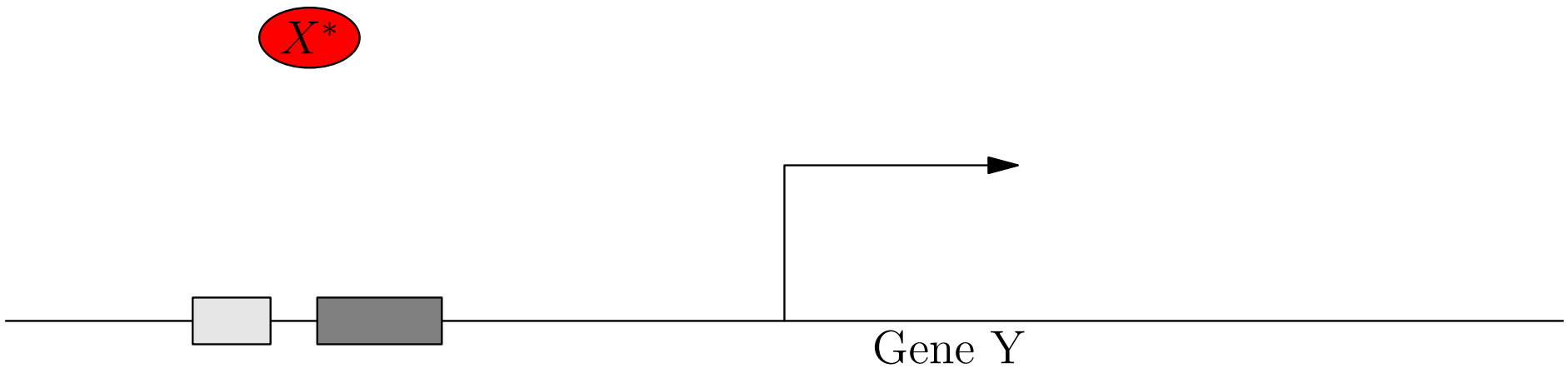

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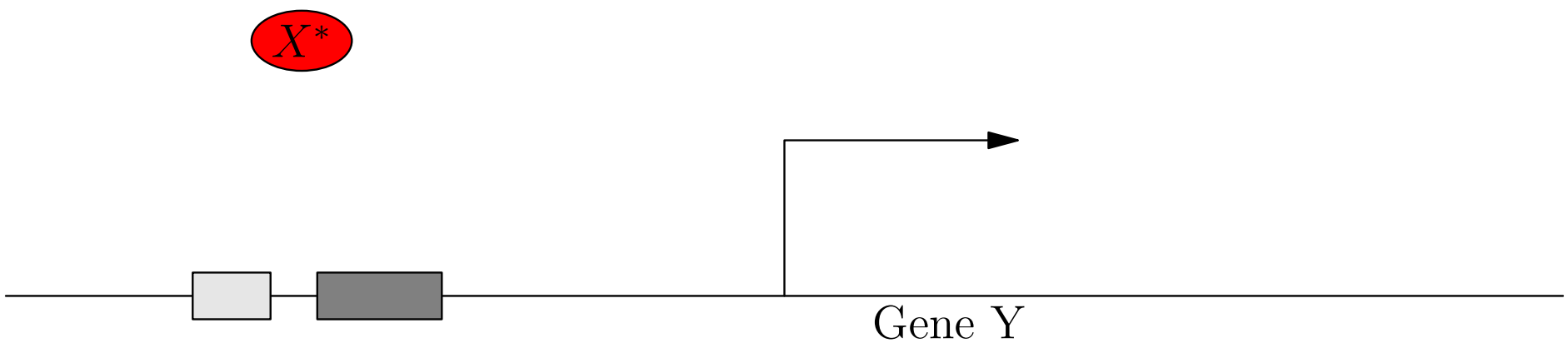

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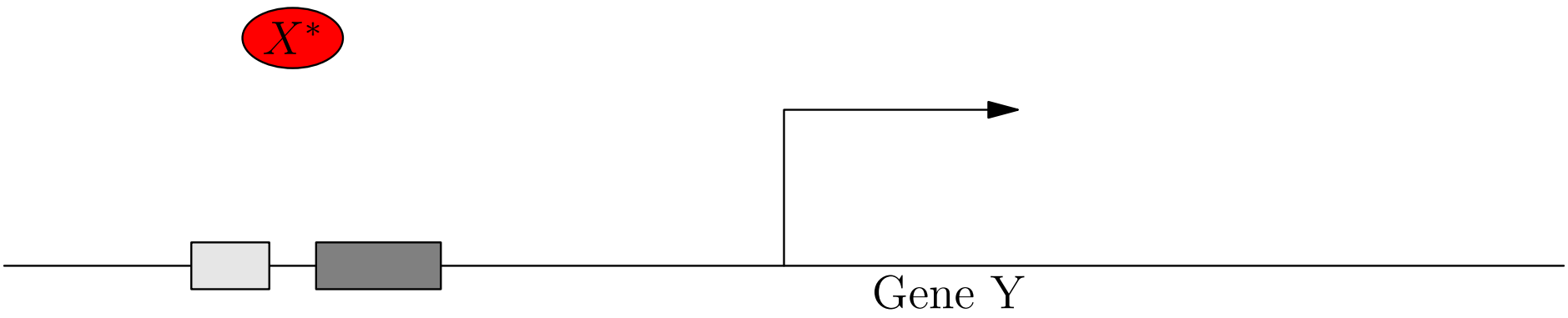

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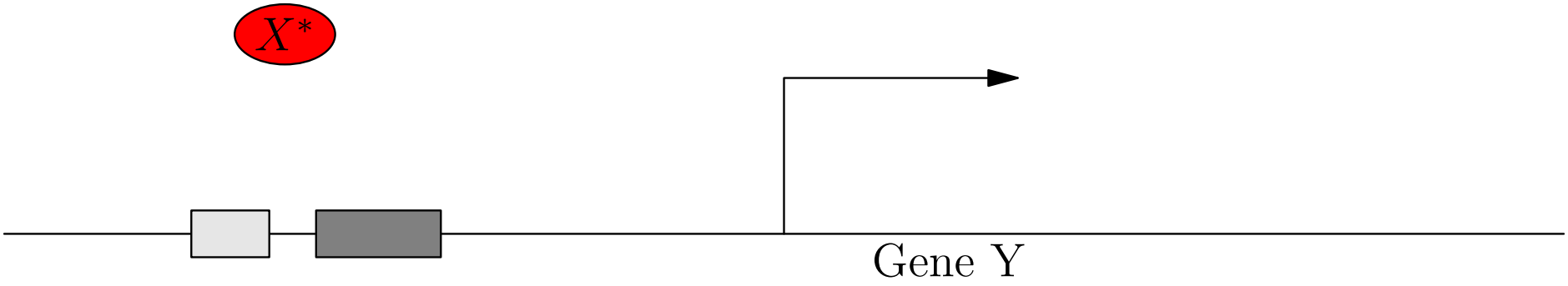

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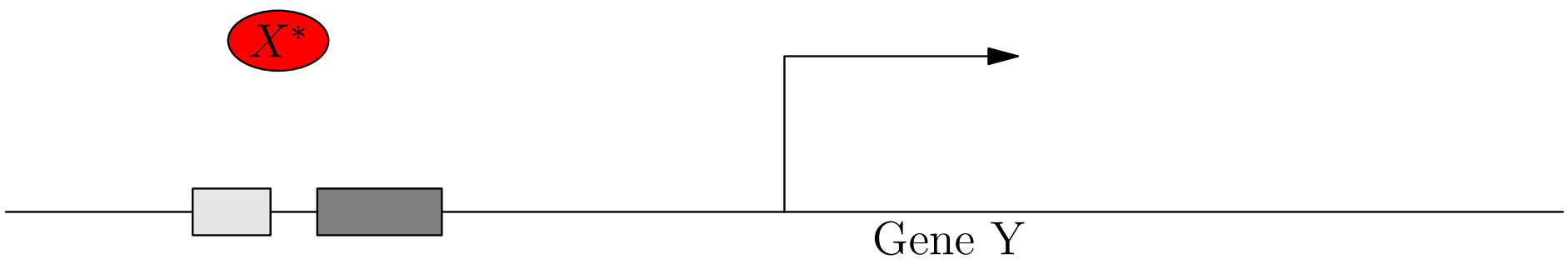

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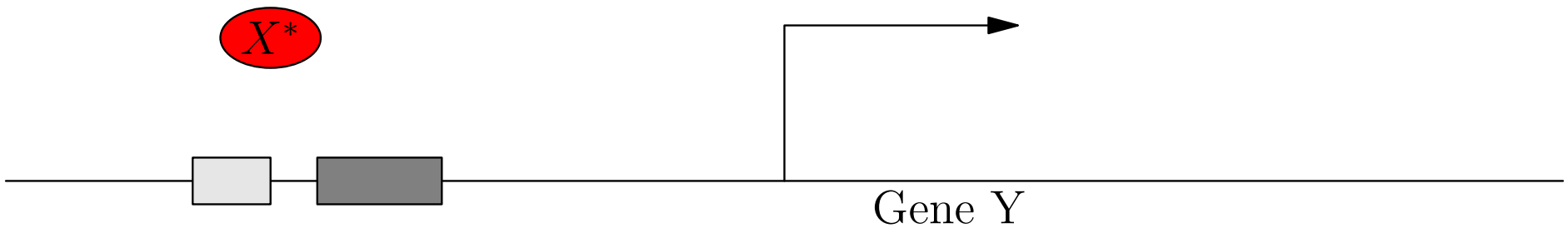

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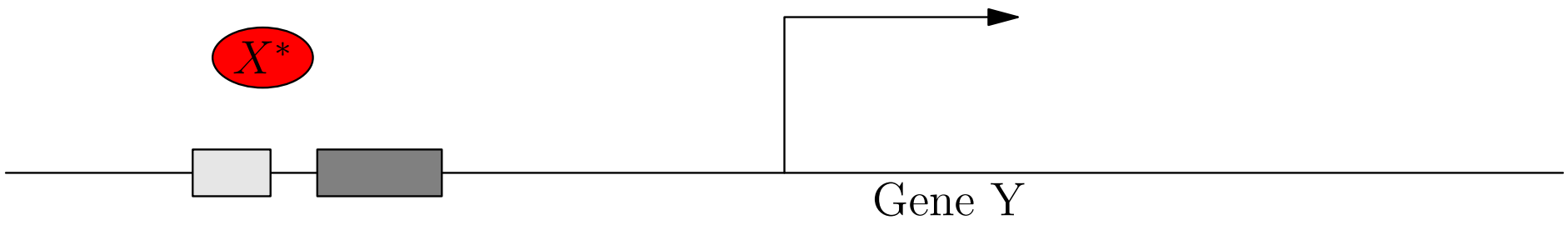

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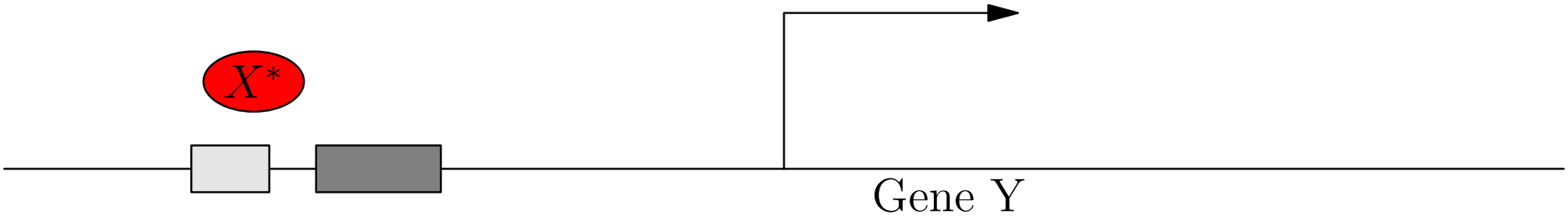

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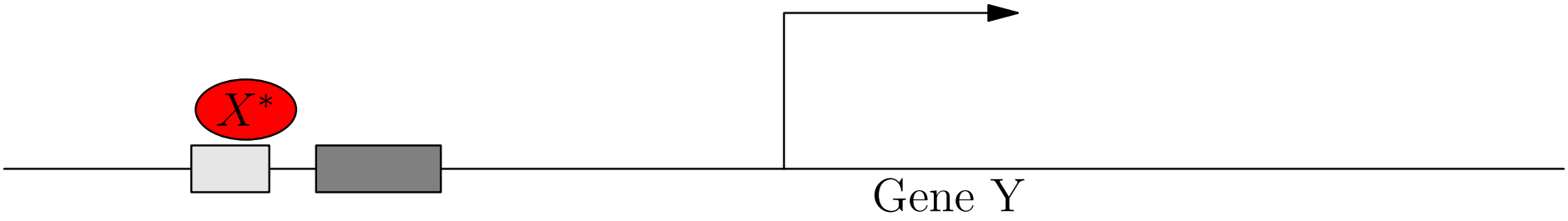

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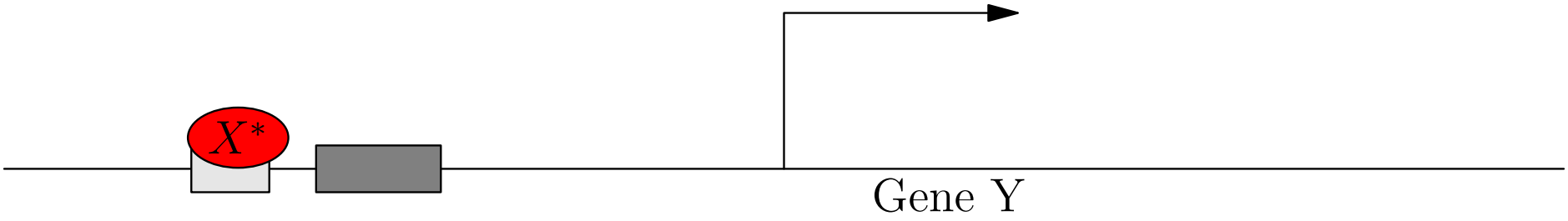
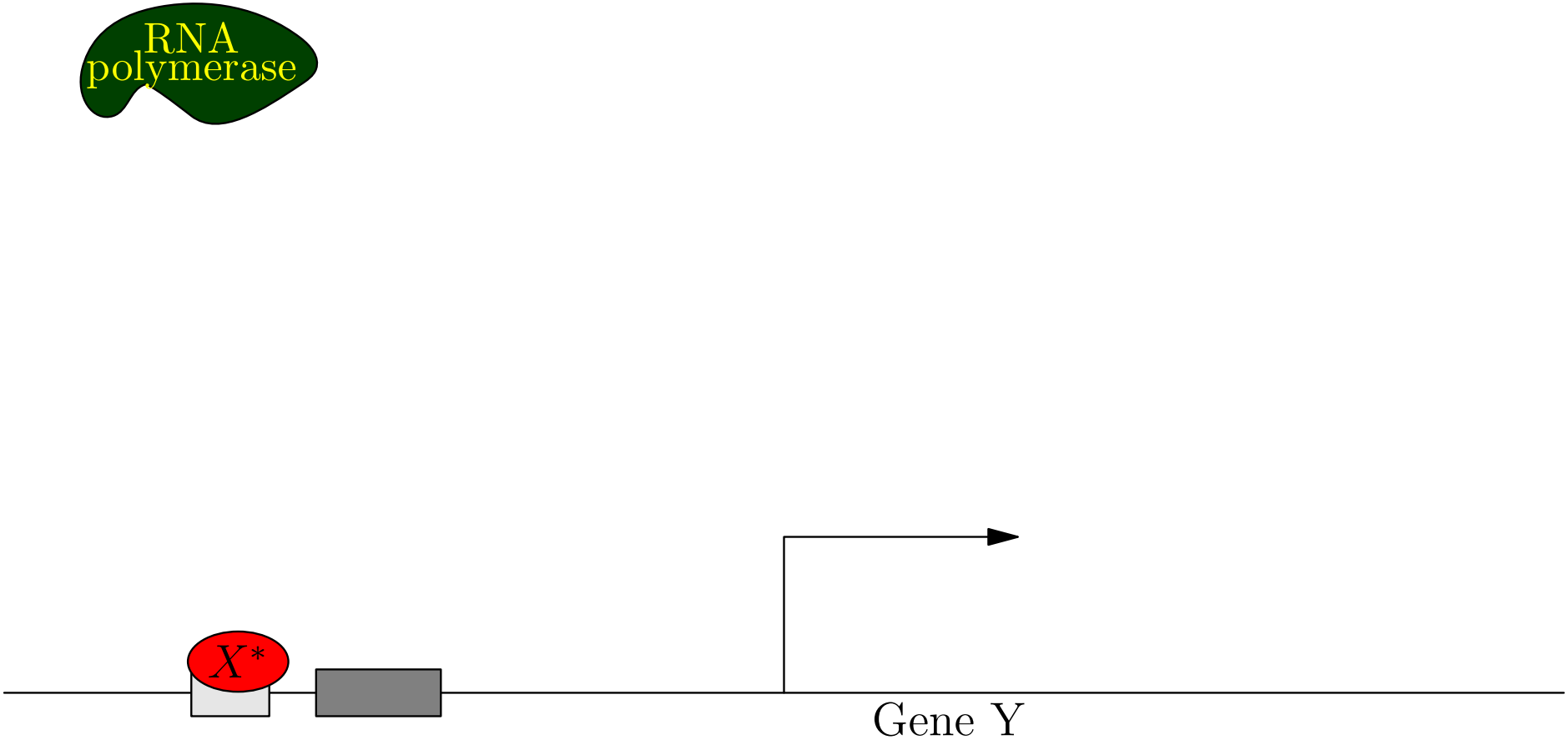


# Gene Transcription Regulation

Excititory Transcription Factor



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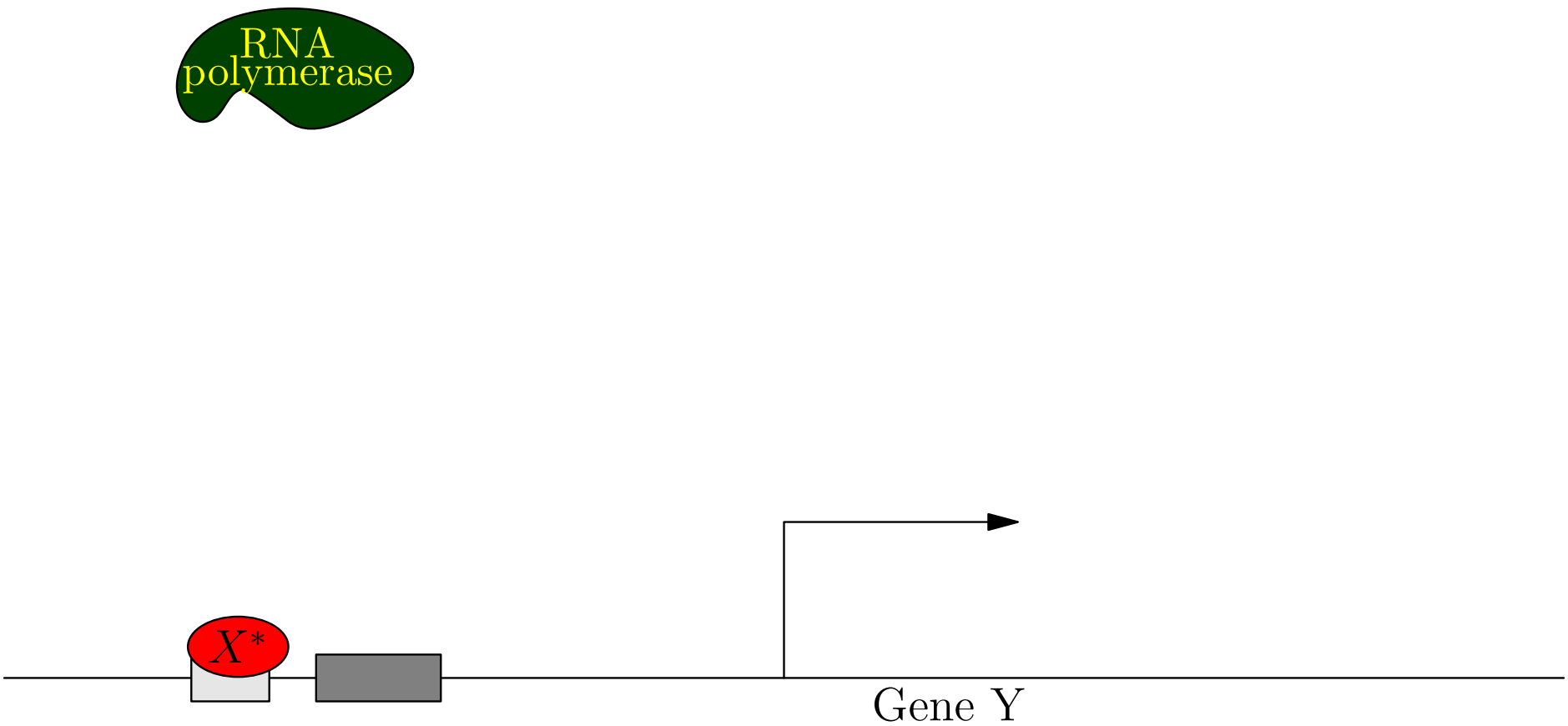

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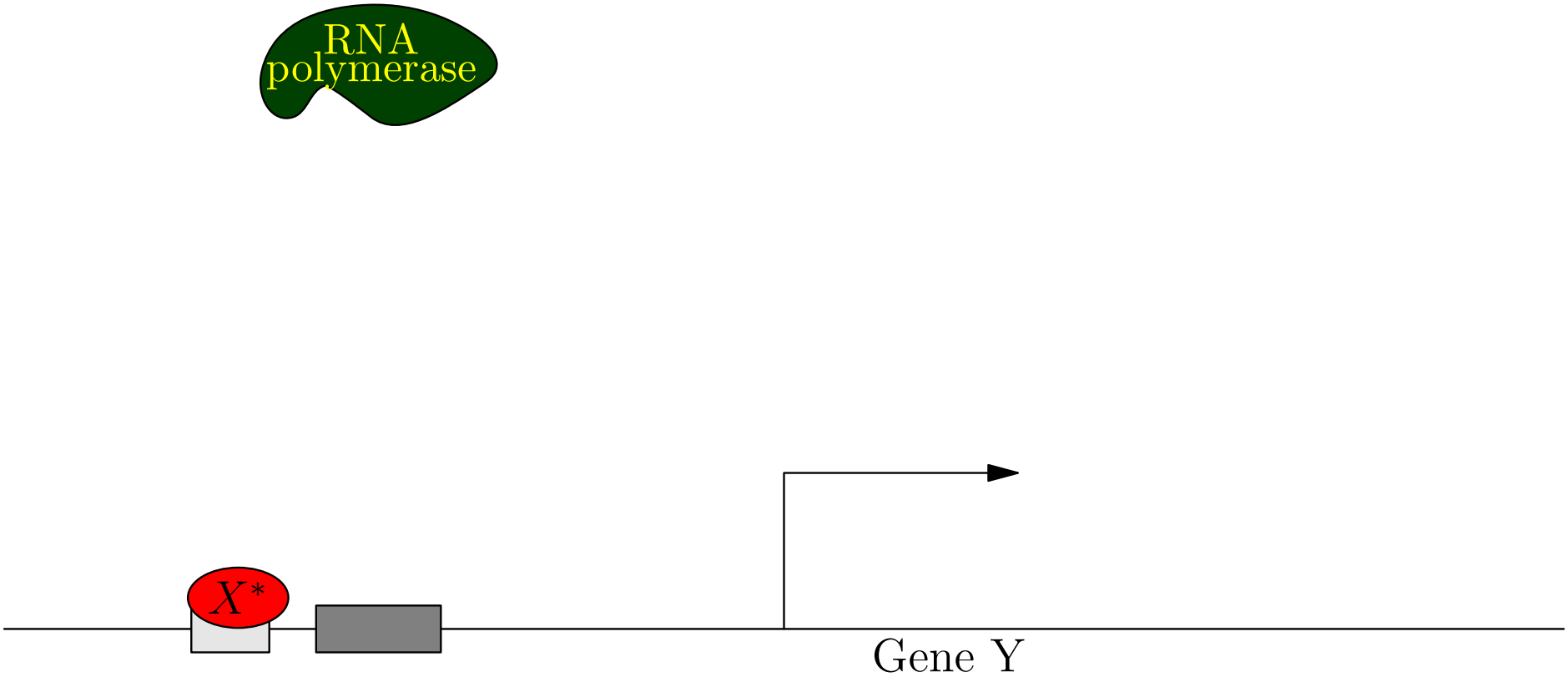

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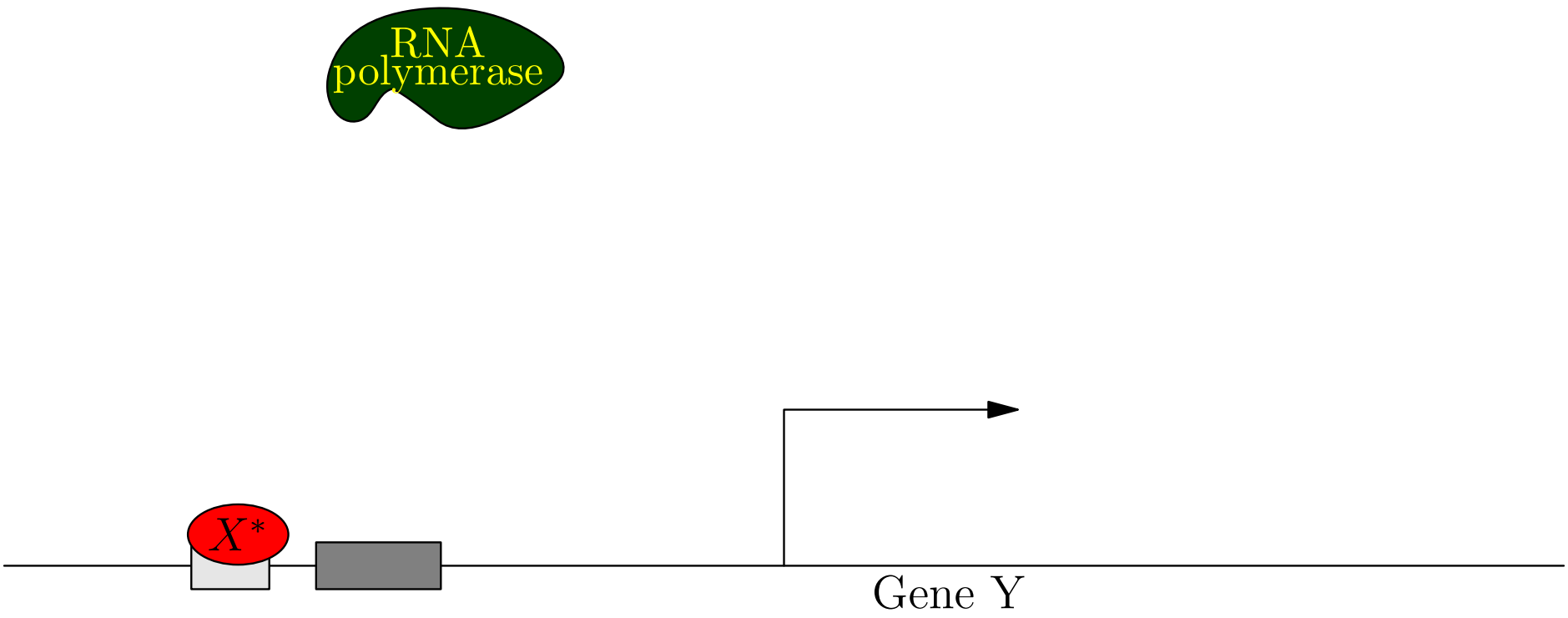

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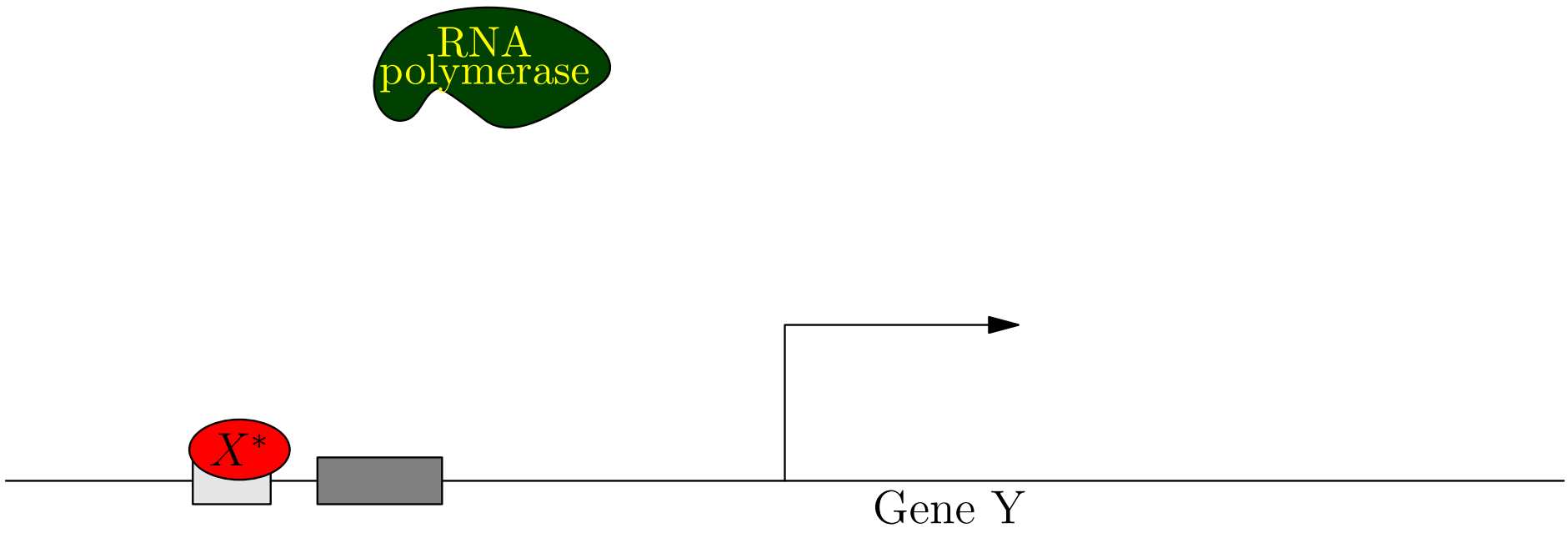
Excititory Transcription Factor



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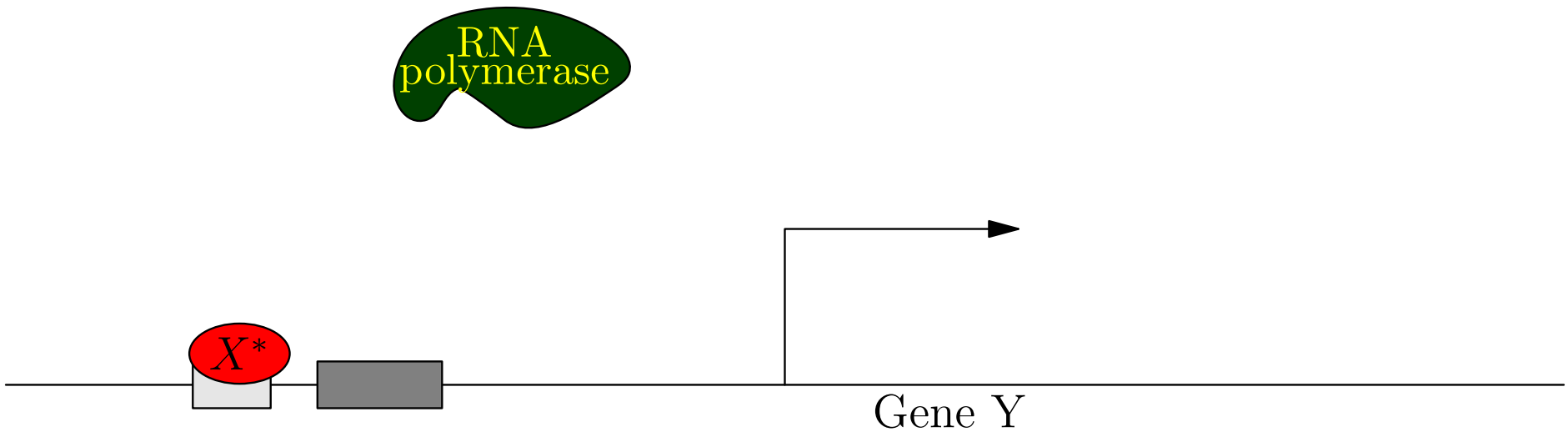

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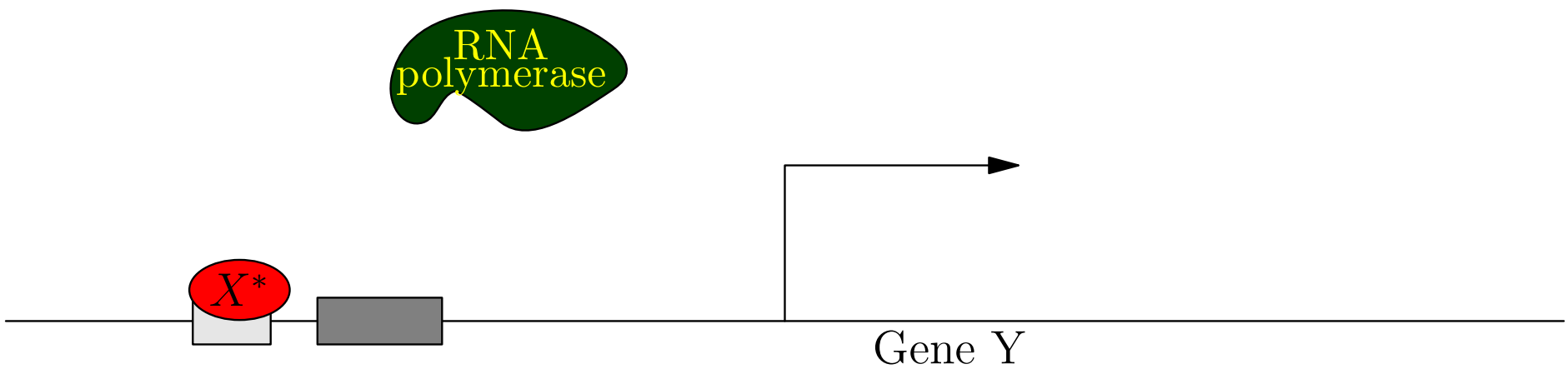

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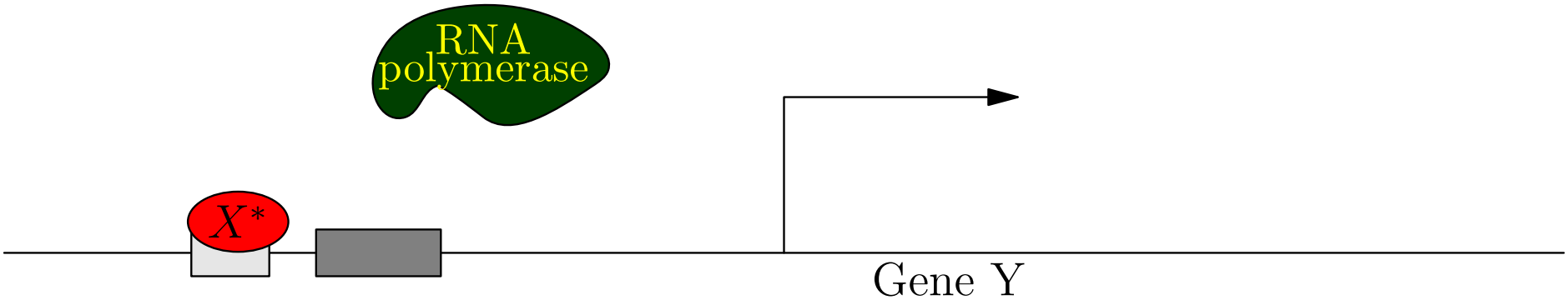

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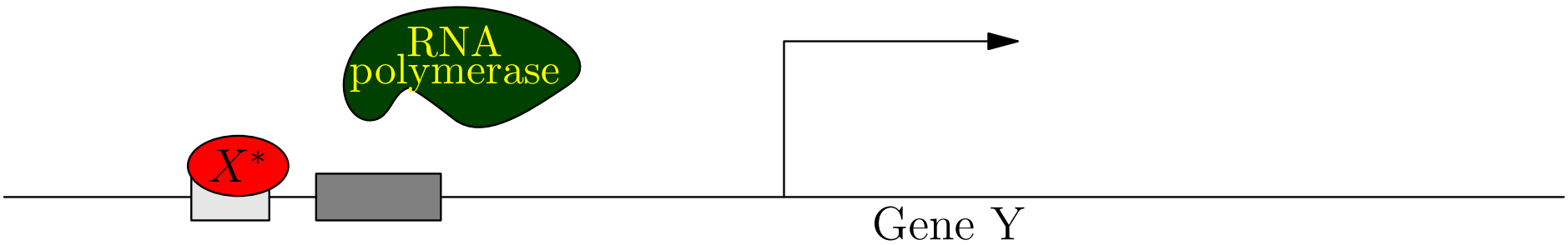

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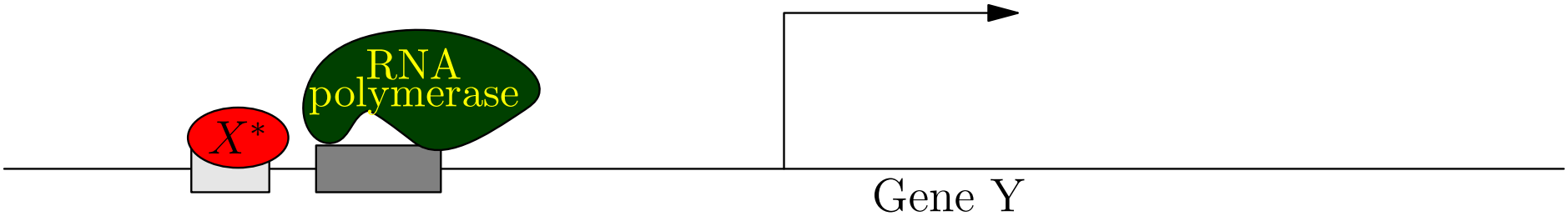

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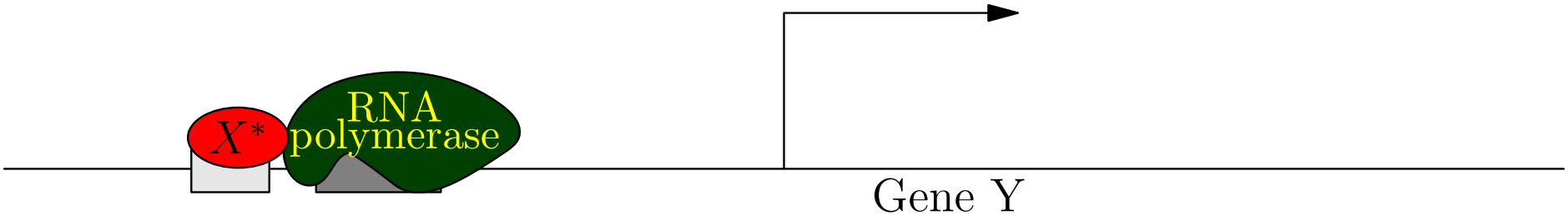

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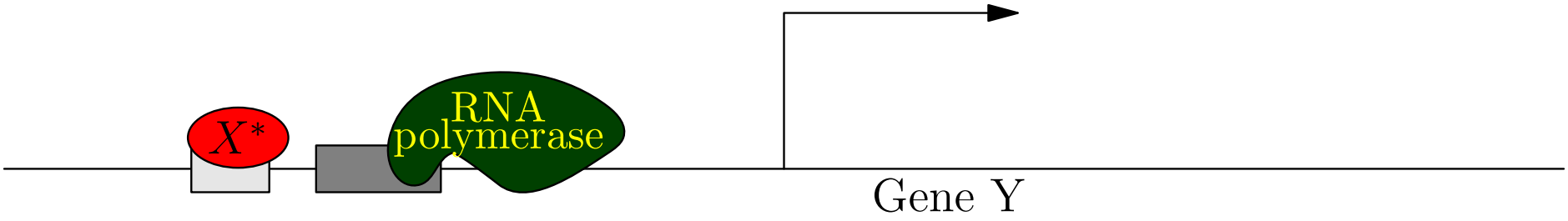

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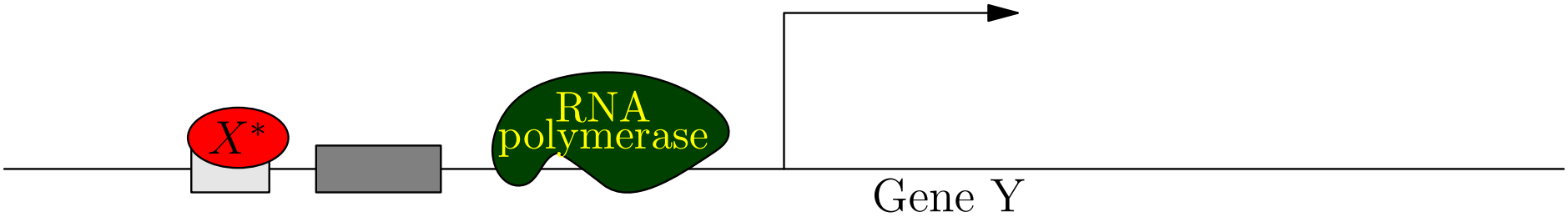

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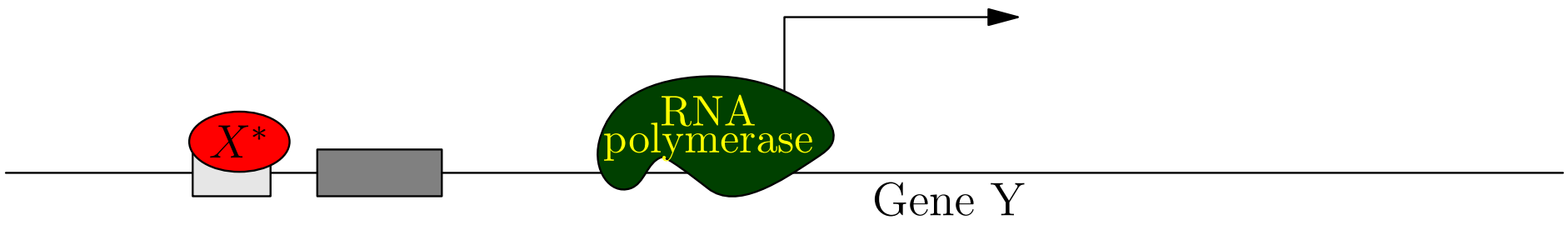

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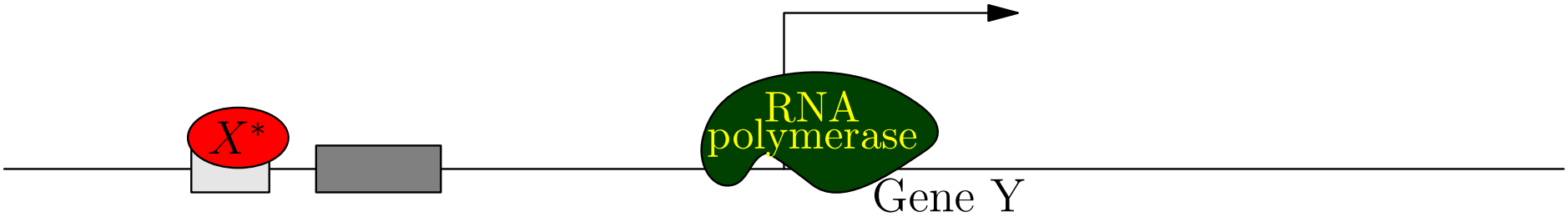

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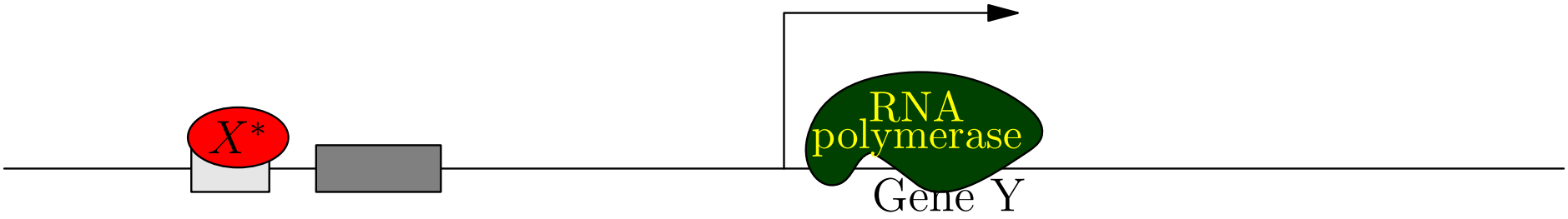

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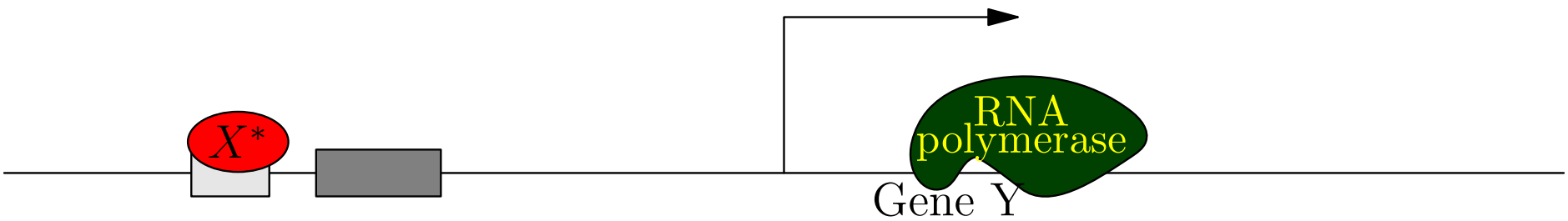

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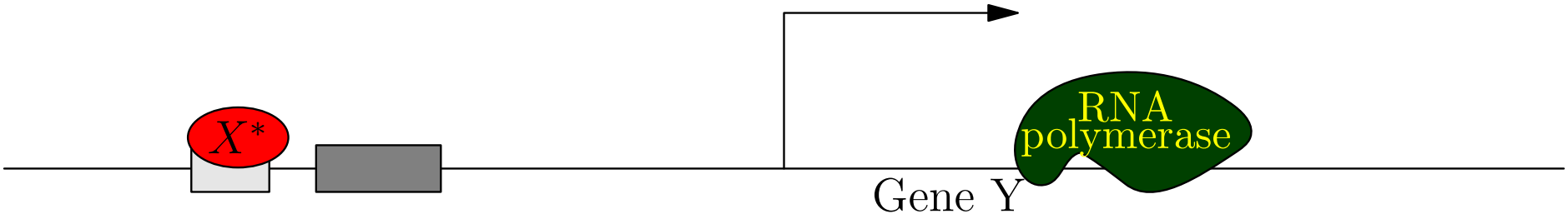

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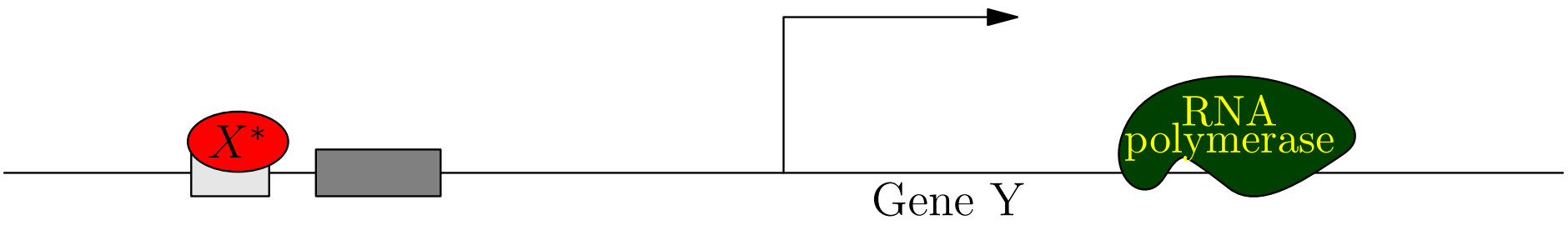

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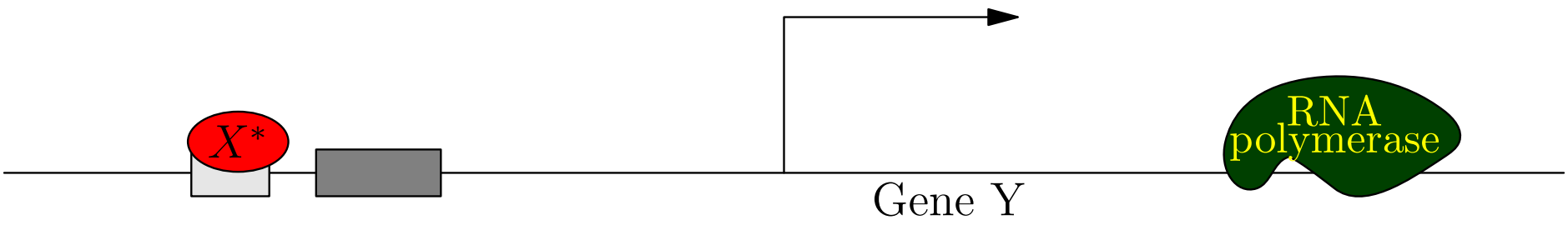

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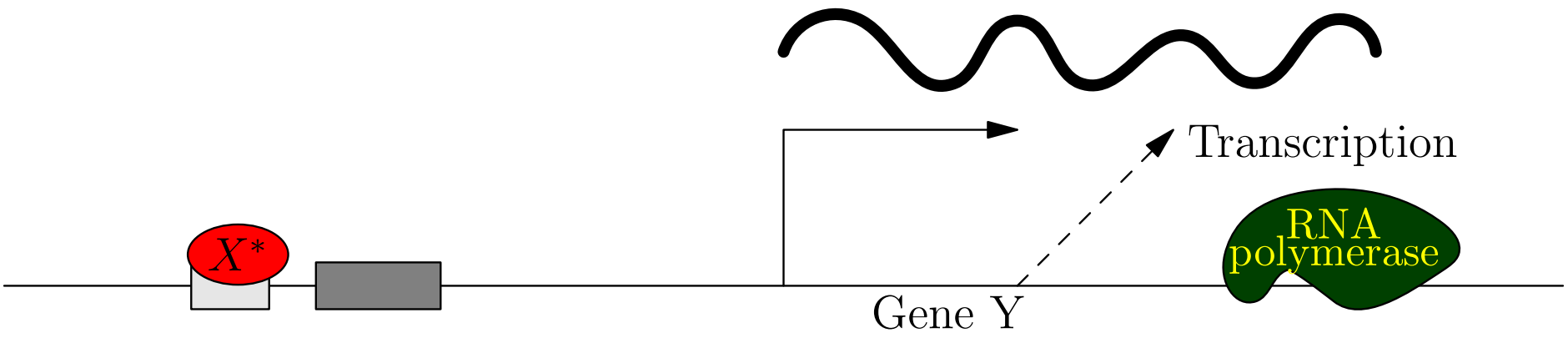

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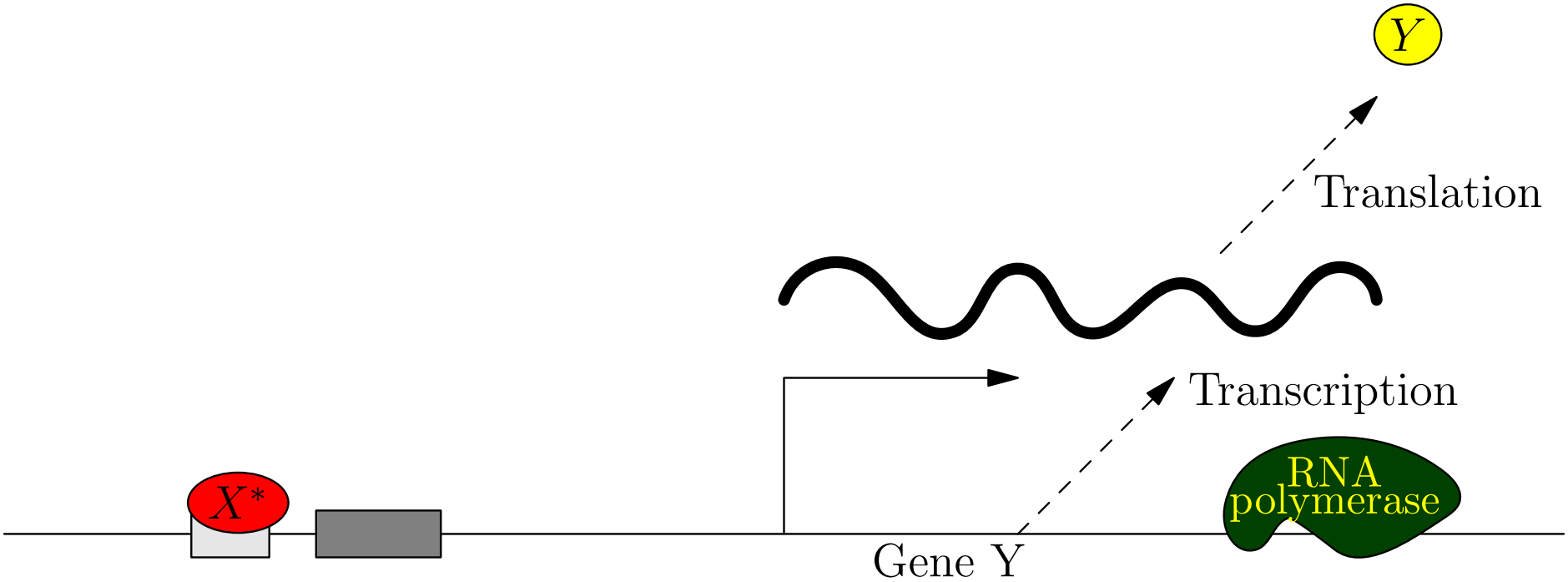

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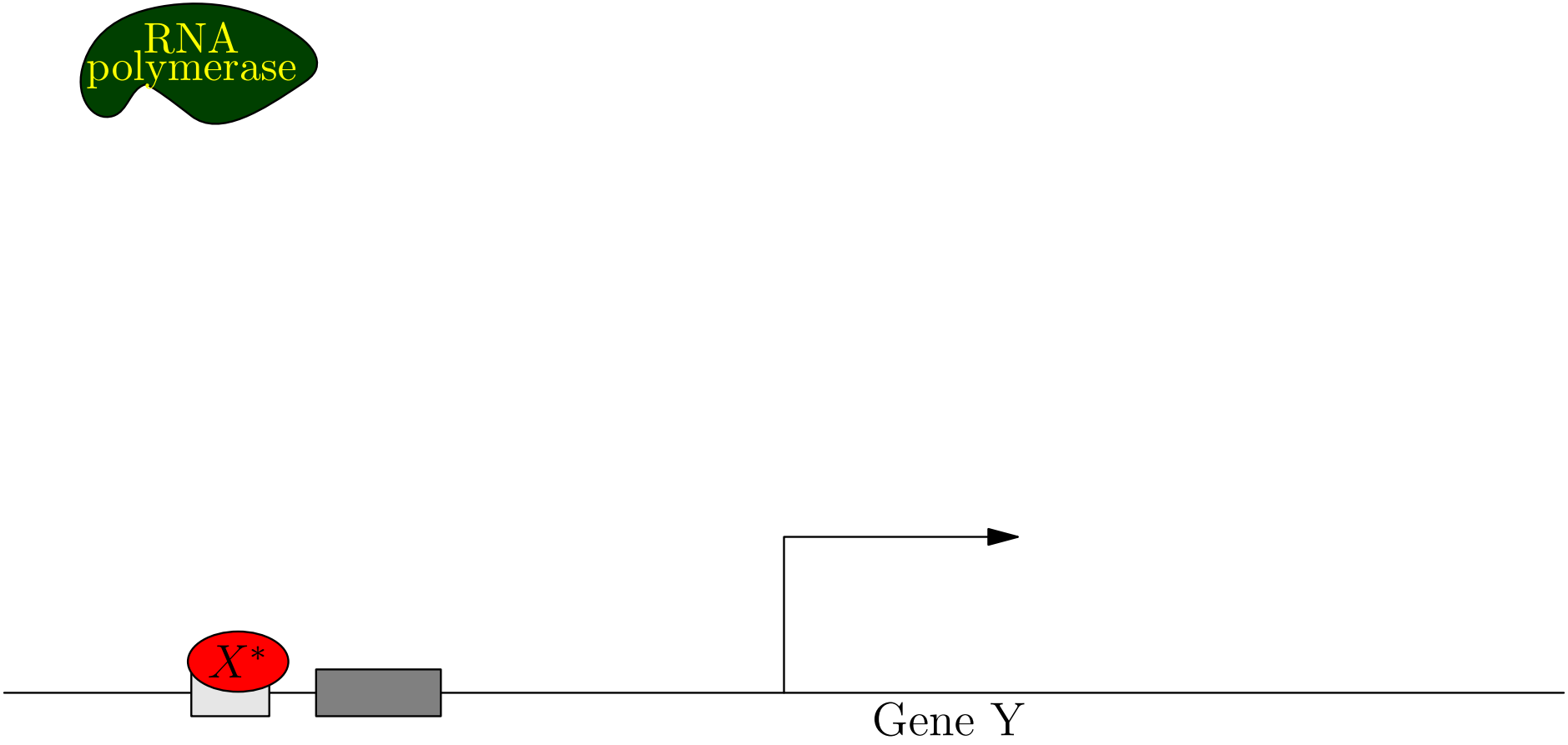

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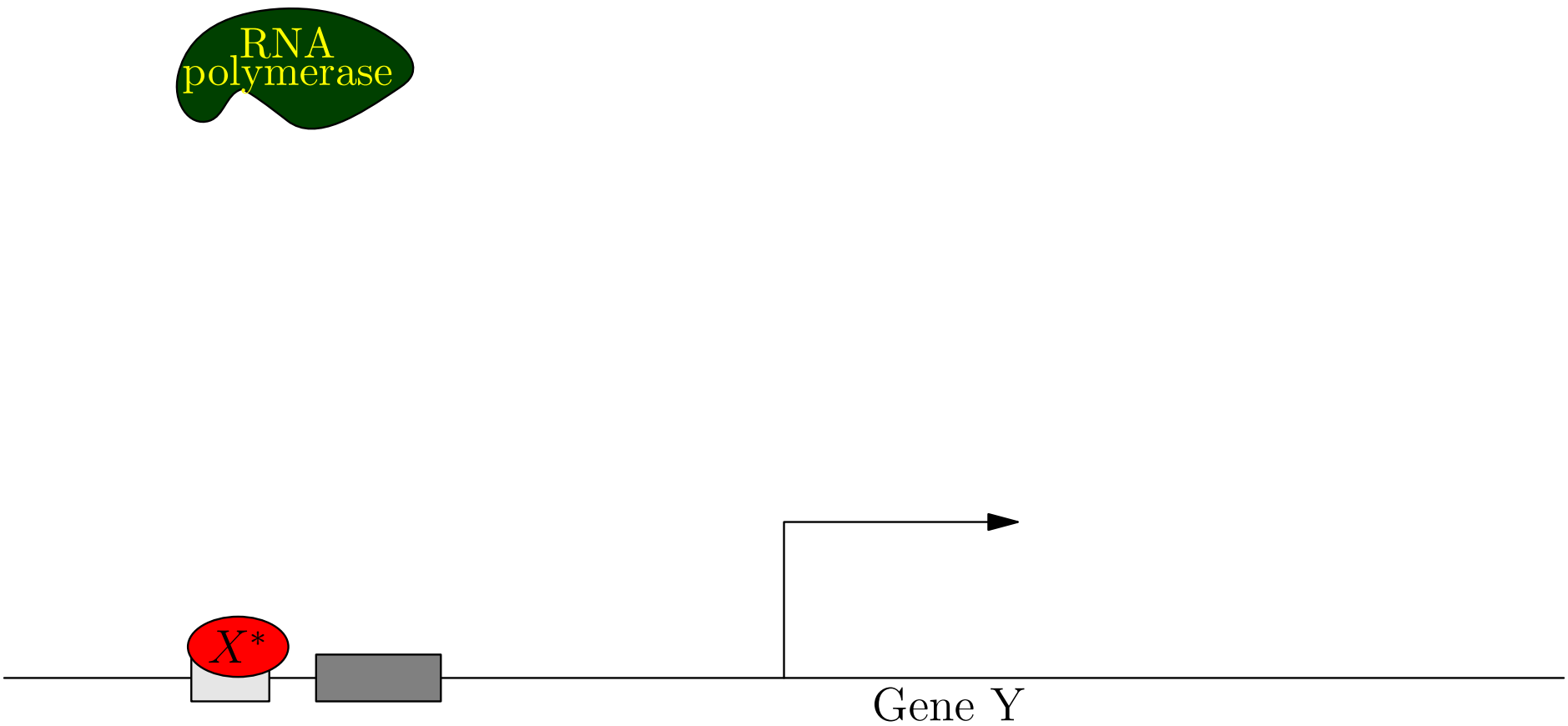

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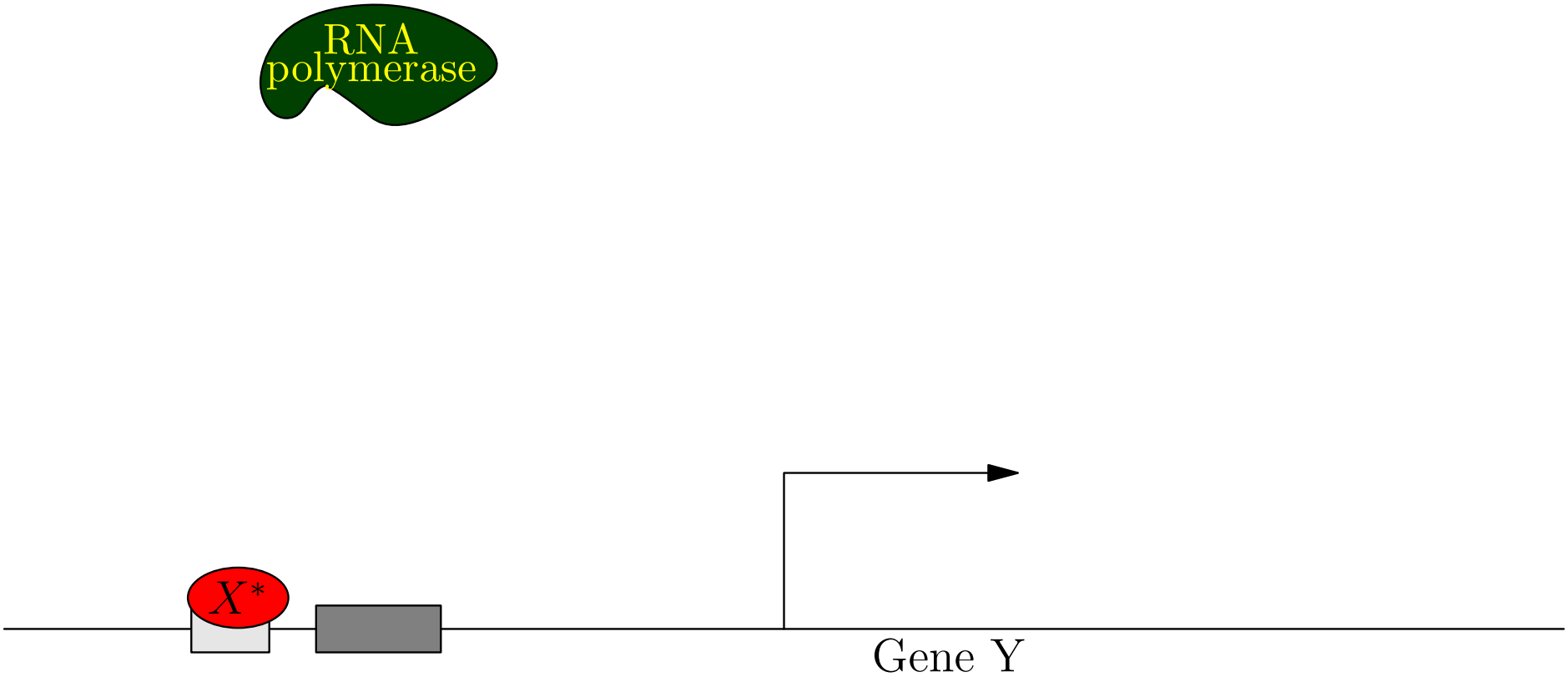

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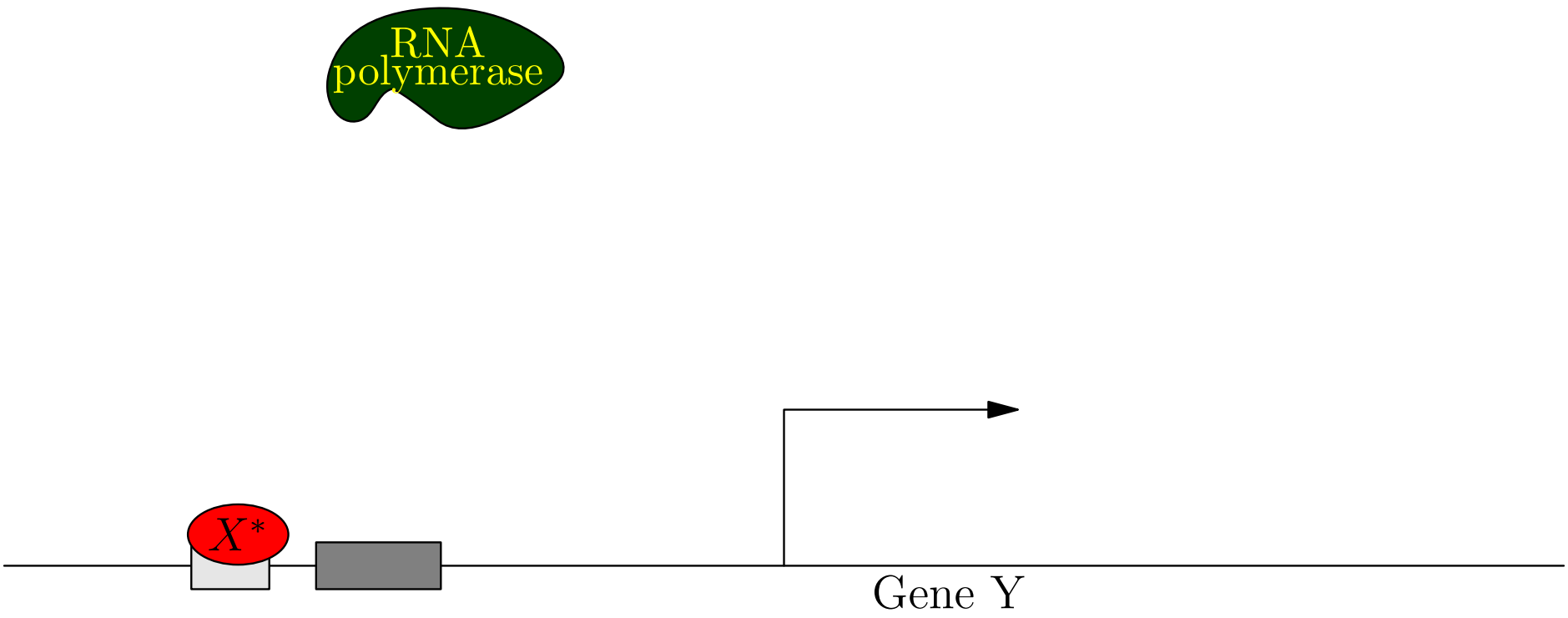

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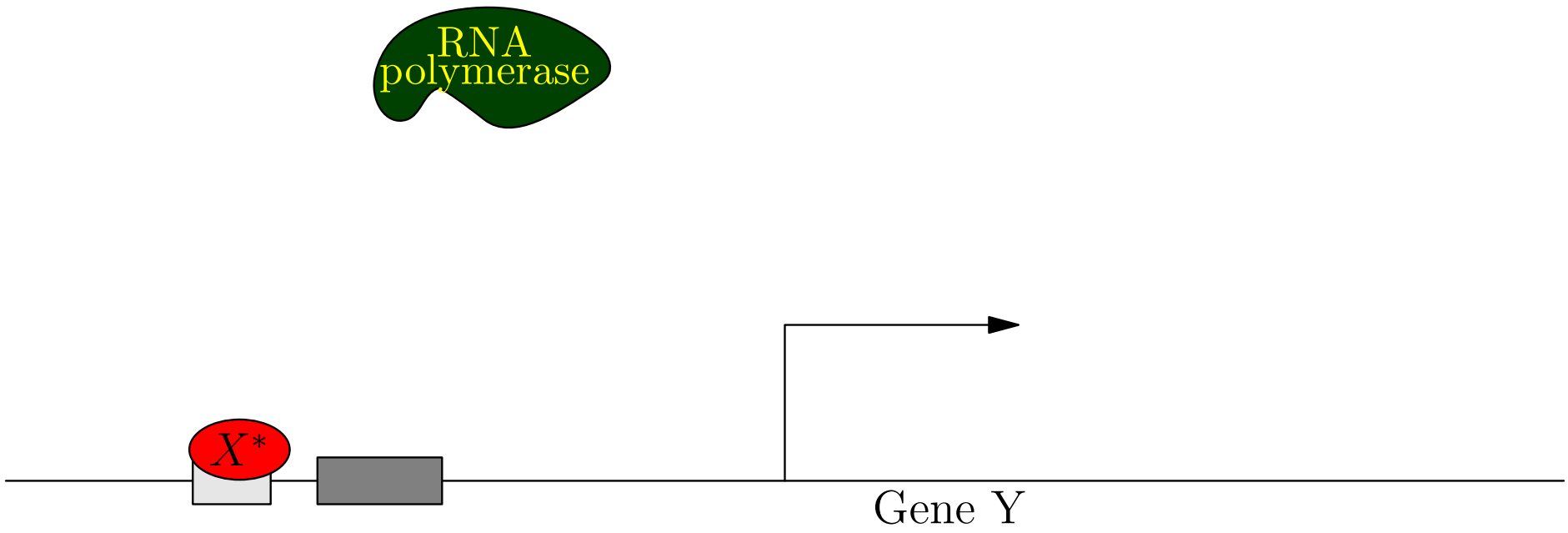

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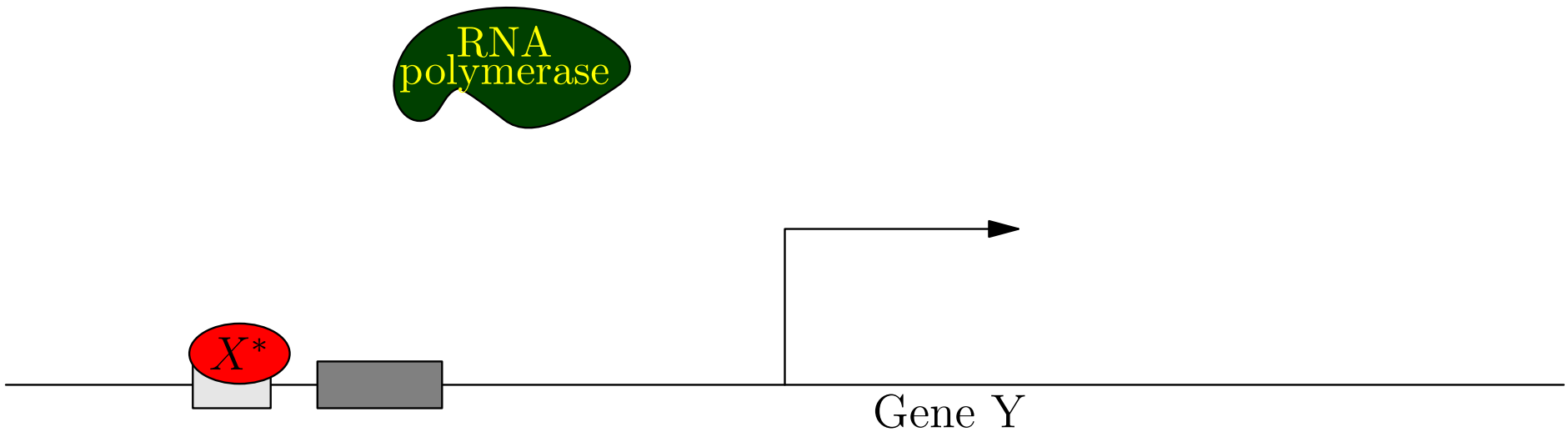

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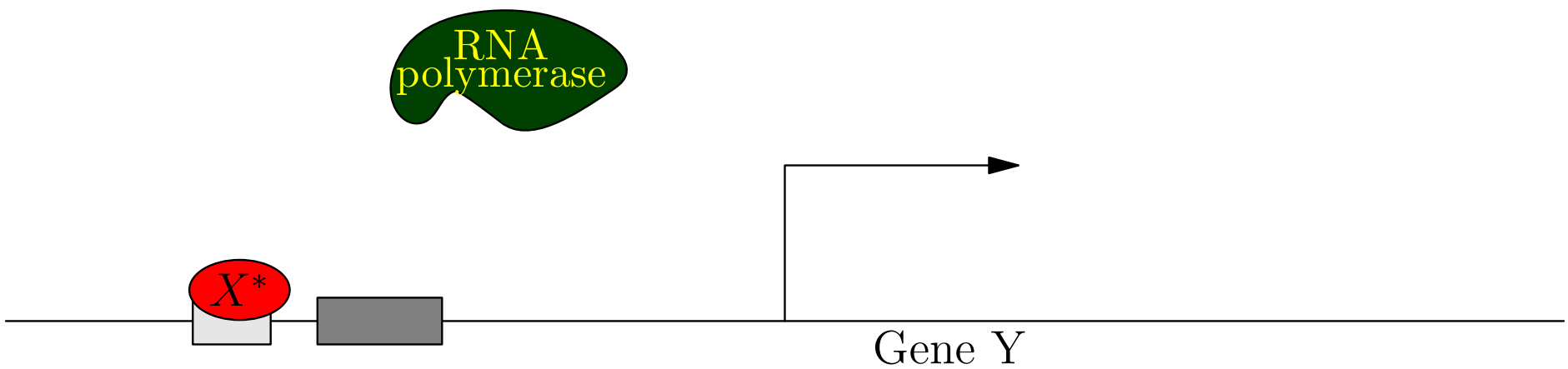

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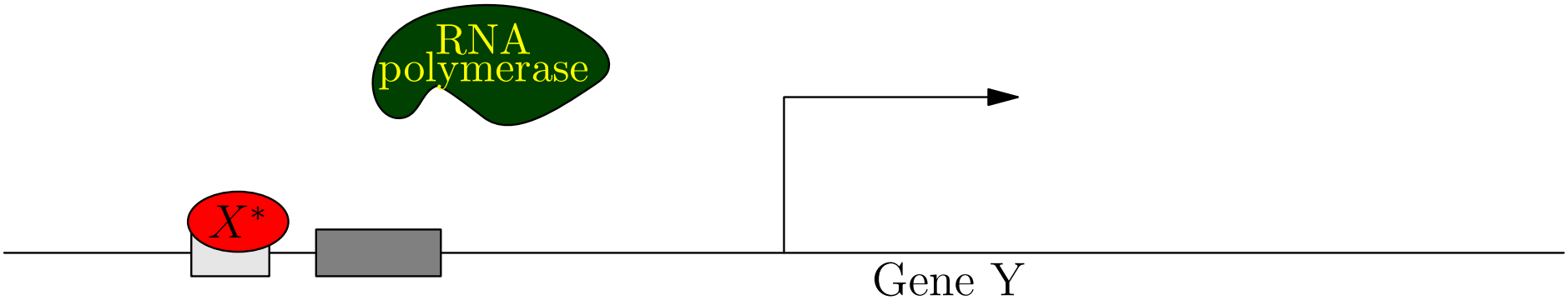

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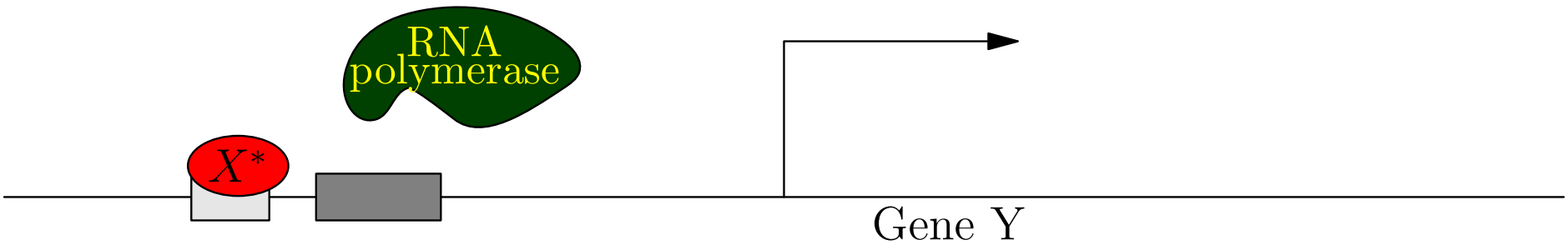

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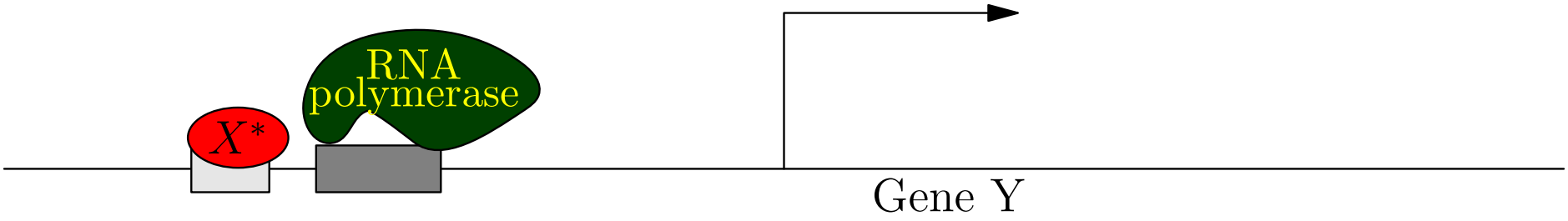

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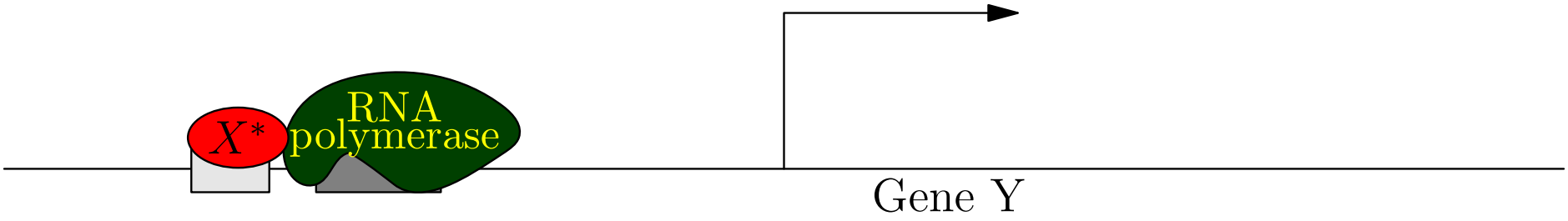

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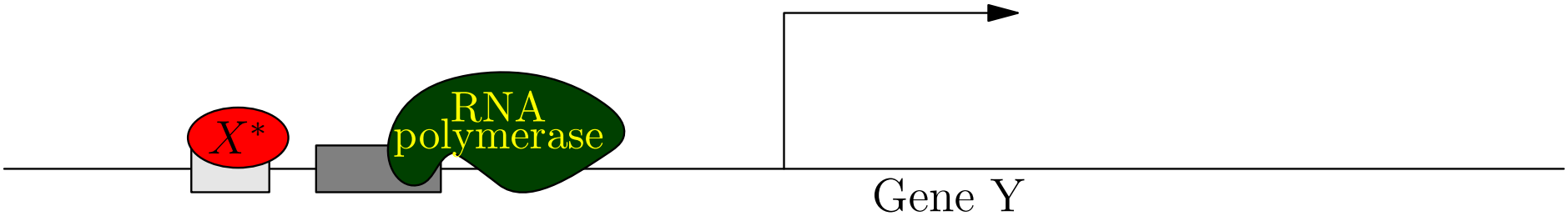

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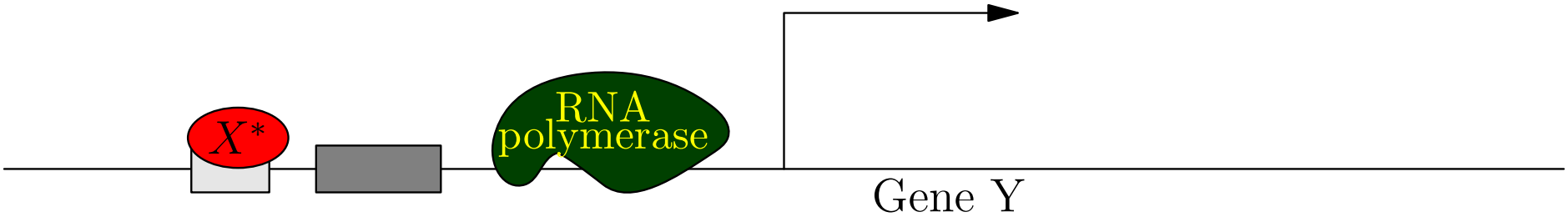

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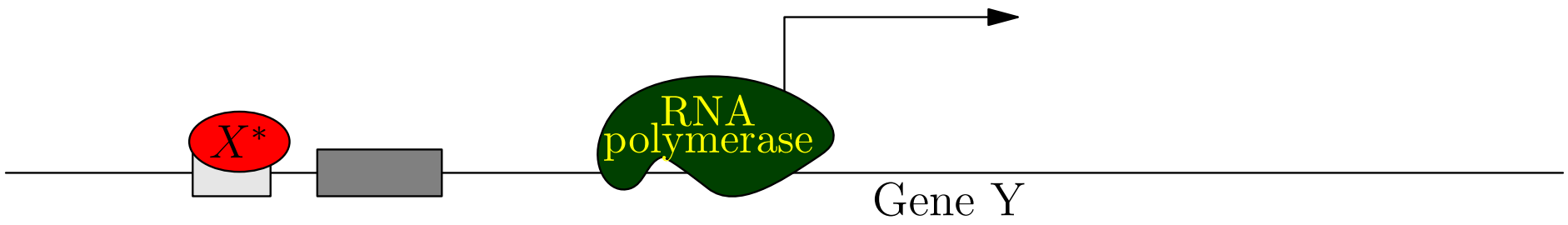

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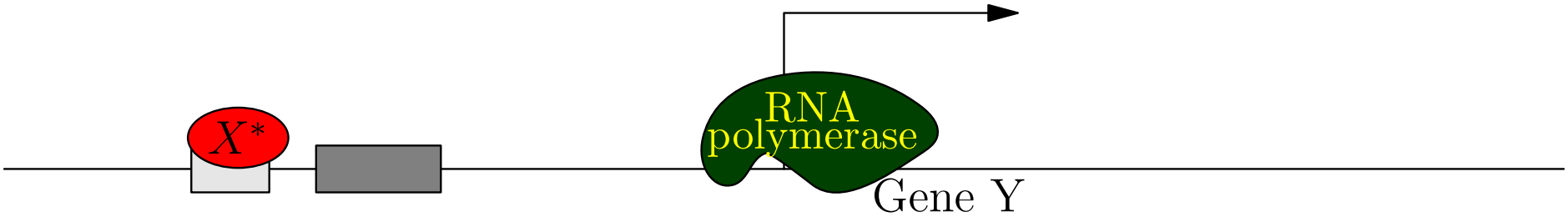

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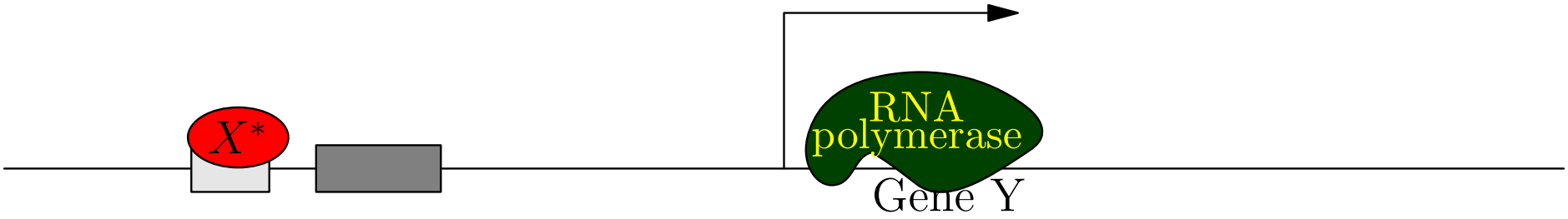

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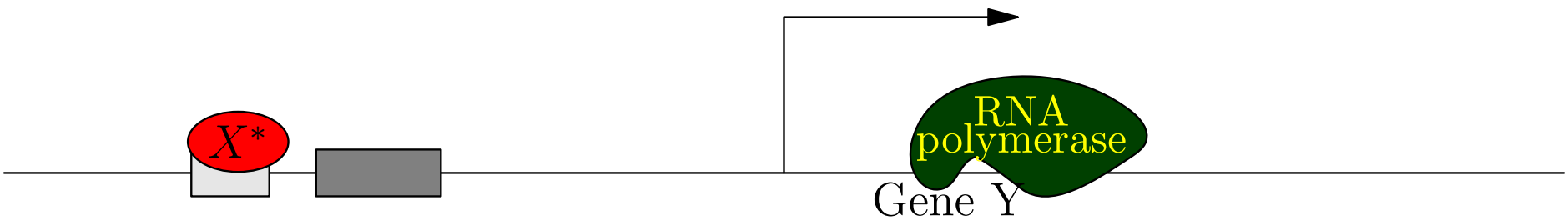

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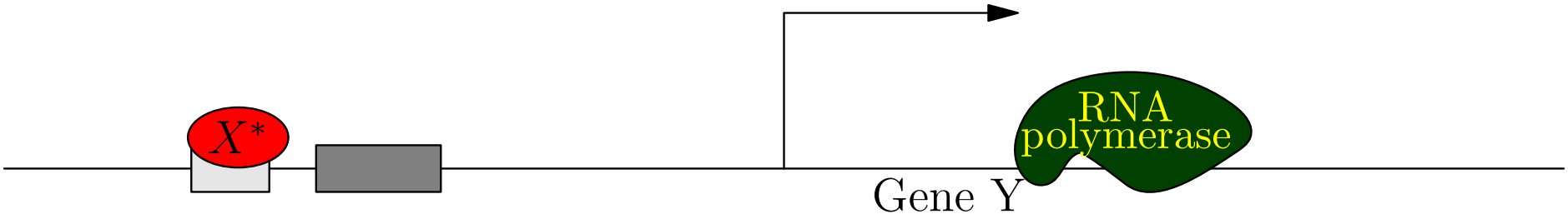

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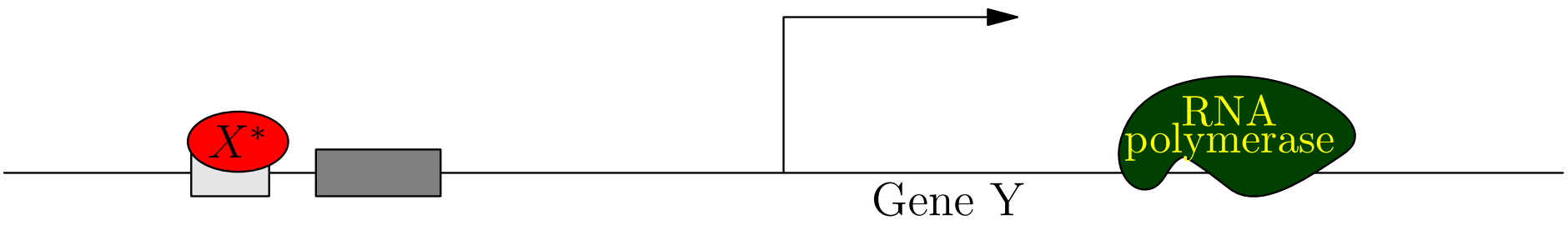

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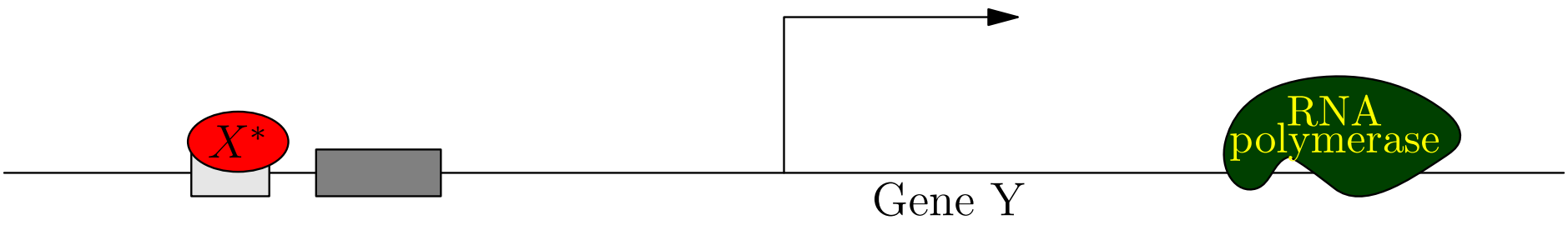

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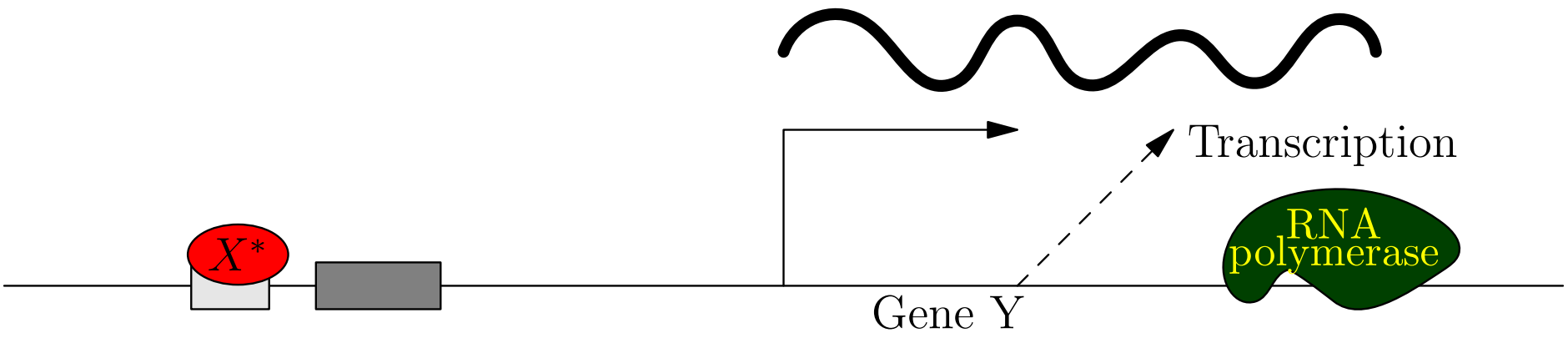

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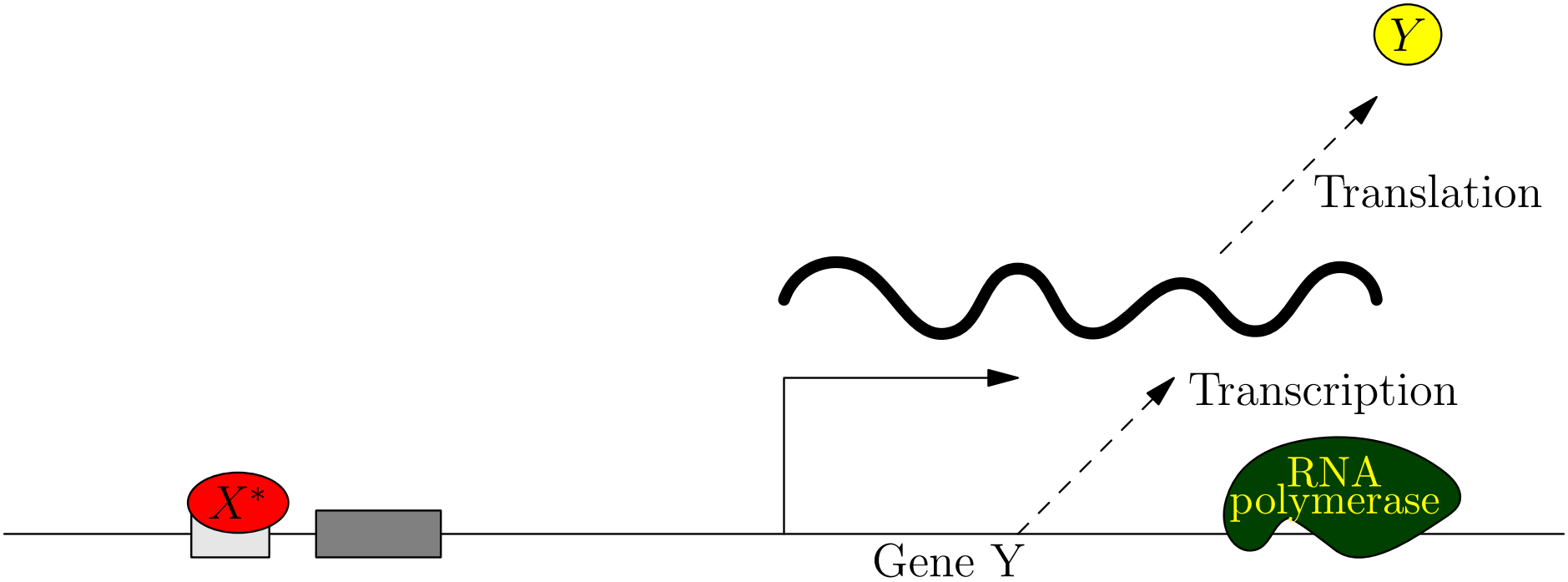

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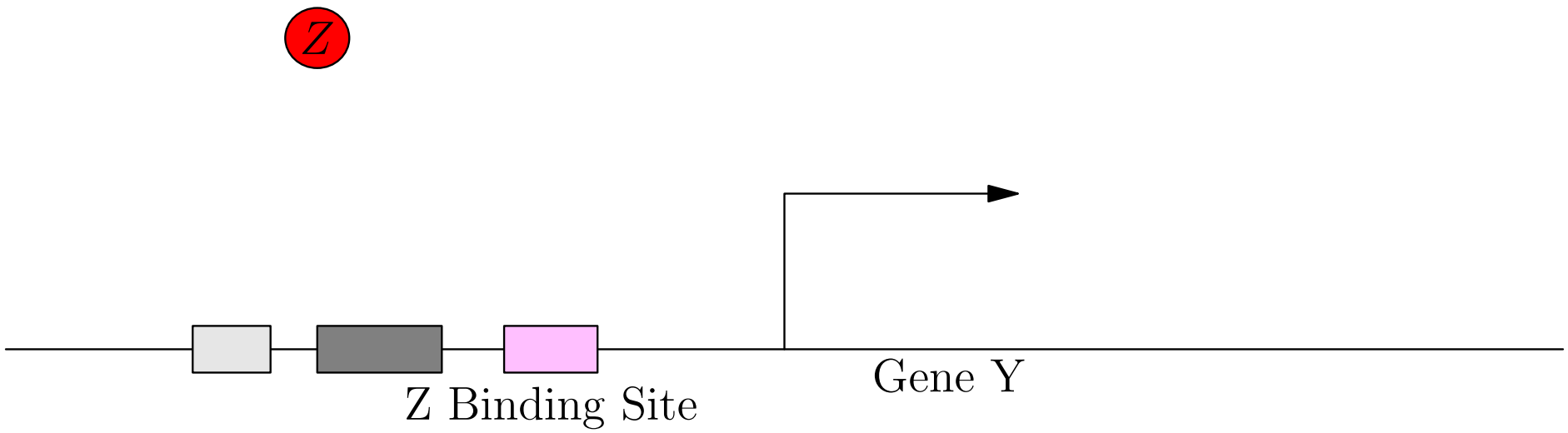

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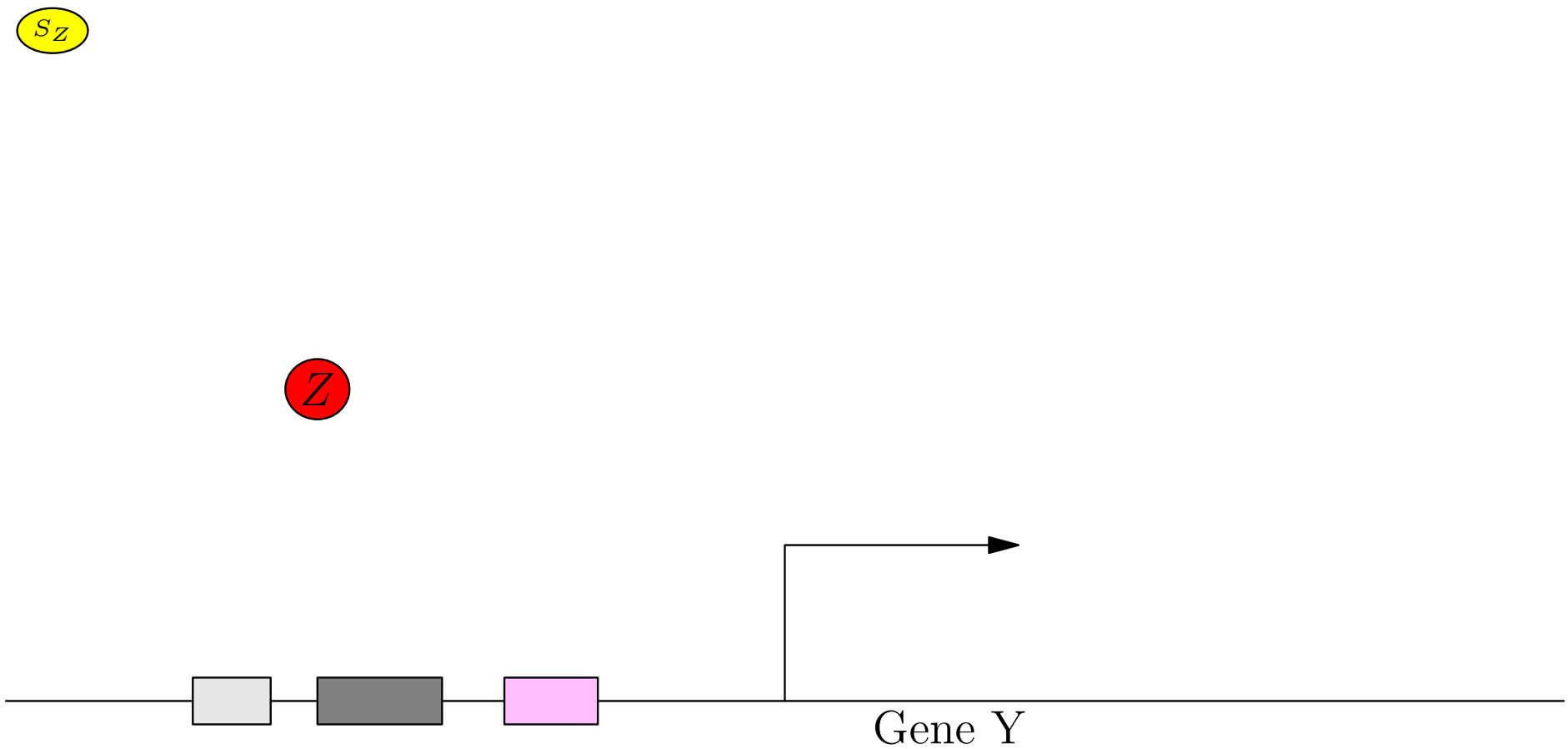

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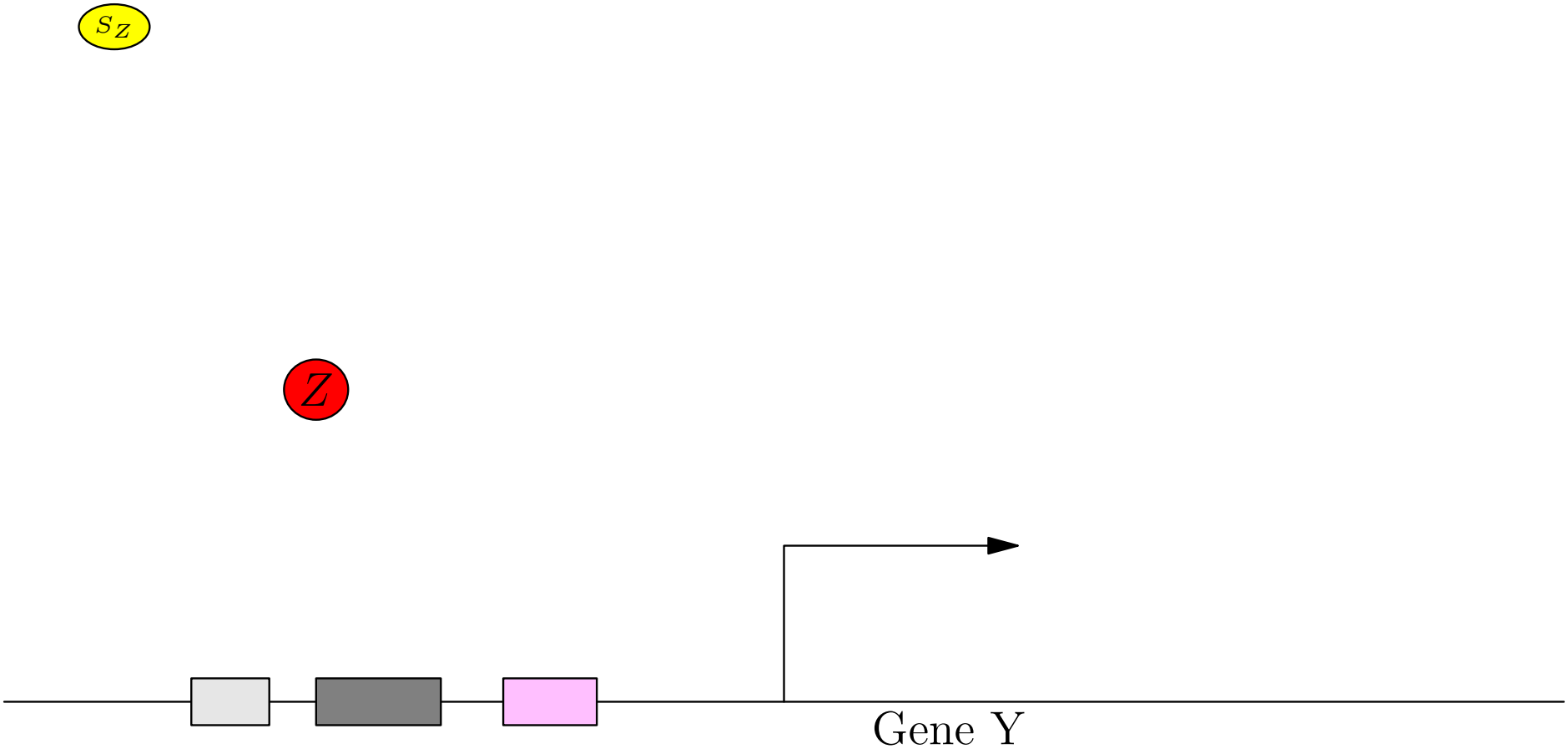

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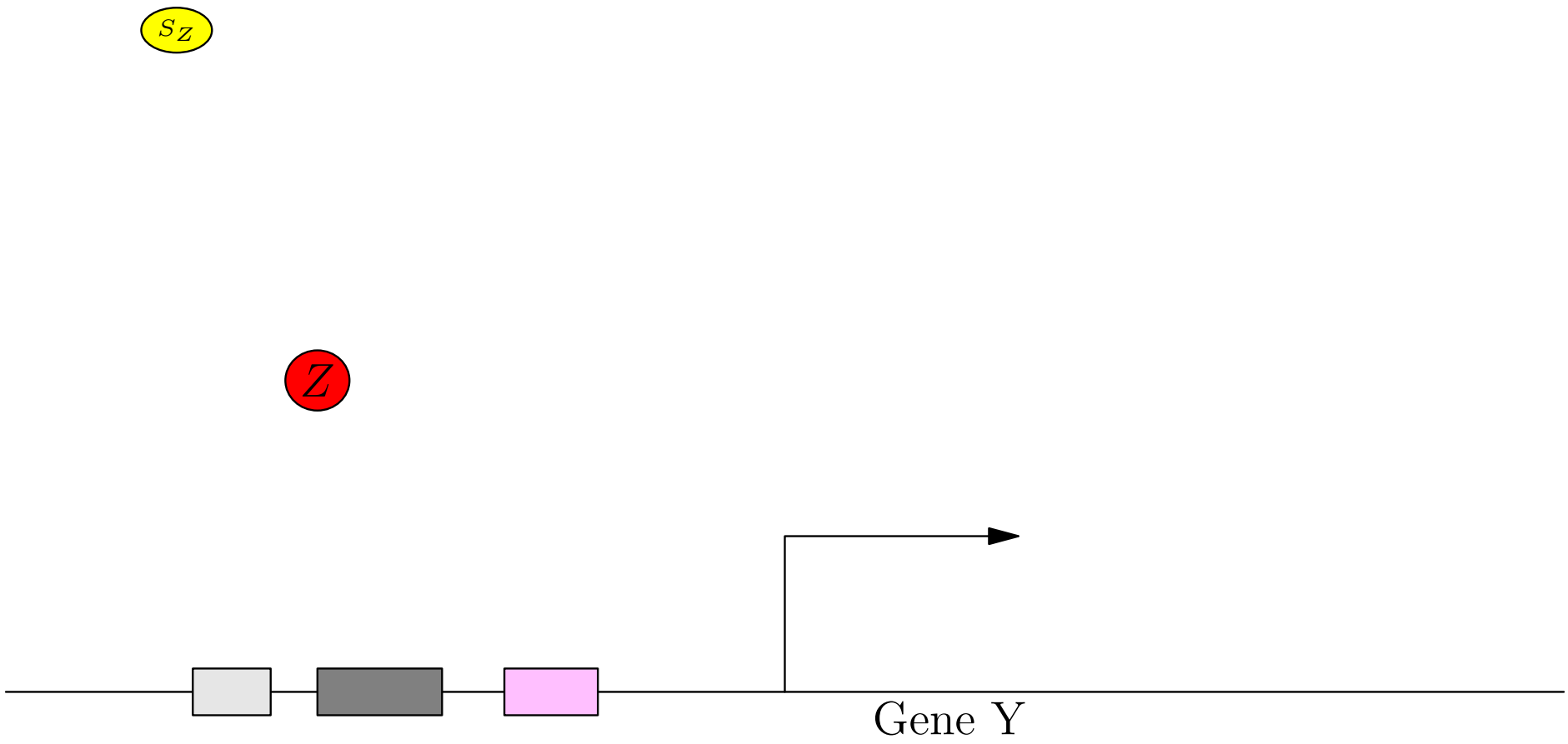

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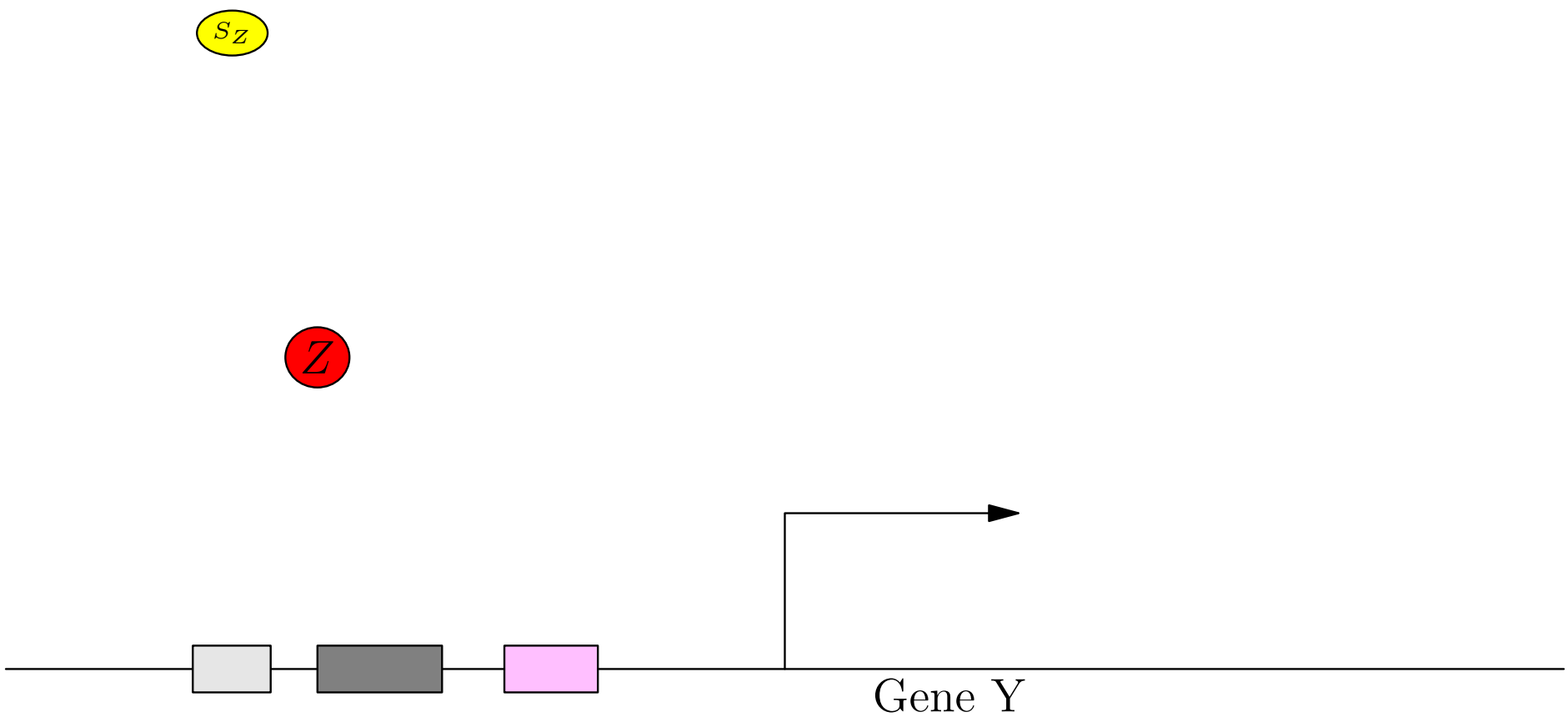

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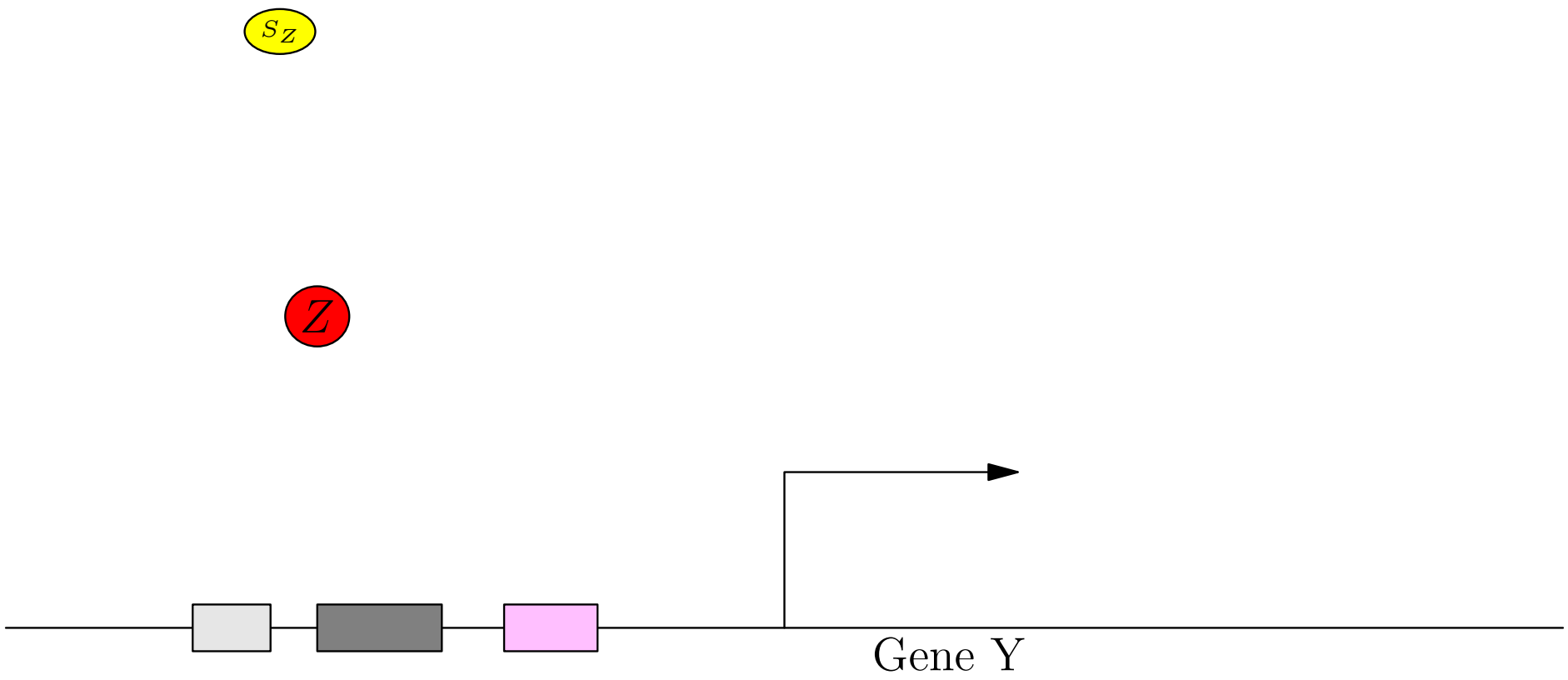

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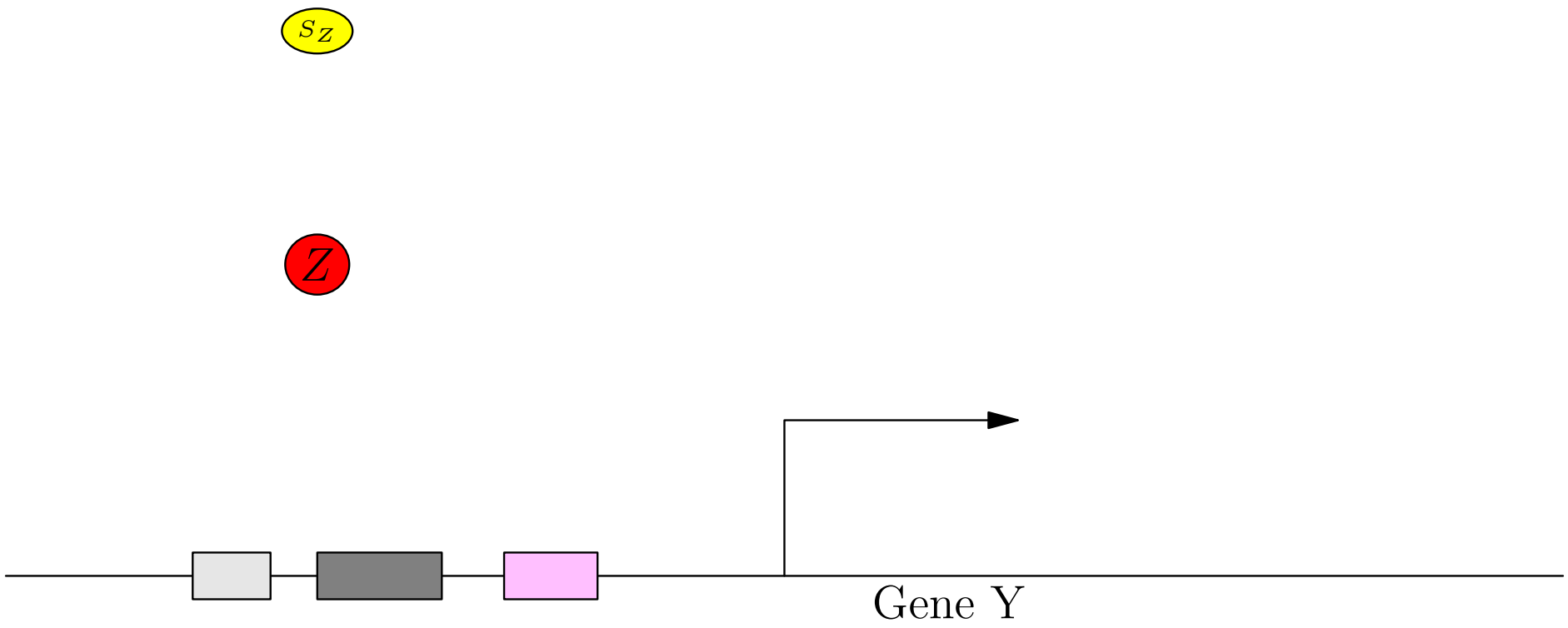

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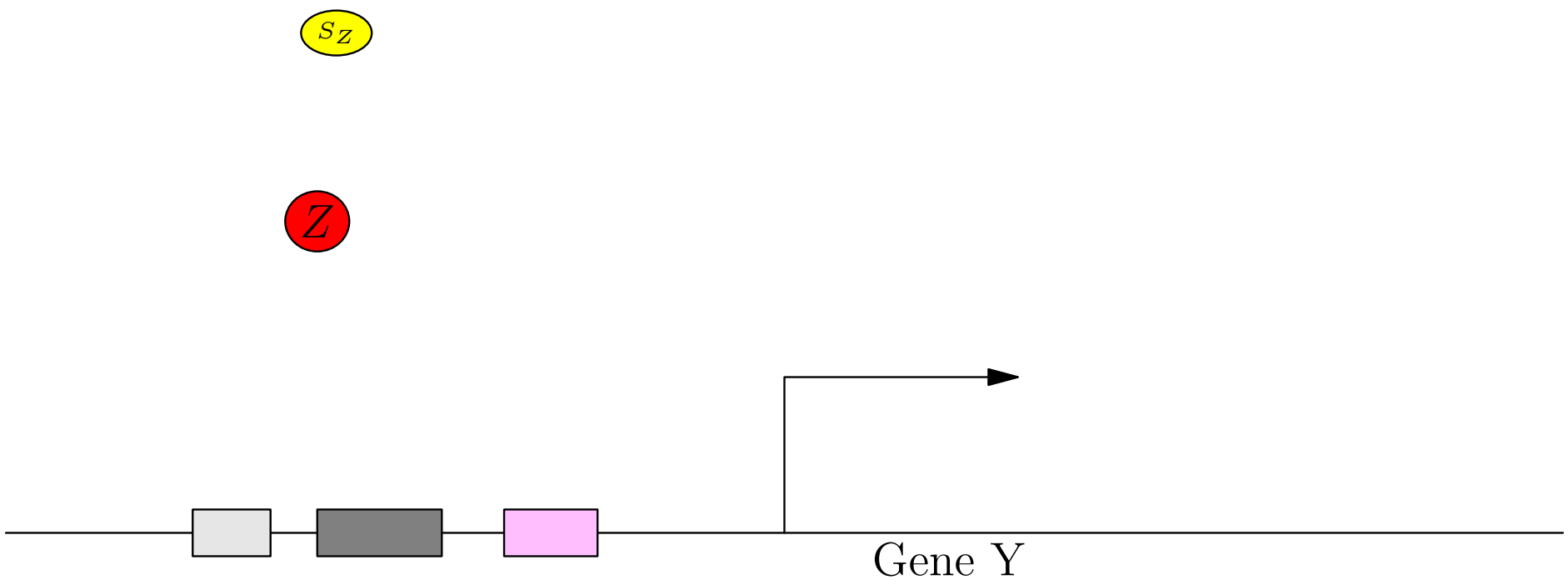

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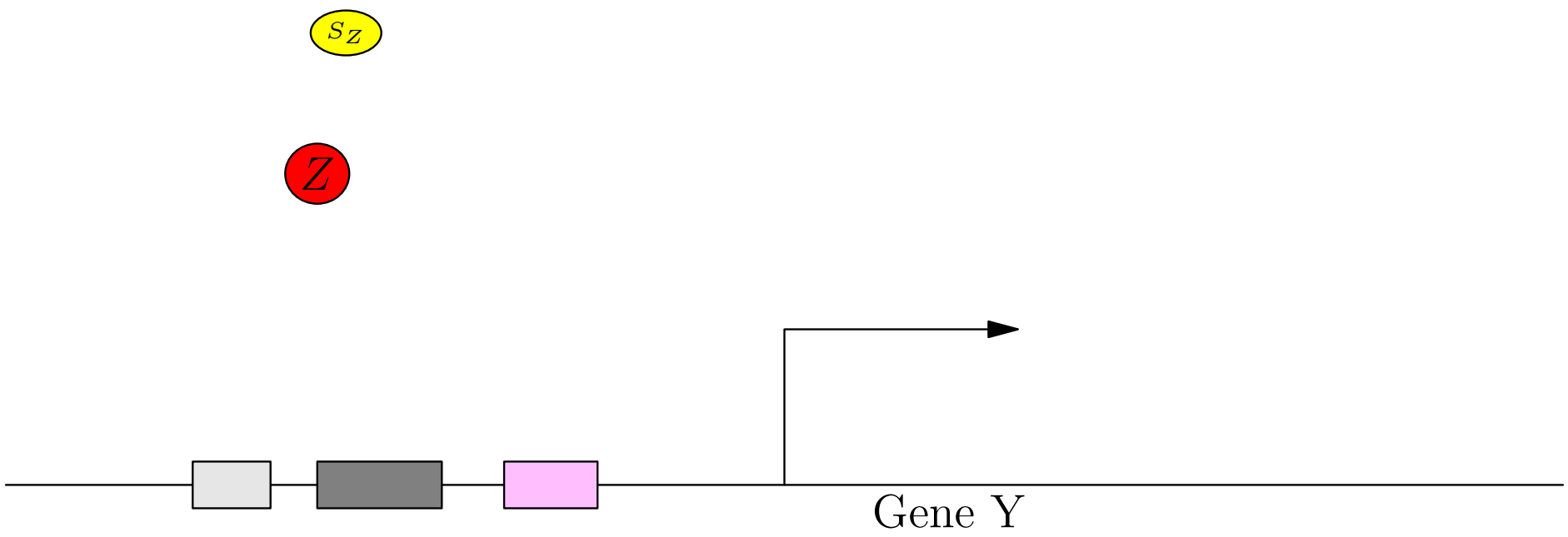

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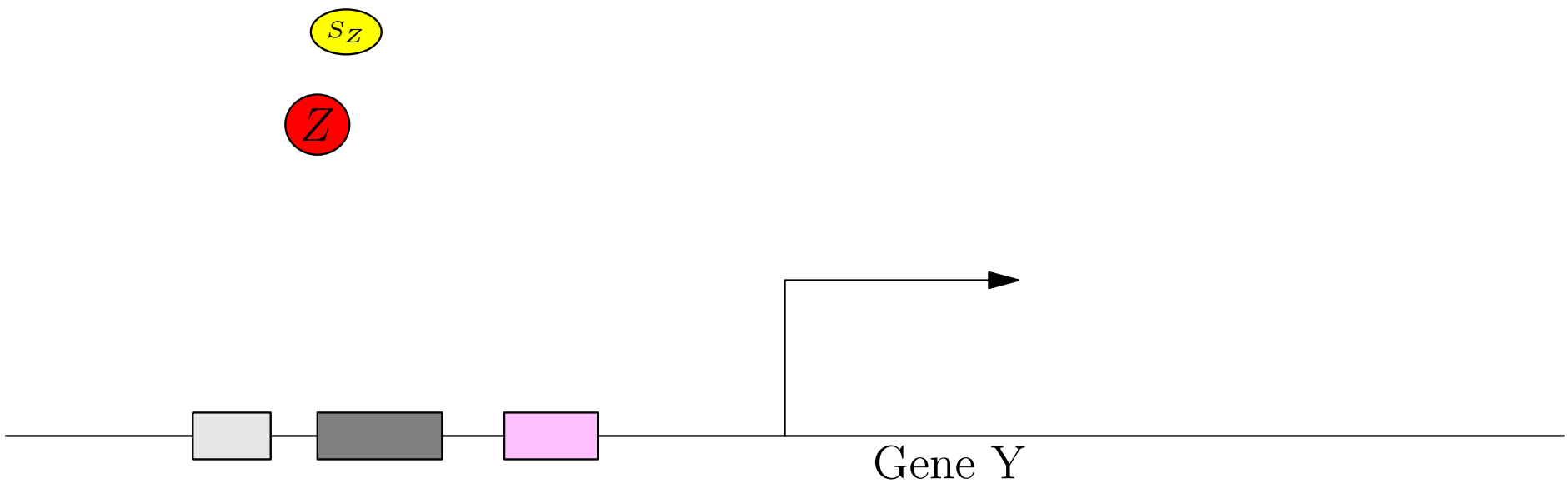

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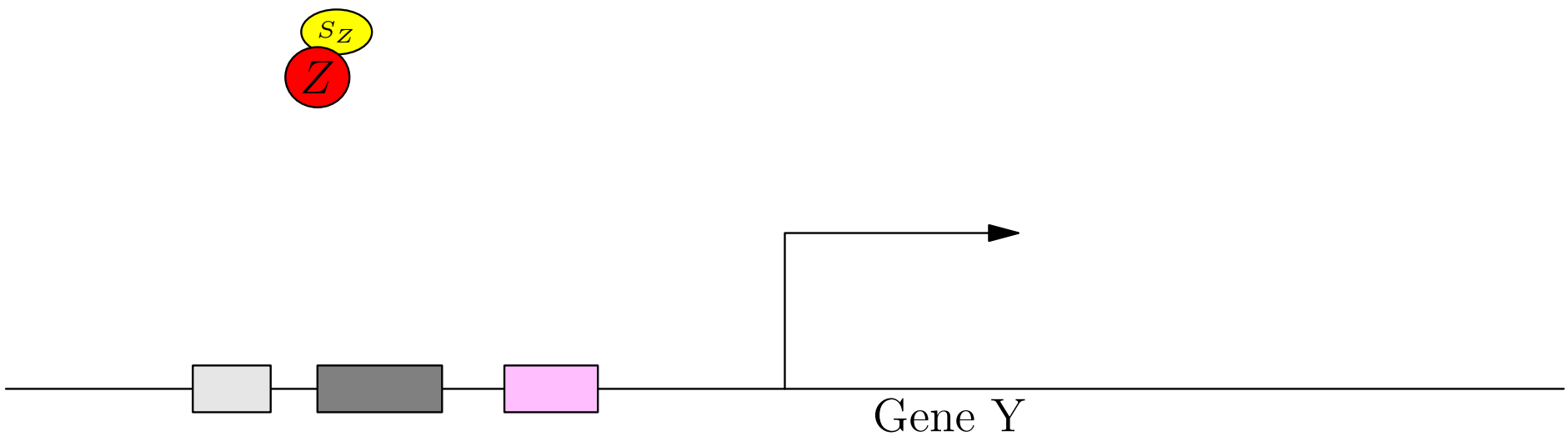

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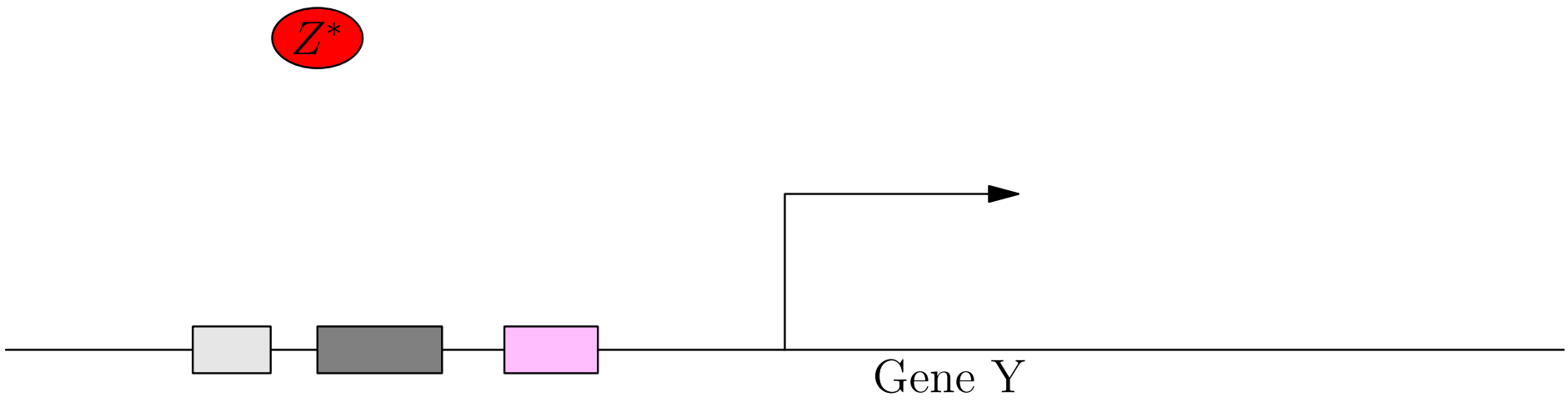

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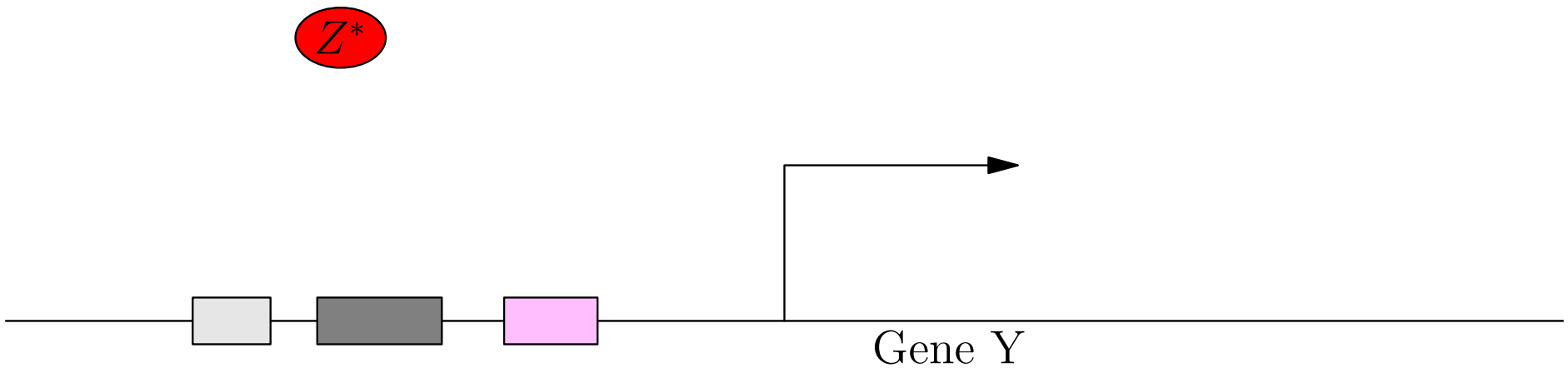

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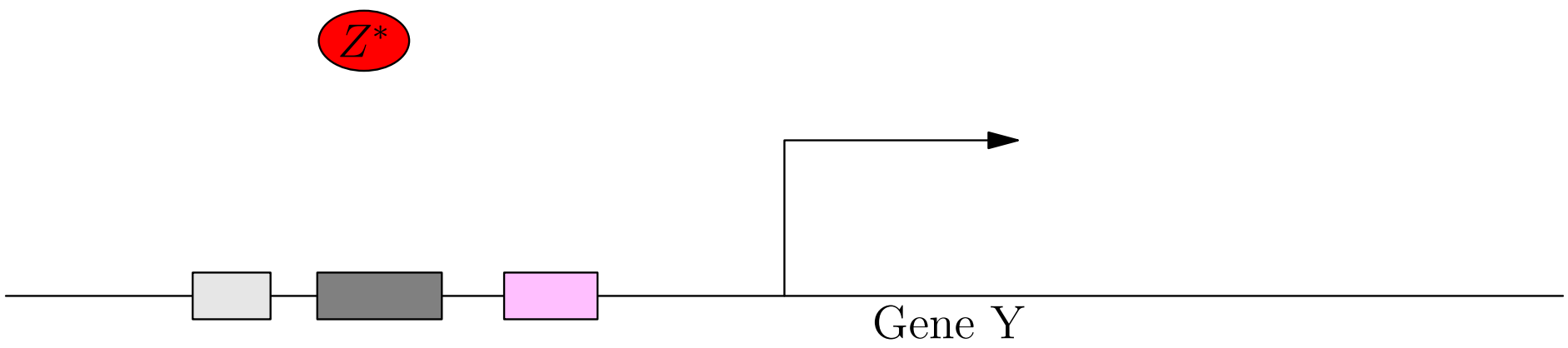

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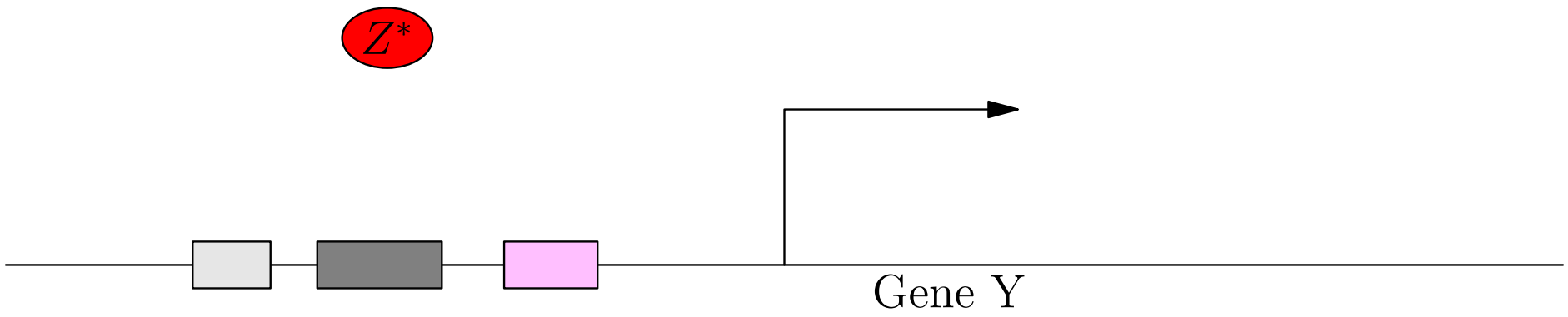

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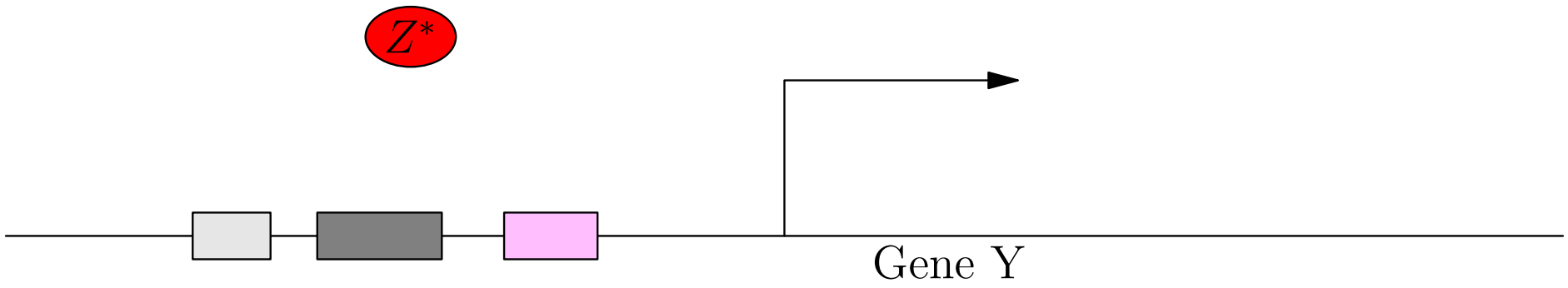

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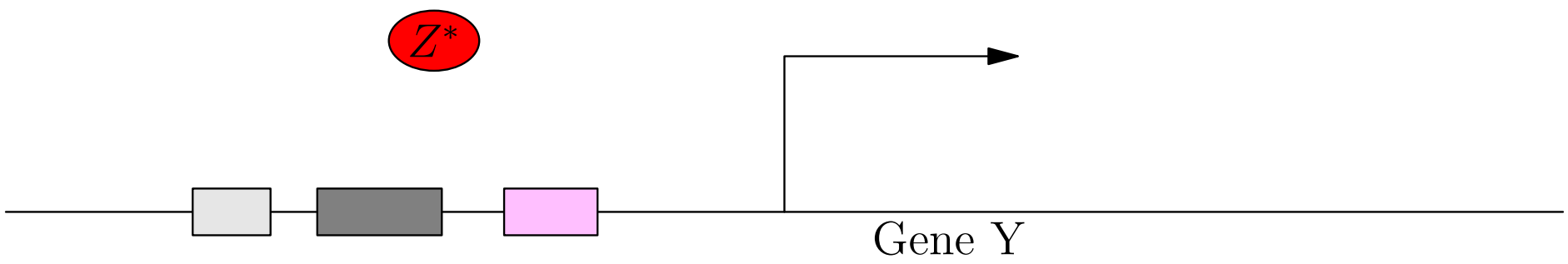

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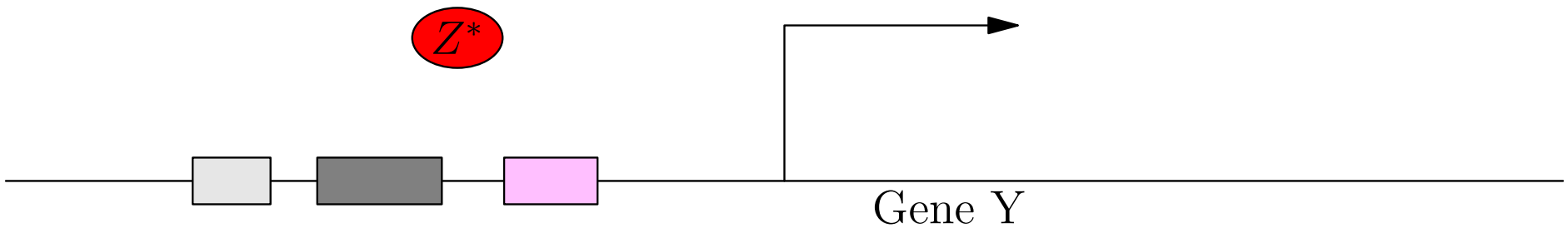

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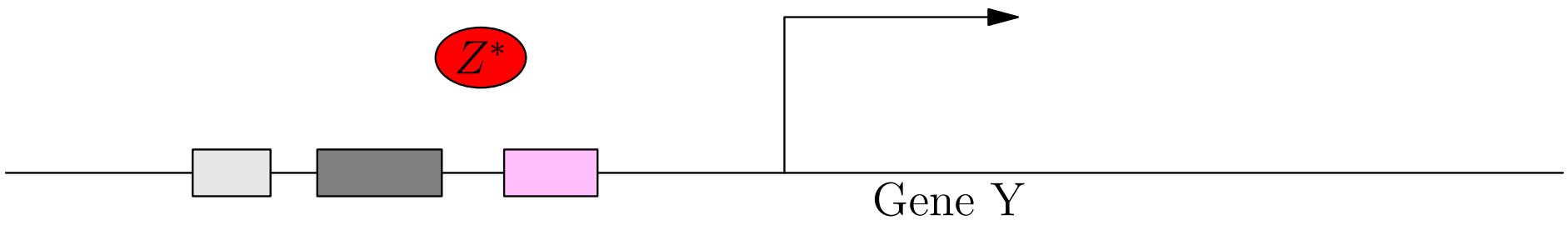

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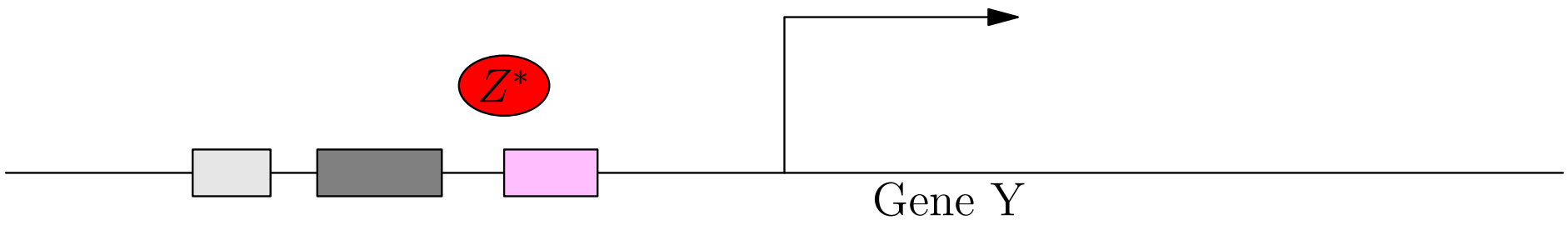

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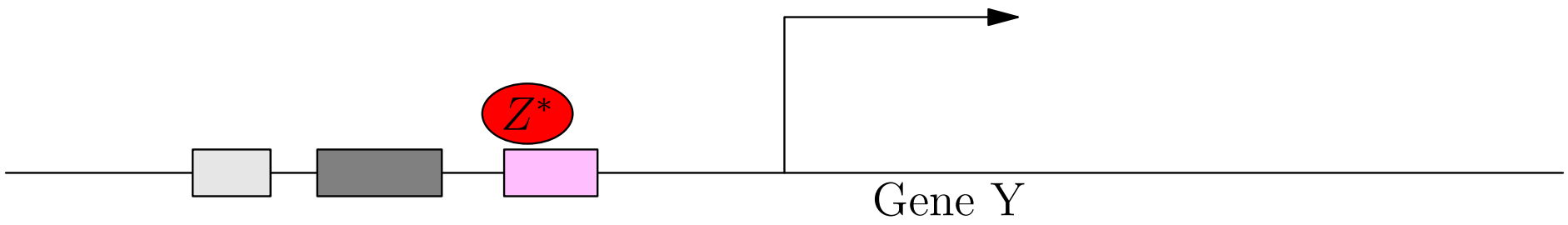

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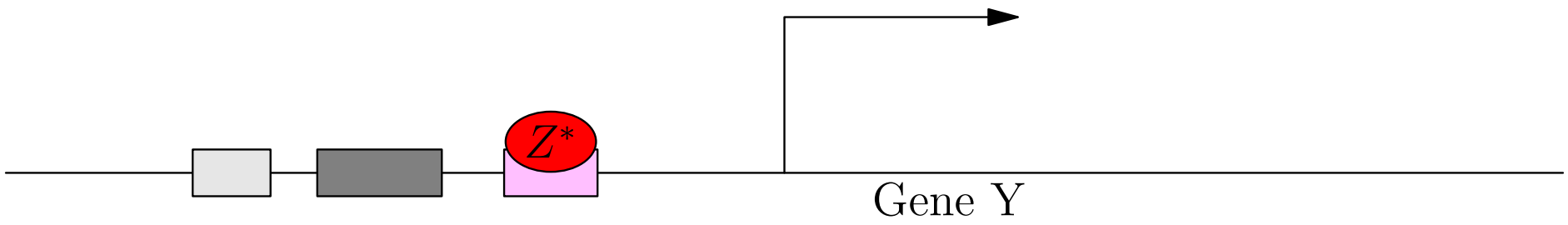

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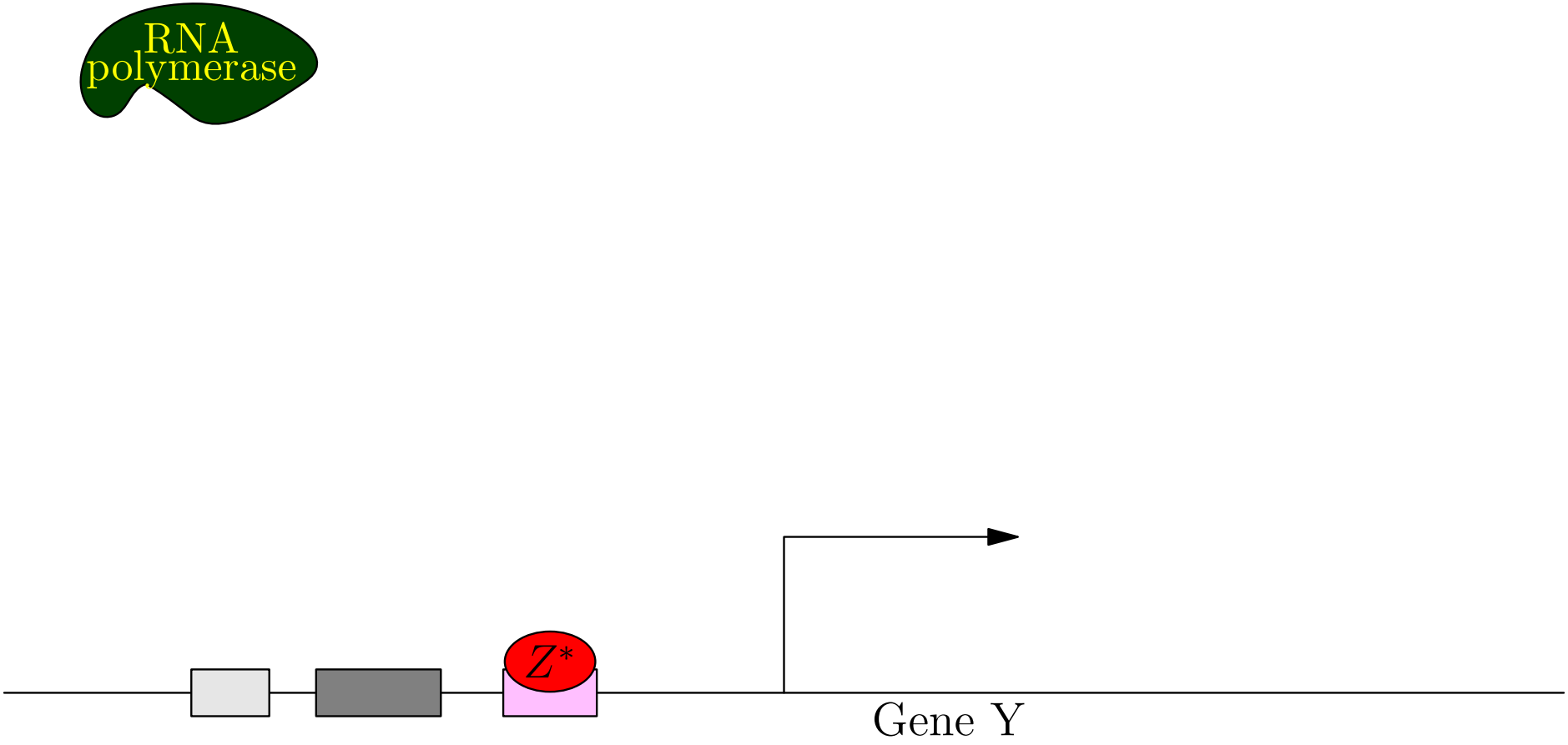
Inhibitory Transcription Factor



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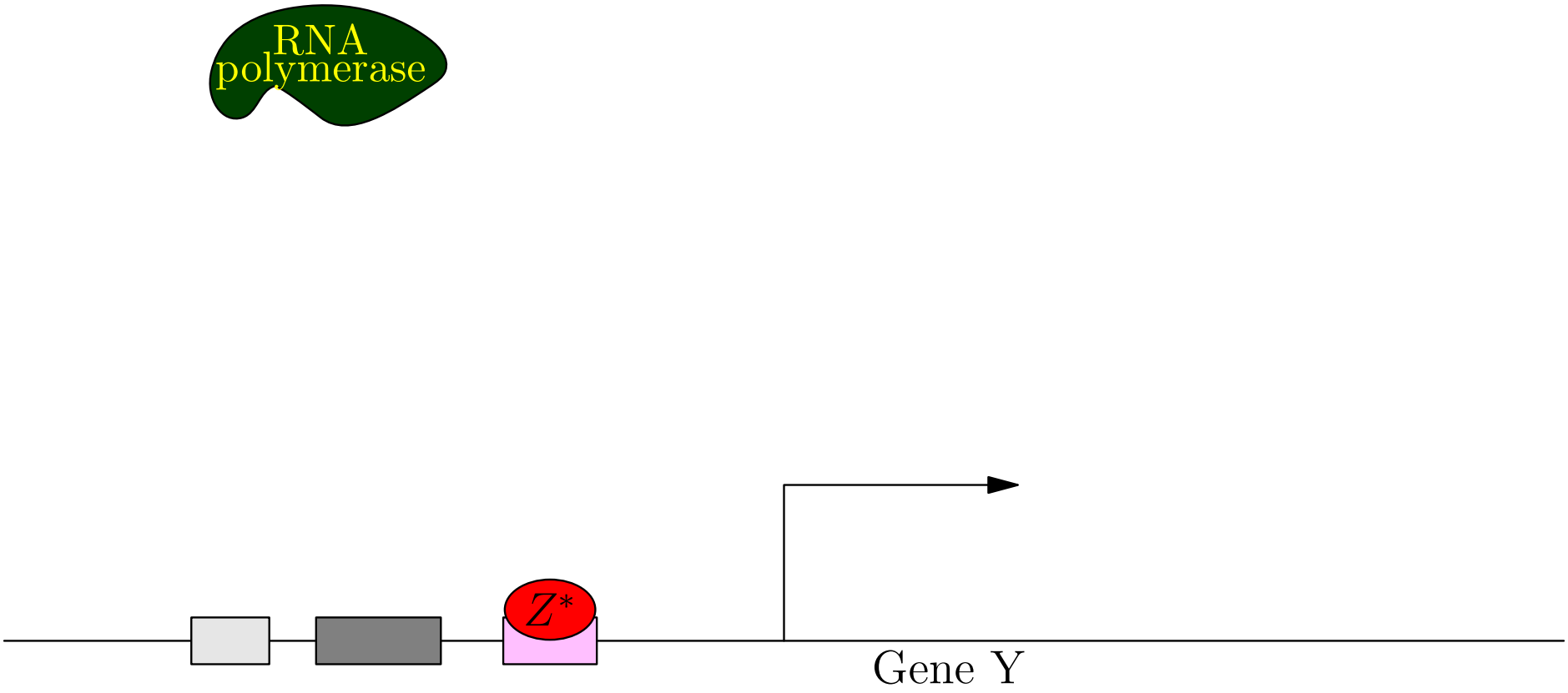

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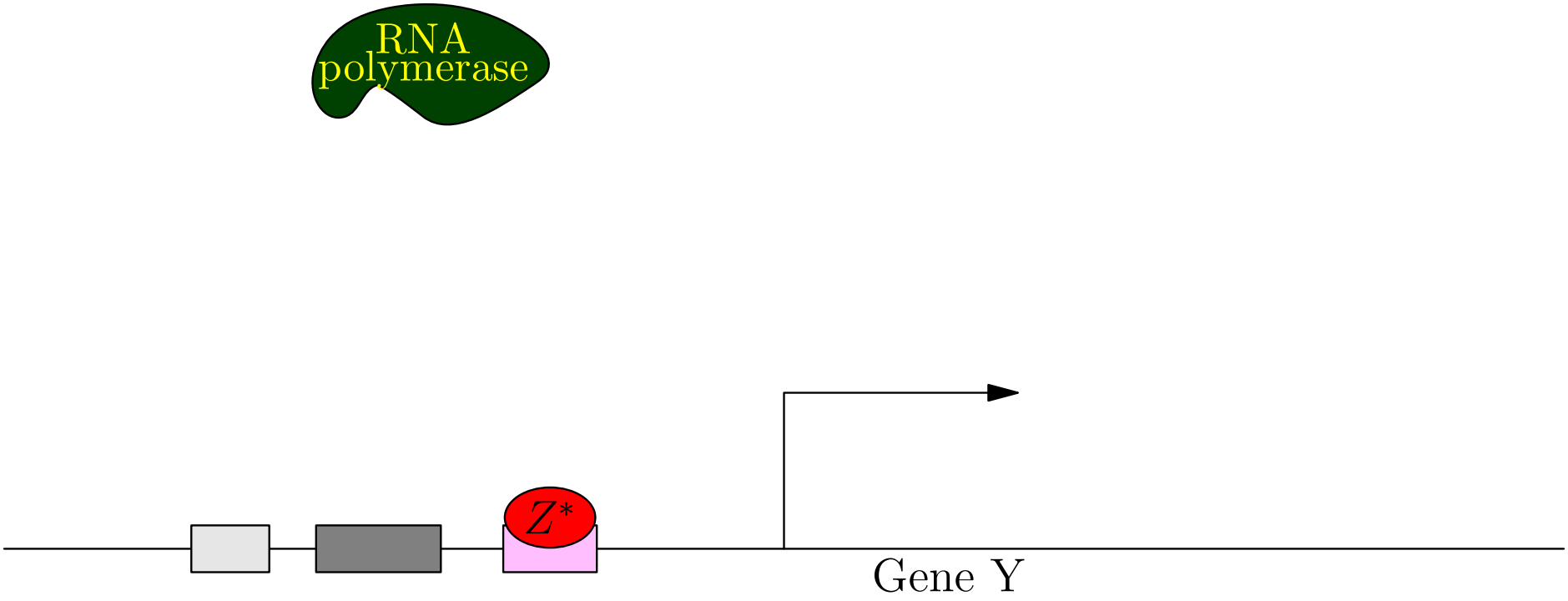

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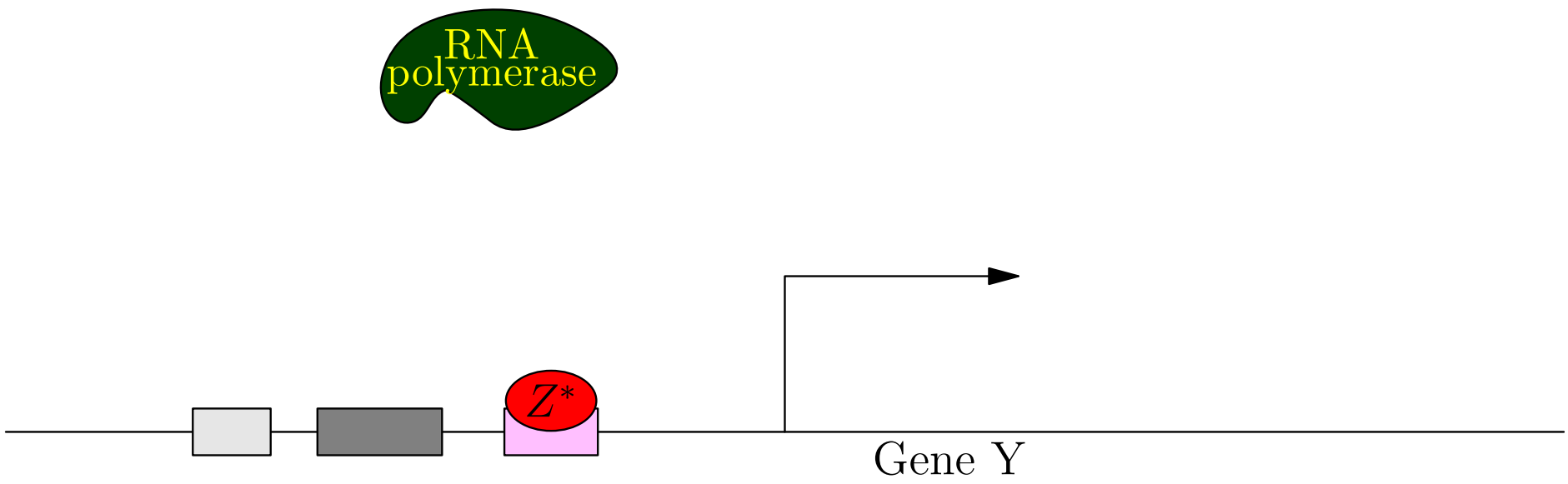

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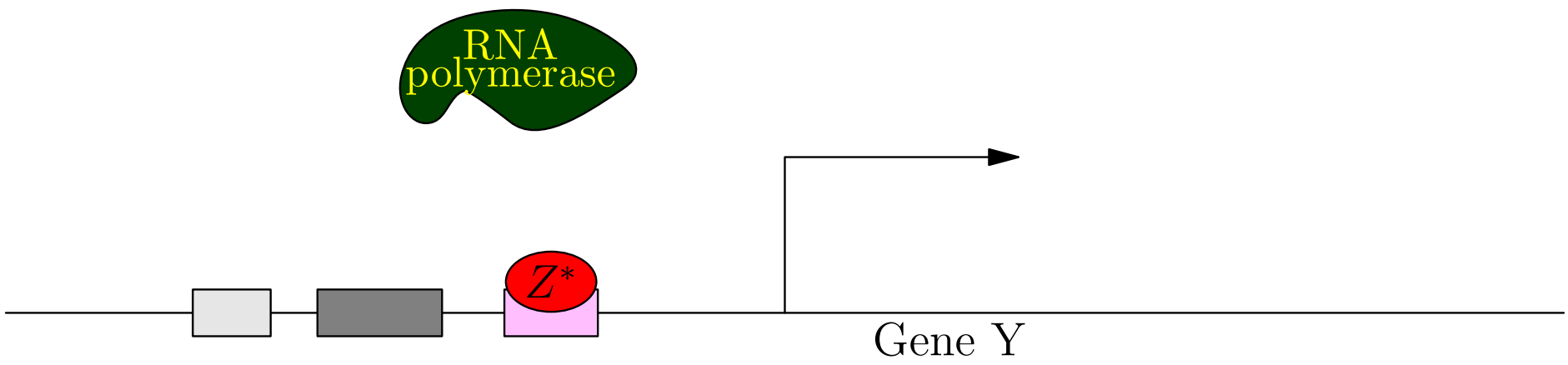

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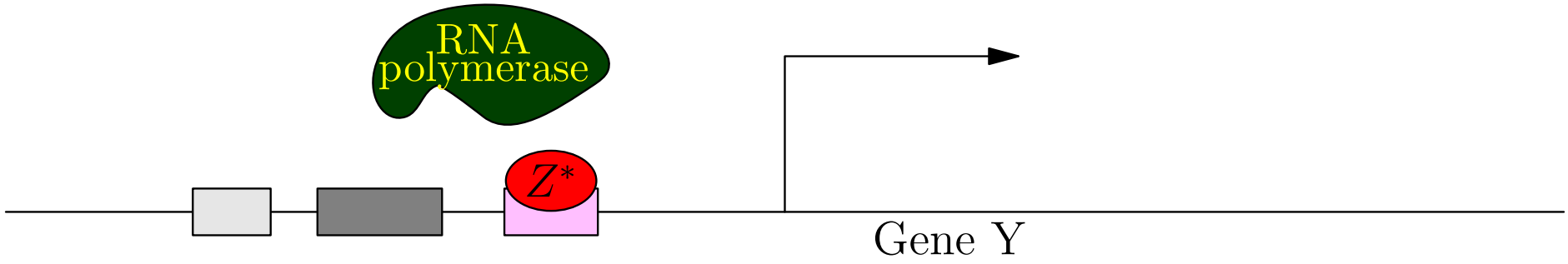

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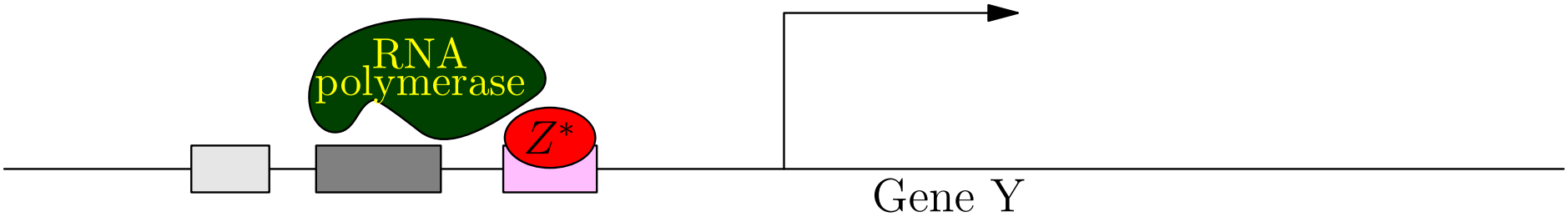

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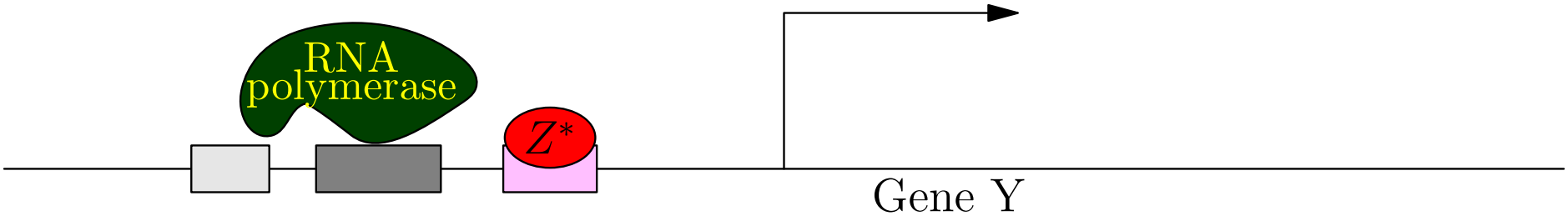

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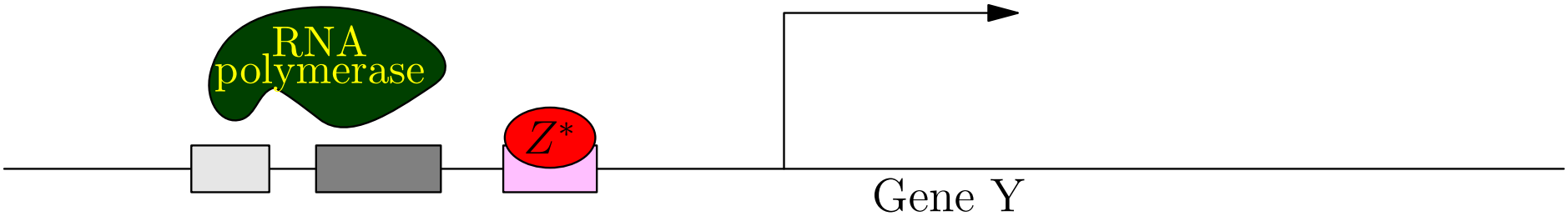

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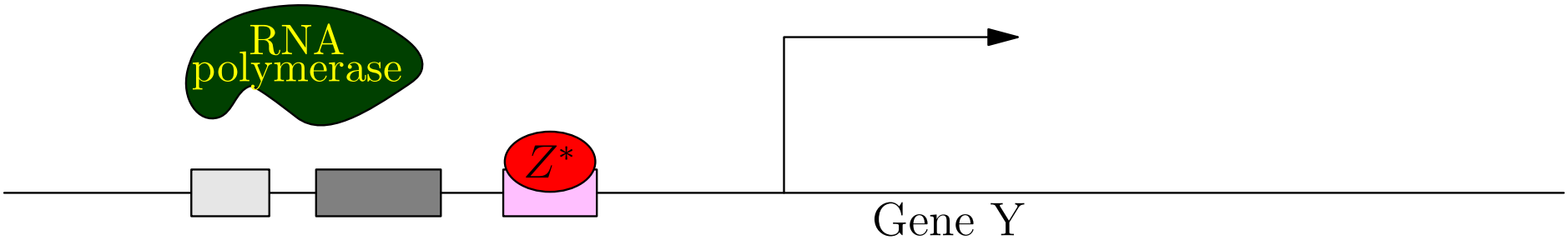

# Time Scales

- Reactions happen on different time scales
  - ⋆ Binding of small (signalling) molecules: $\sim 1$ msec
  - ⋆ Binding of active transcription factor: $\sim 1$ sec
  - ⋆ Transciption + translation of gene: $\sim 5$ min
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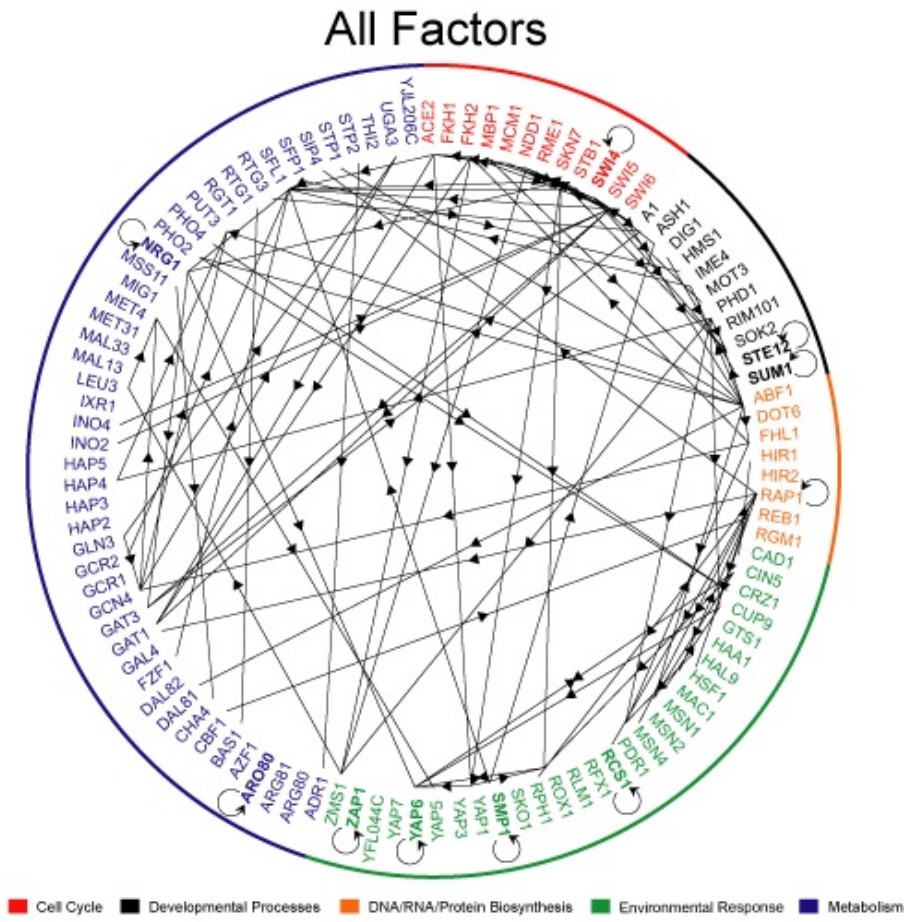
# Outline

1. Gene Regulation
2. **Chemical Equations**
3. Motifs



# Chemical Equations

- We can write down the mechanism that underlie gene regulation in terms of chemical equations

$$X \xrightarrow{k_1} X + RNA_Y$$

$$RNA_Y \xrightarrow{k_2} RNA_Y + Y$$

$$RNA_Y \xrightarrow{k_3} \varnothing$$

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- The reaction rates depend on concentrations of $X$ and $Y$ (how likely is it for two molecules to collide)
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$$\frac{\mathrm{d}[C]}{\mathrm{d}t} = k_4\,[X]\,[Y] \qquad \frac{\mathrm{d}C}{\mathrm{d}t} = \frac{k_4}{\Omega}\,X\,Y$$

- $\Omega = V\,A$, $[X] = X/\Omega$ and $[Y] = Y/\Omega$

# Fluctuations

- In chemistry we are used to there being so many molecules that the average rate equations provide a very good approximation of what really happens
- This is often true in biology, but not always
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- Proteins can have multiple sub-units that each bind with a signalling molecule
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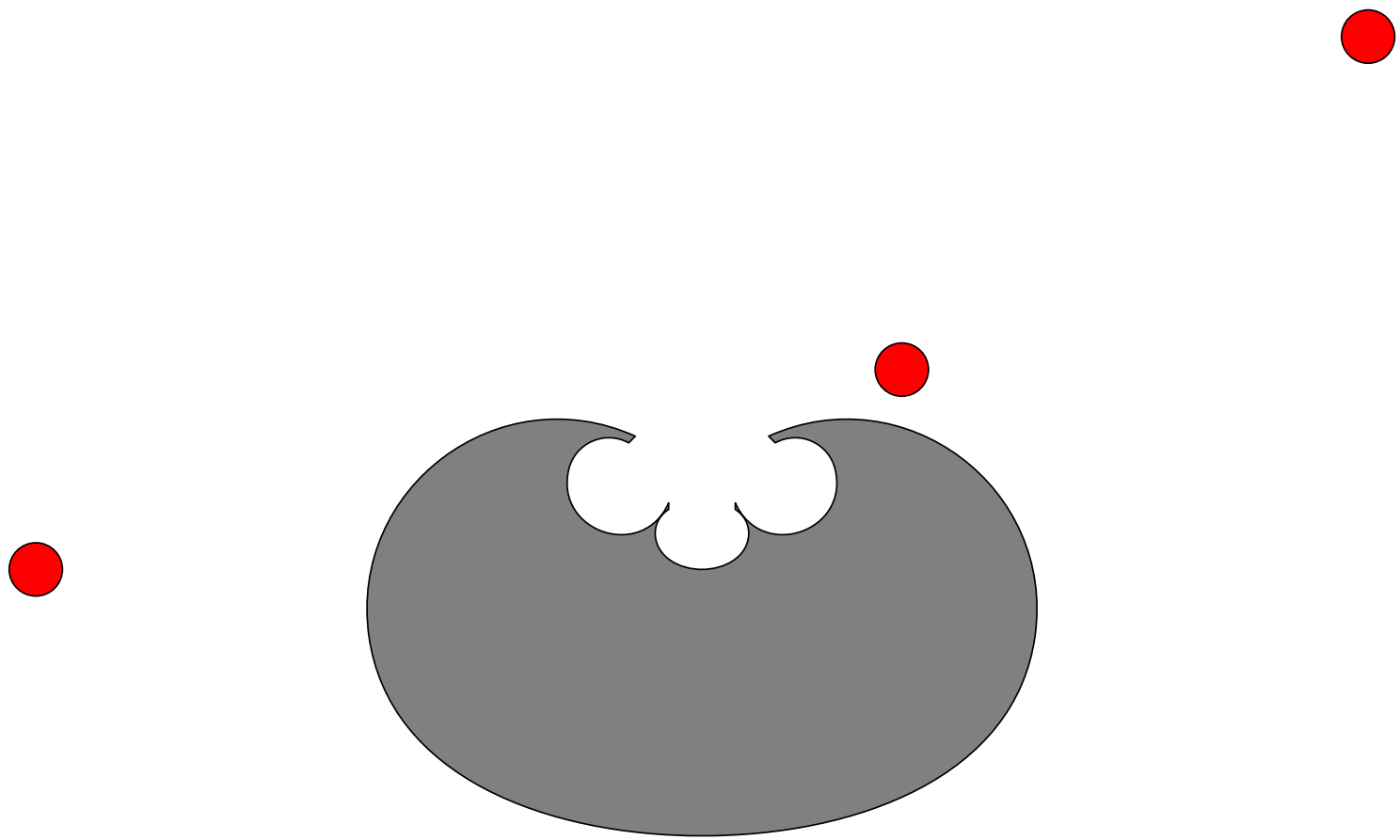

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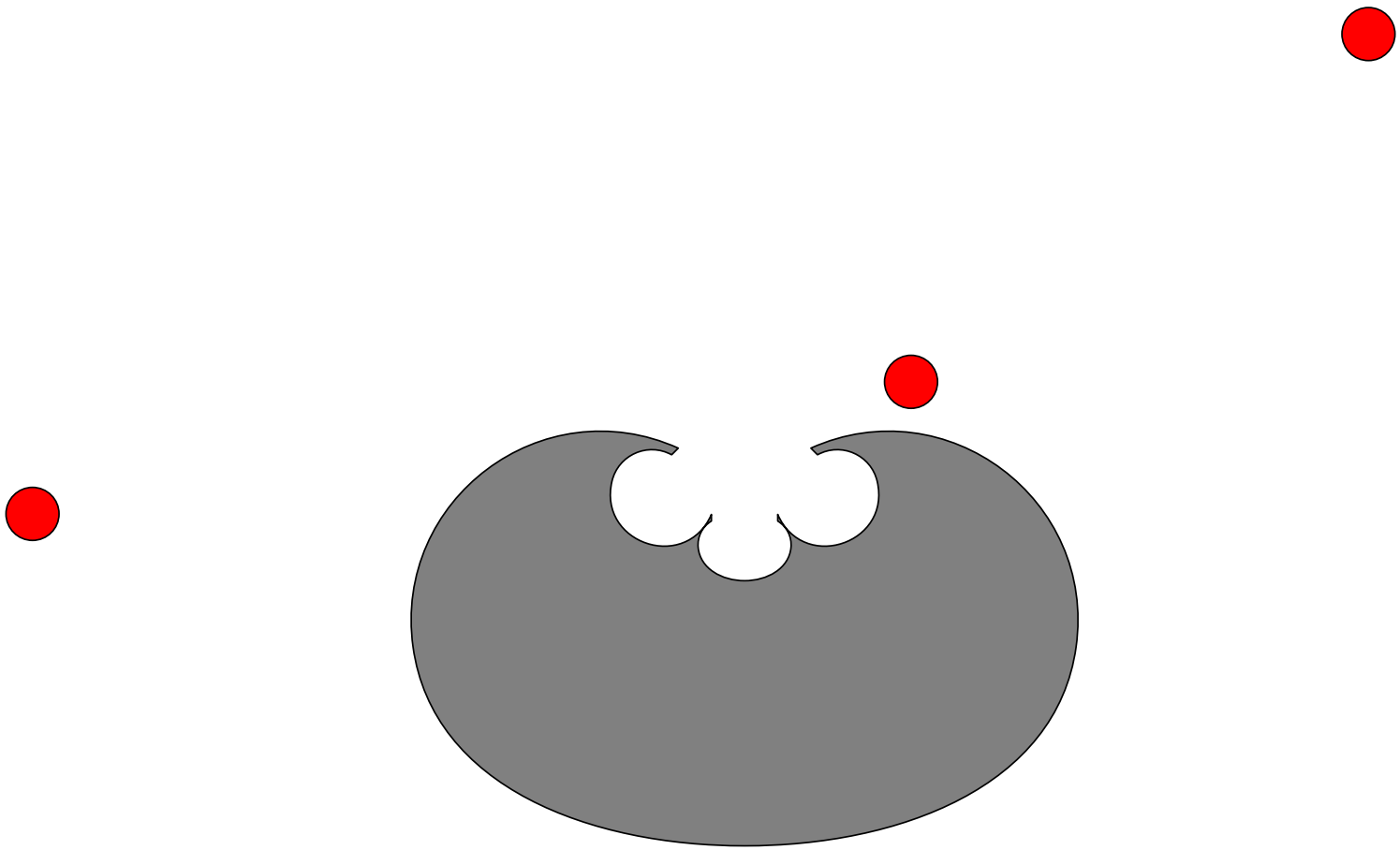

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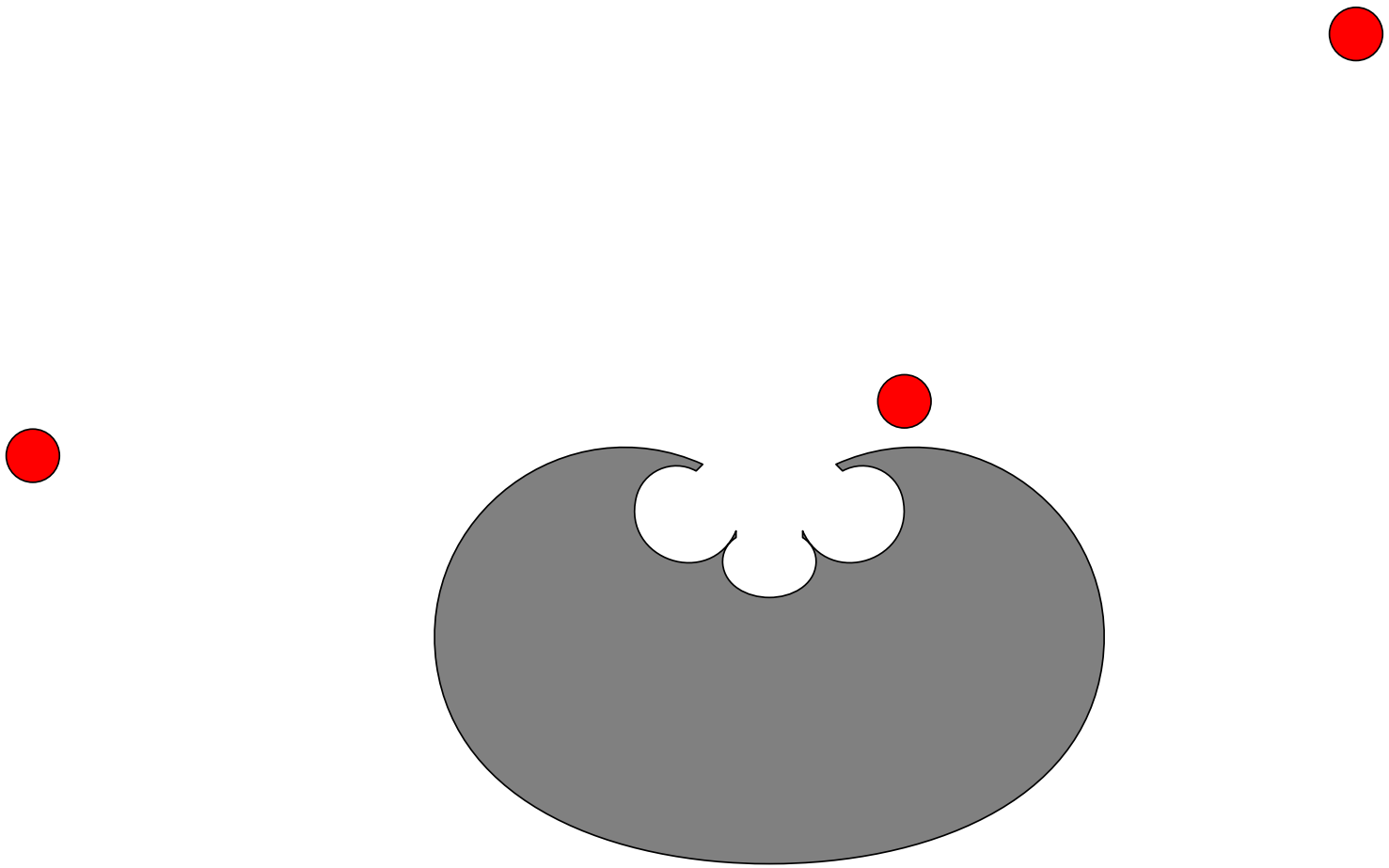

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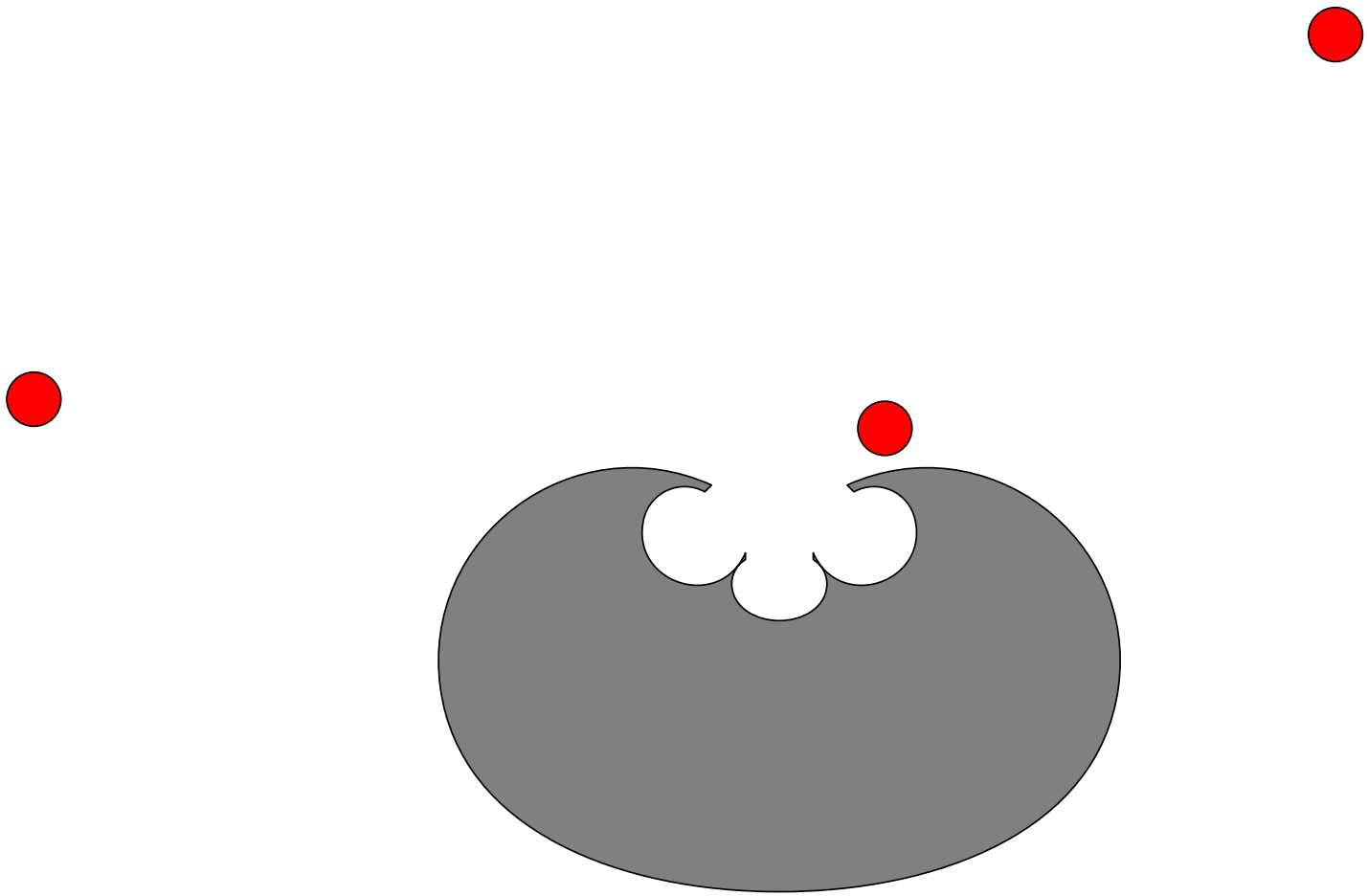

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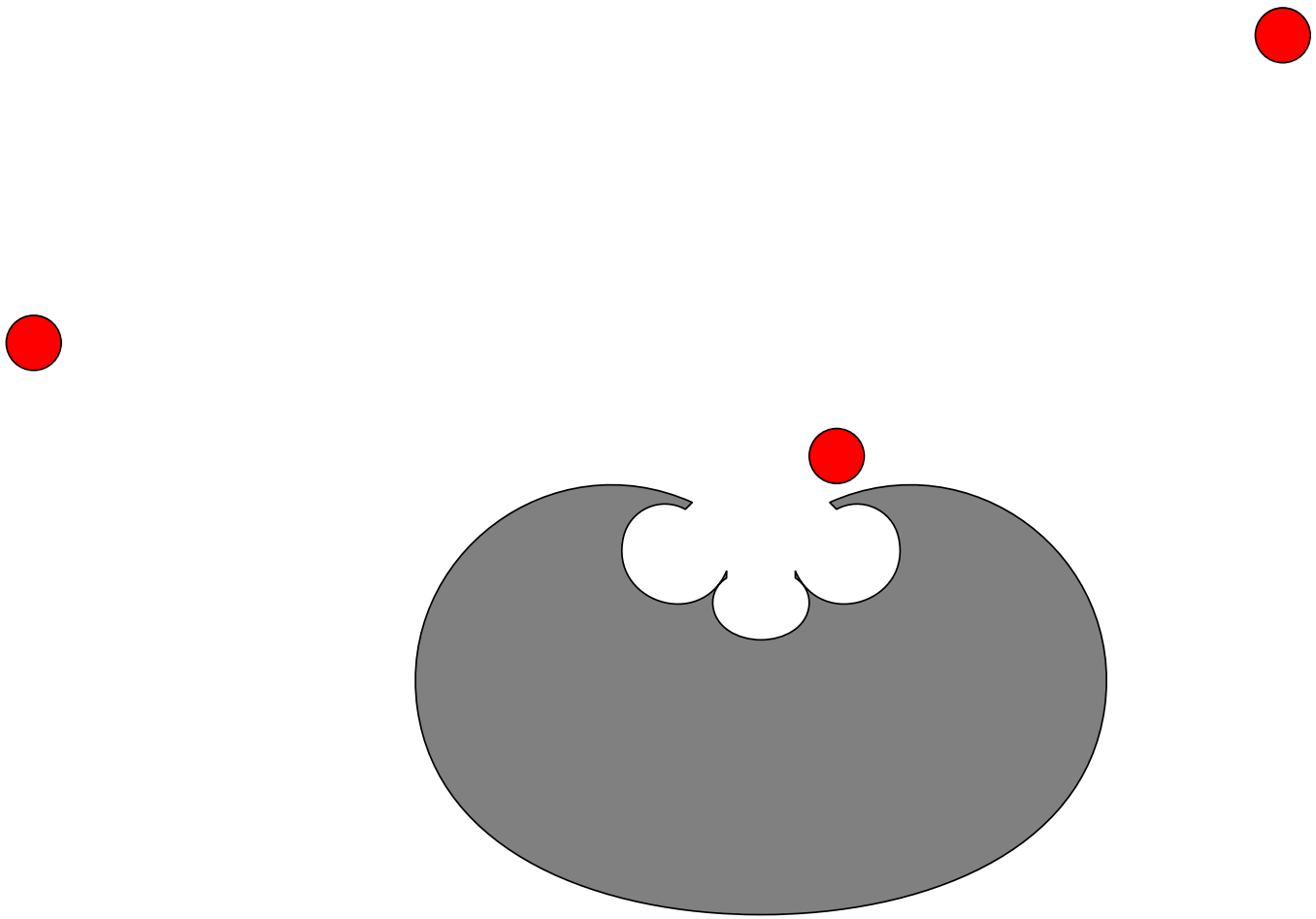

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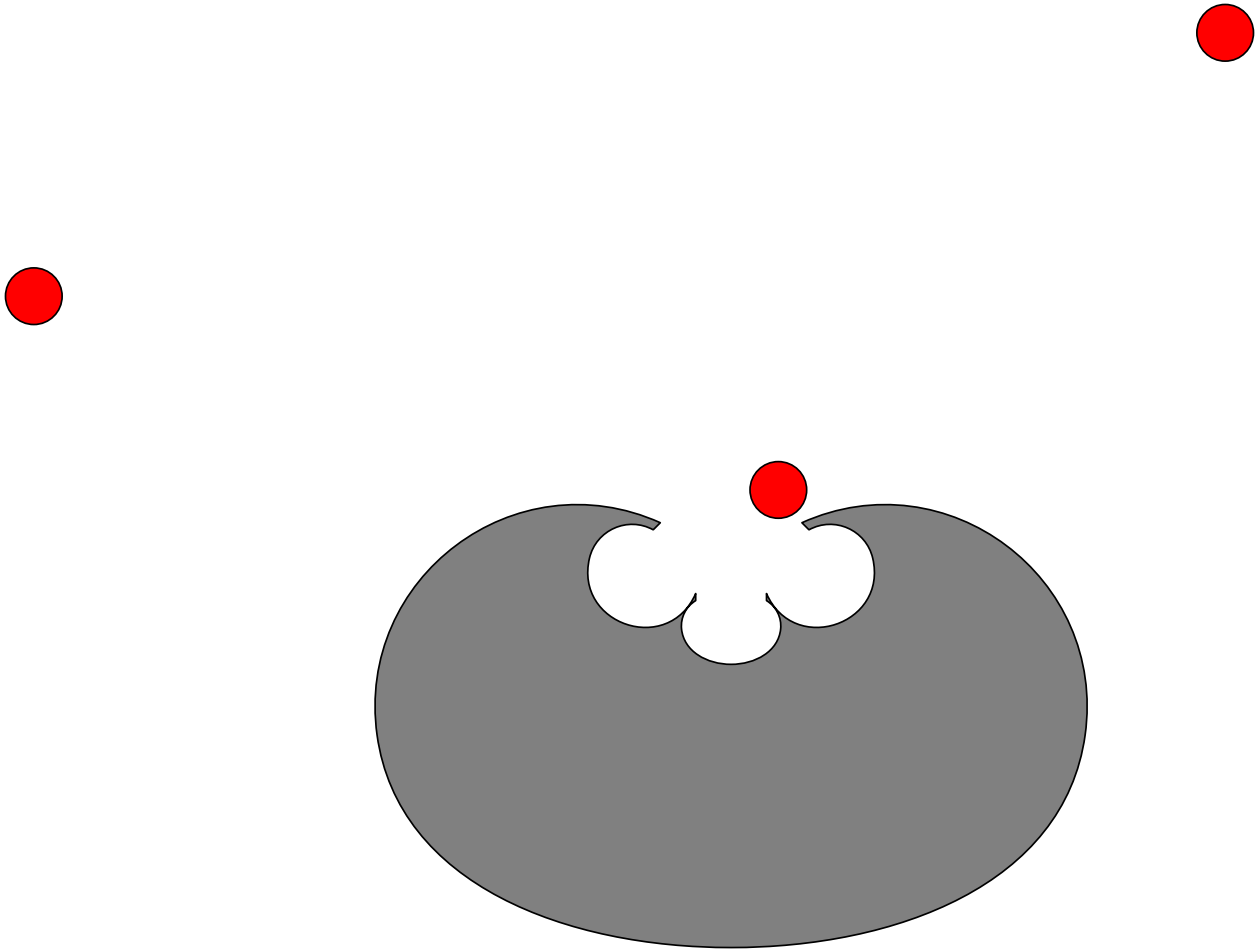

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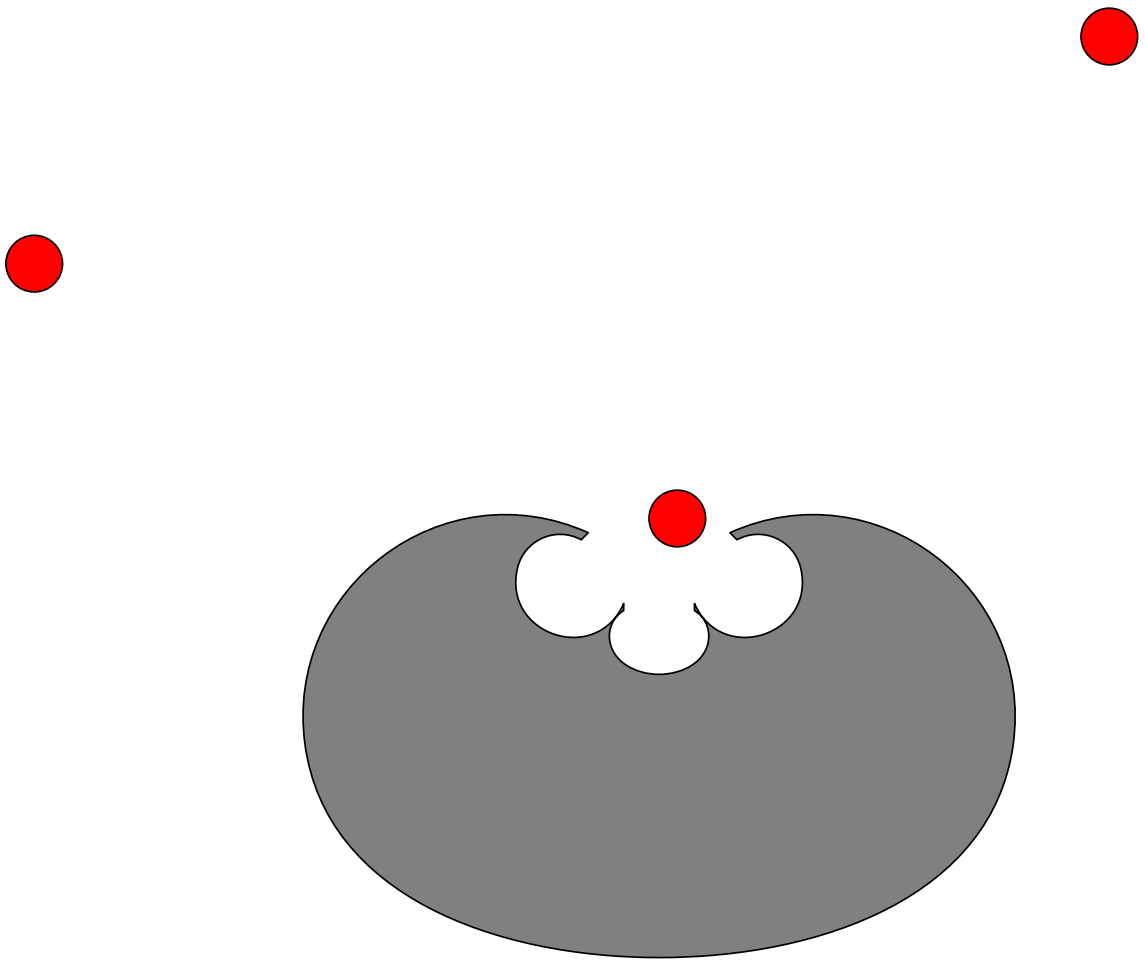

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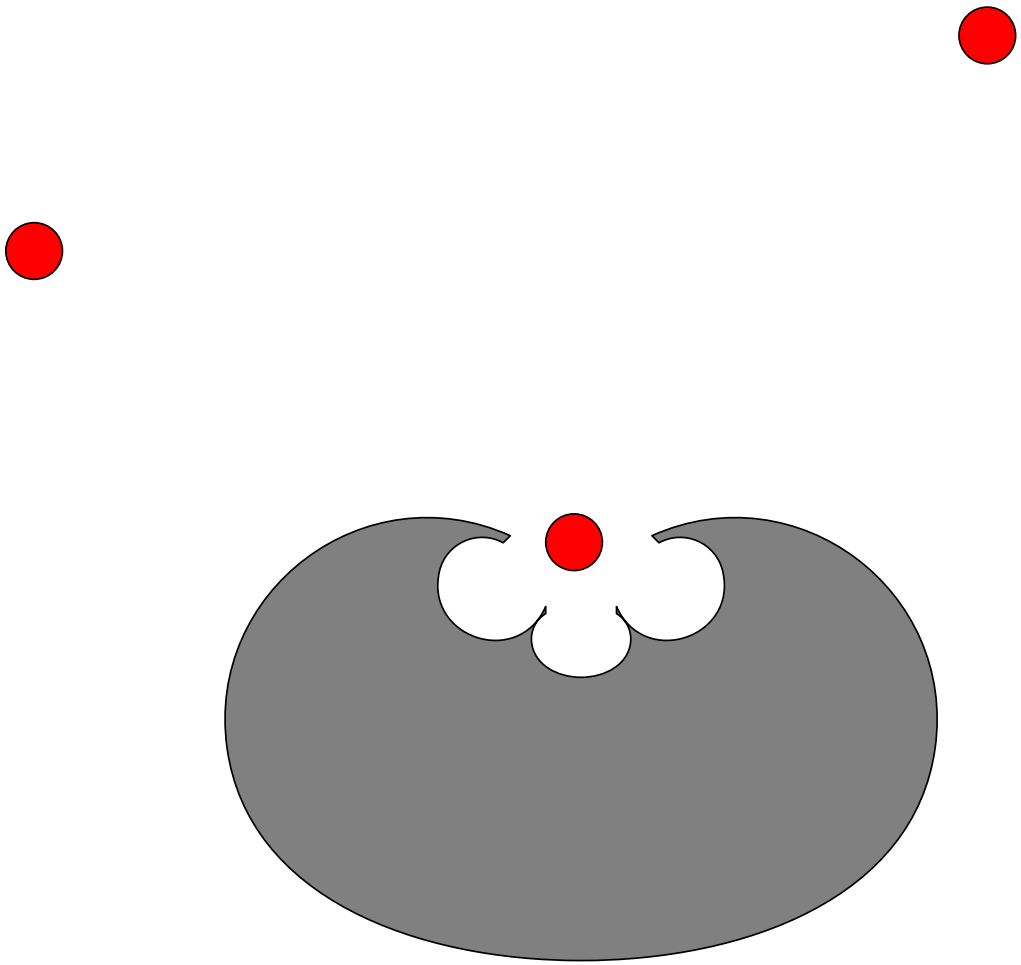

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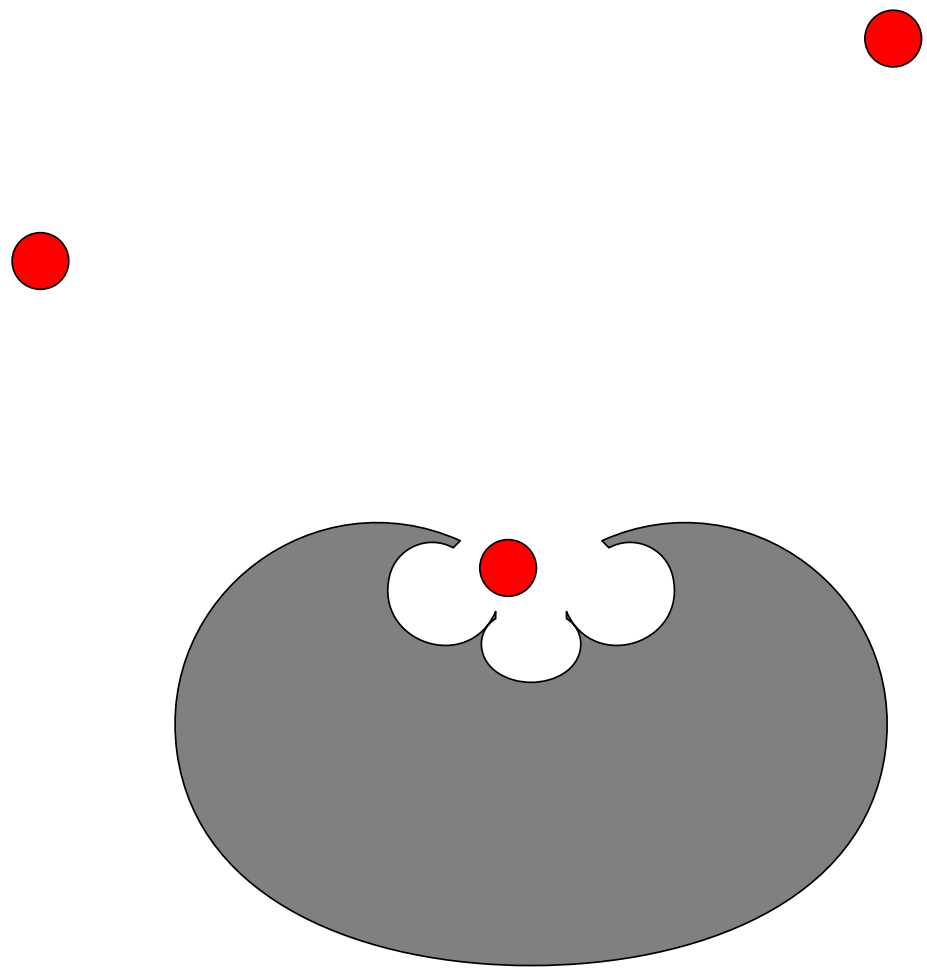

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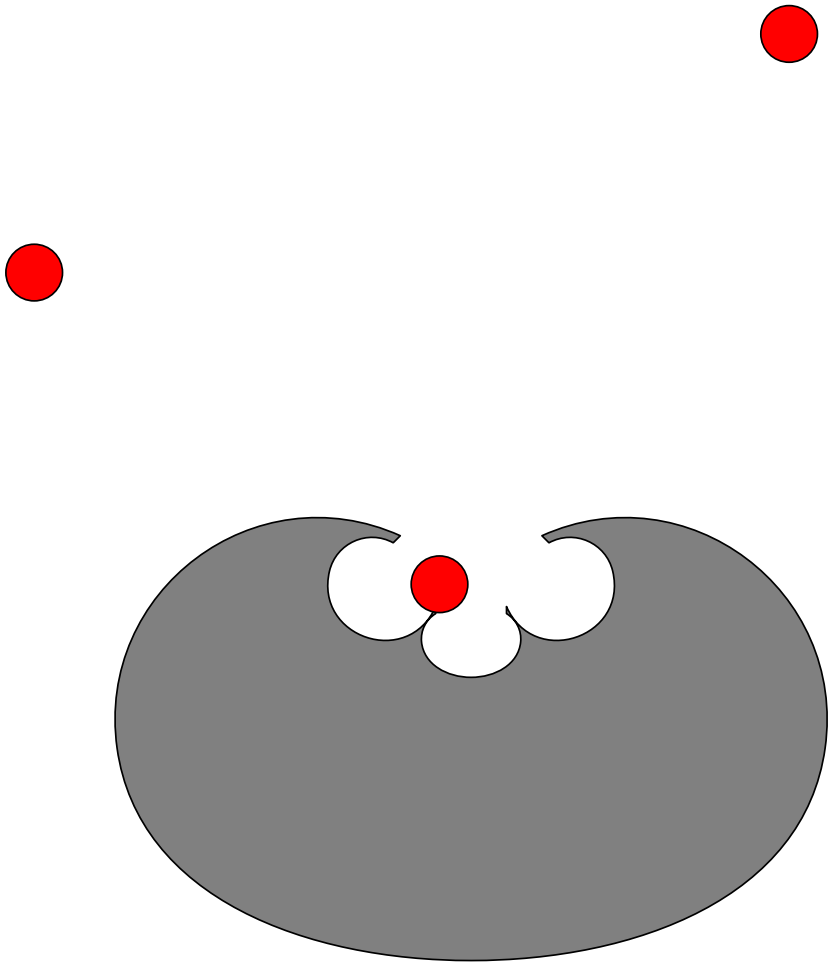

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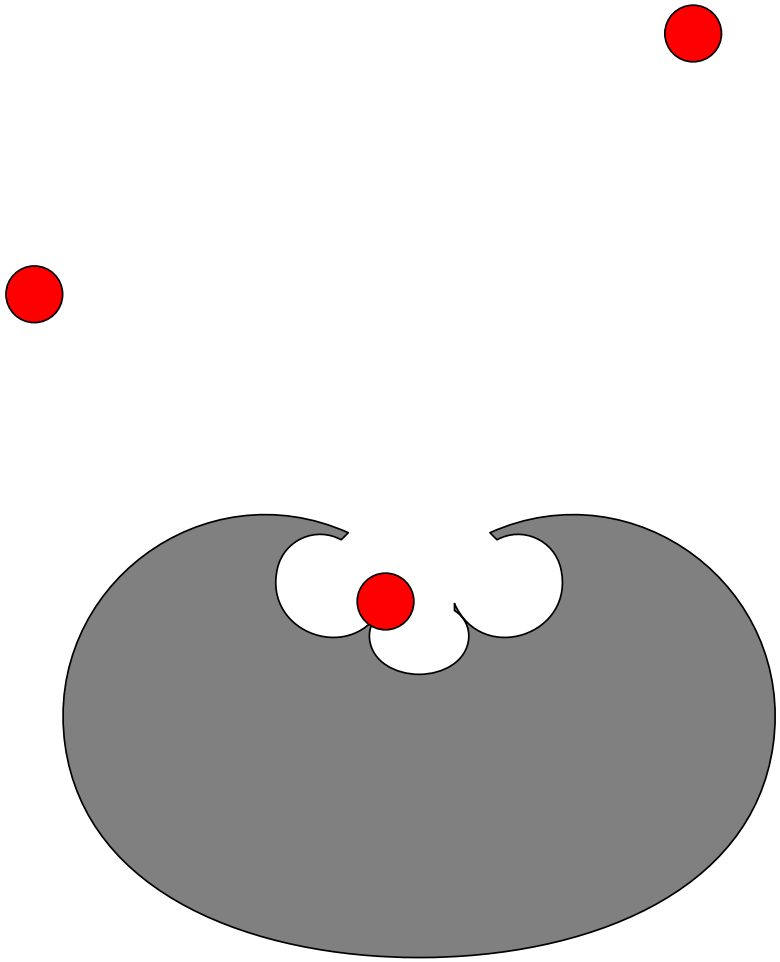

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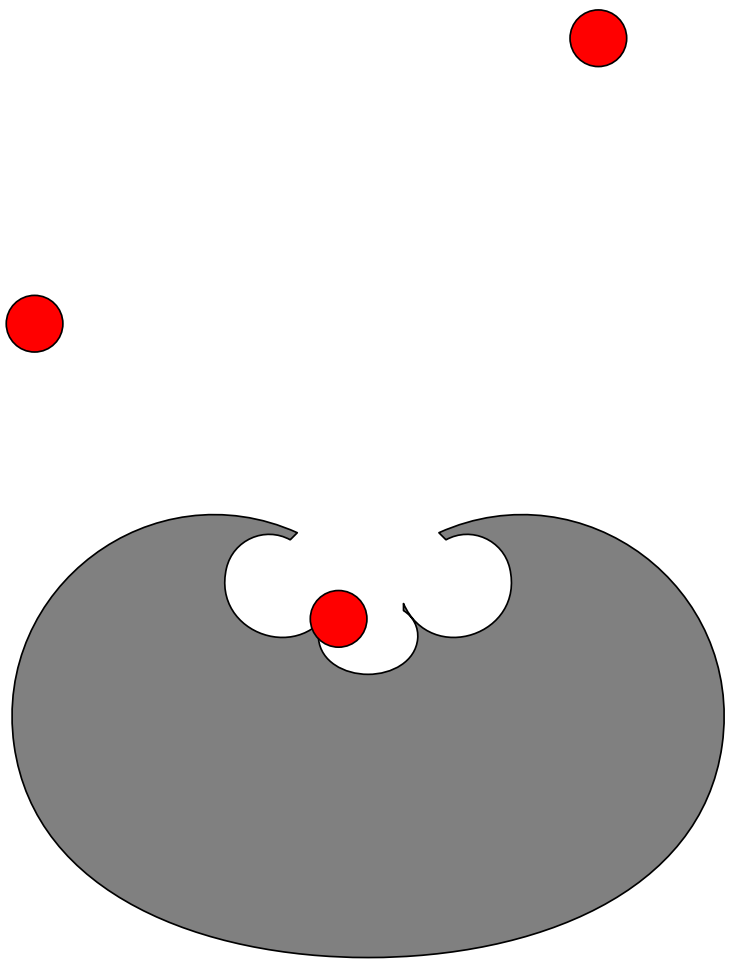

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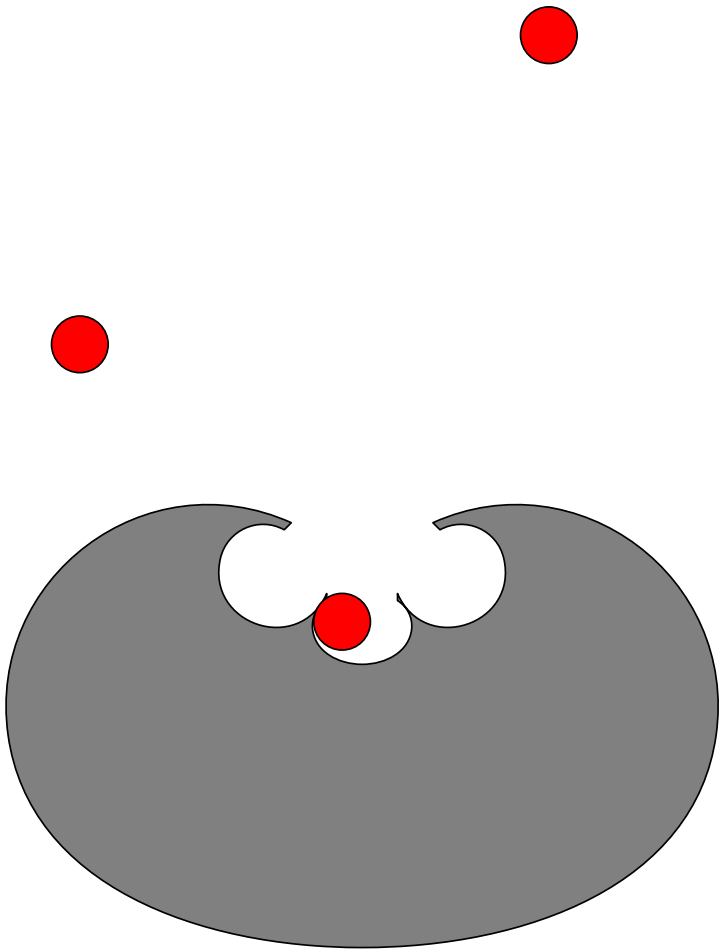

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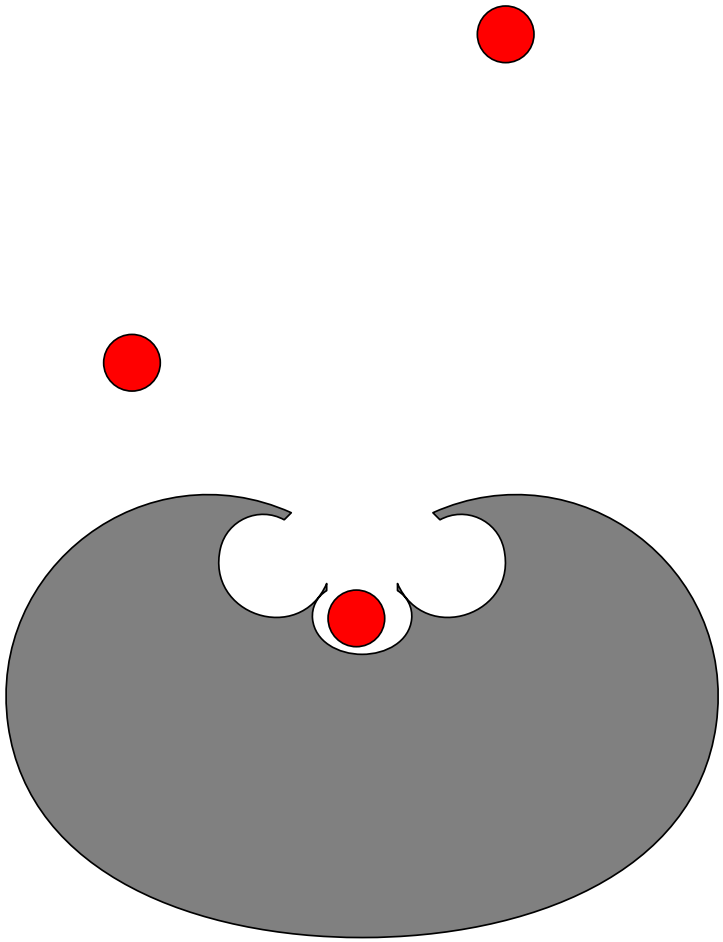

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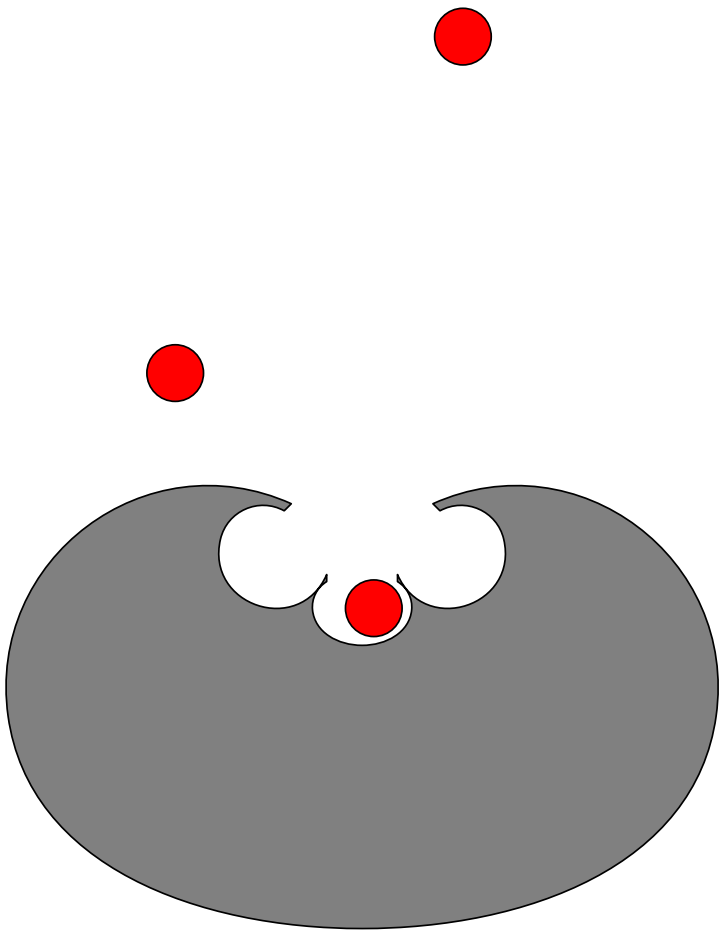

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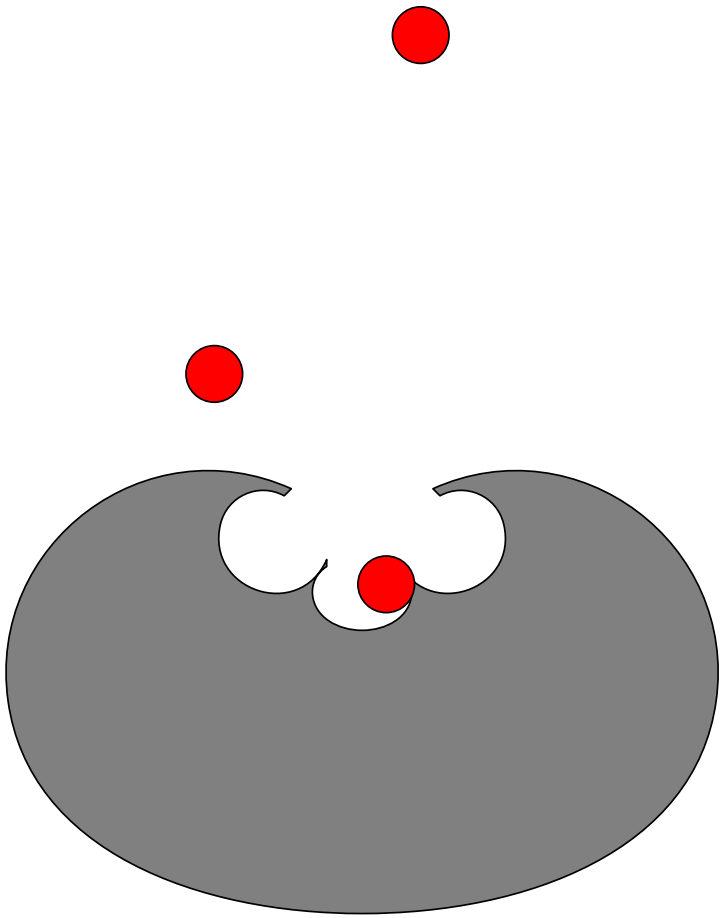

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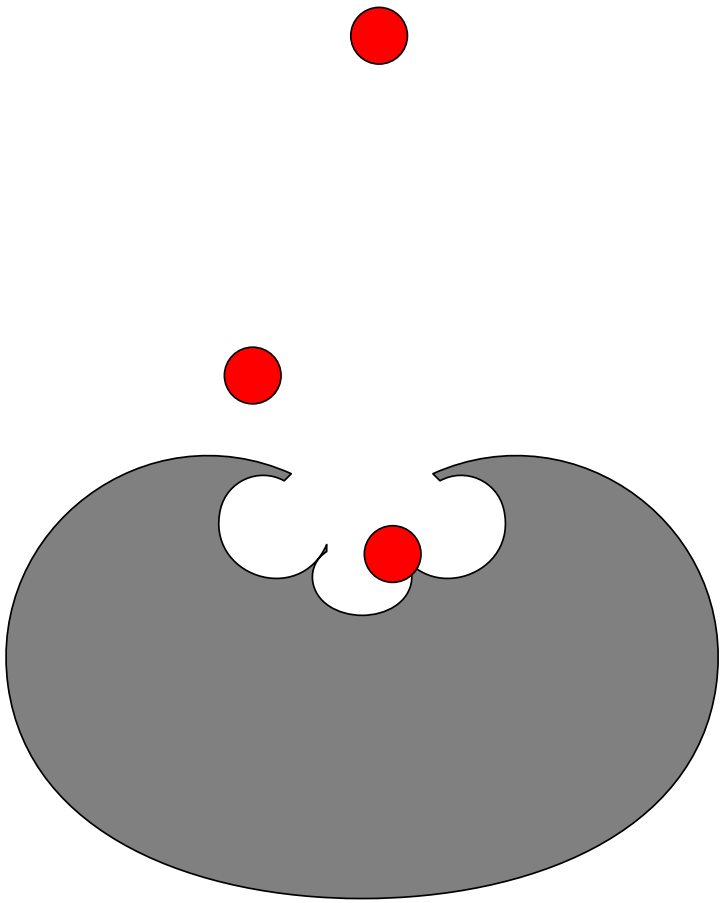

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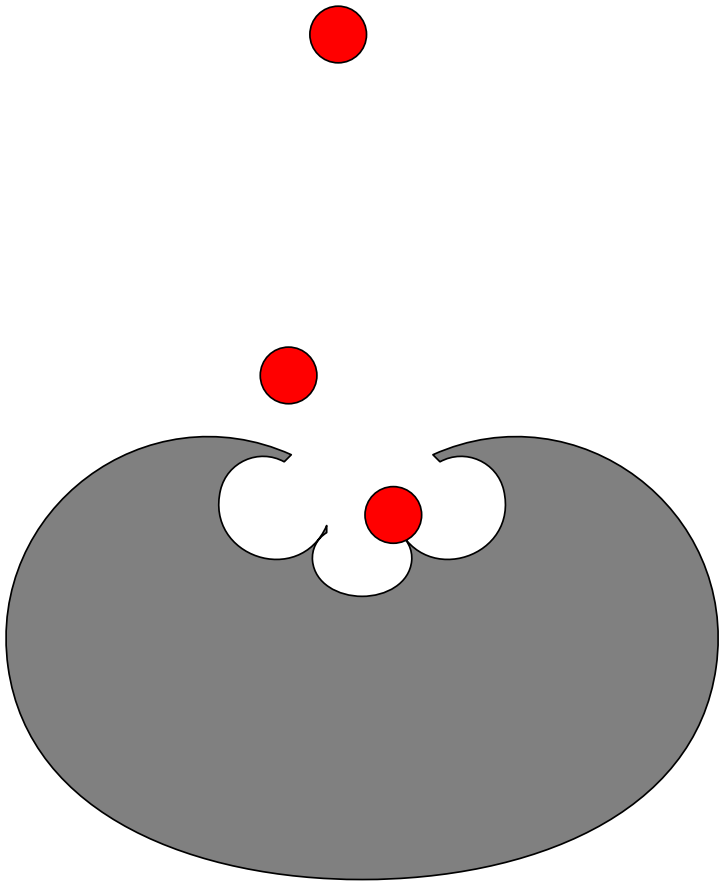

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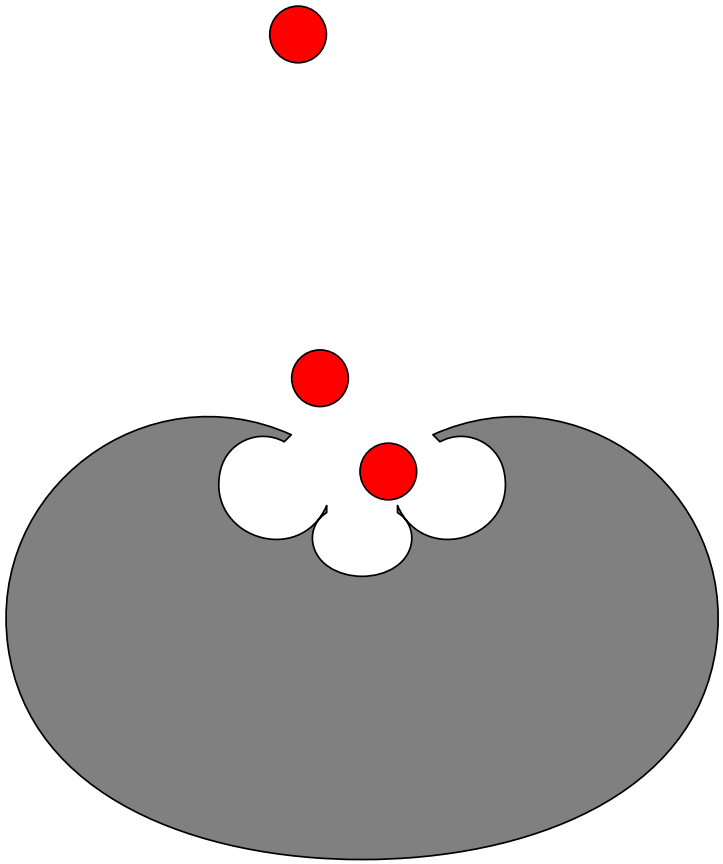

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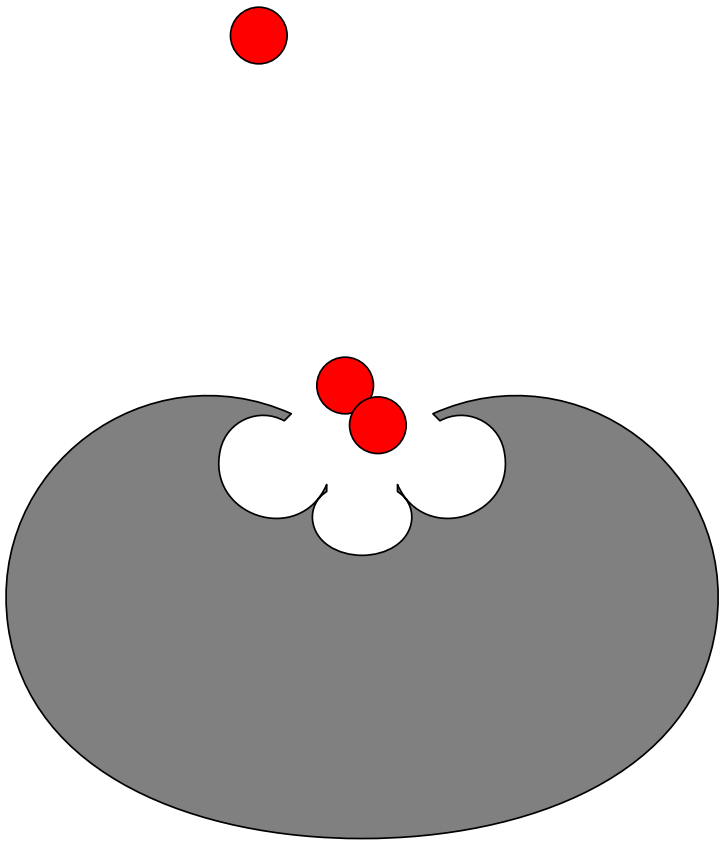

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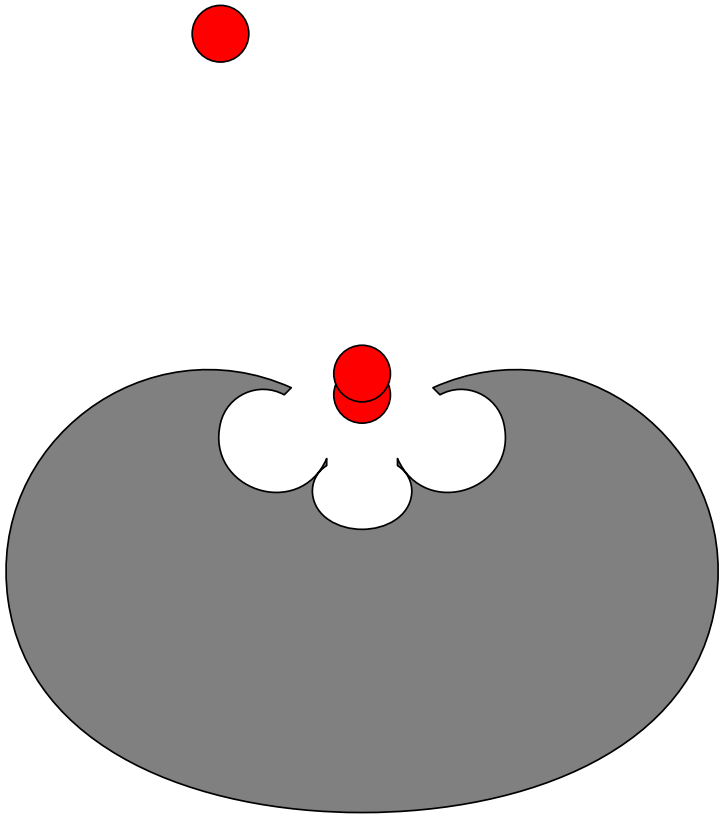

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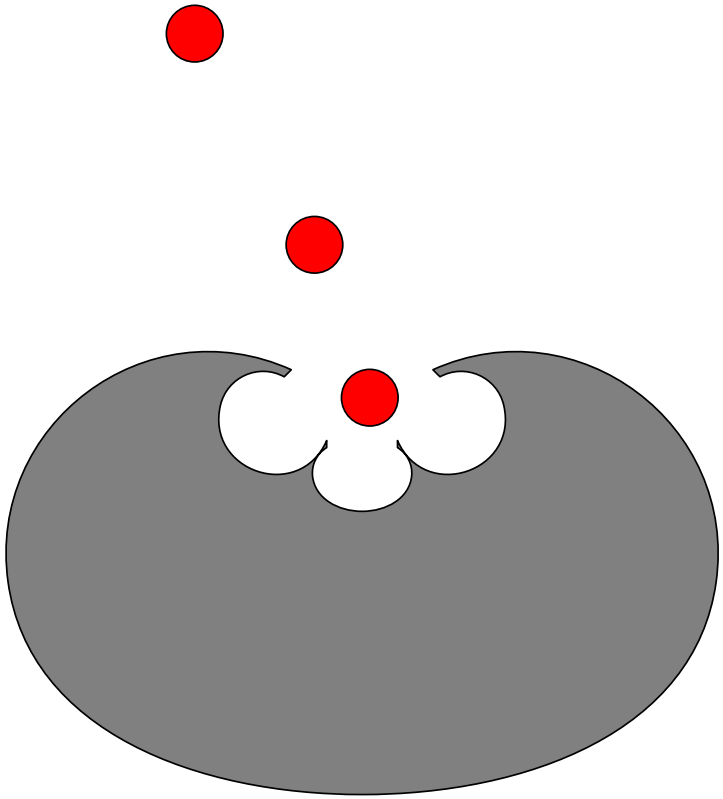

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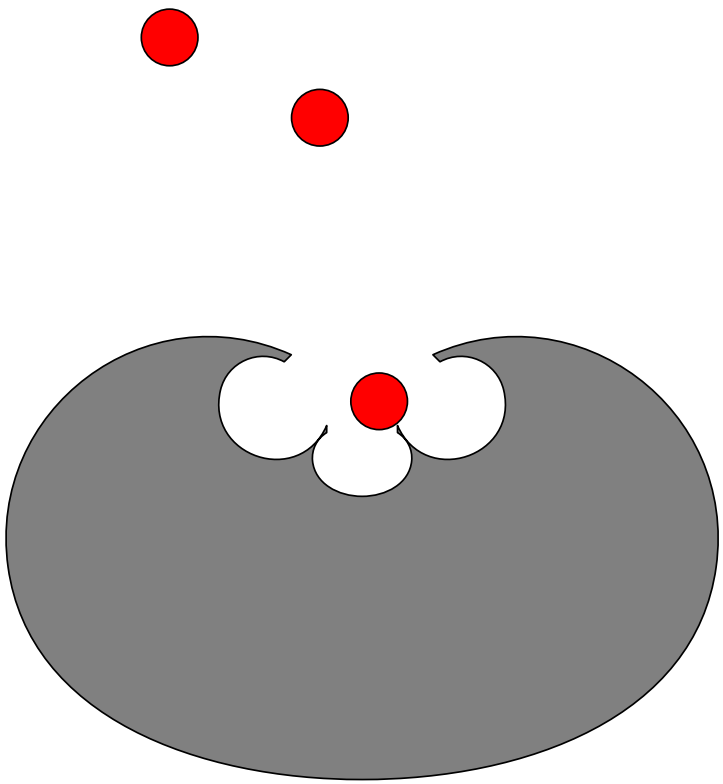

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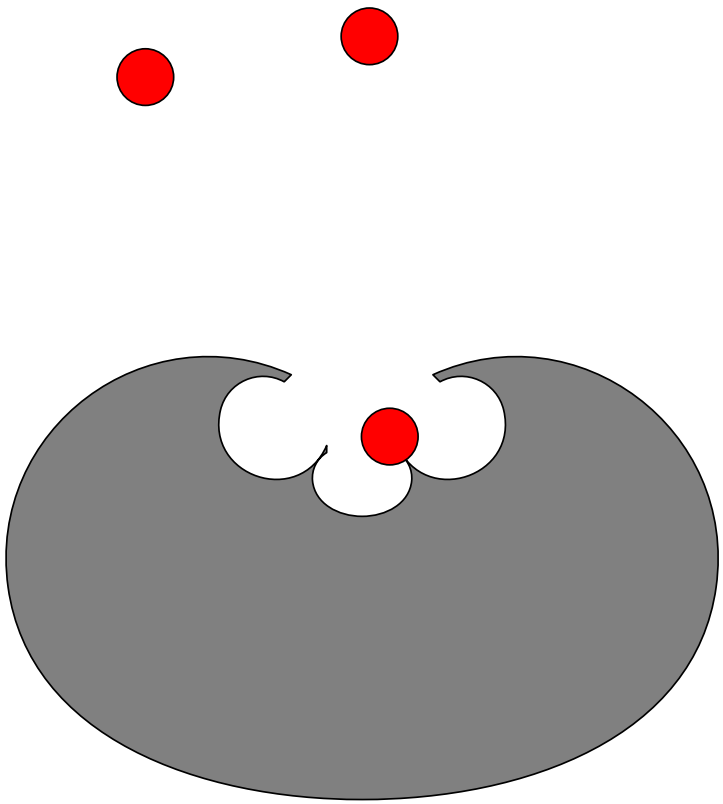

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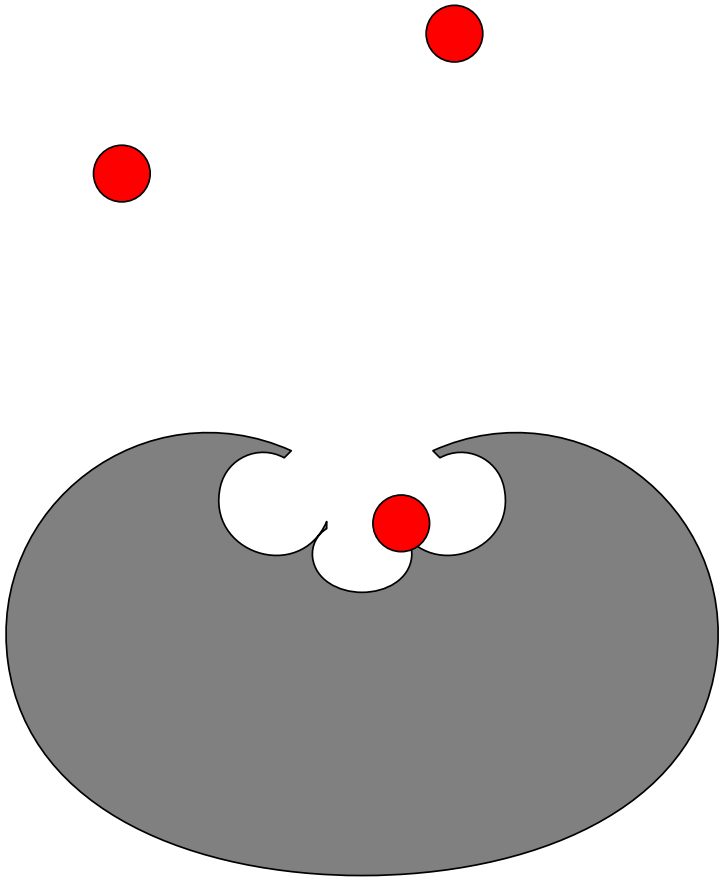

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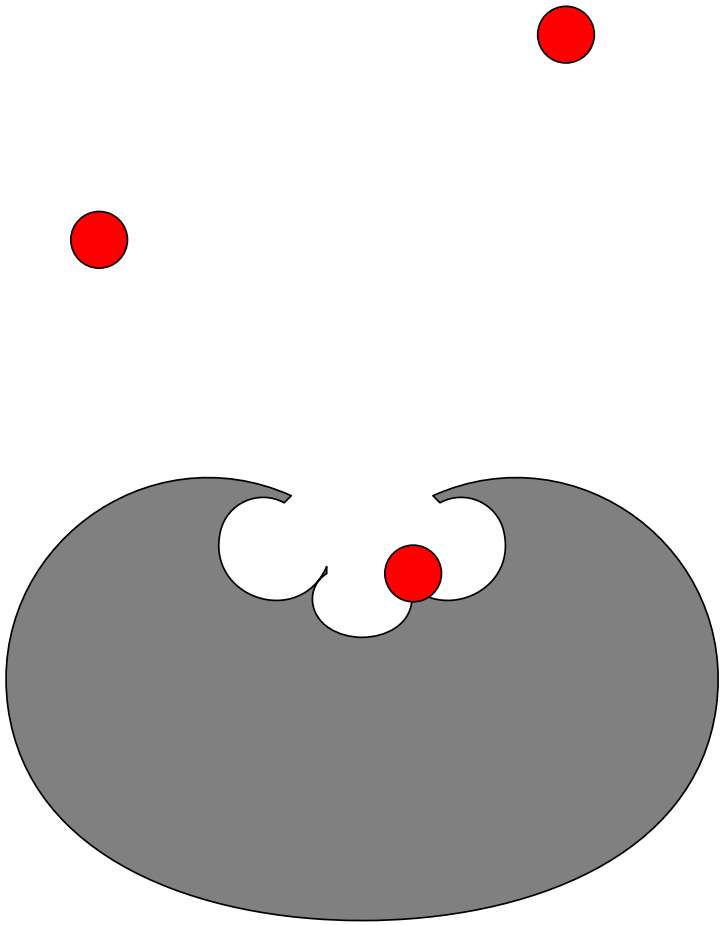

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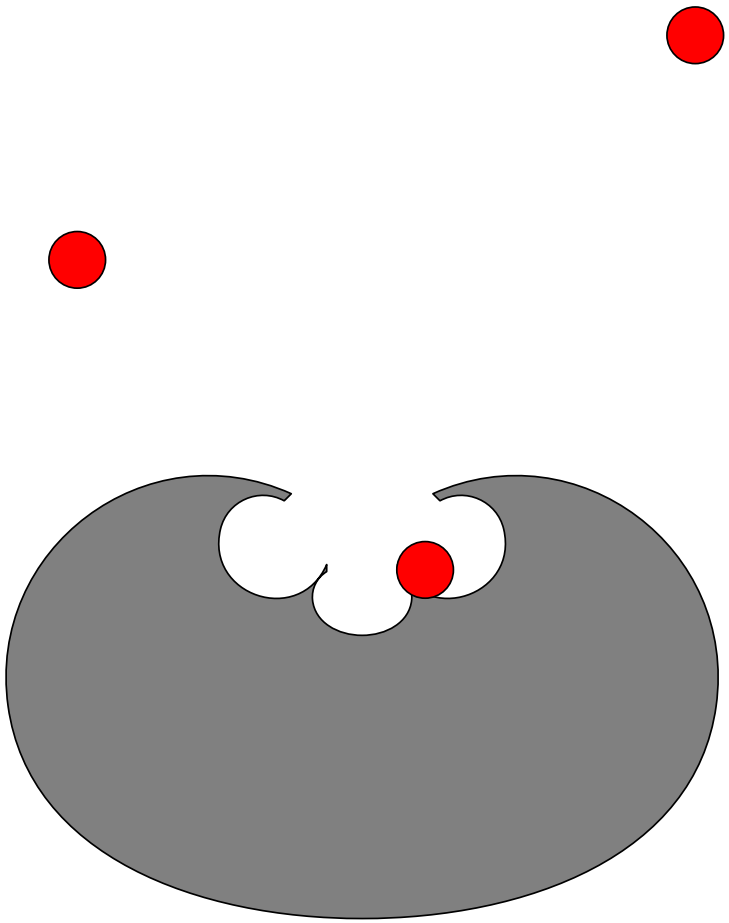

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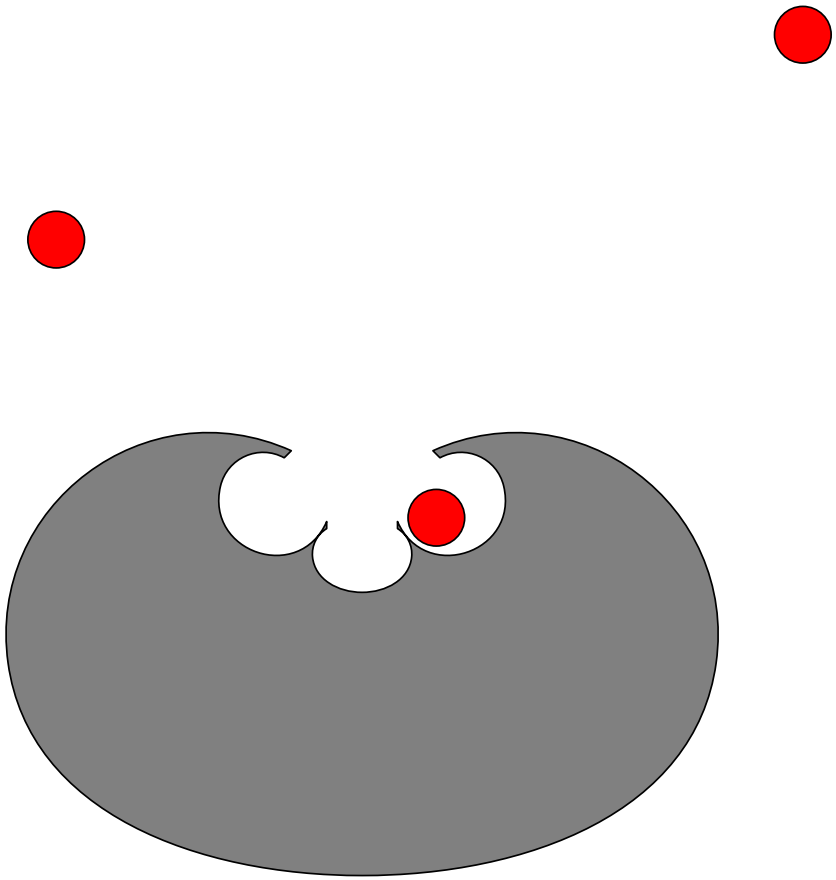

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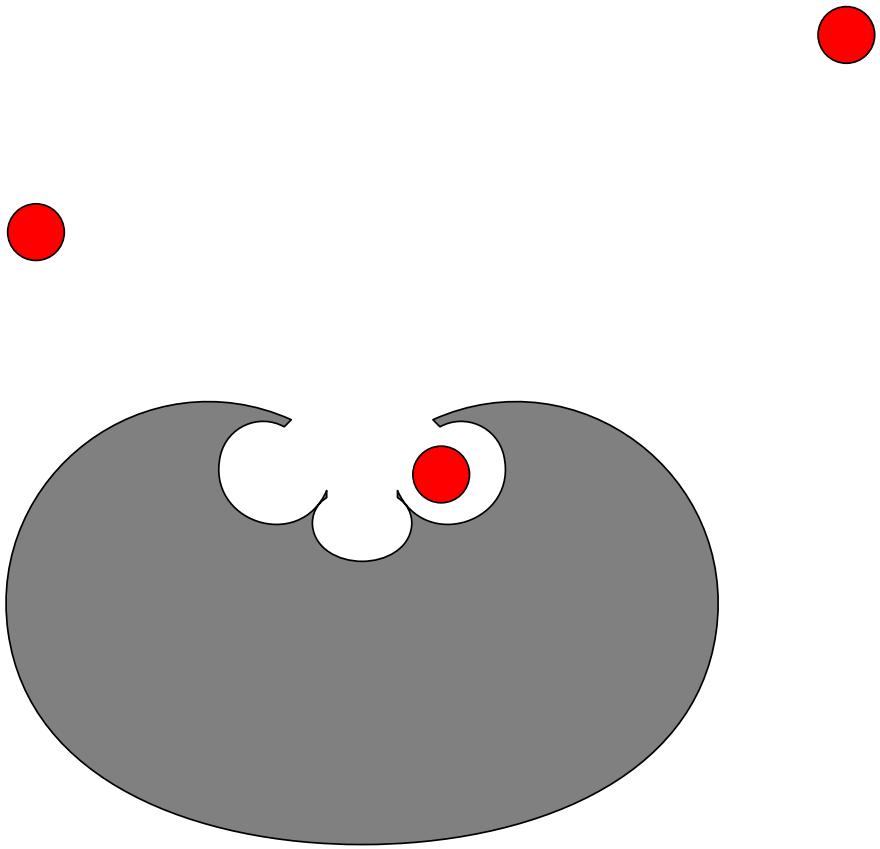

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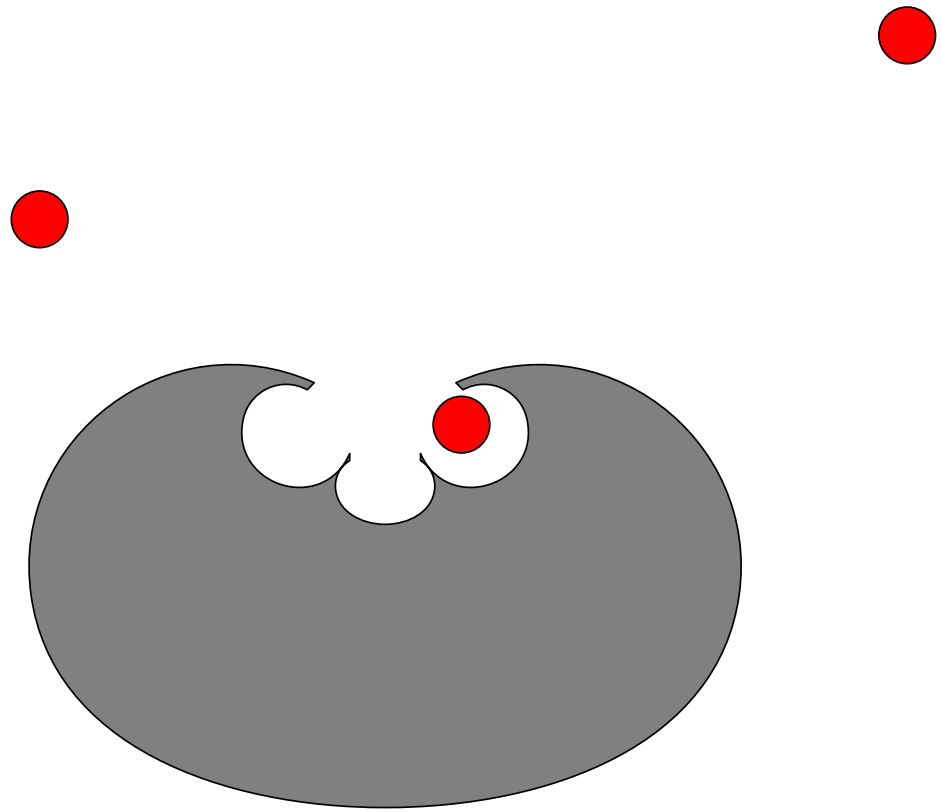

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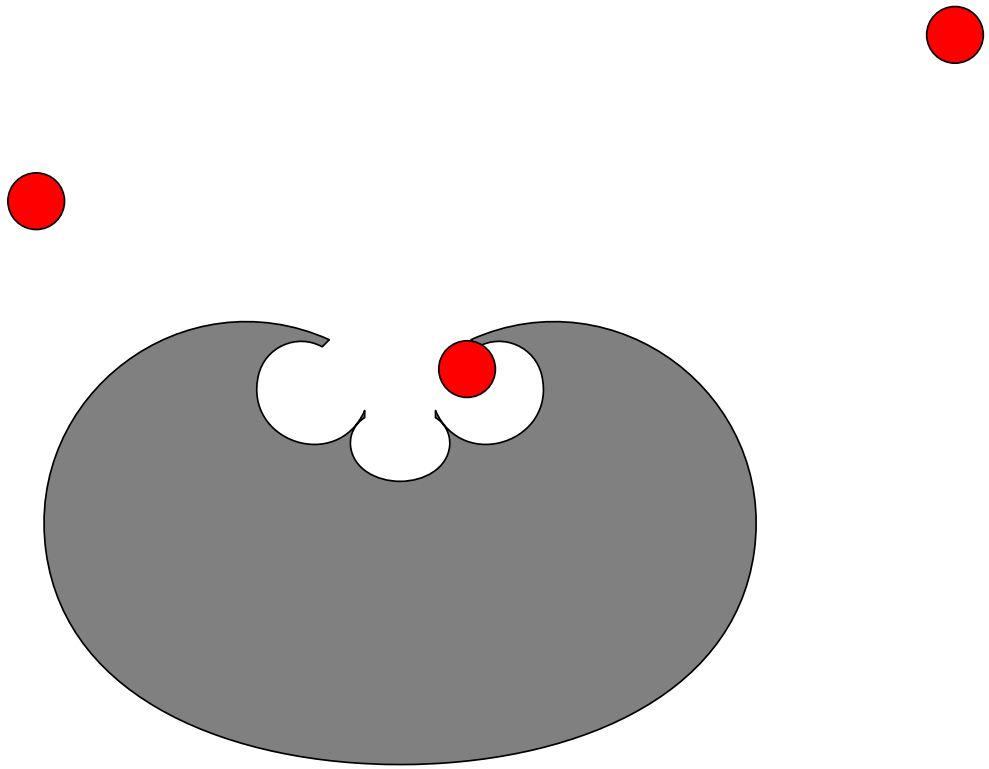

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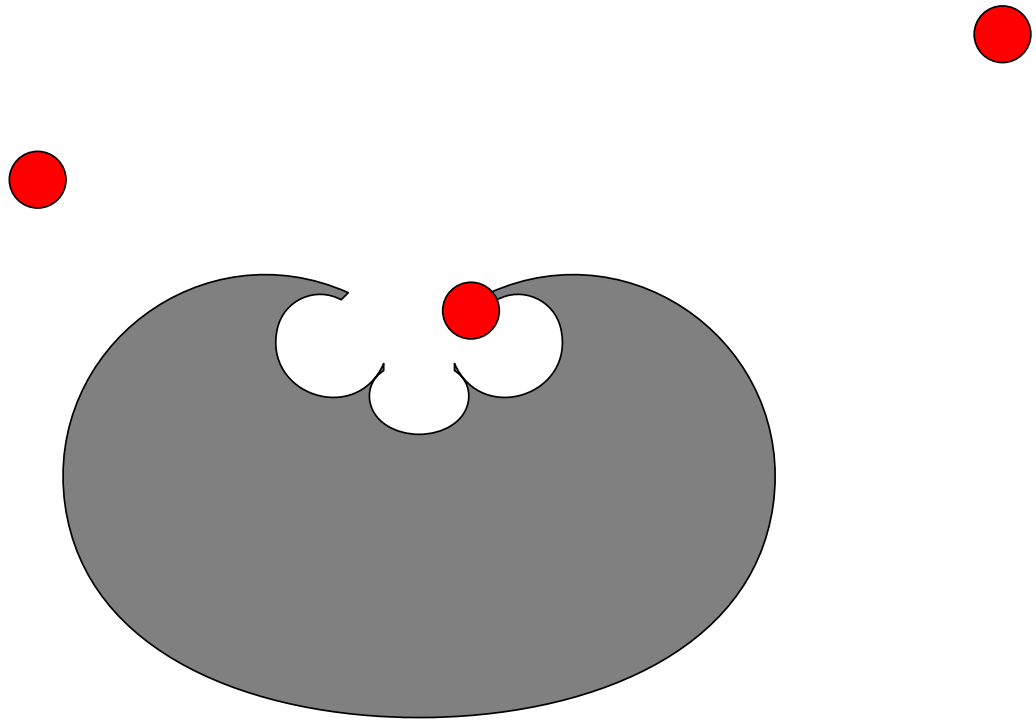

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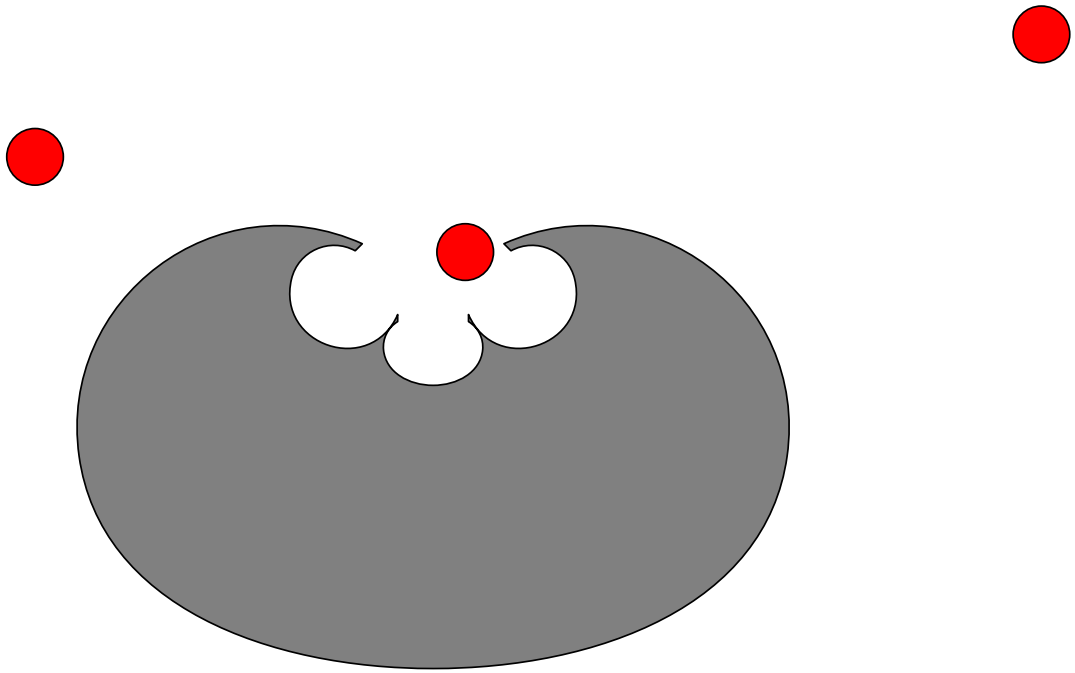

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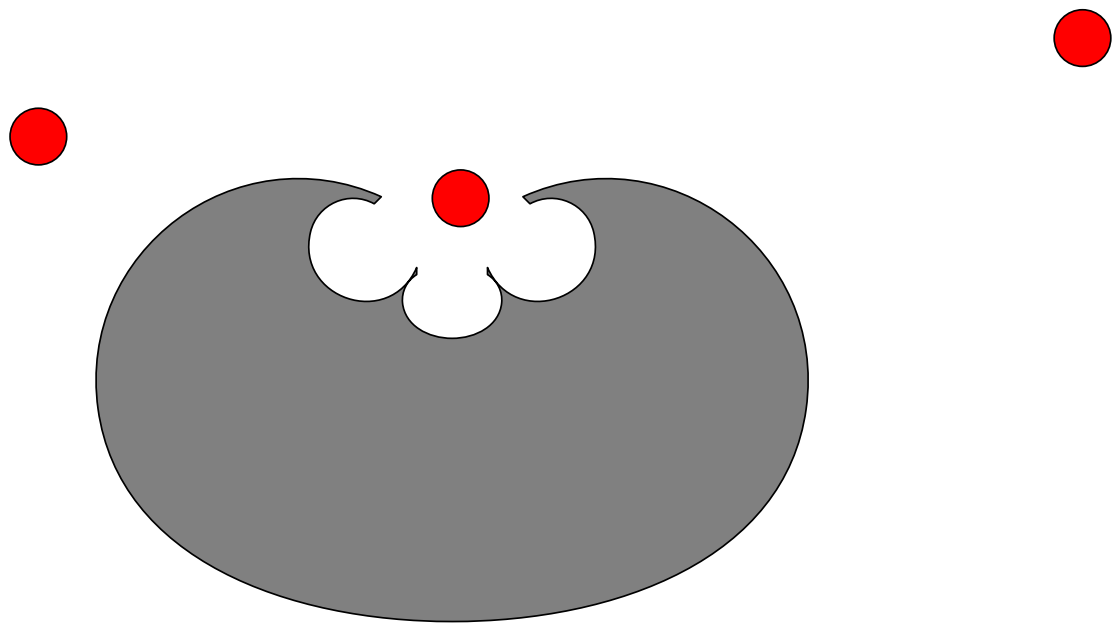

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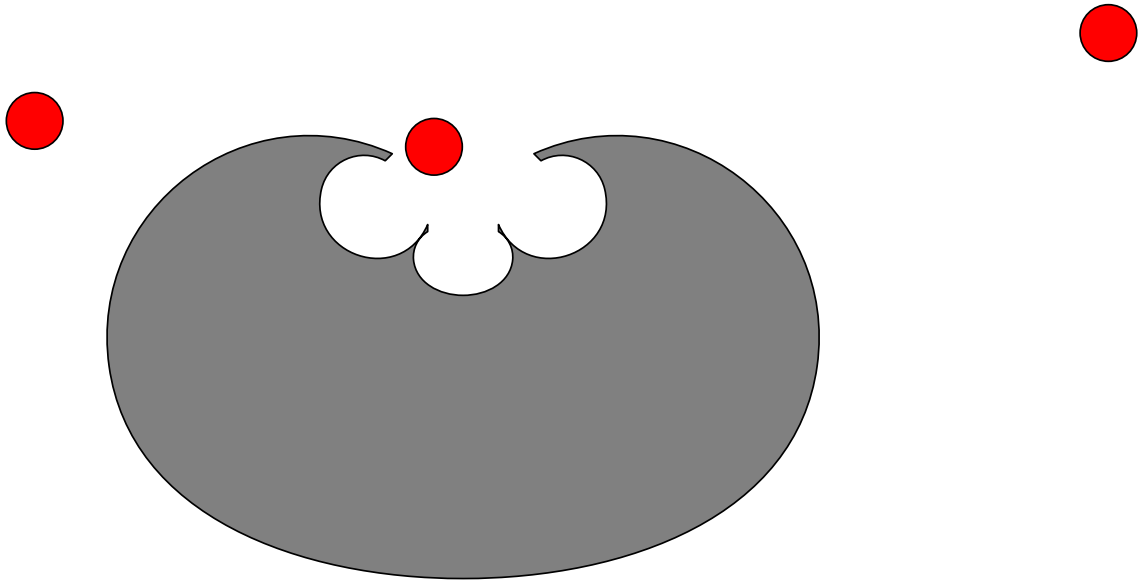

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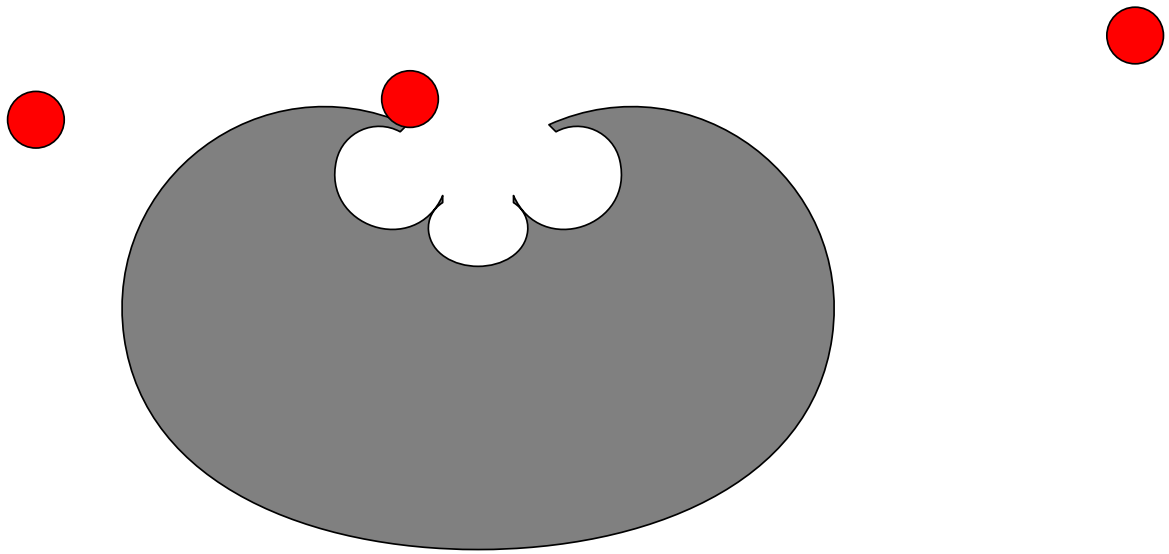

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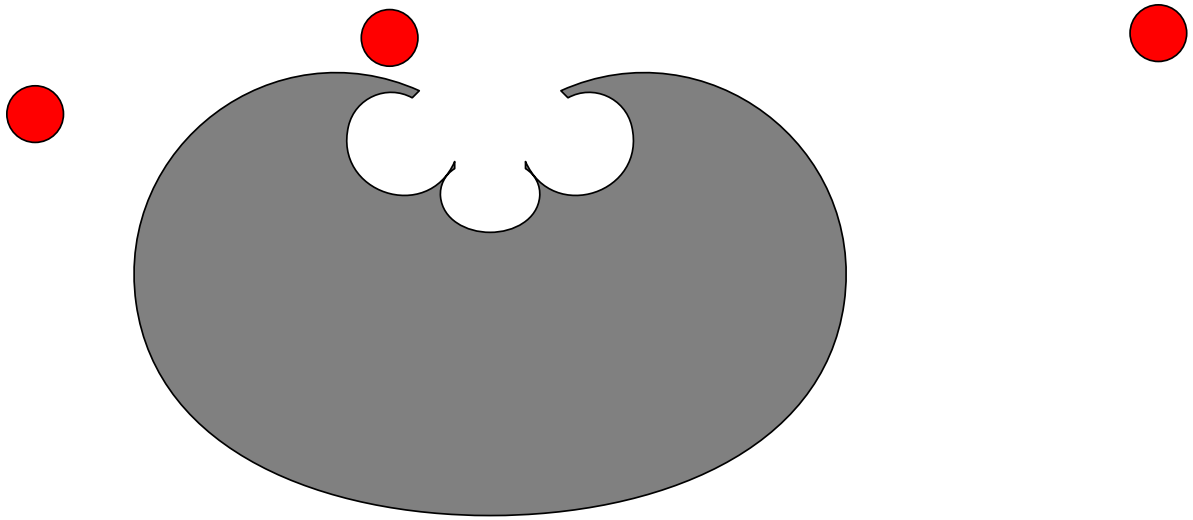

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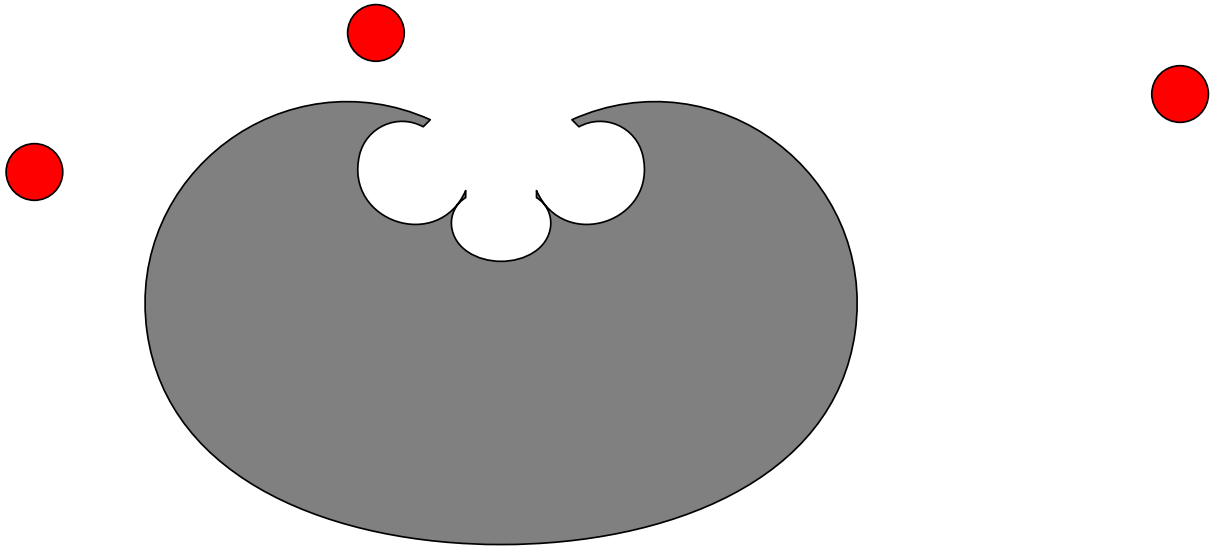

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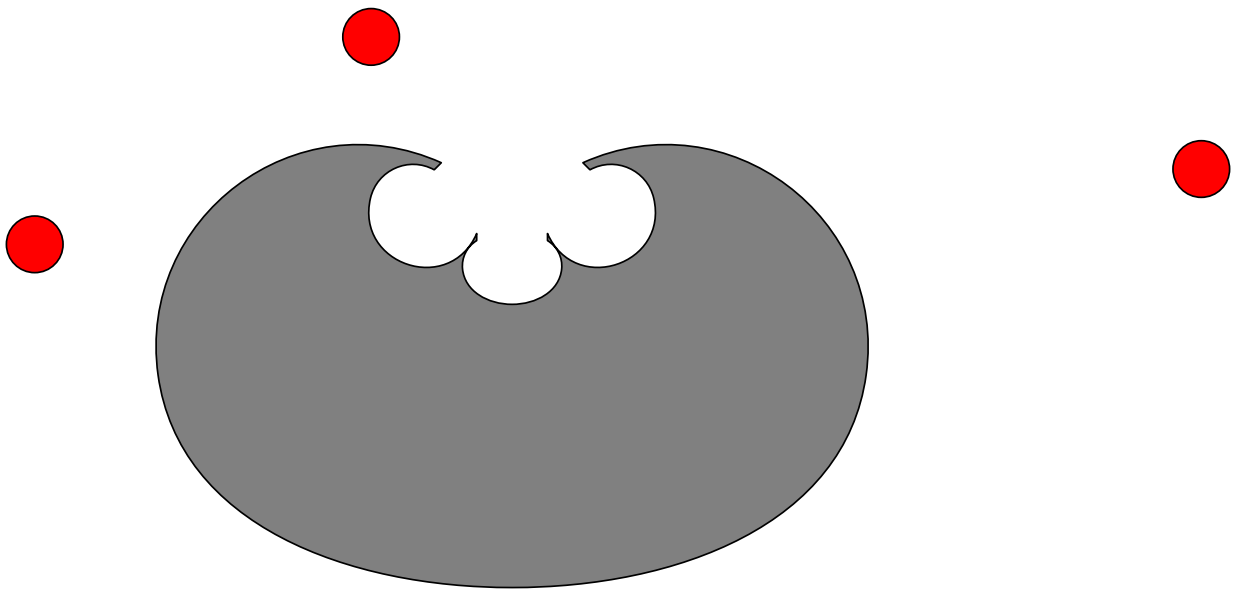

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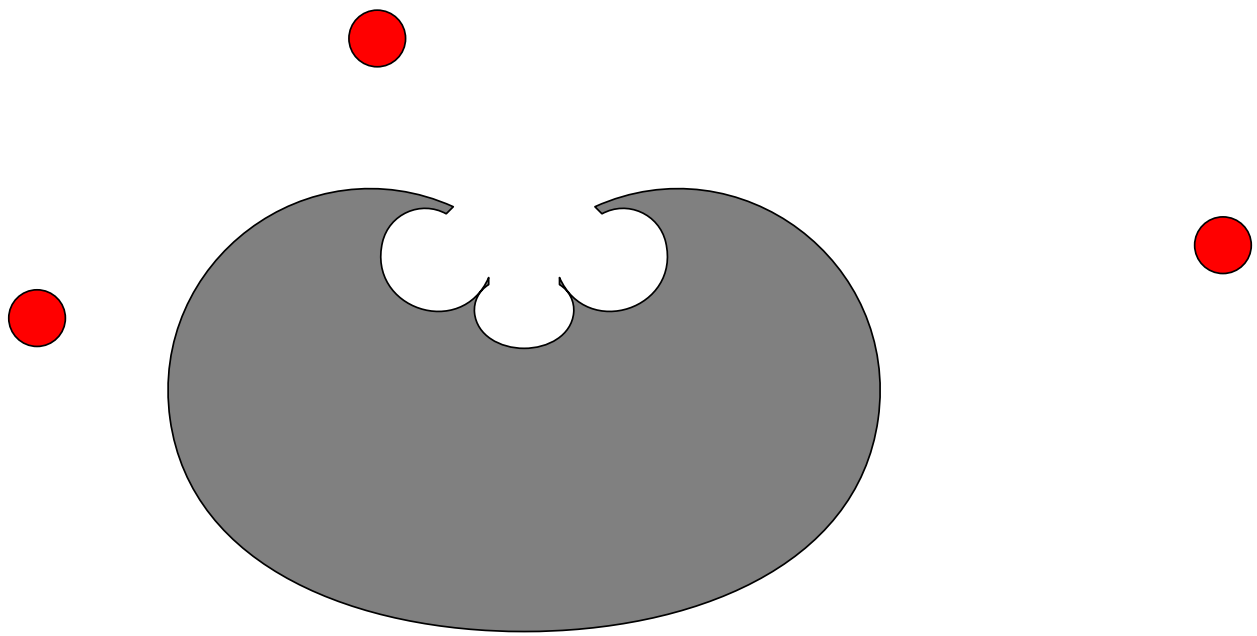

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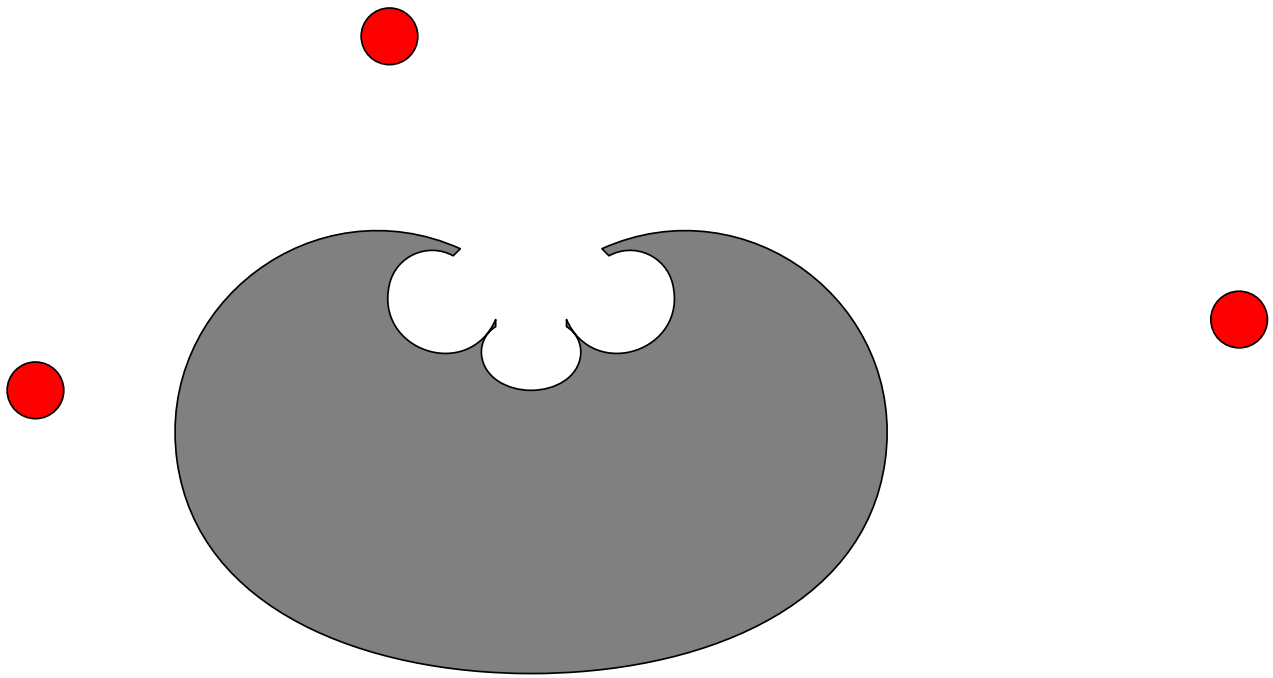

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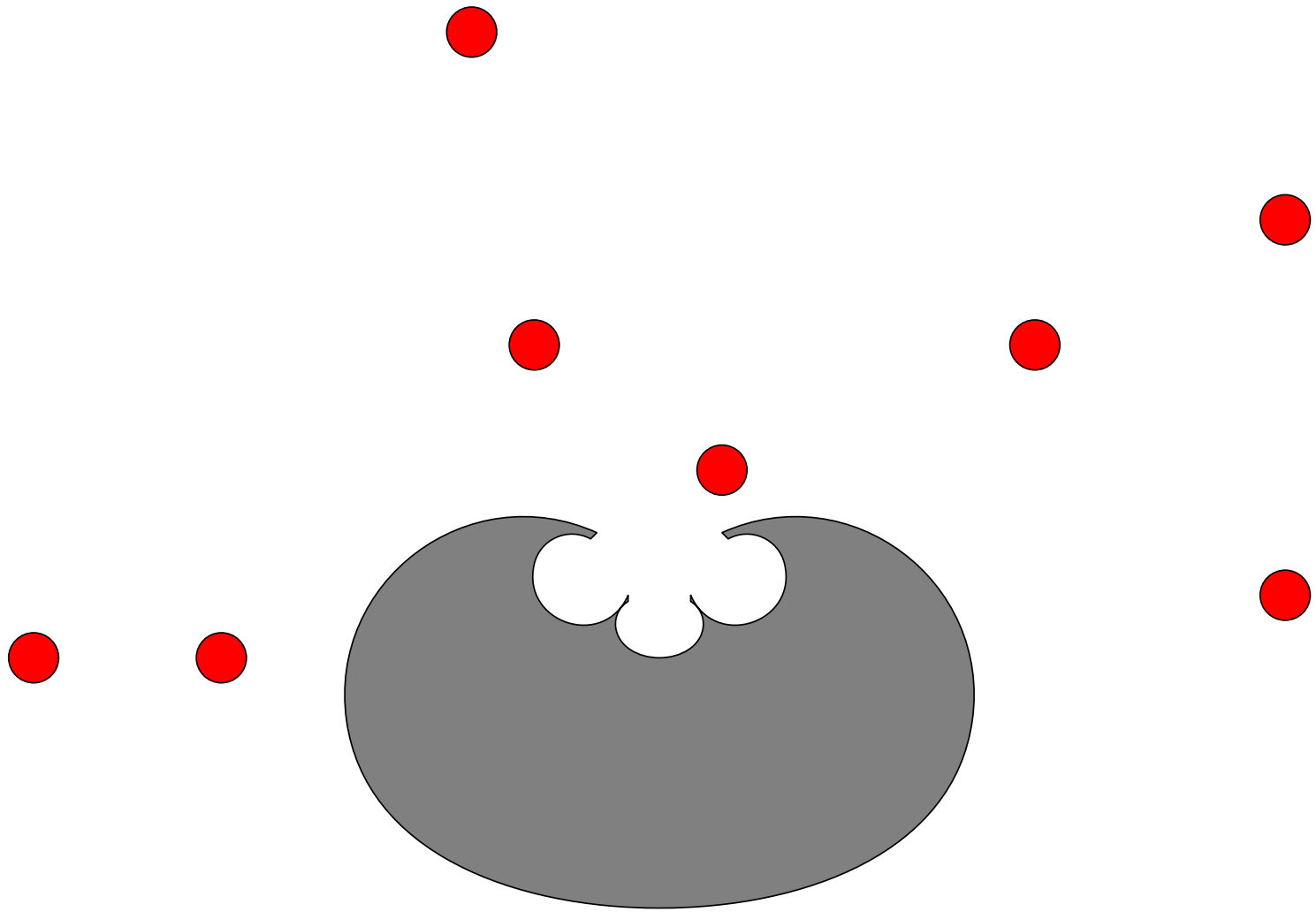

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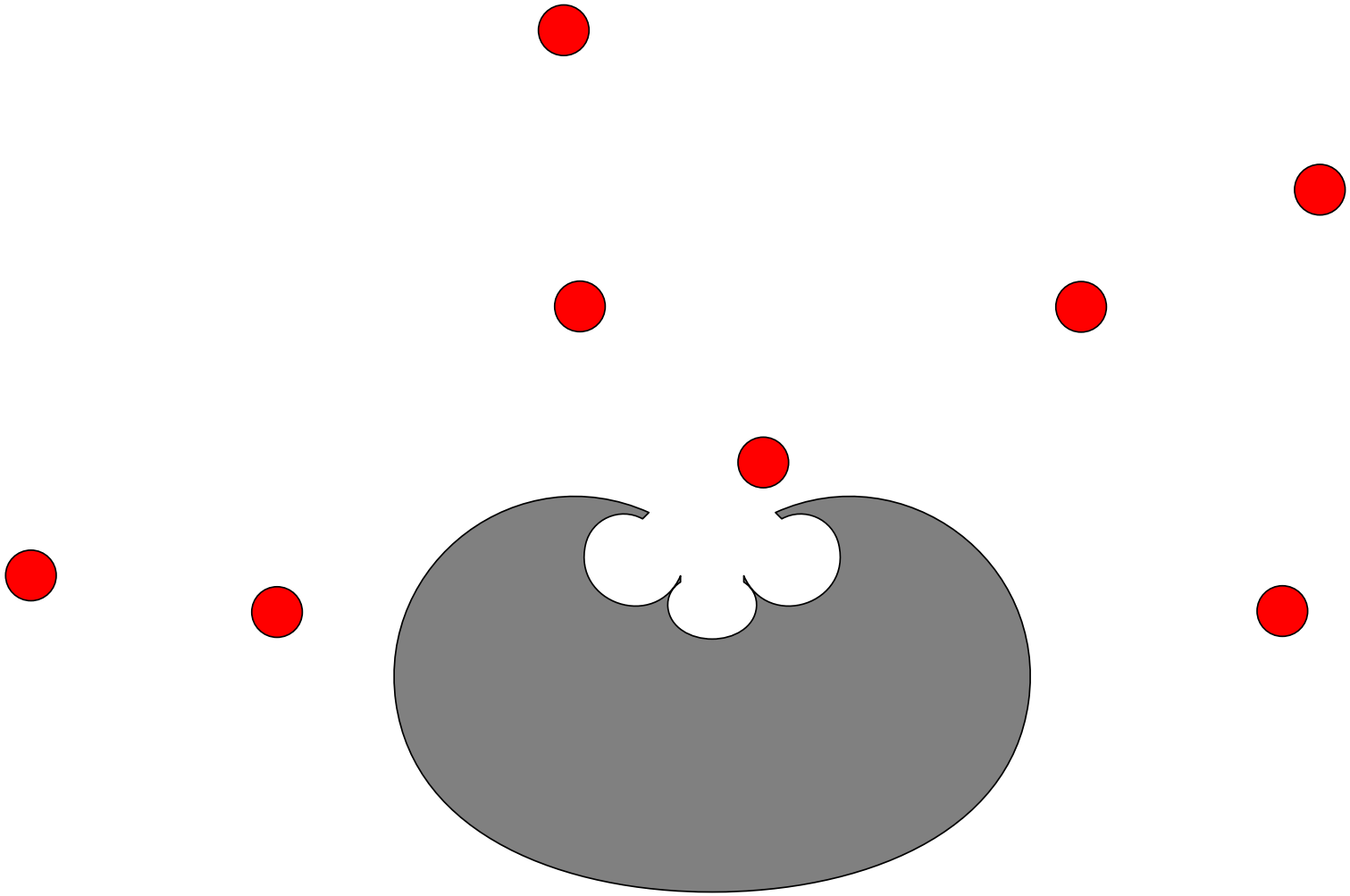

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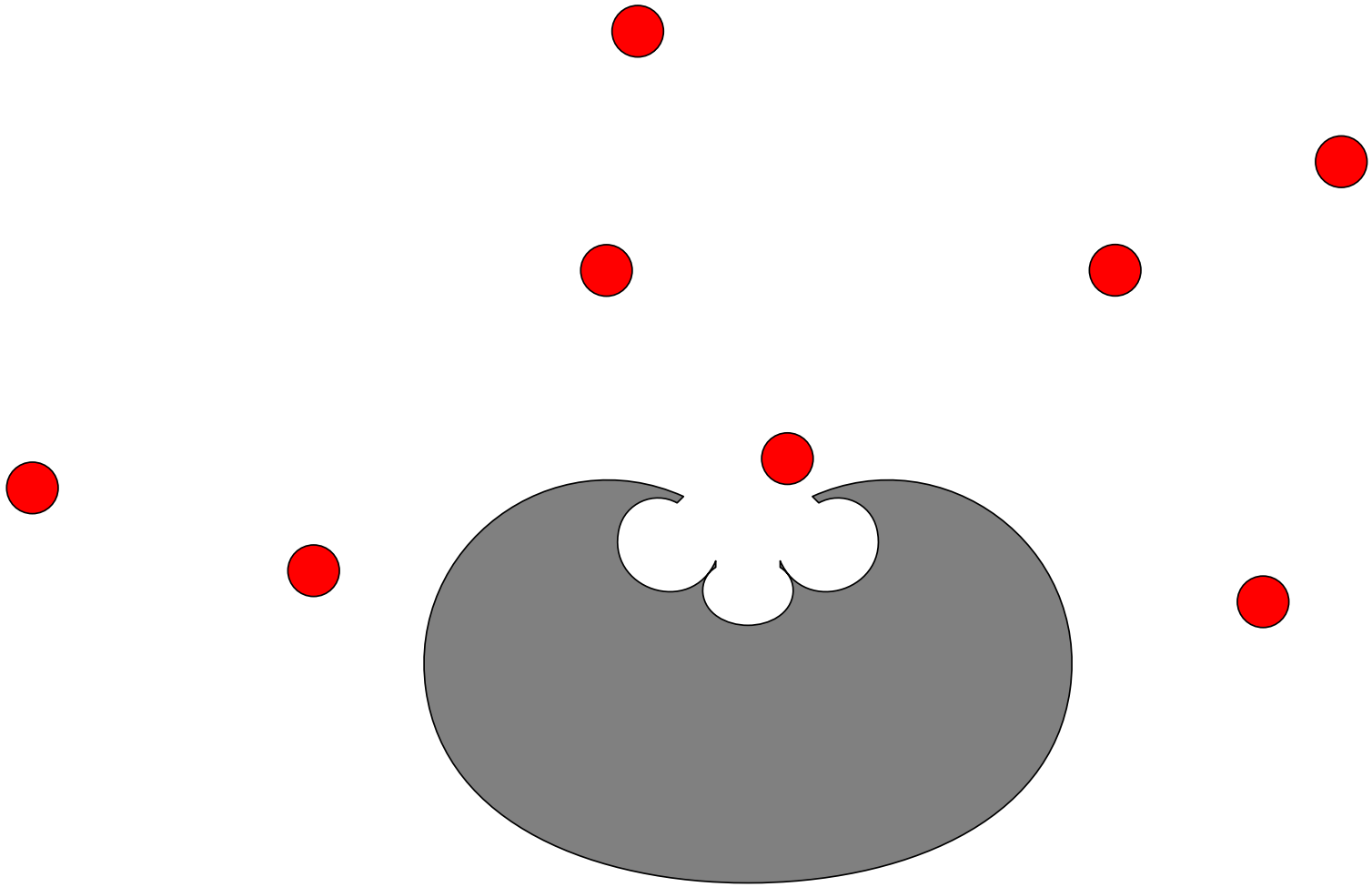

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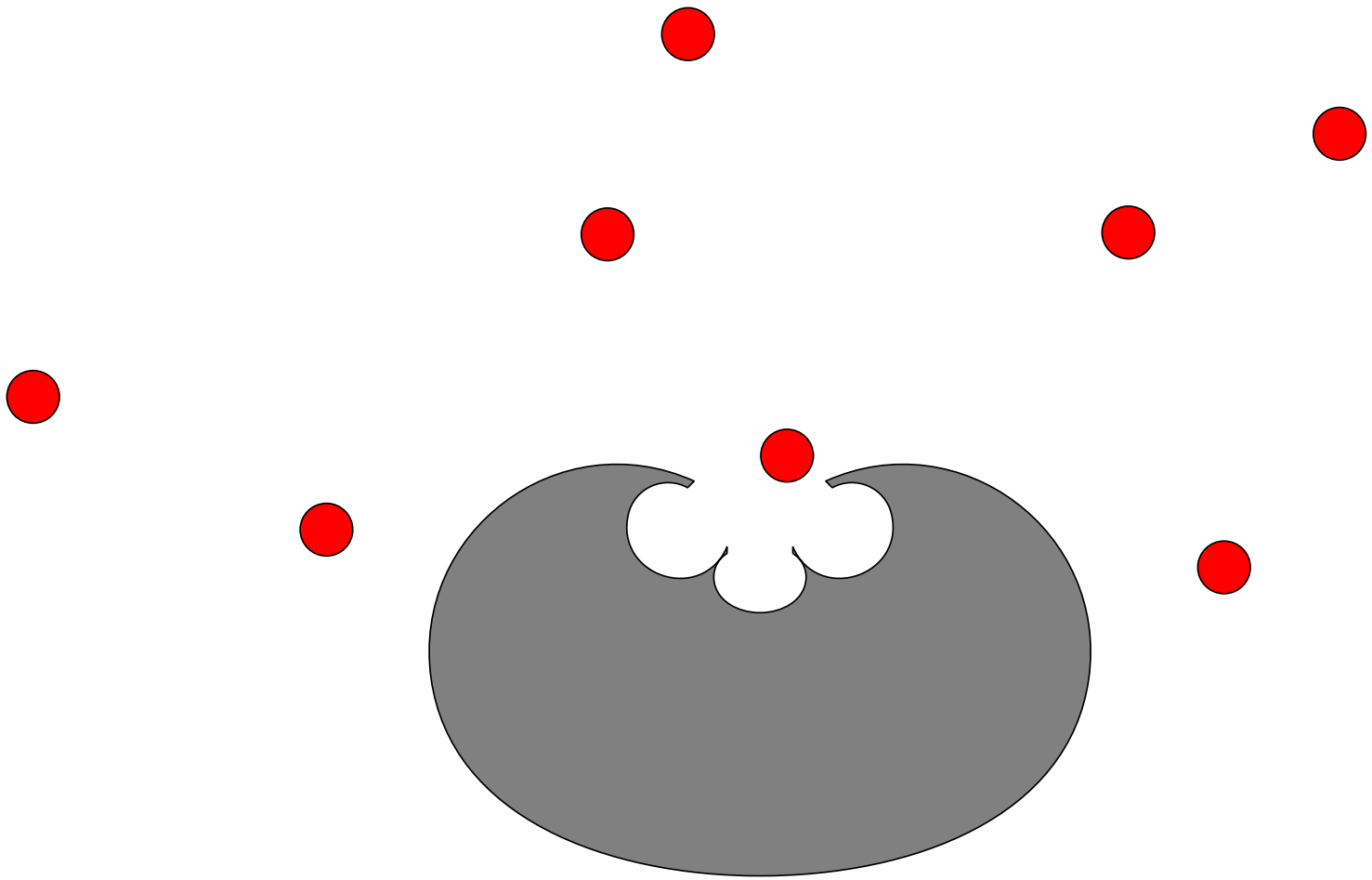

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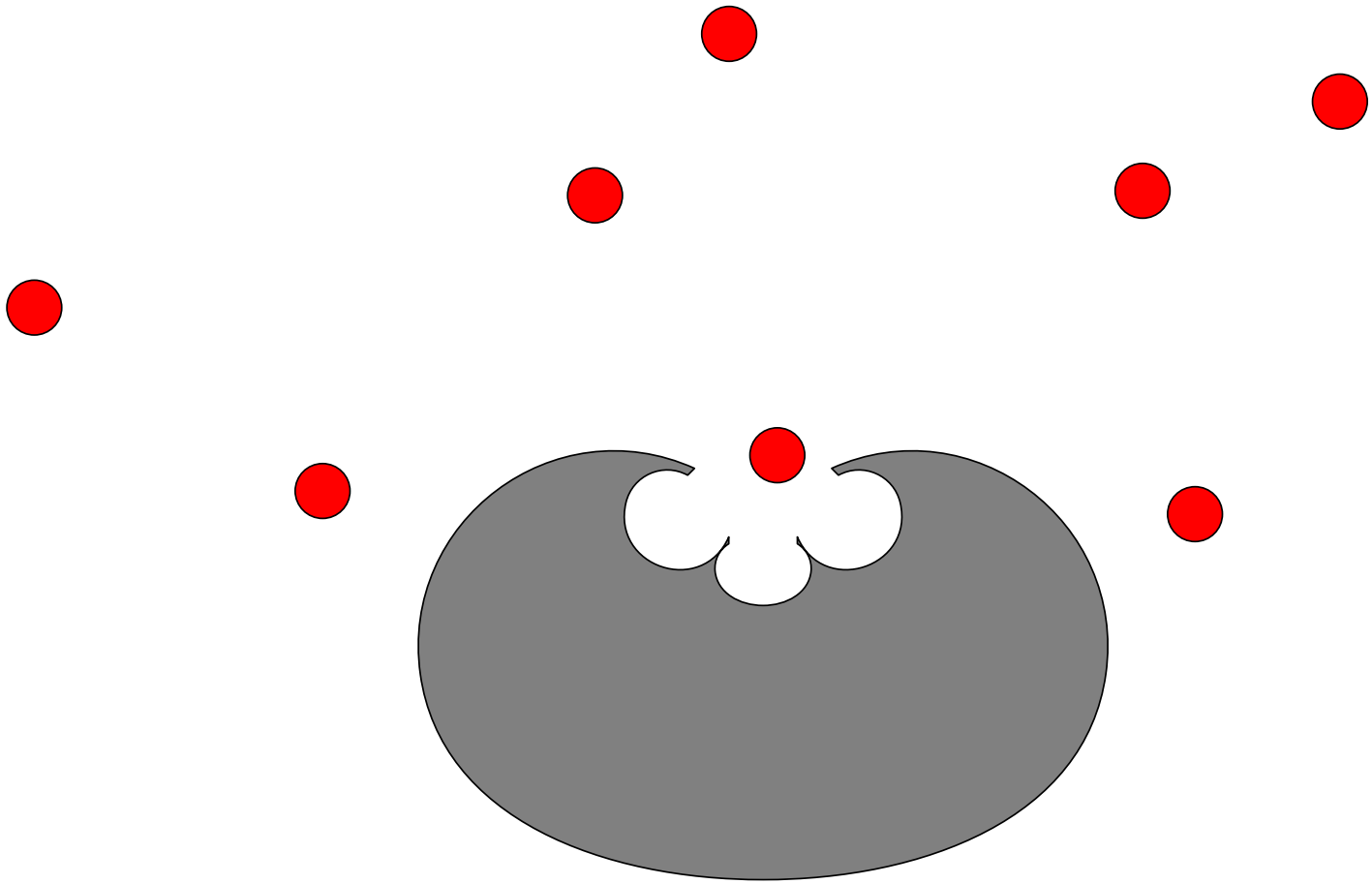

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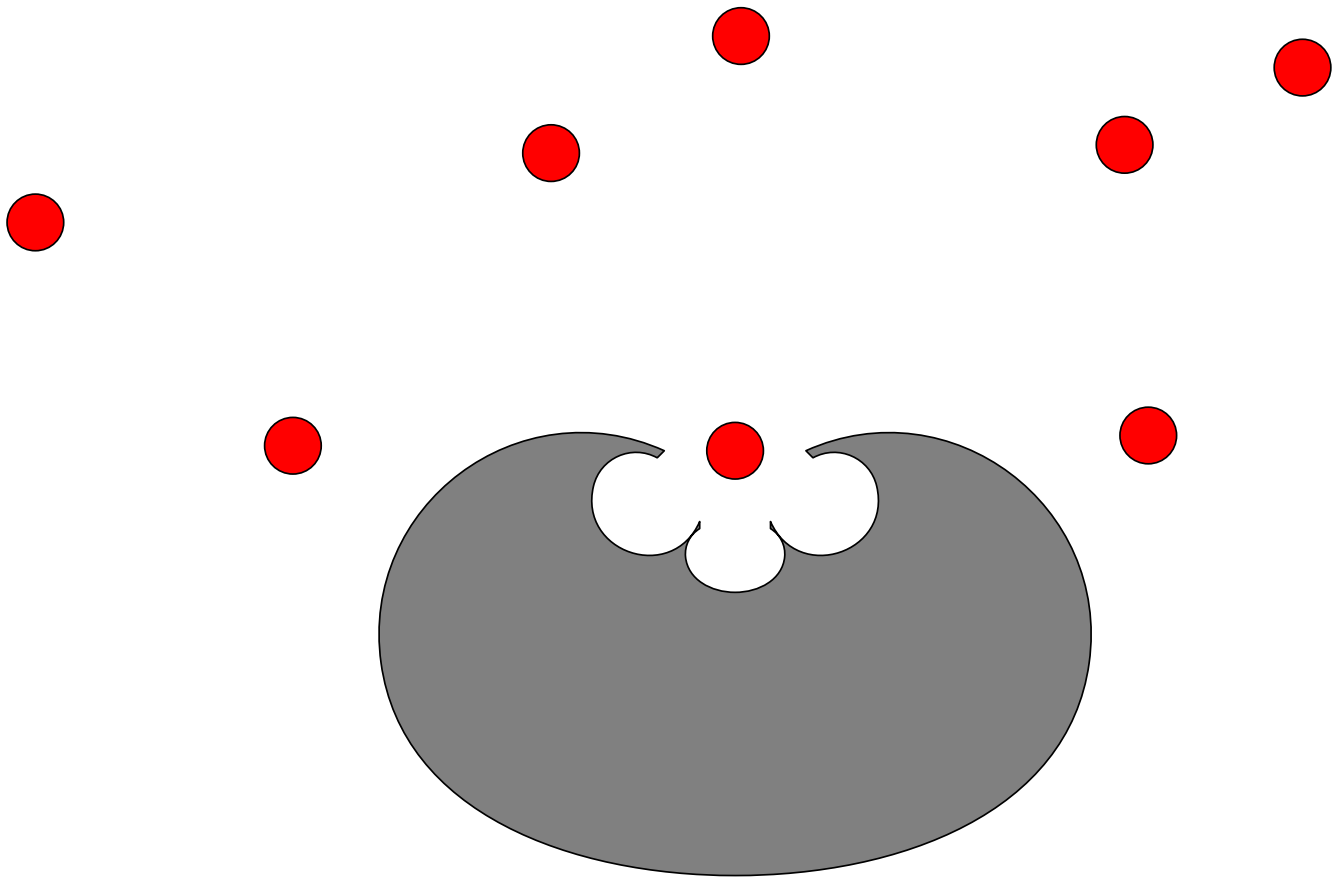

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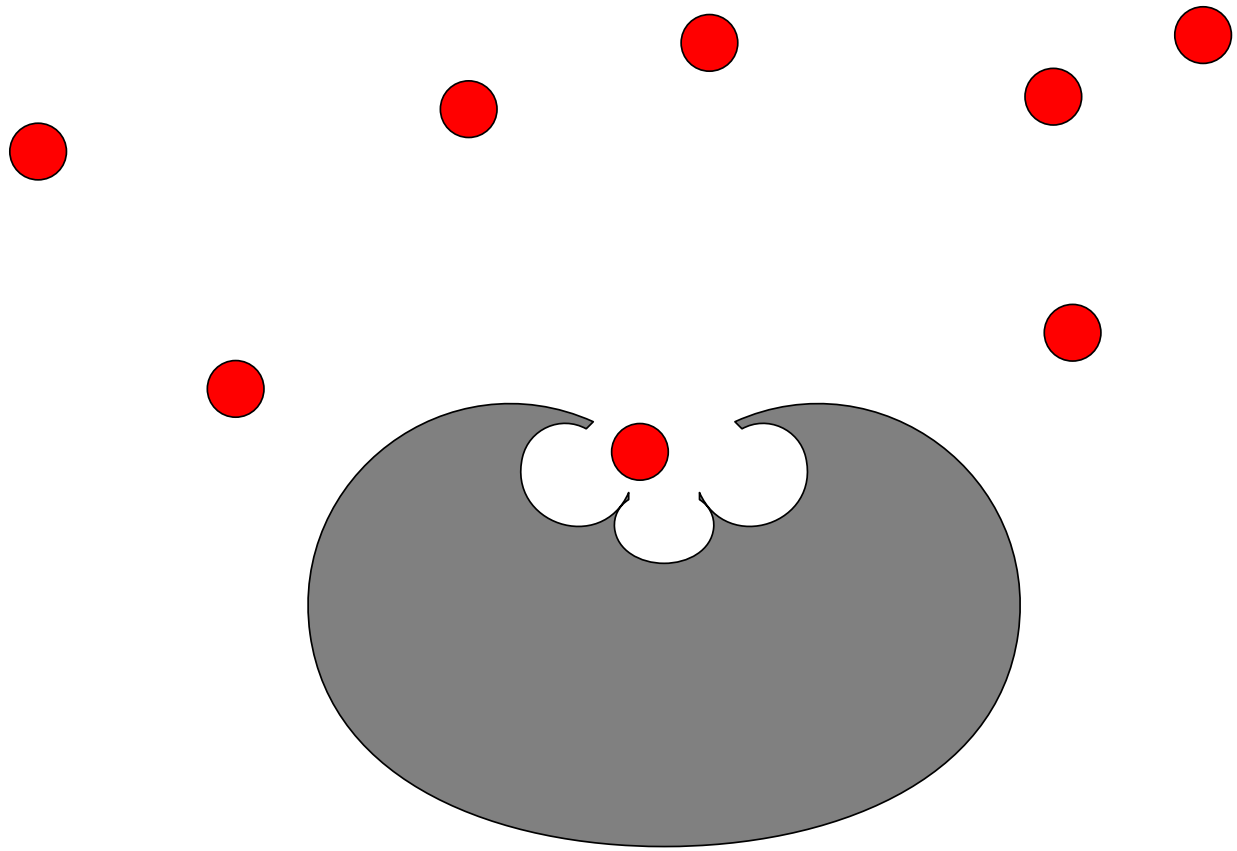

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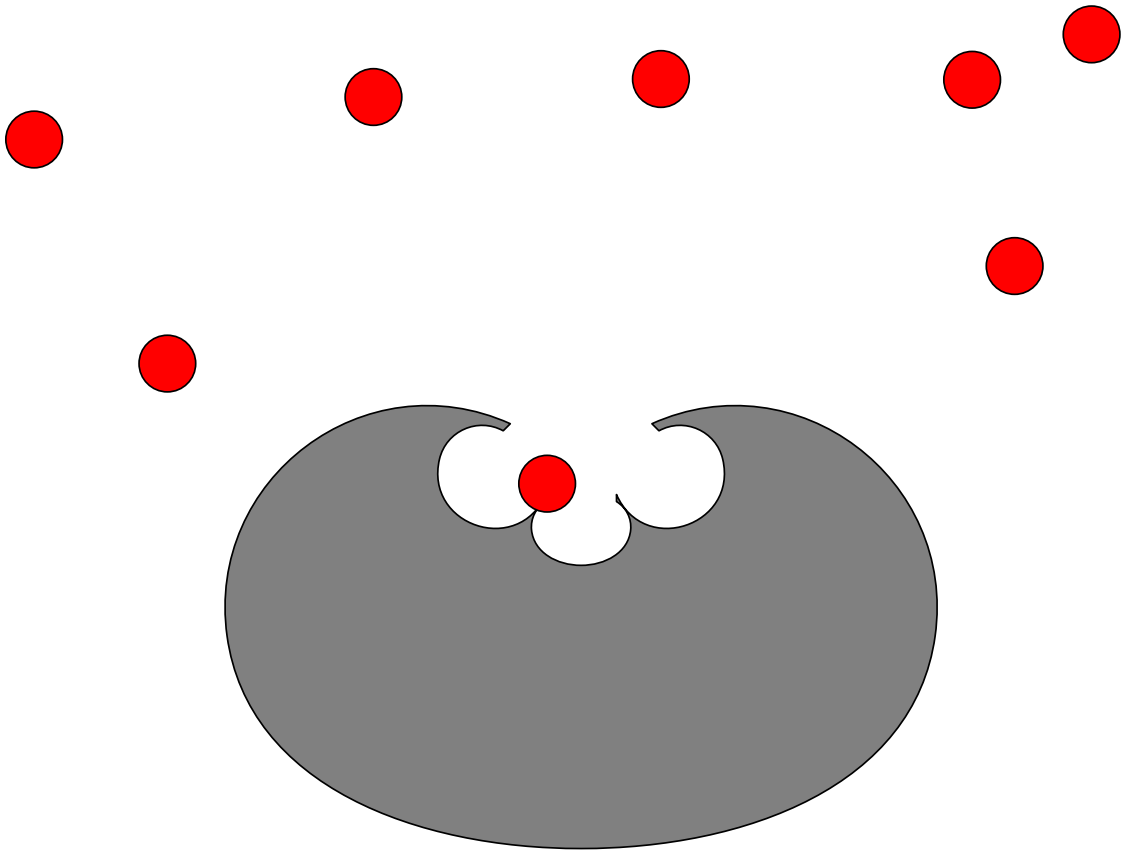

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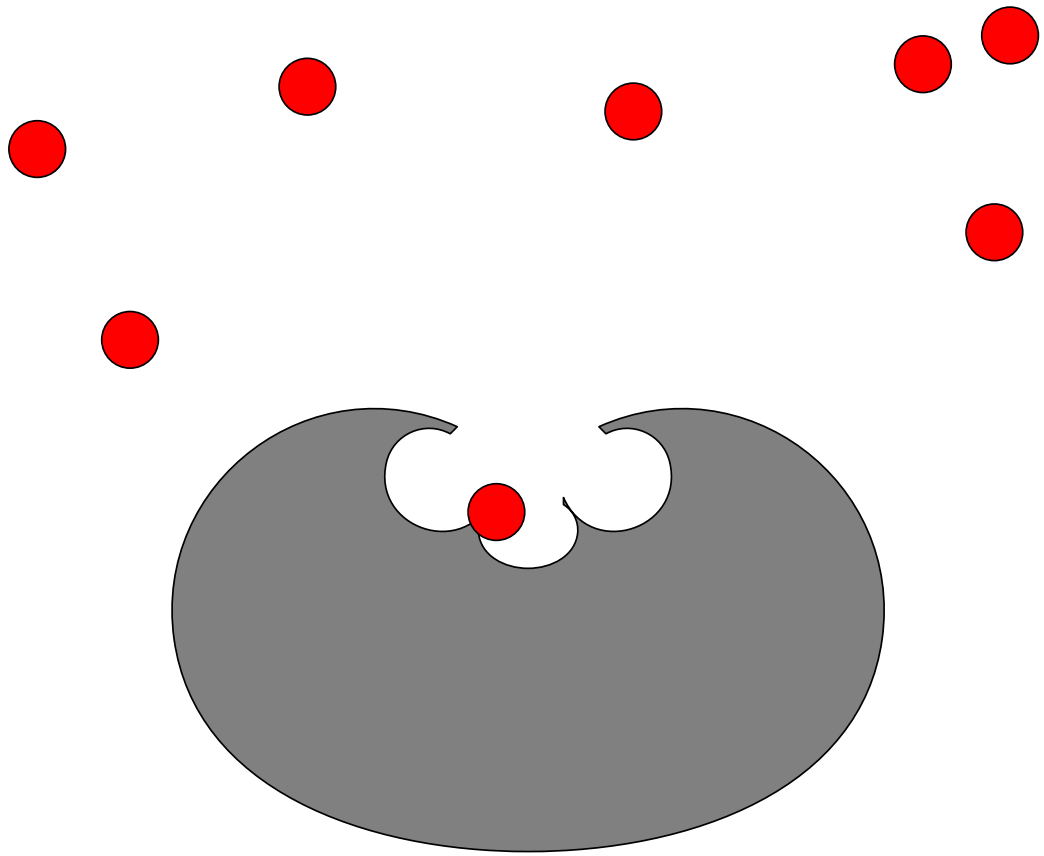

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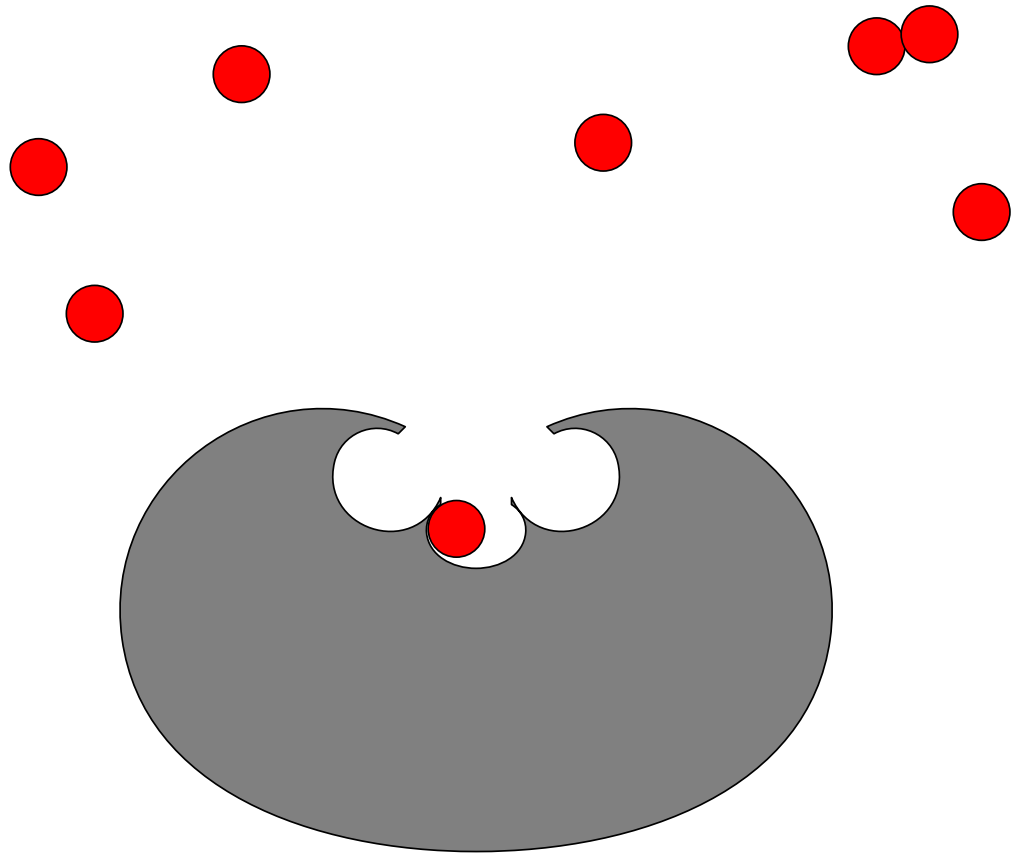

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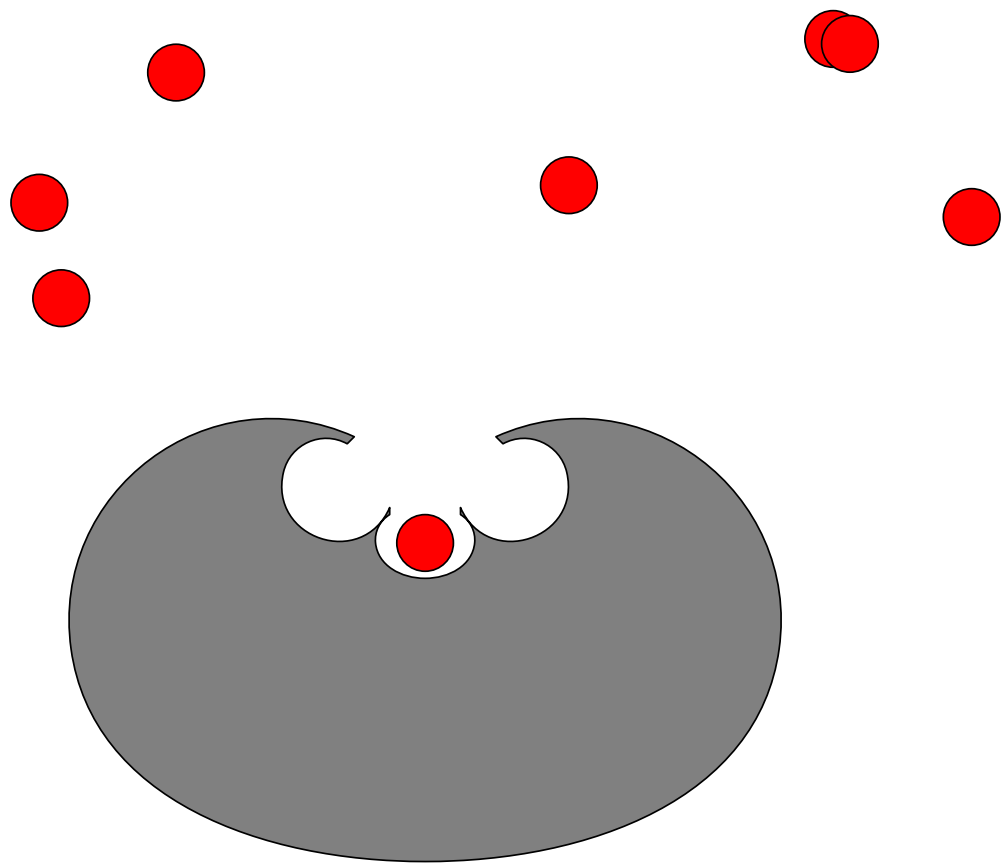

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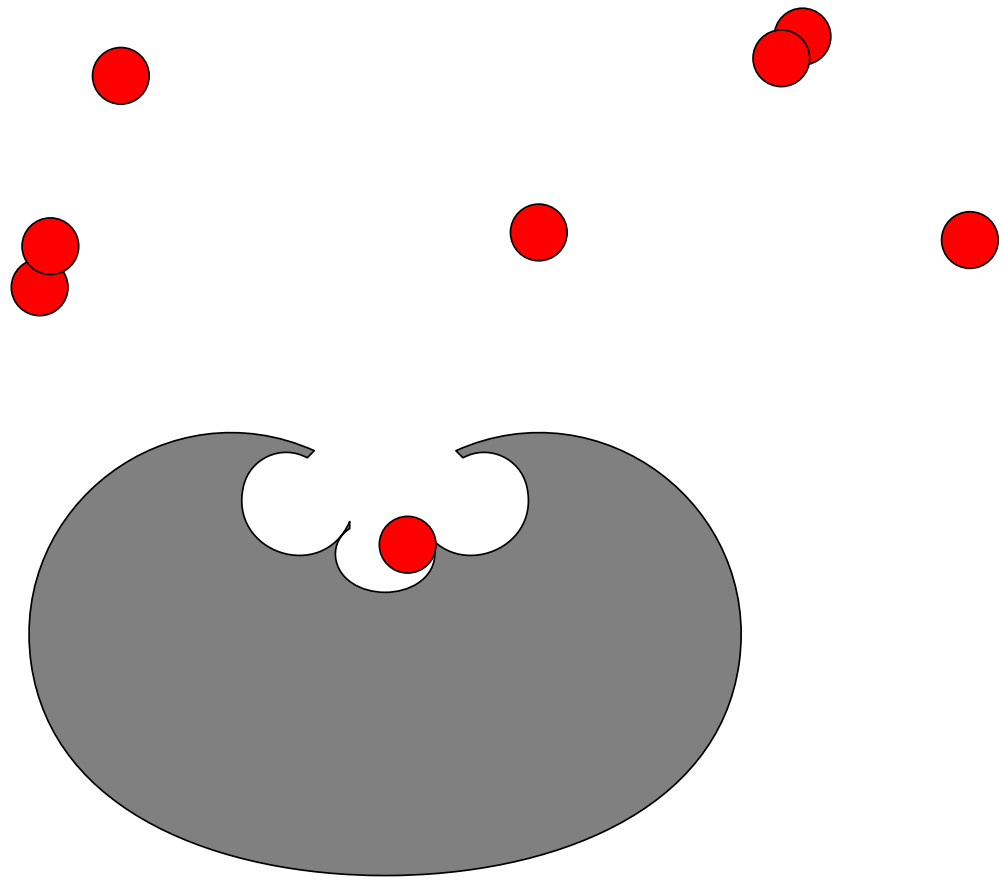

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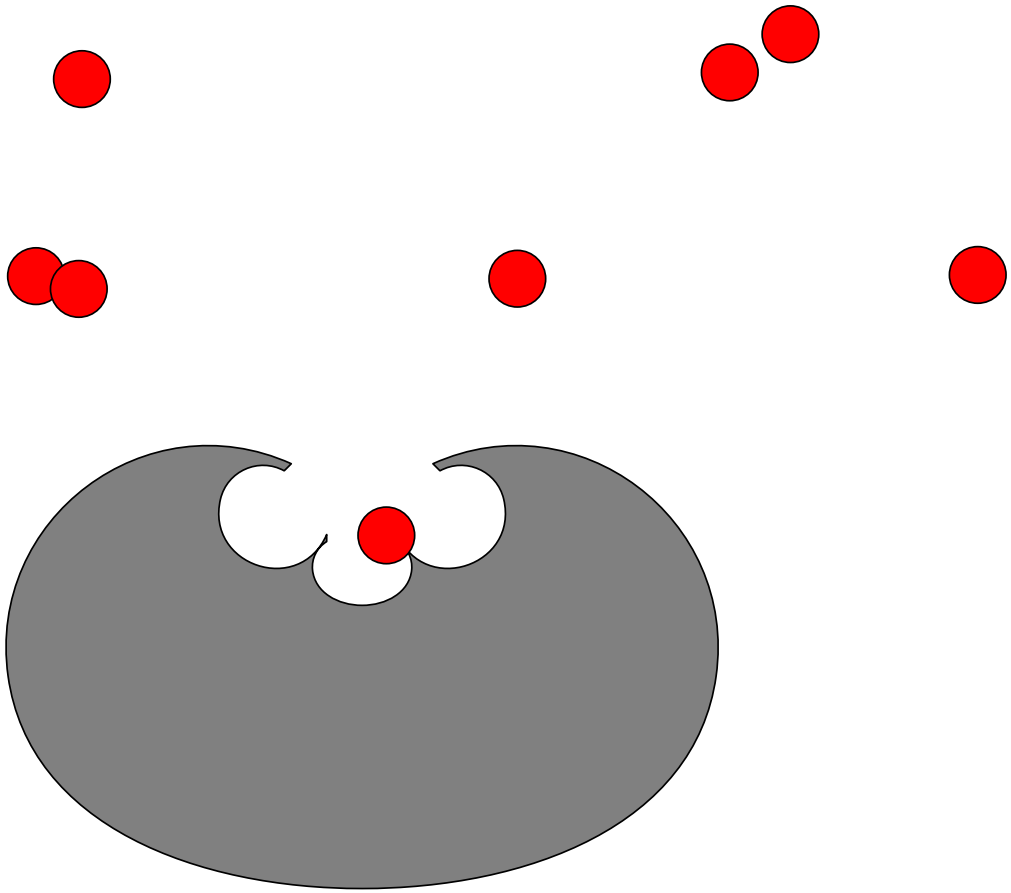

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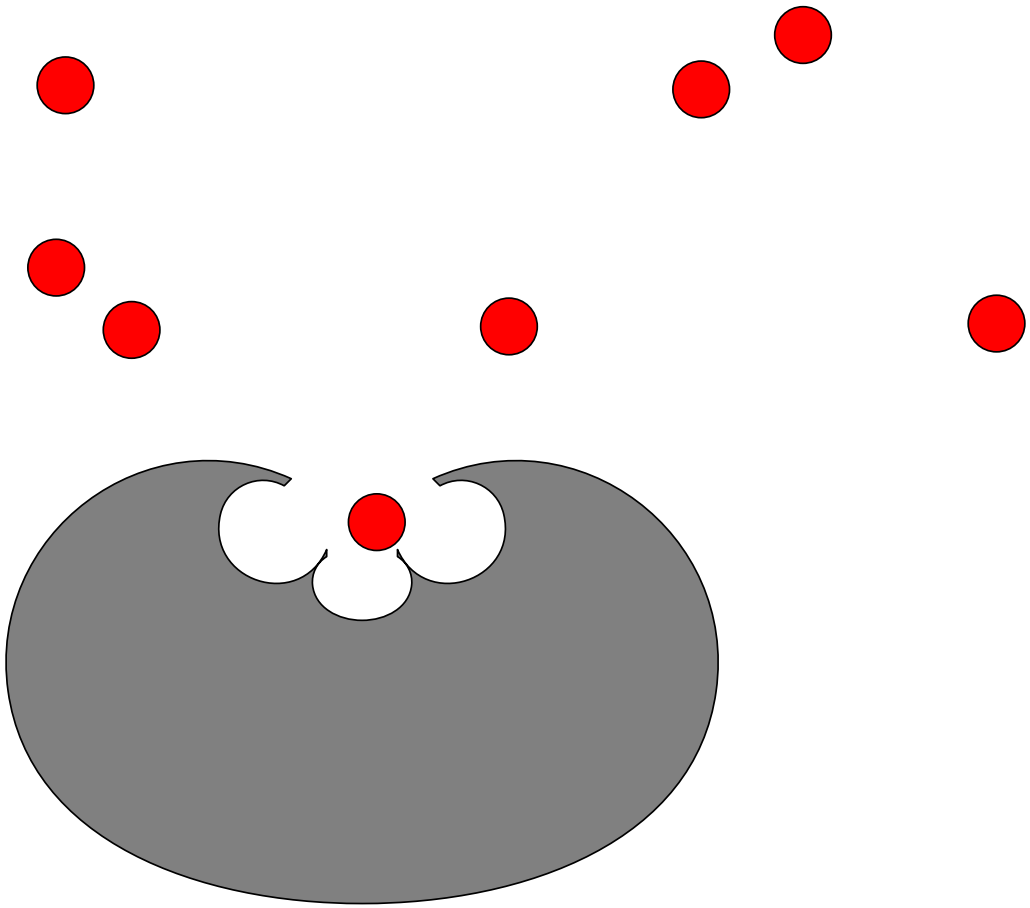

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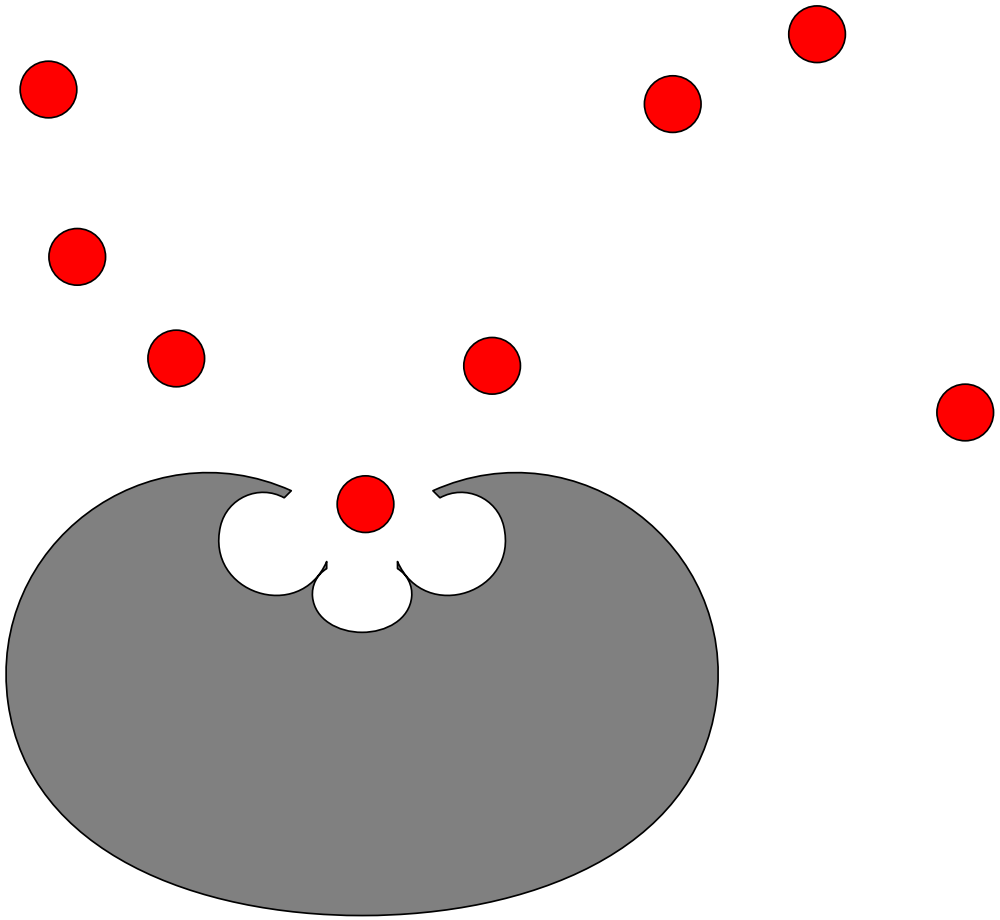

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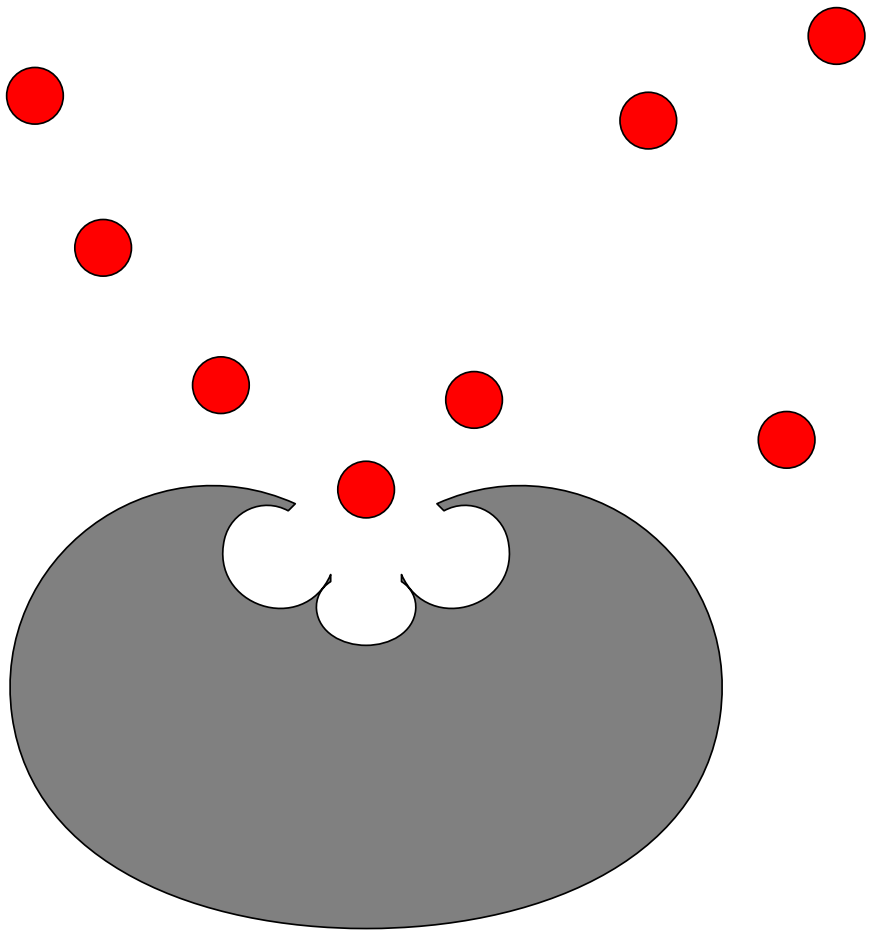

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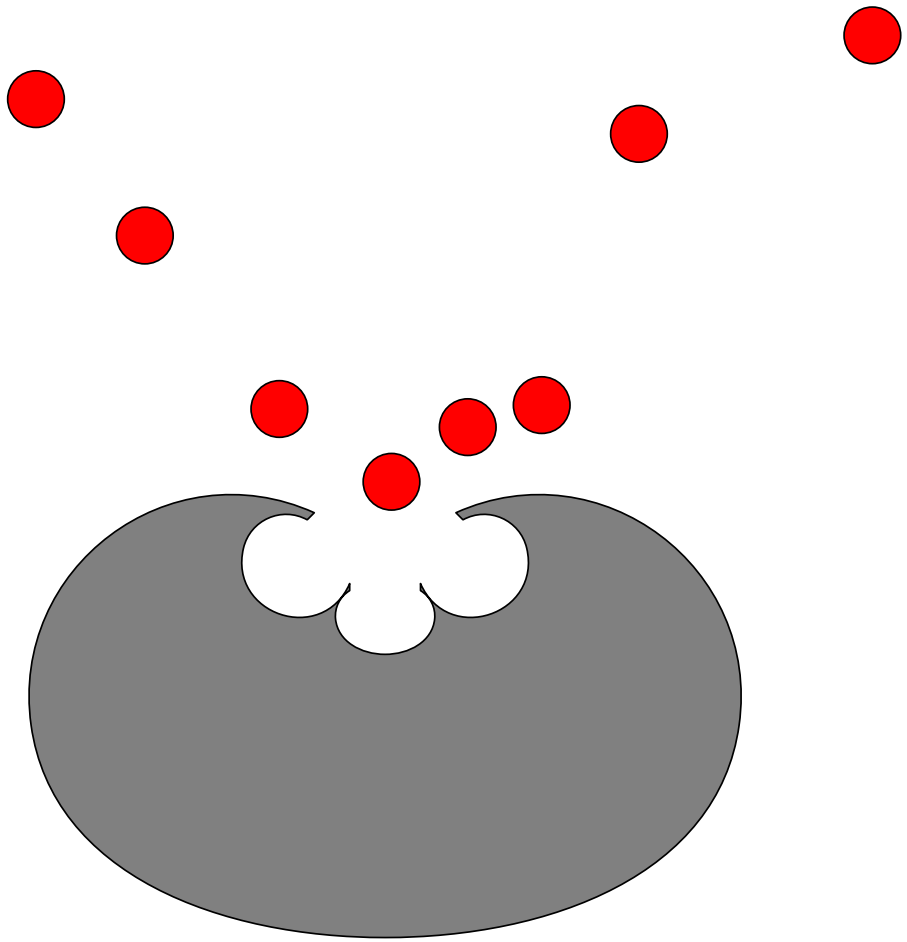

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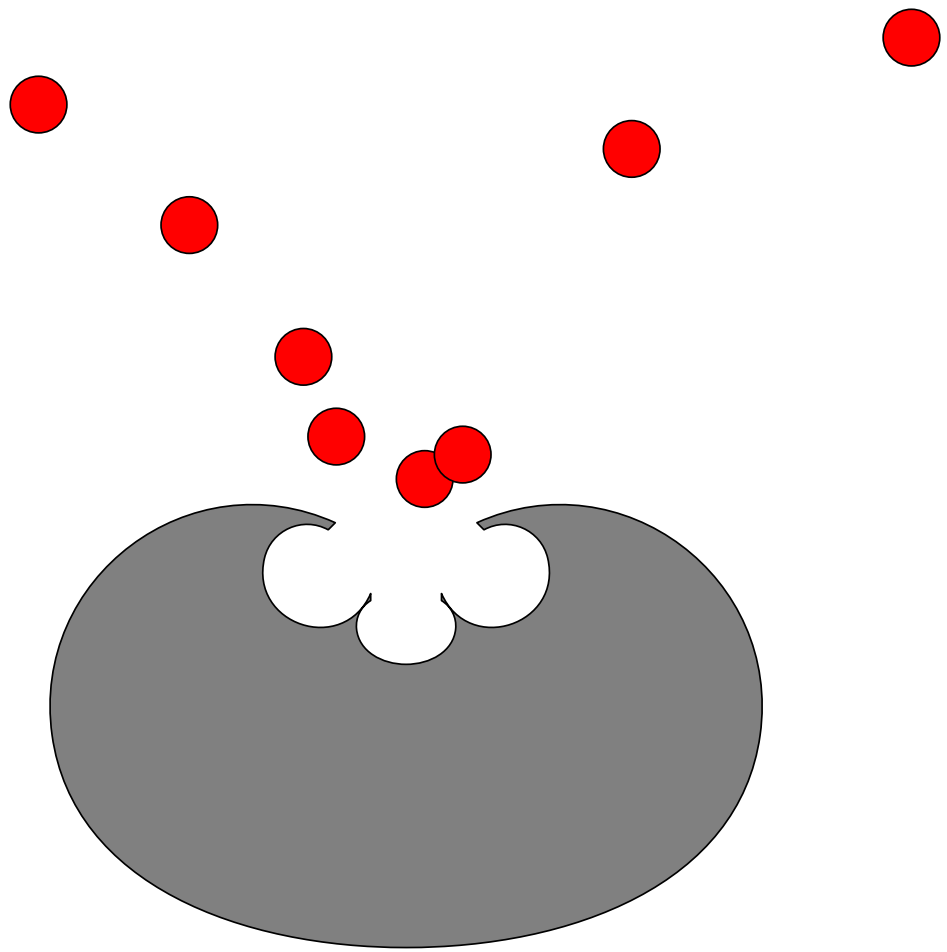

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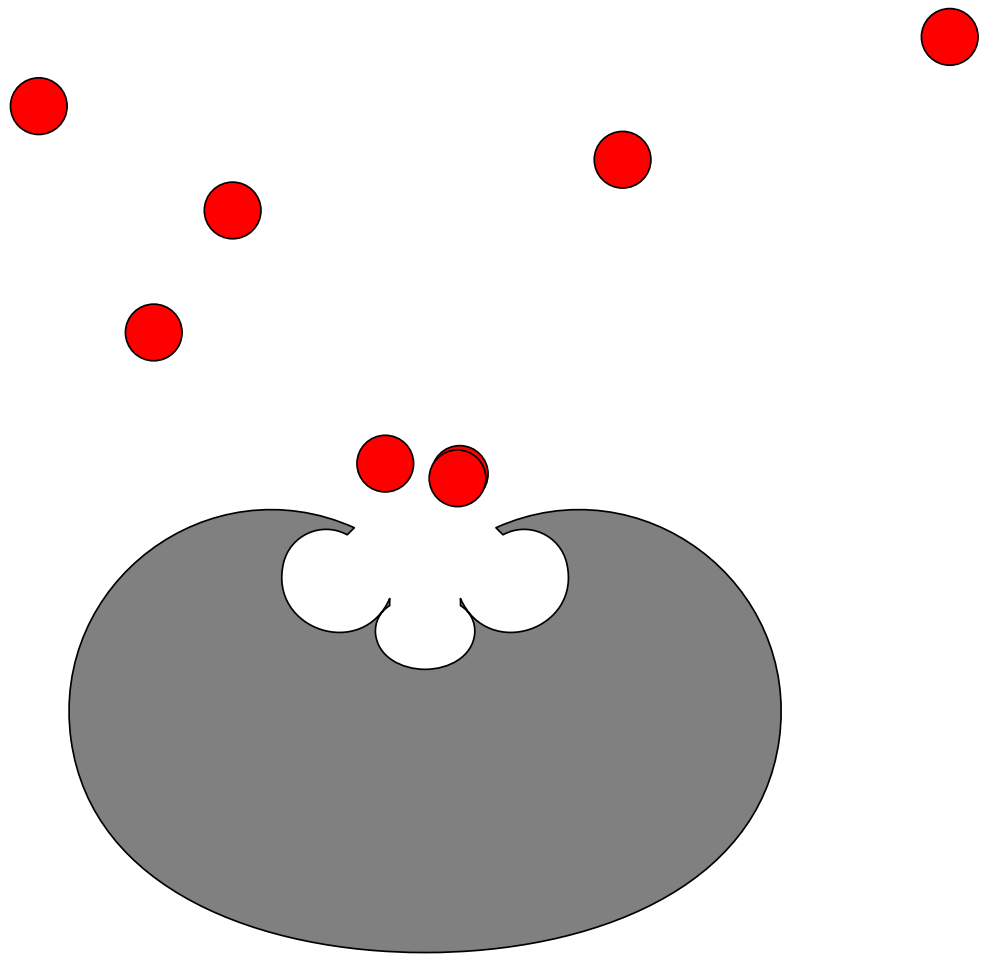

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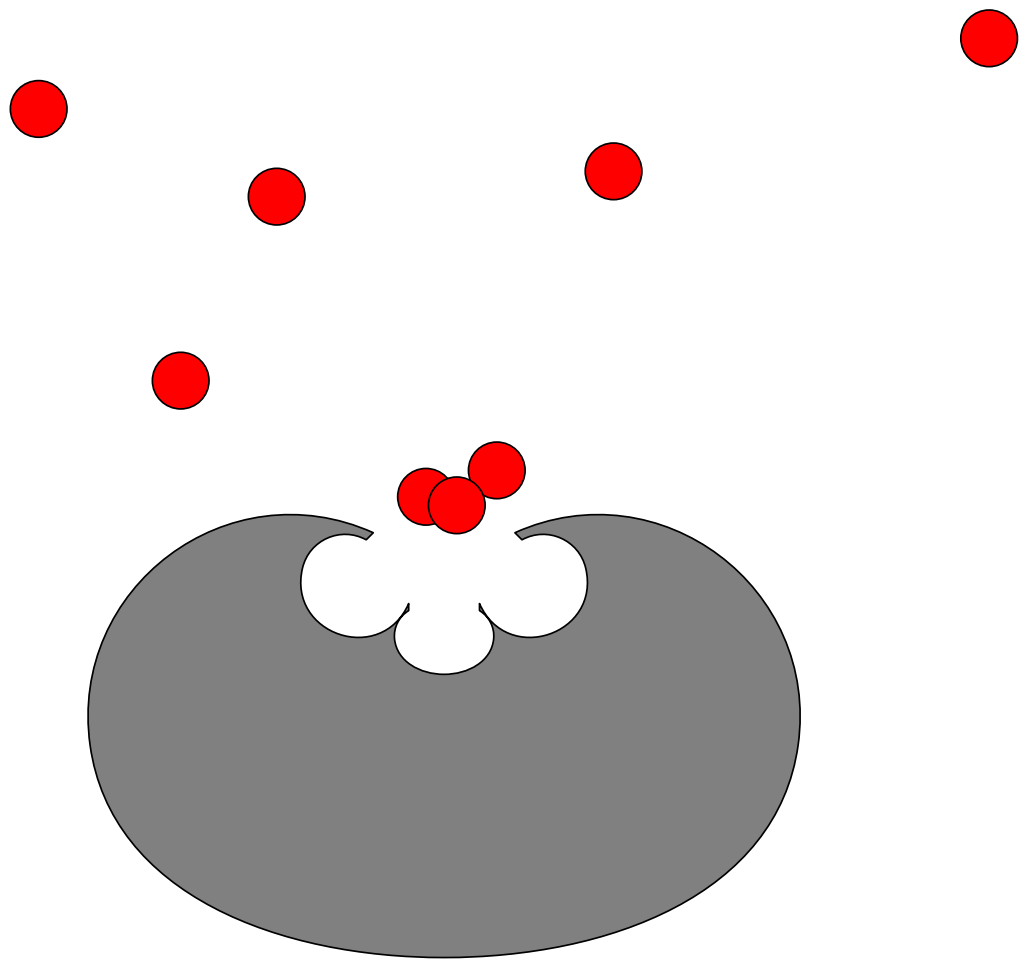

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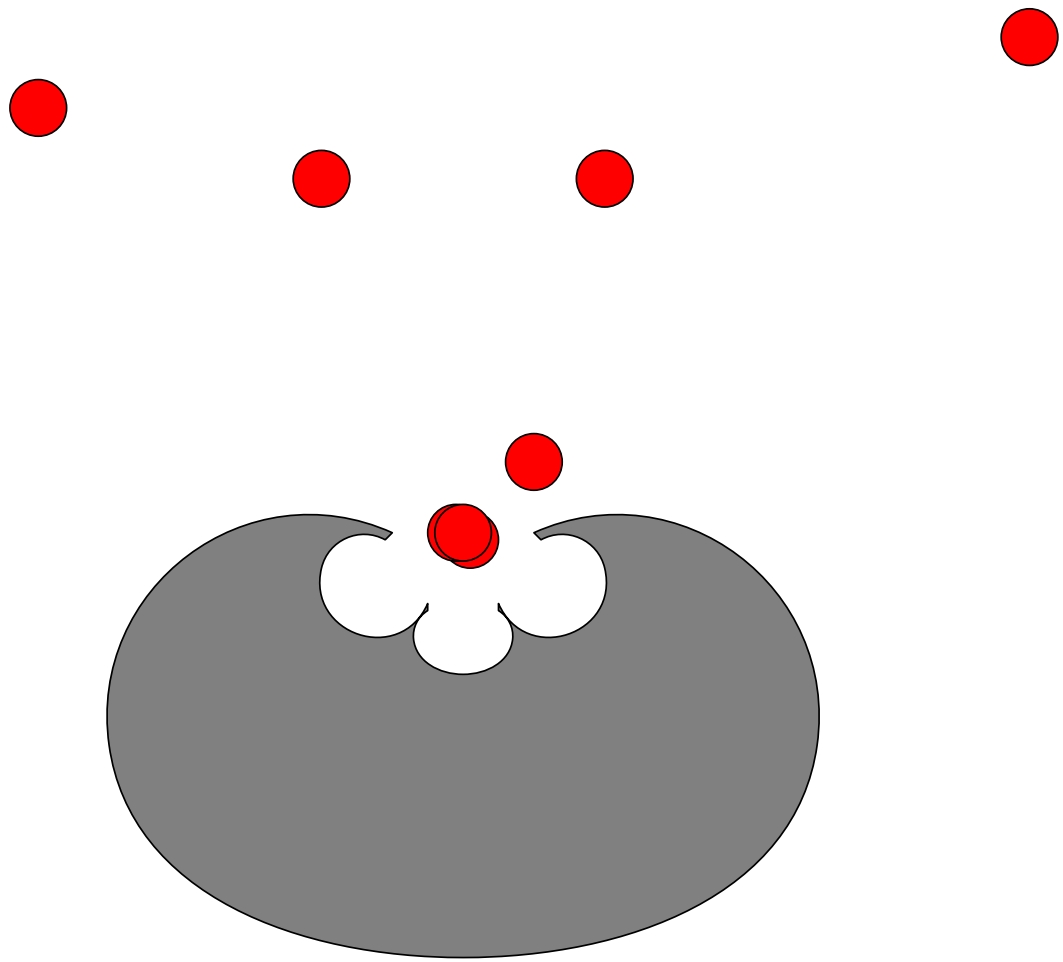

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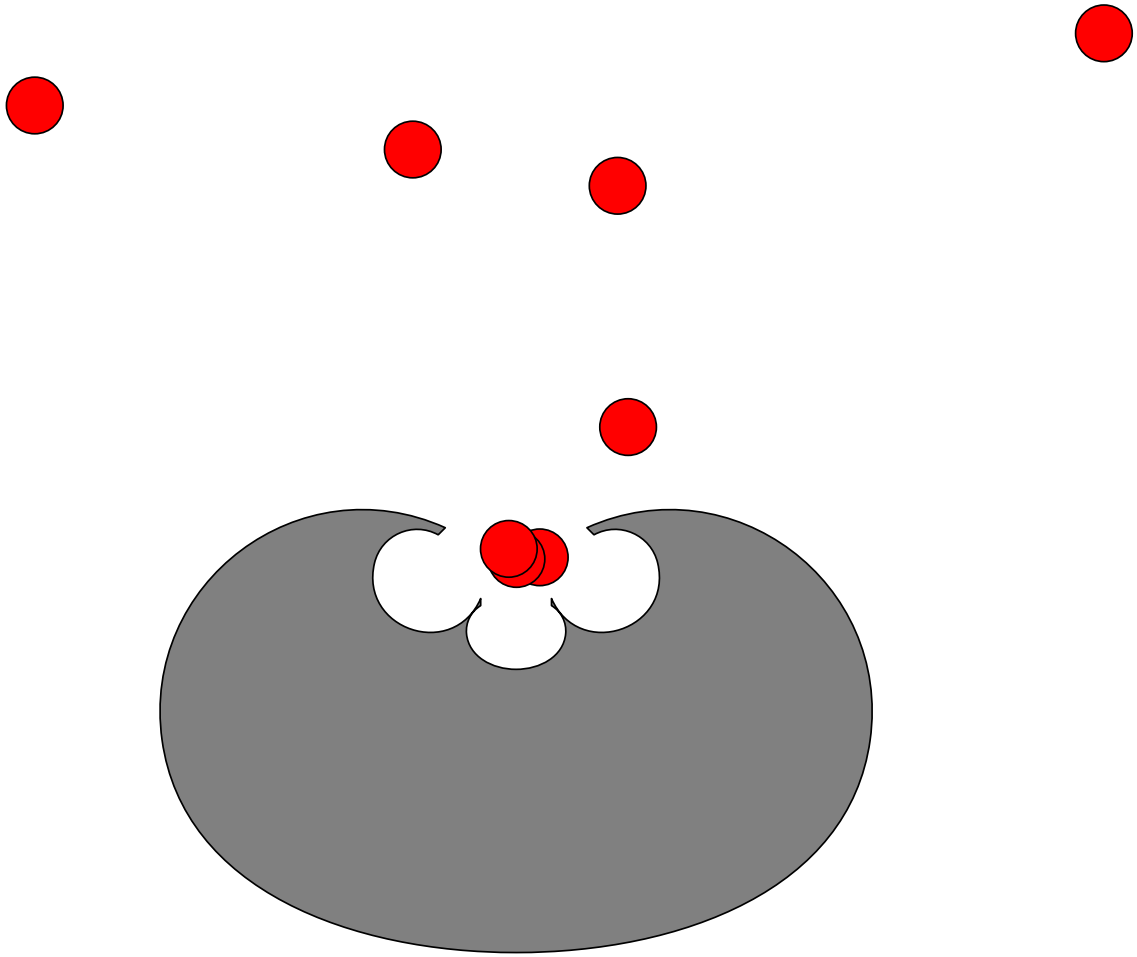

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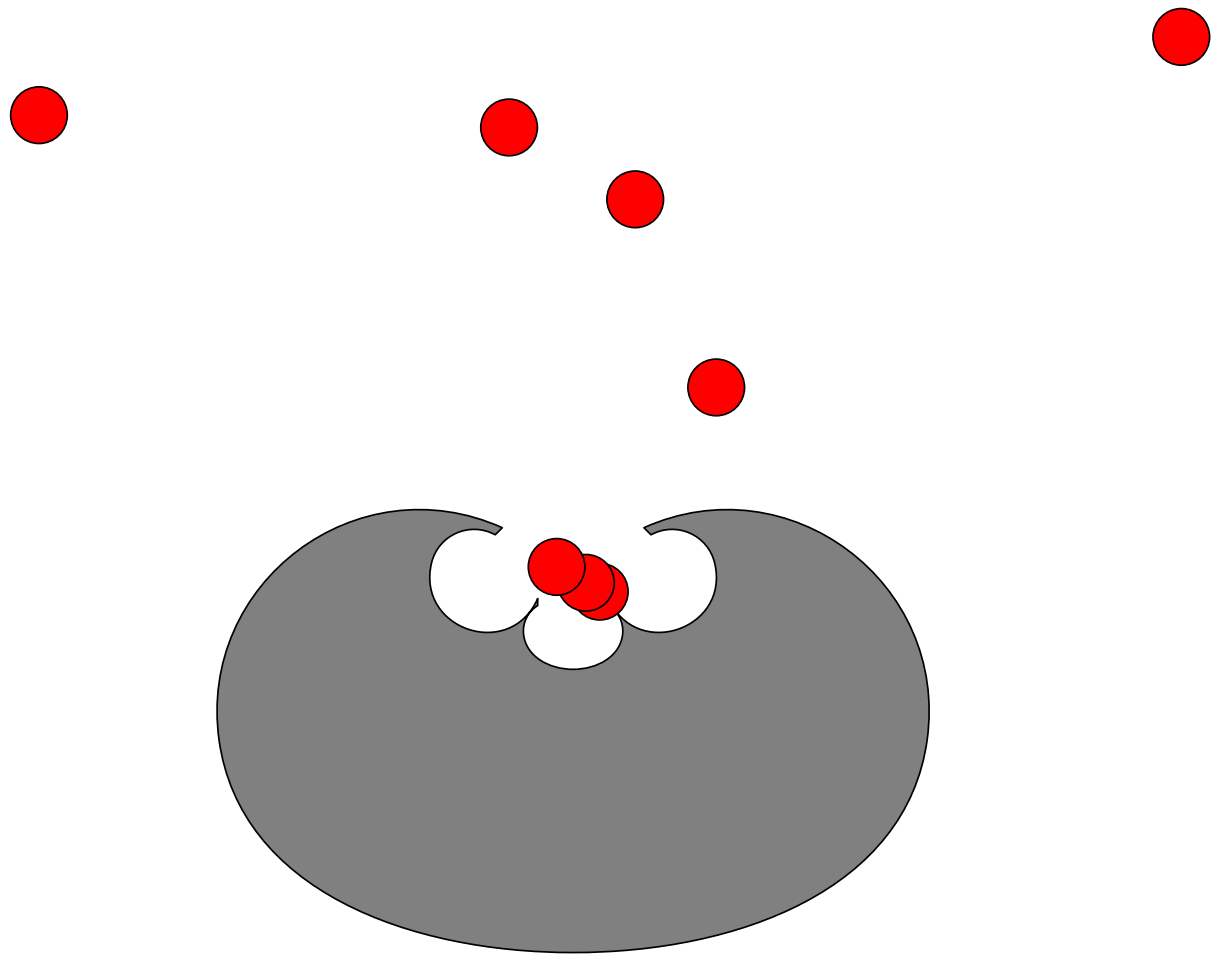

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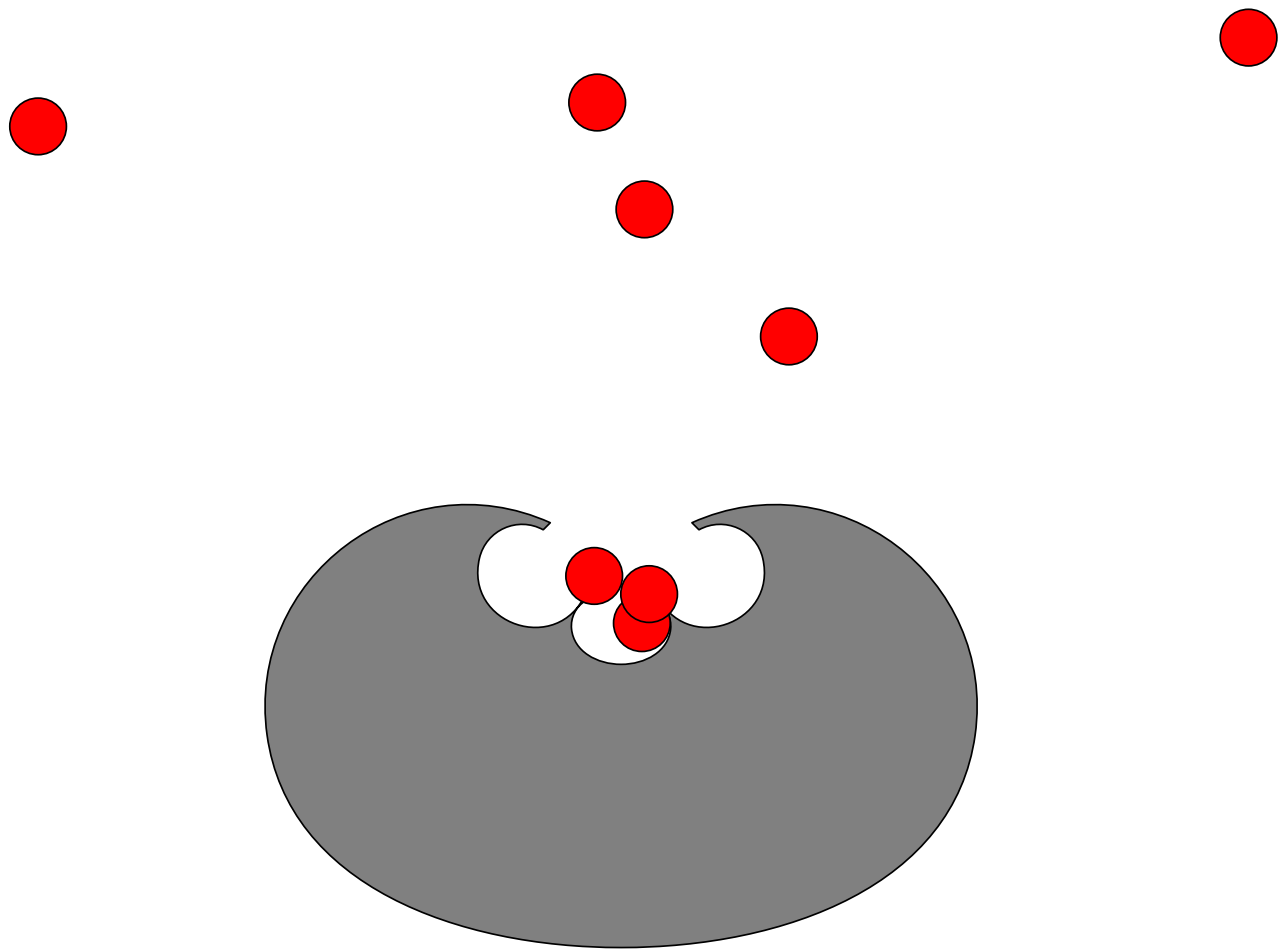

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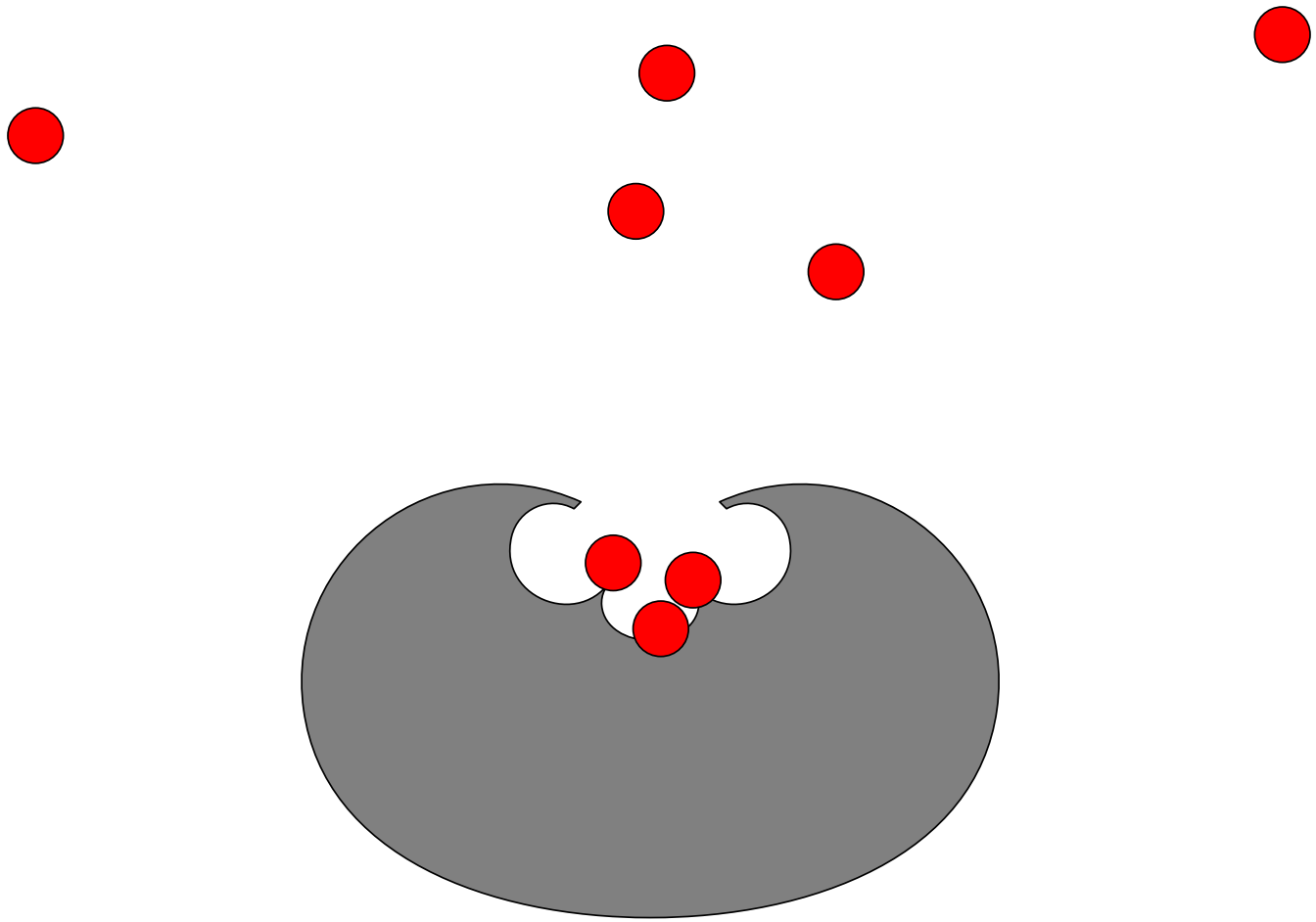

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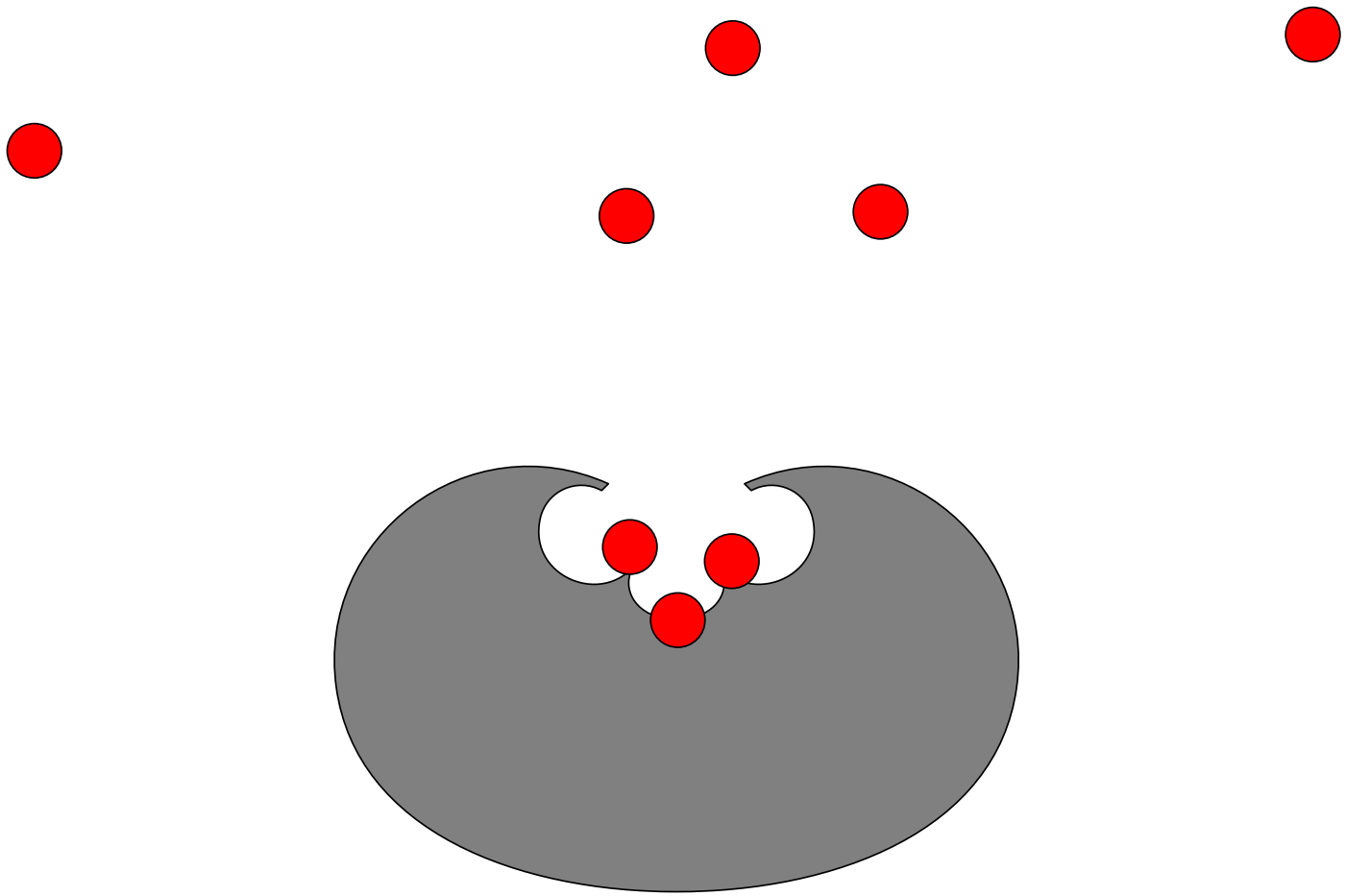

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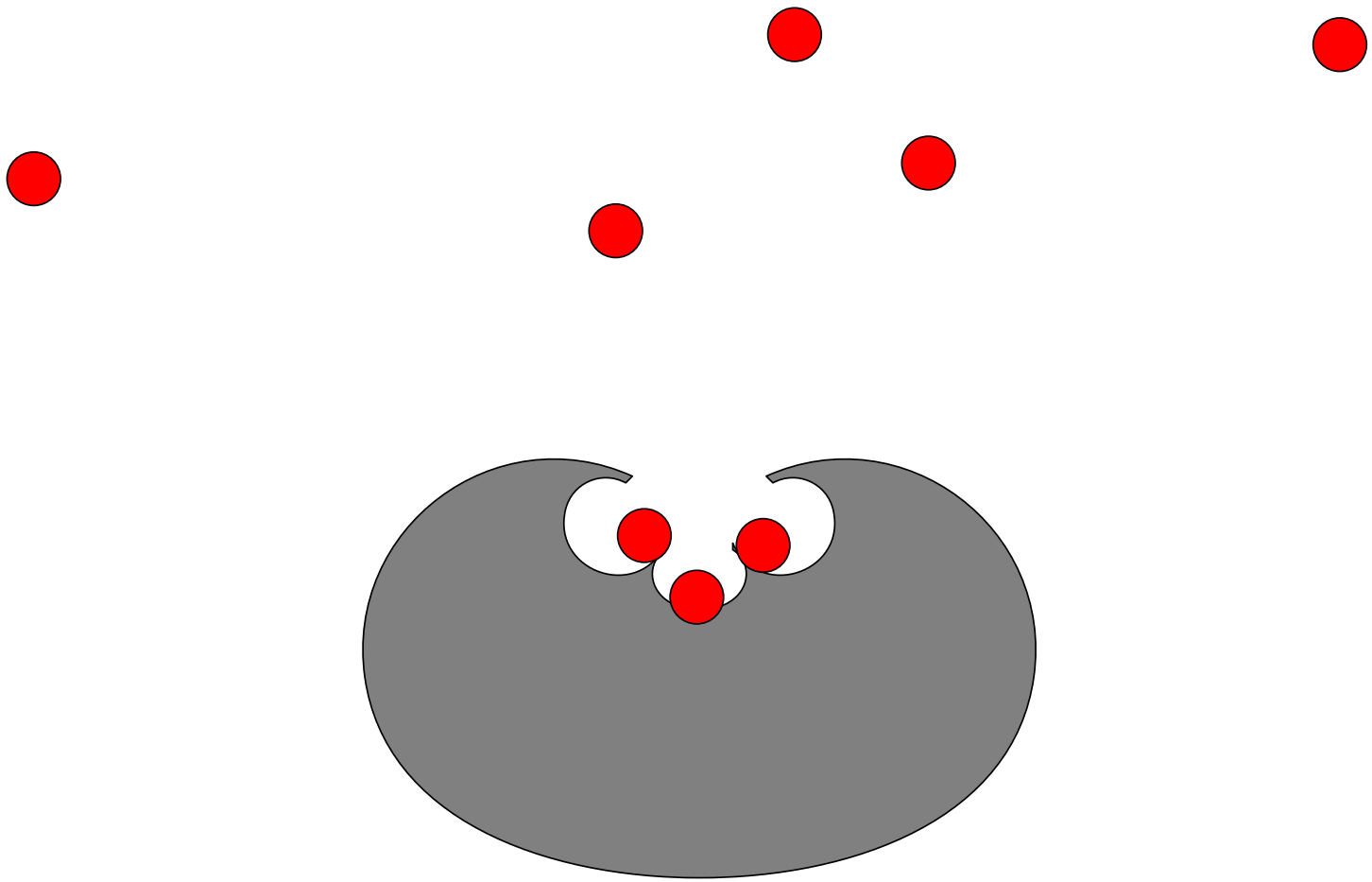

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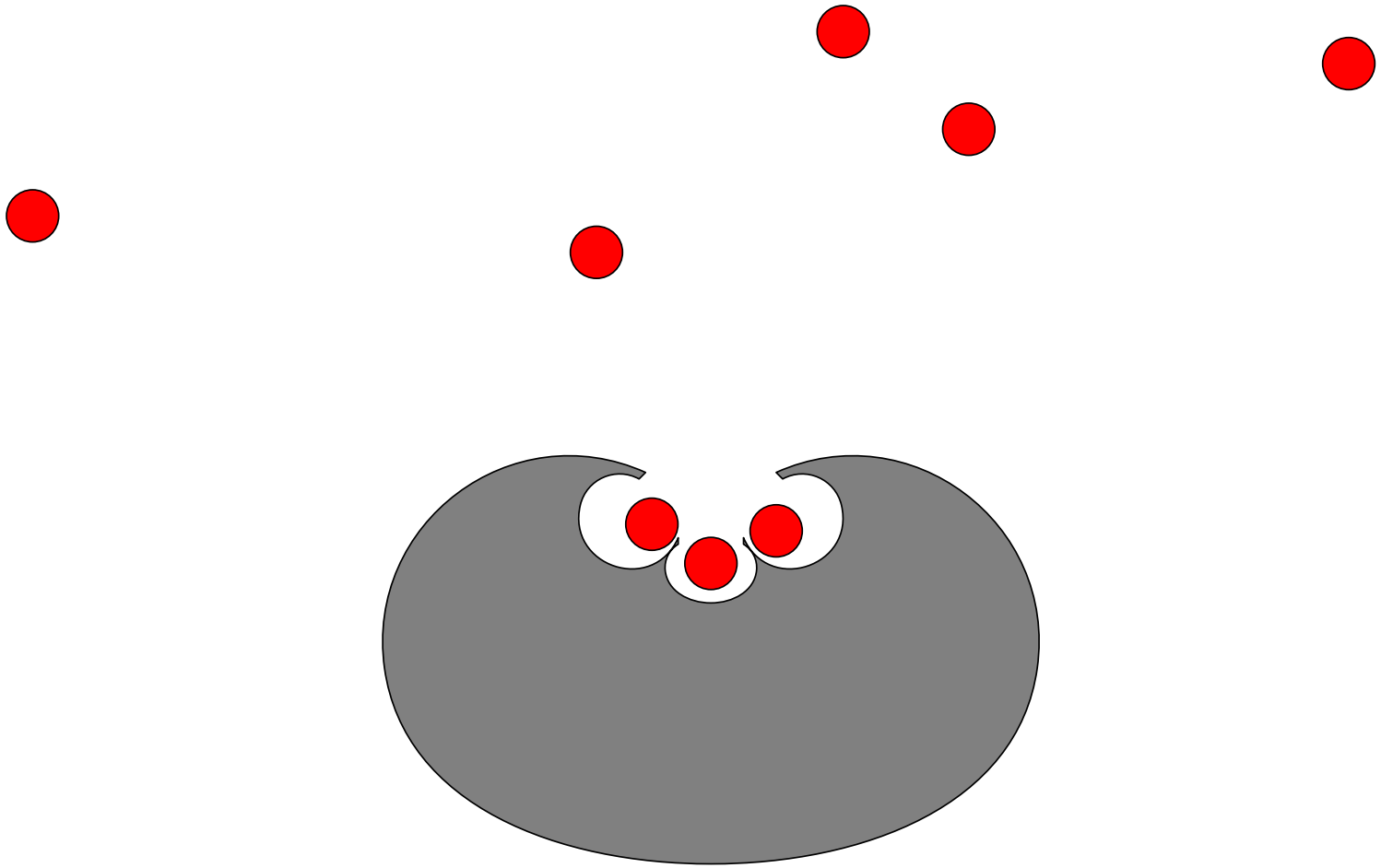

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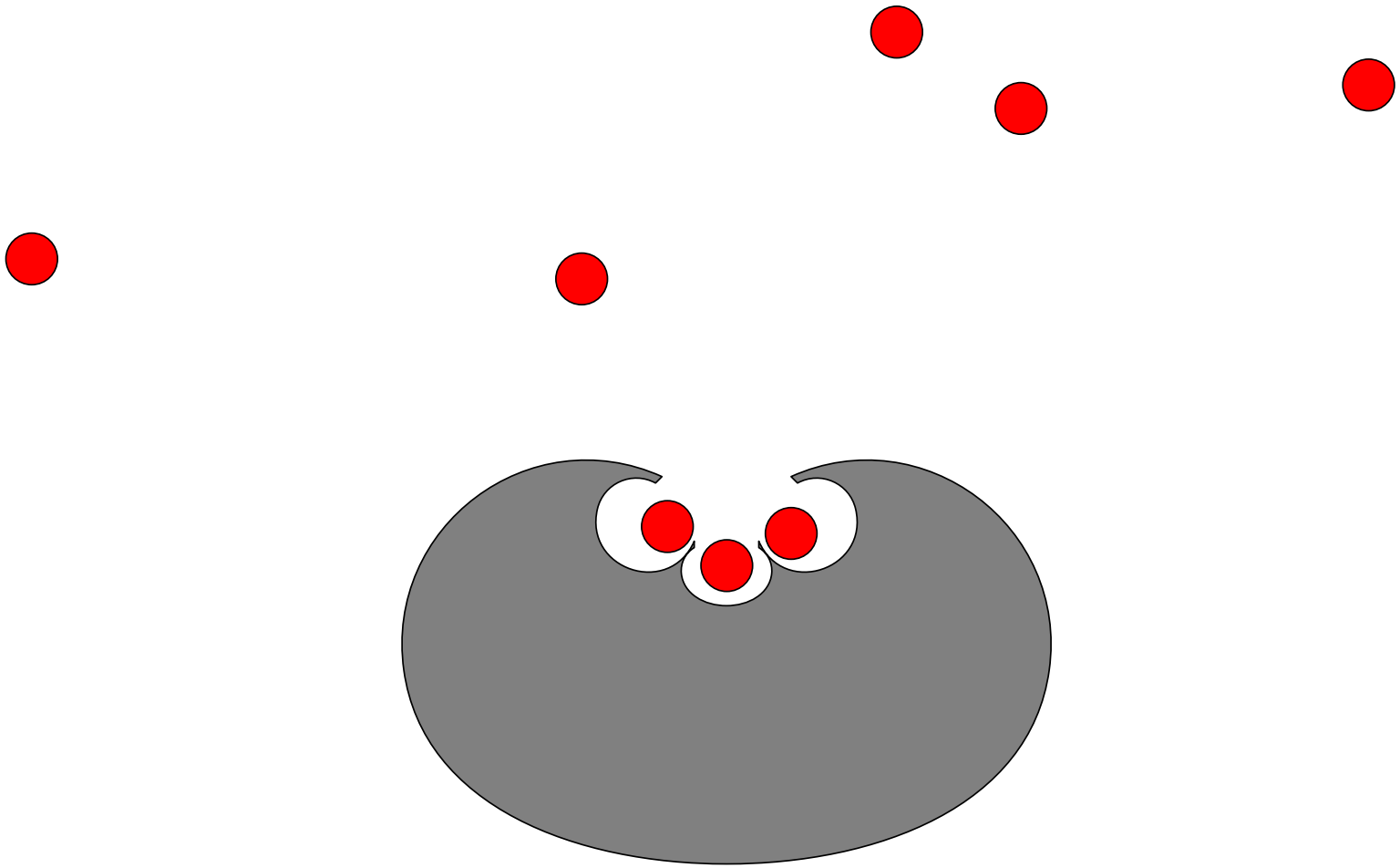

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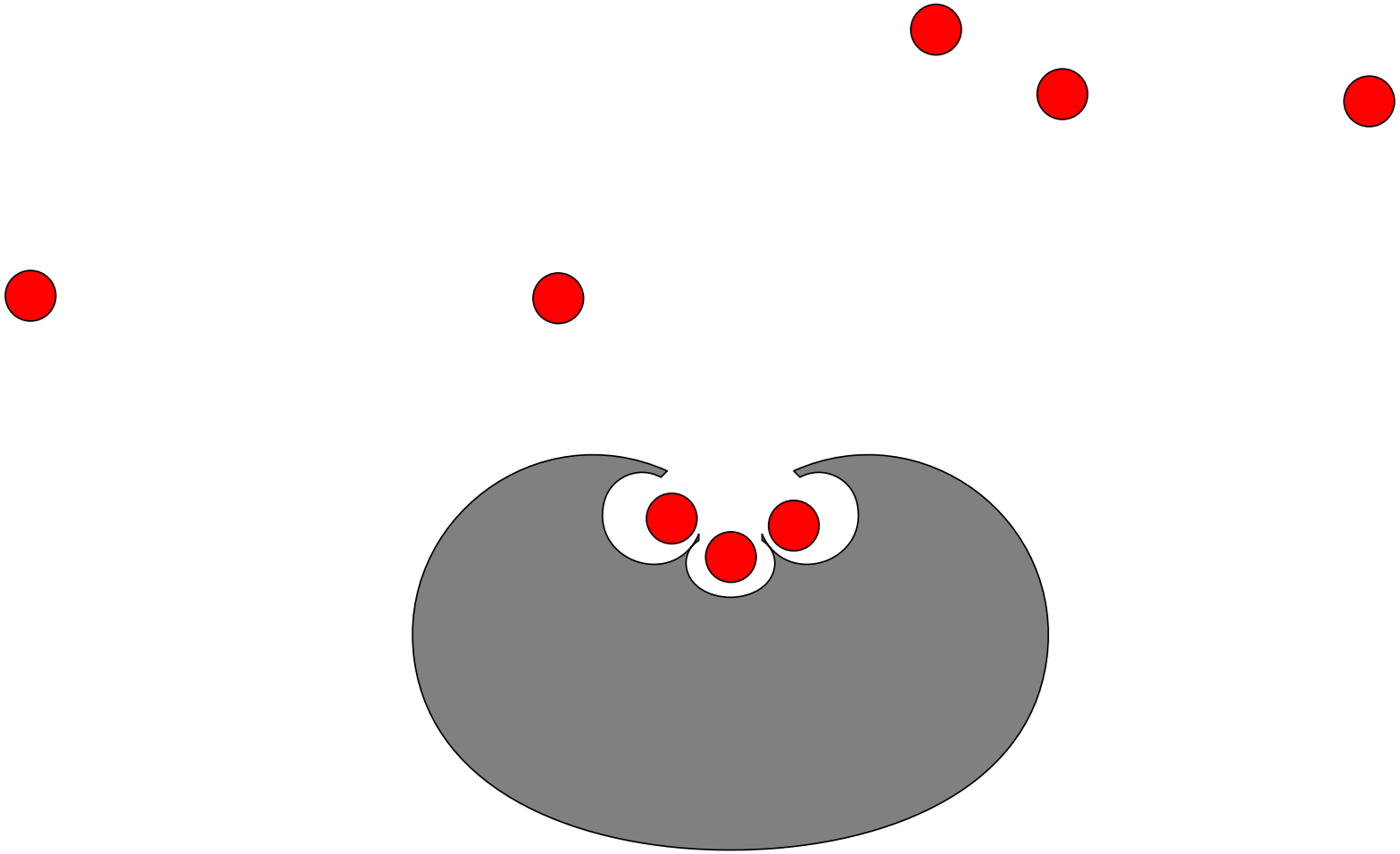

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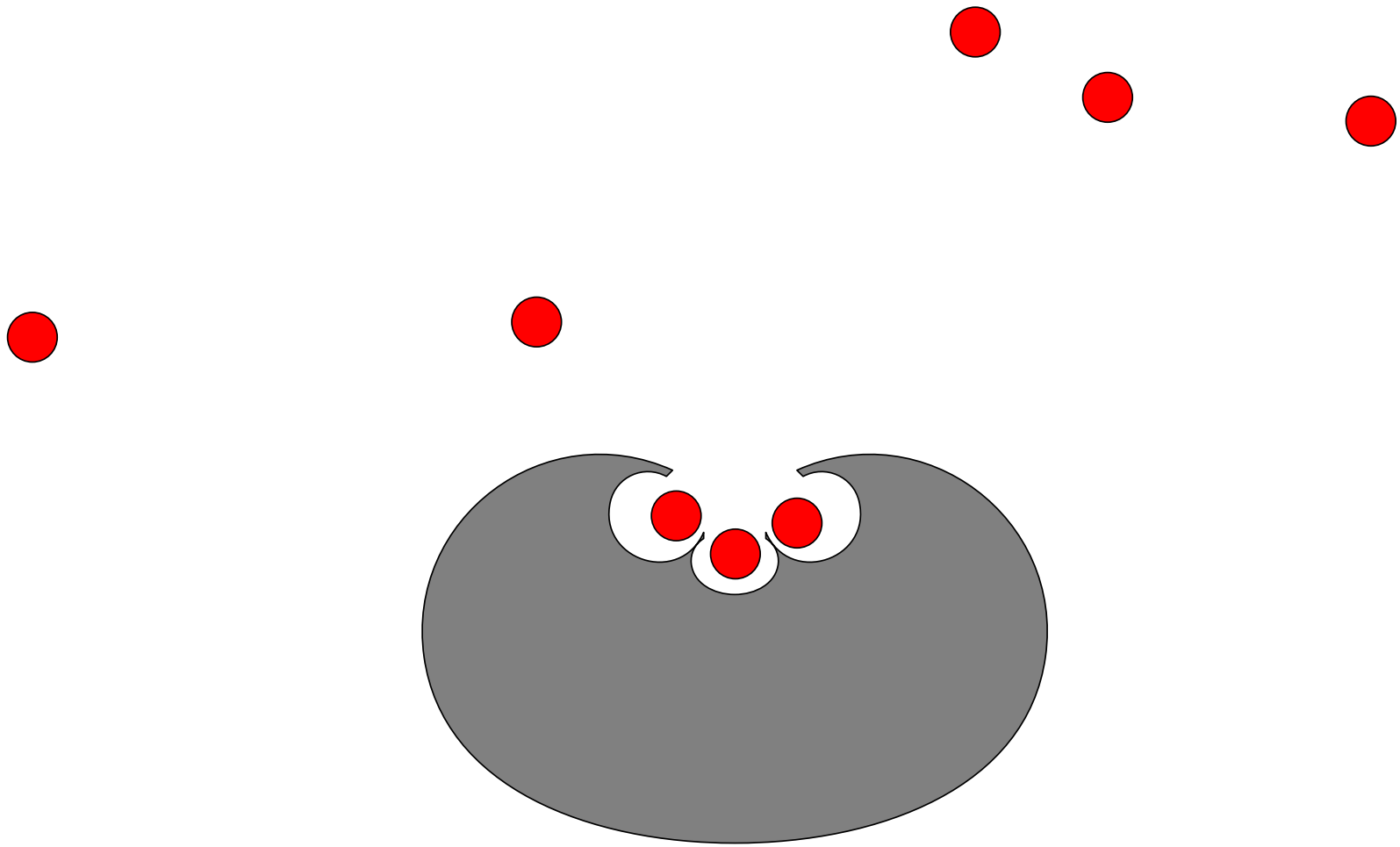

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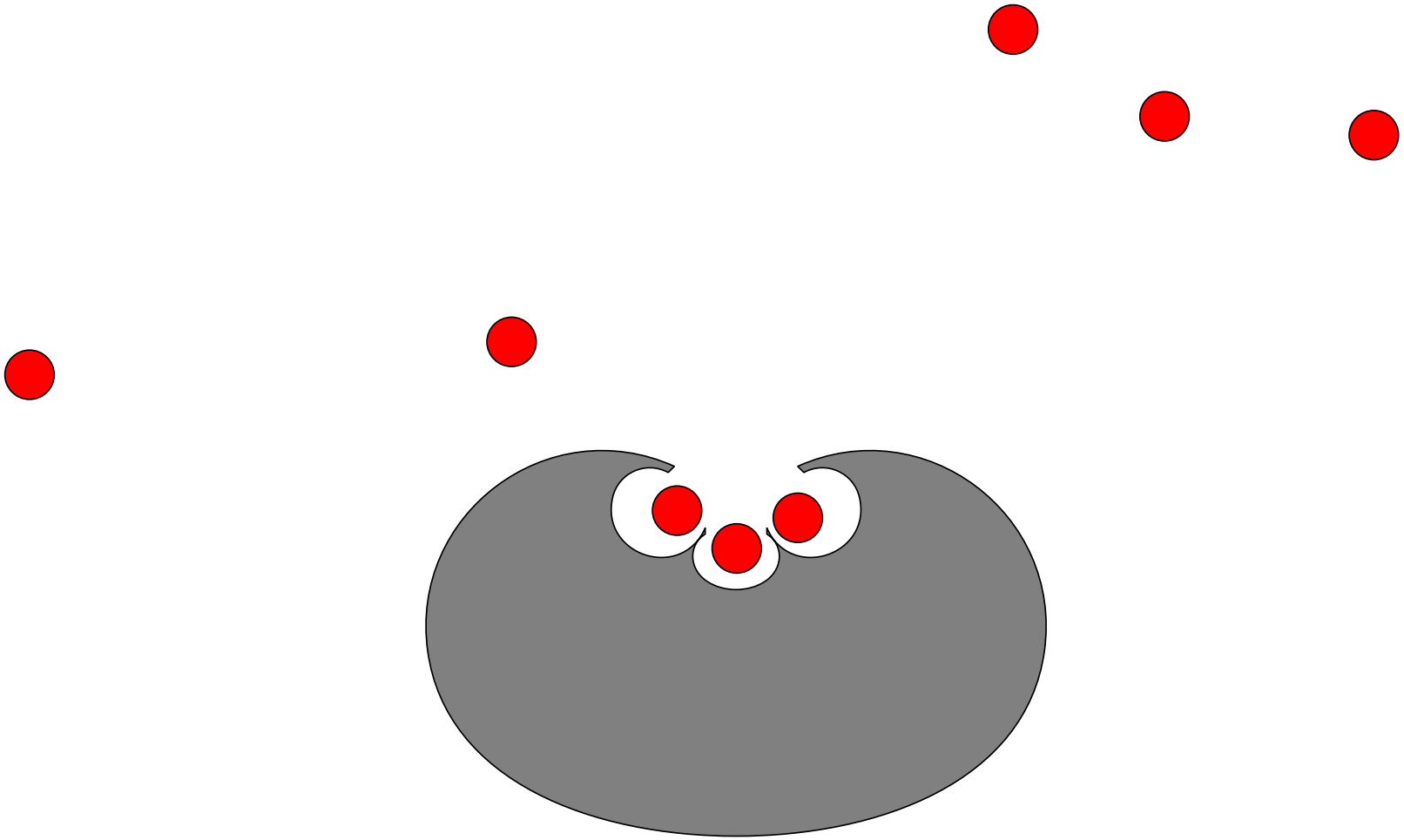

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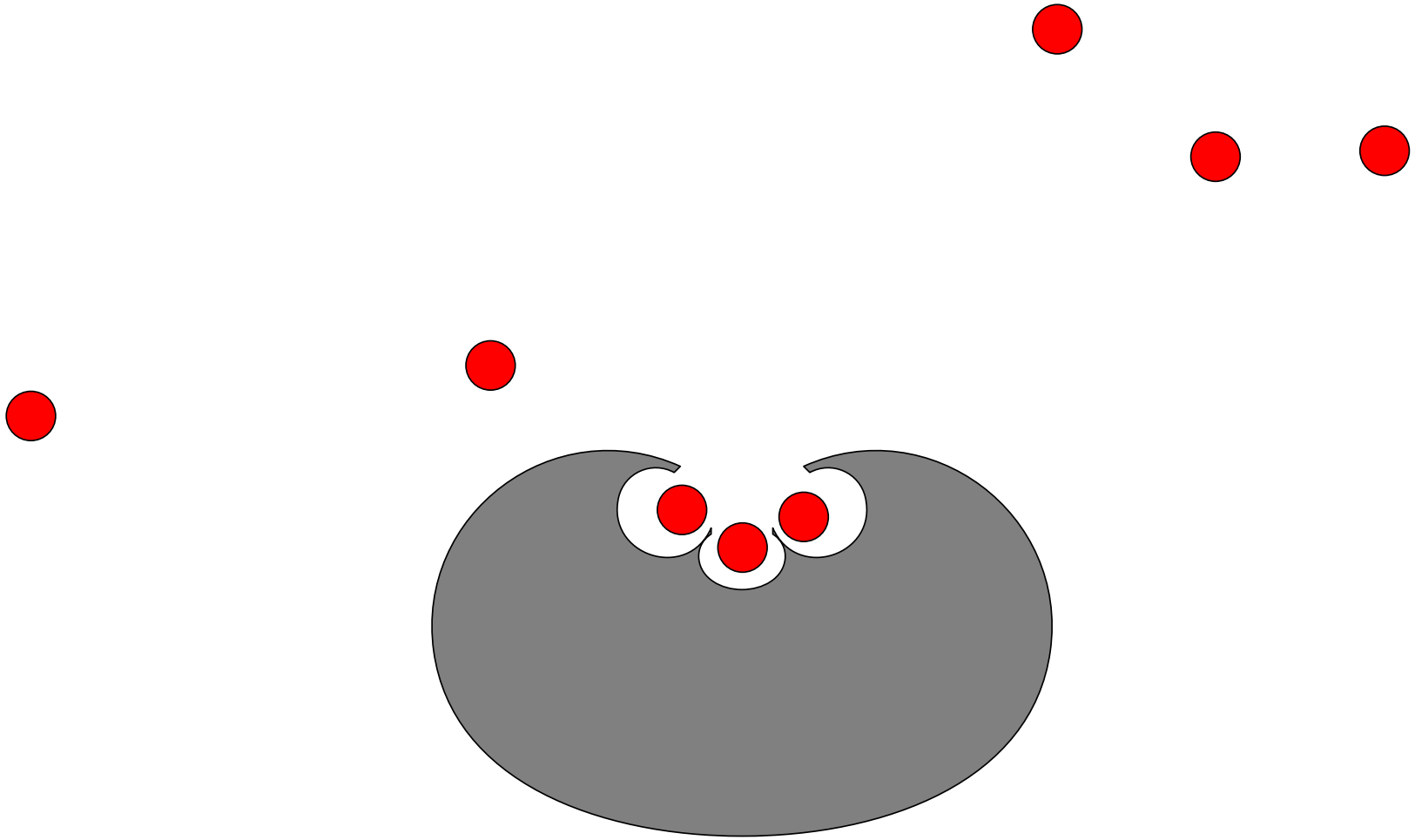

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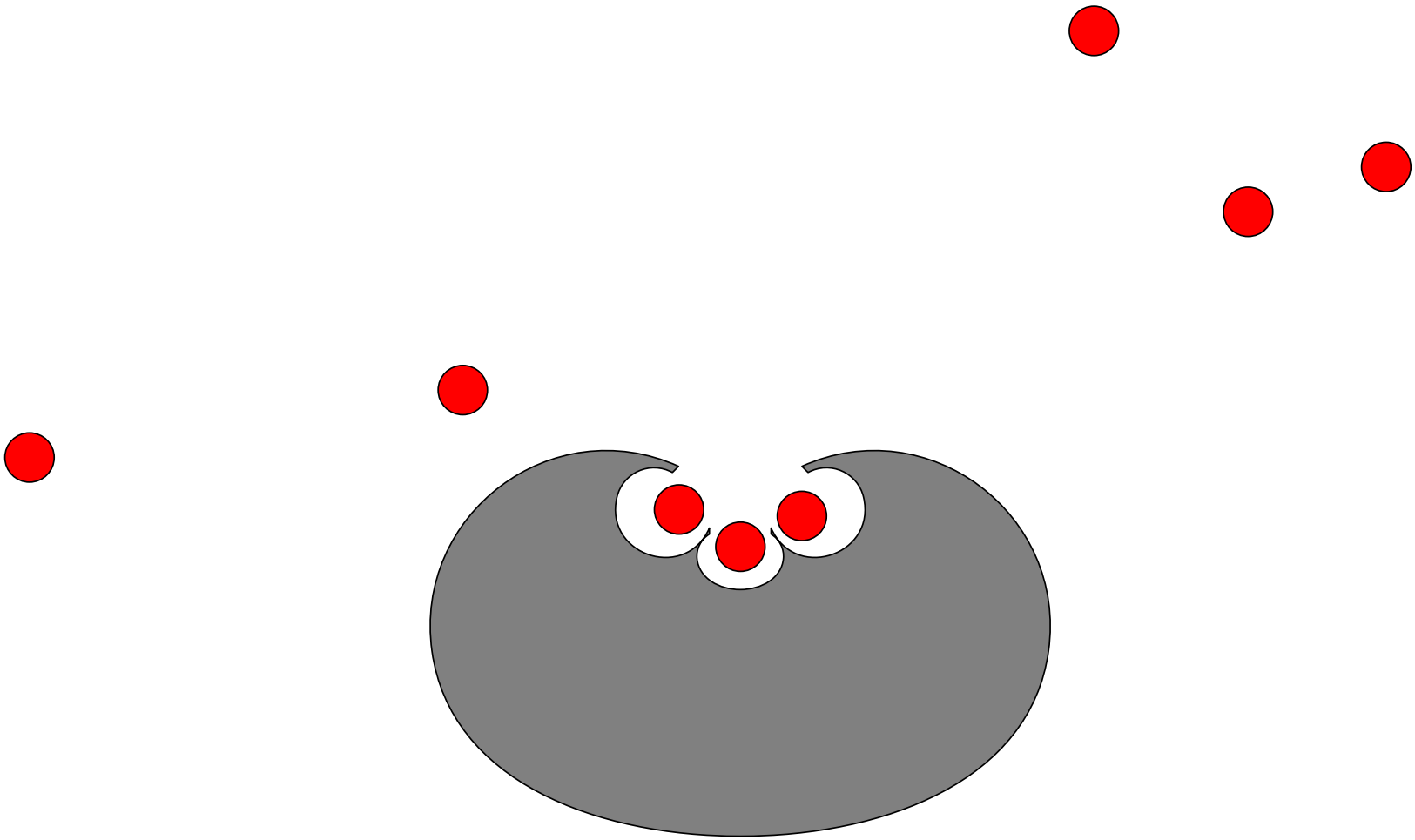

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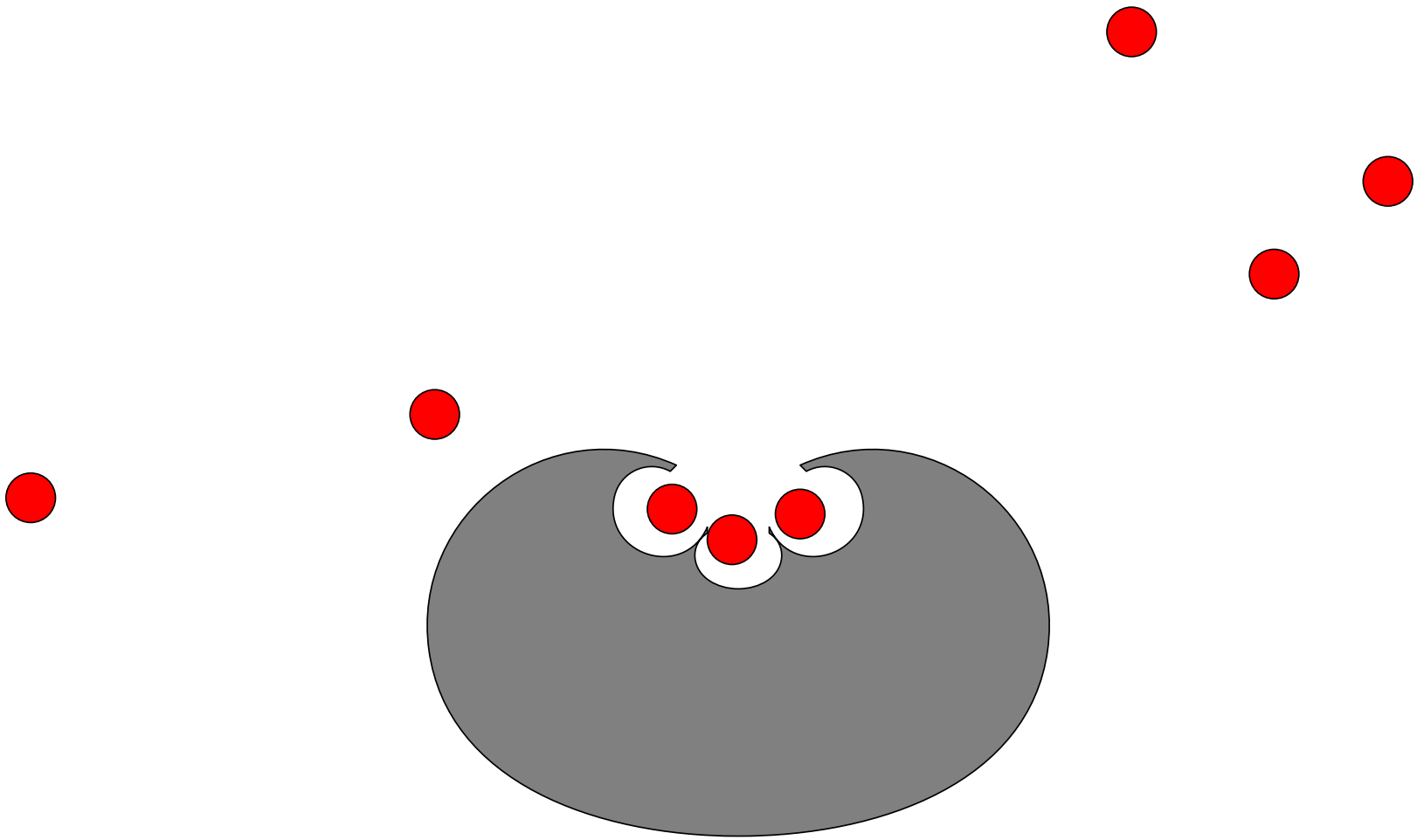

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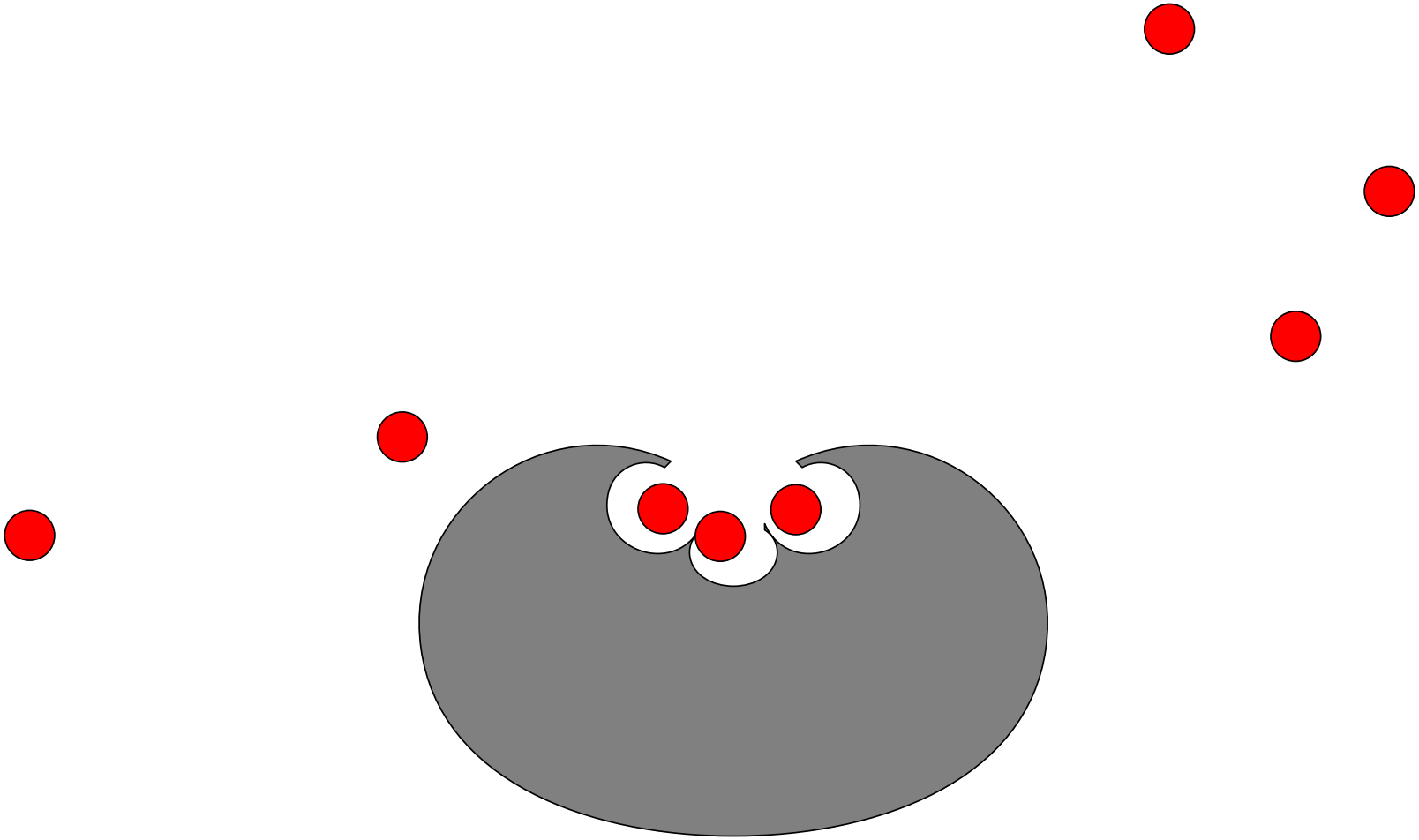

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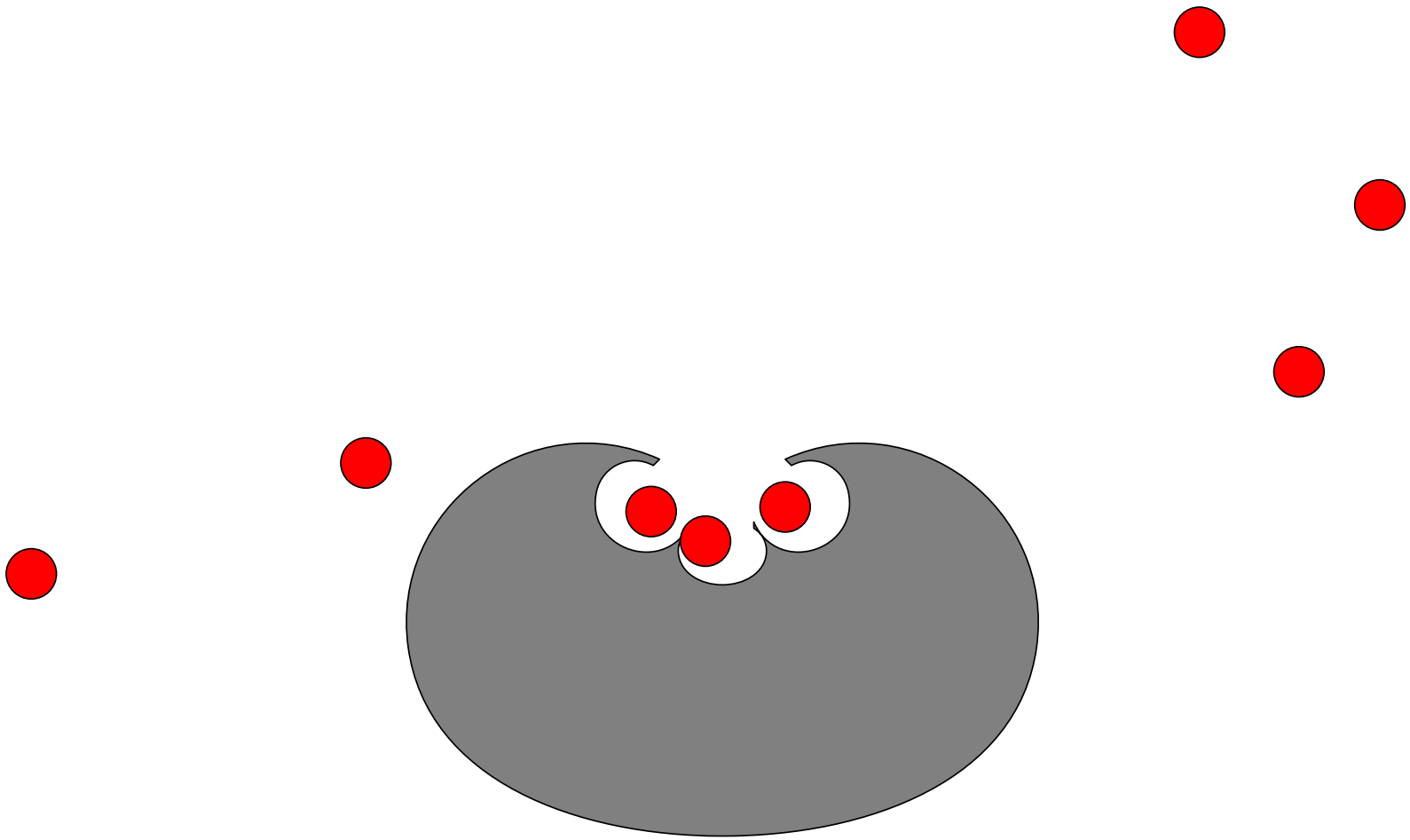

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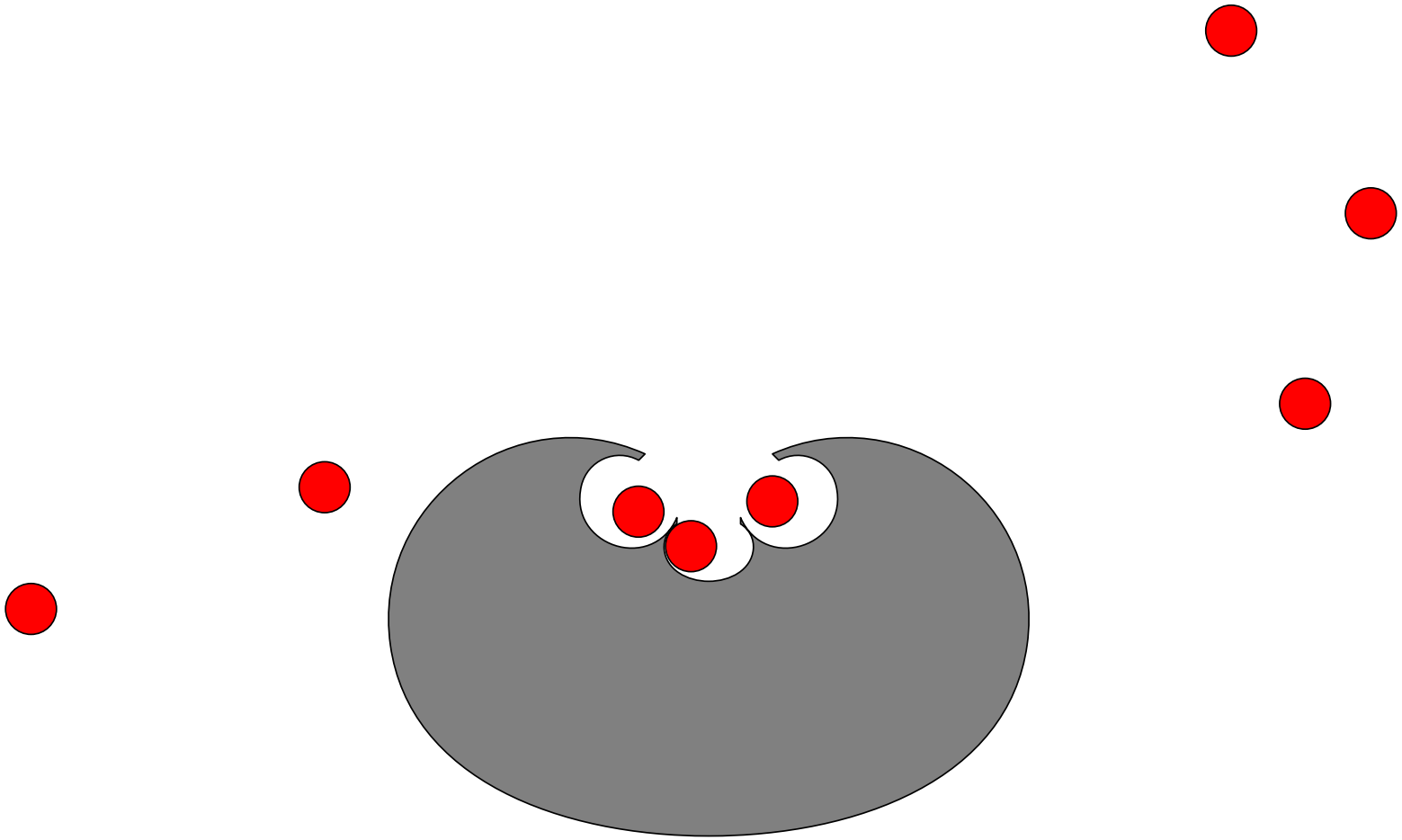

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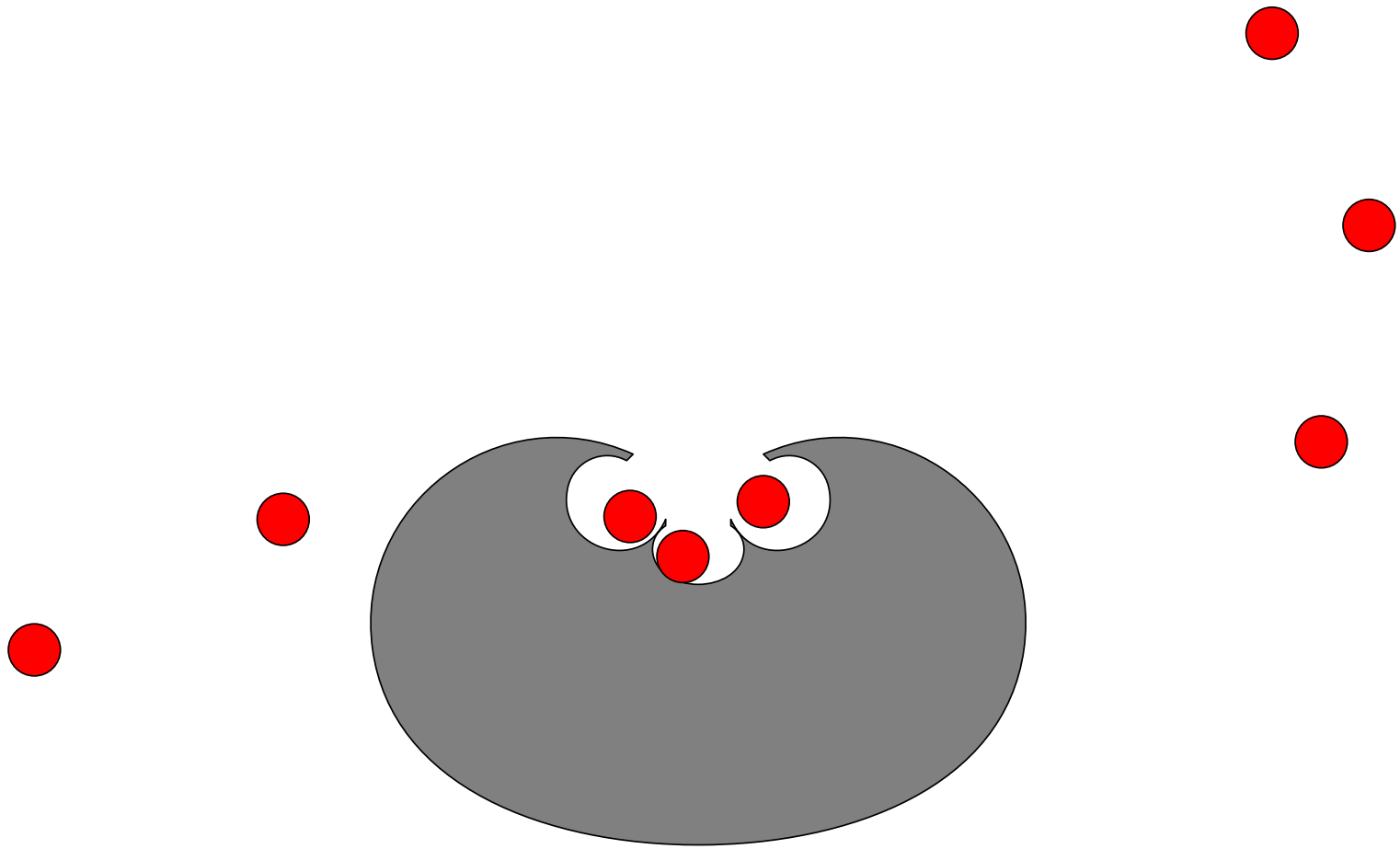

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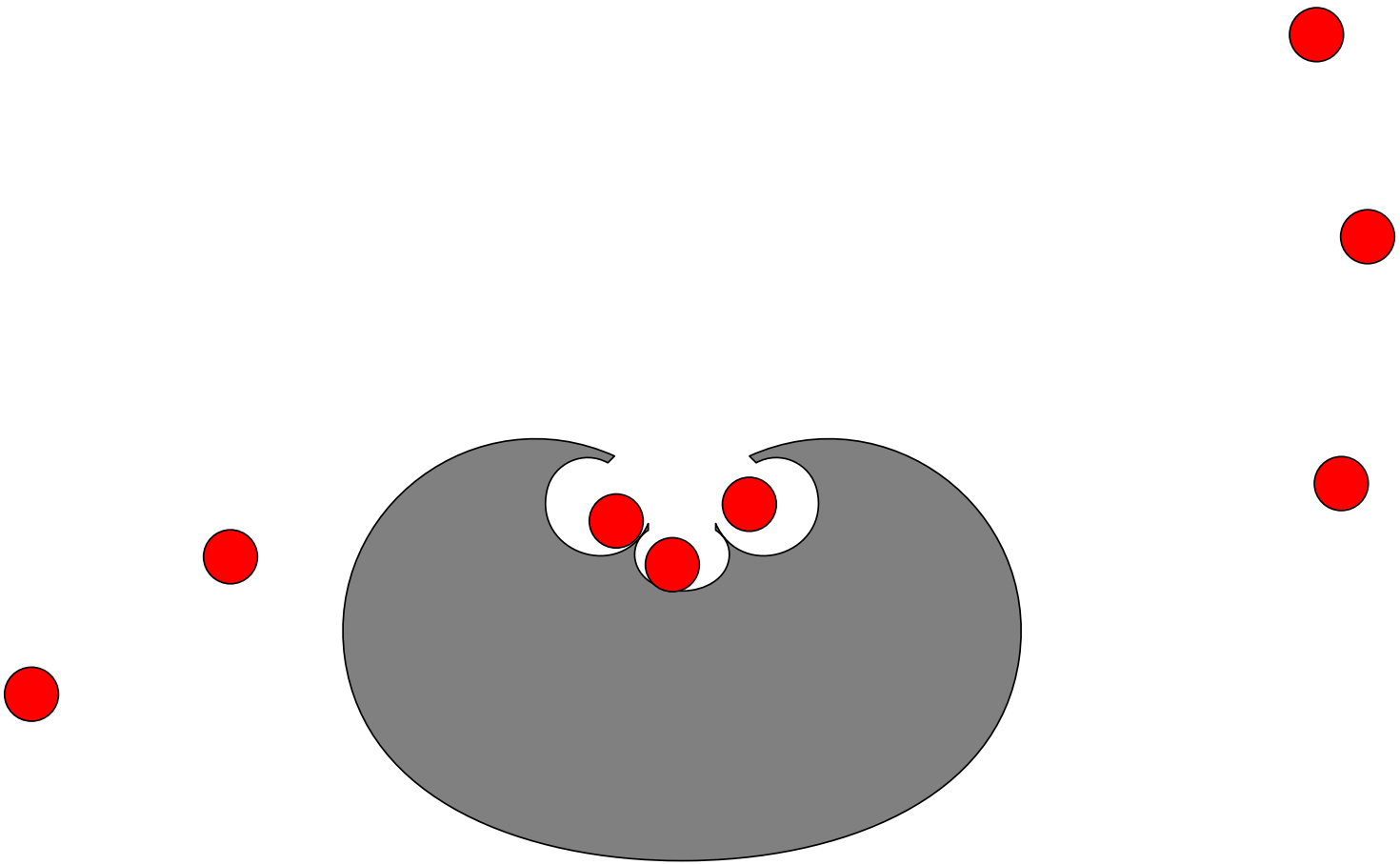

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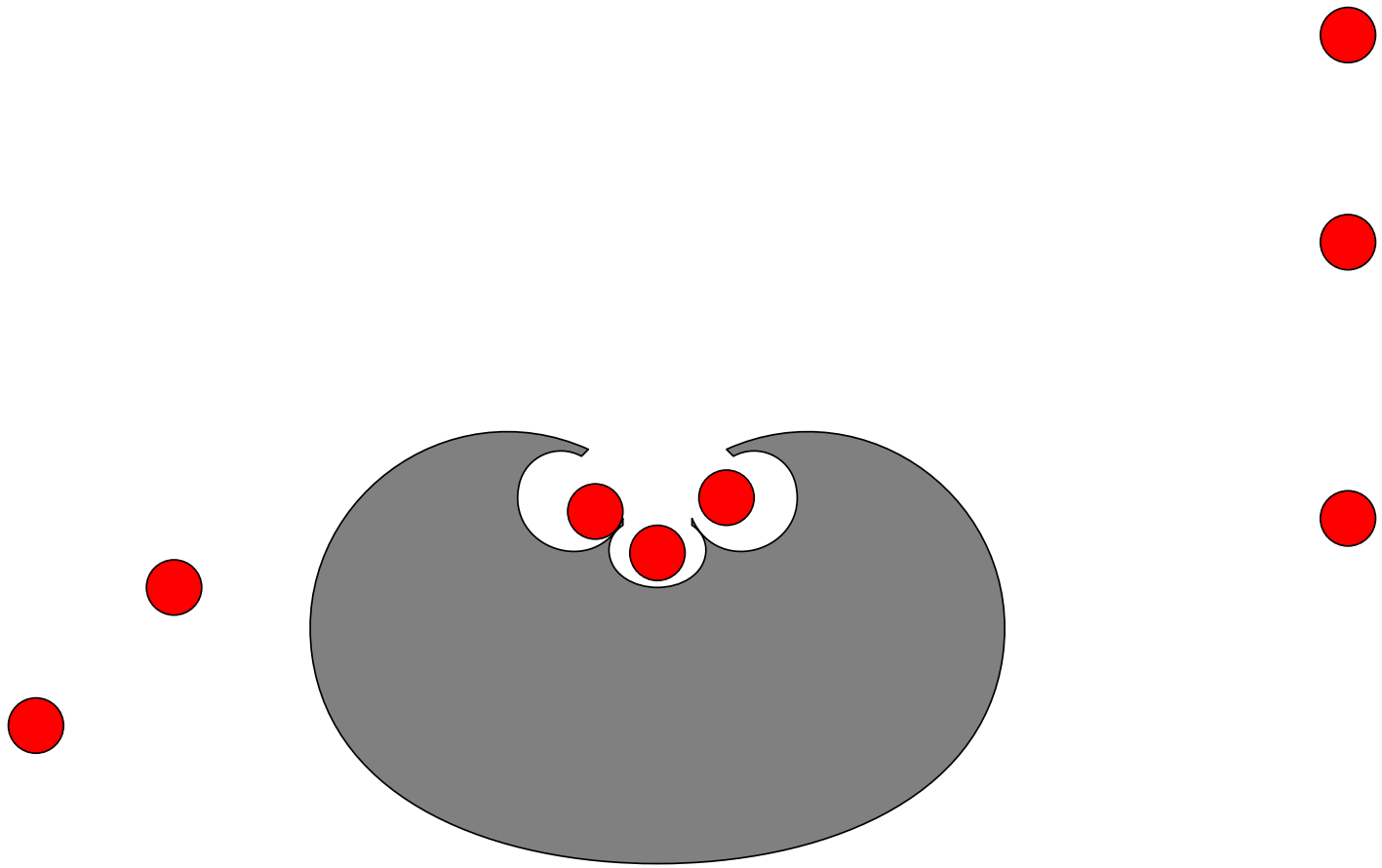

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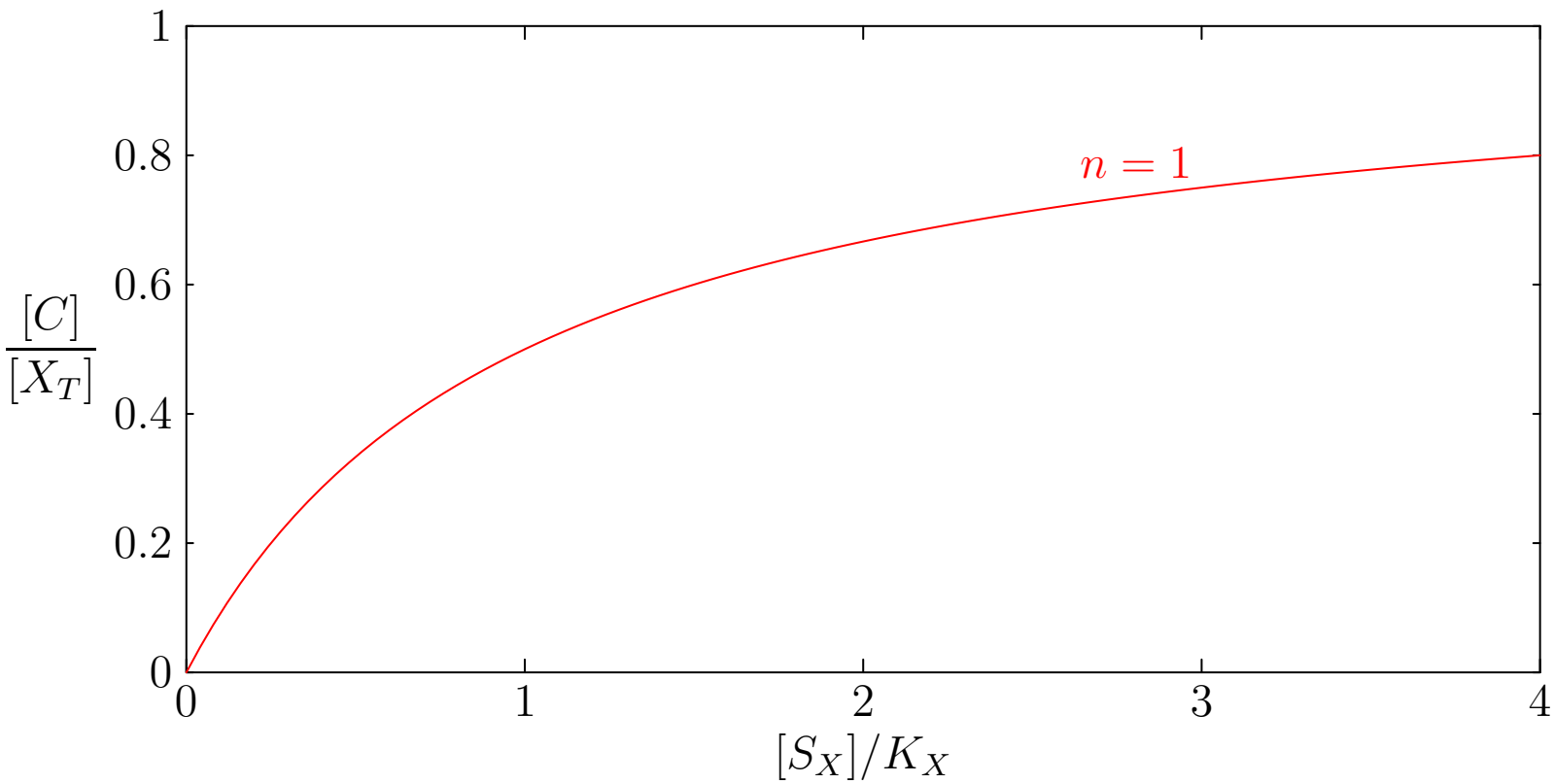

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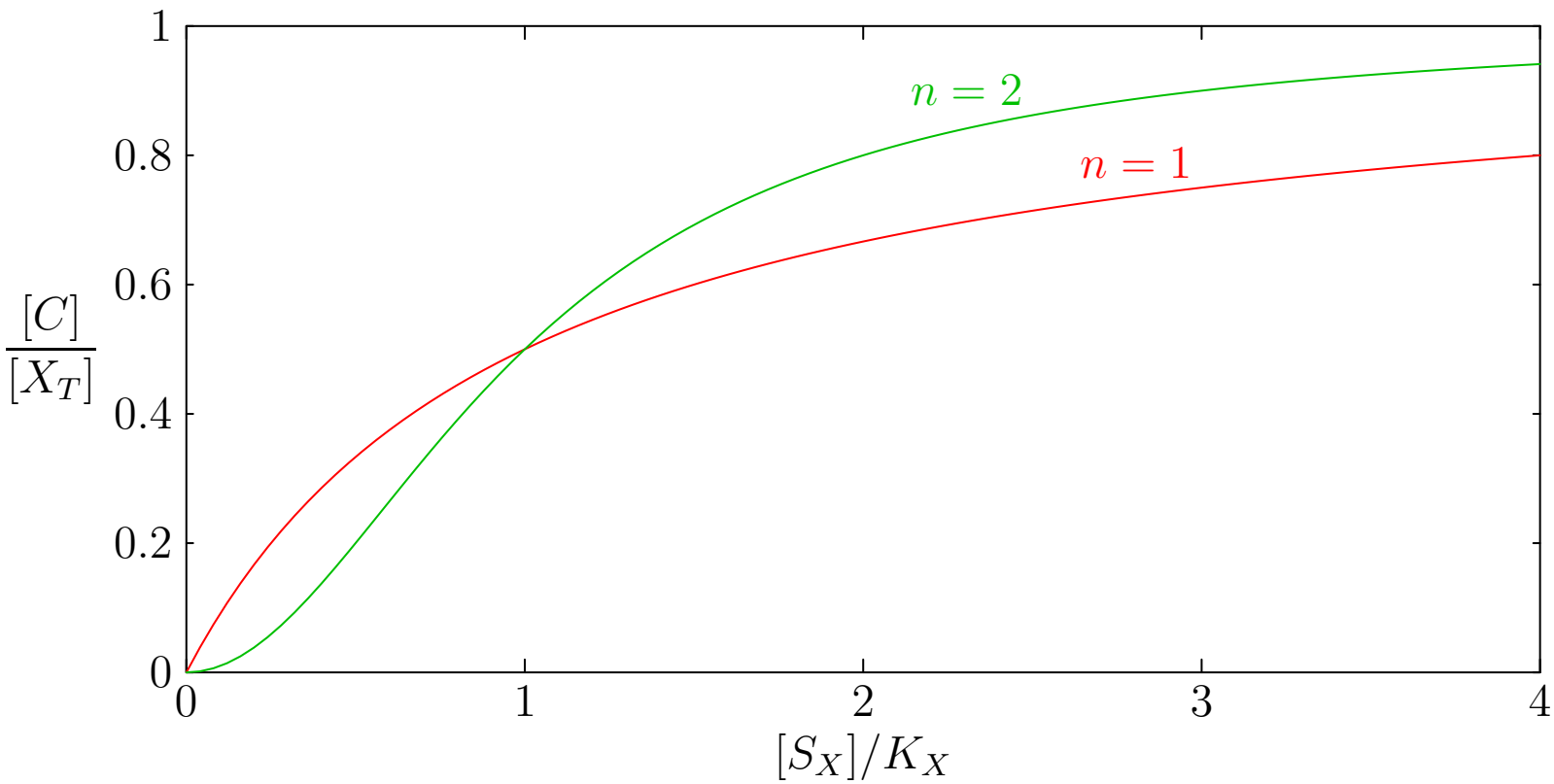

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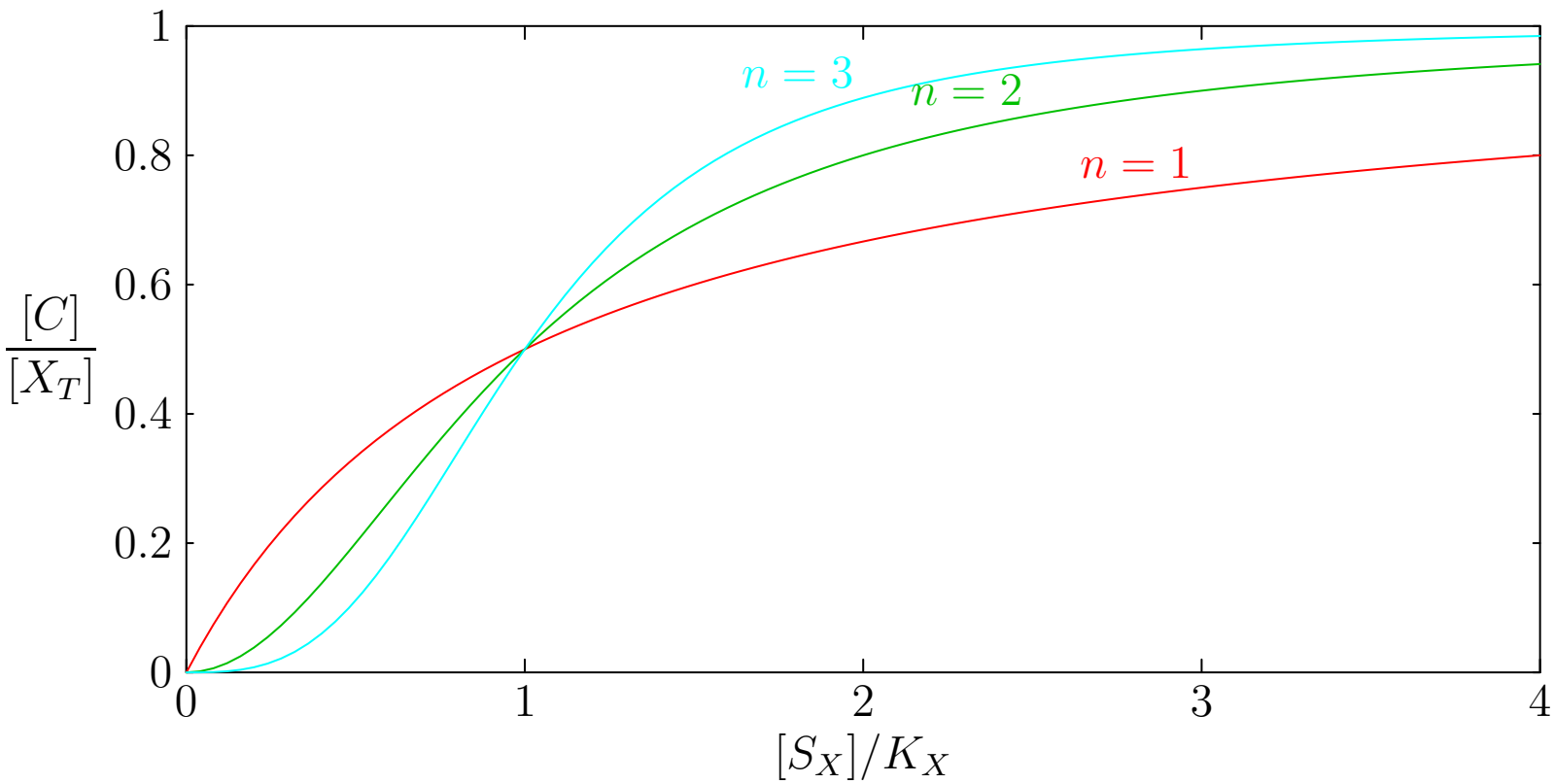

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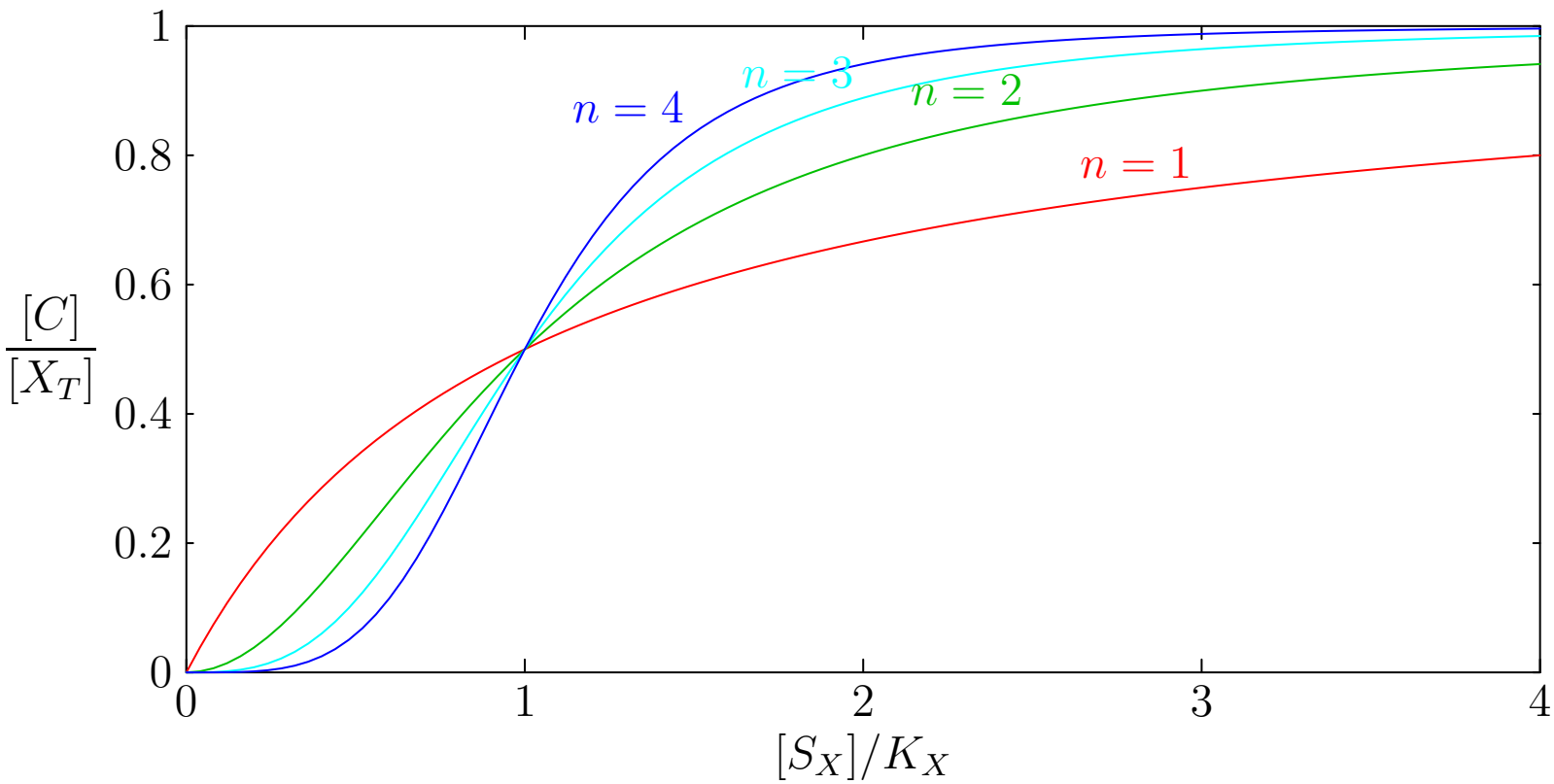

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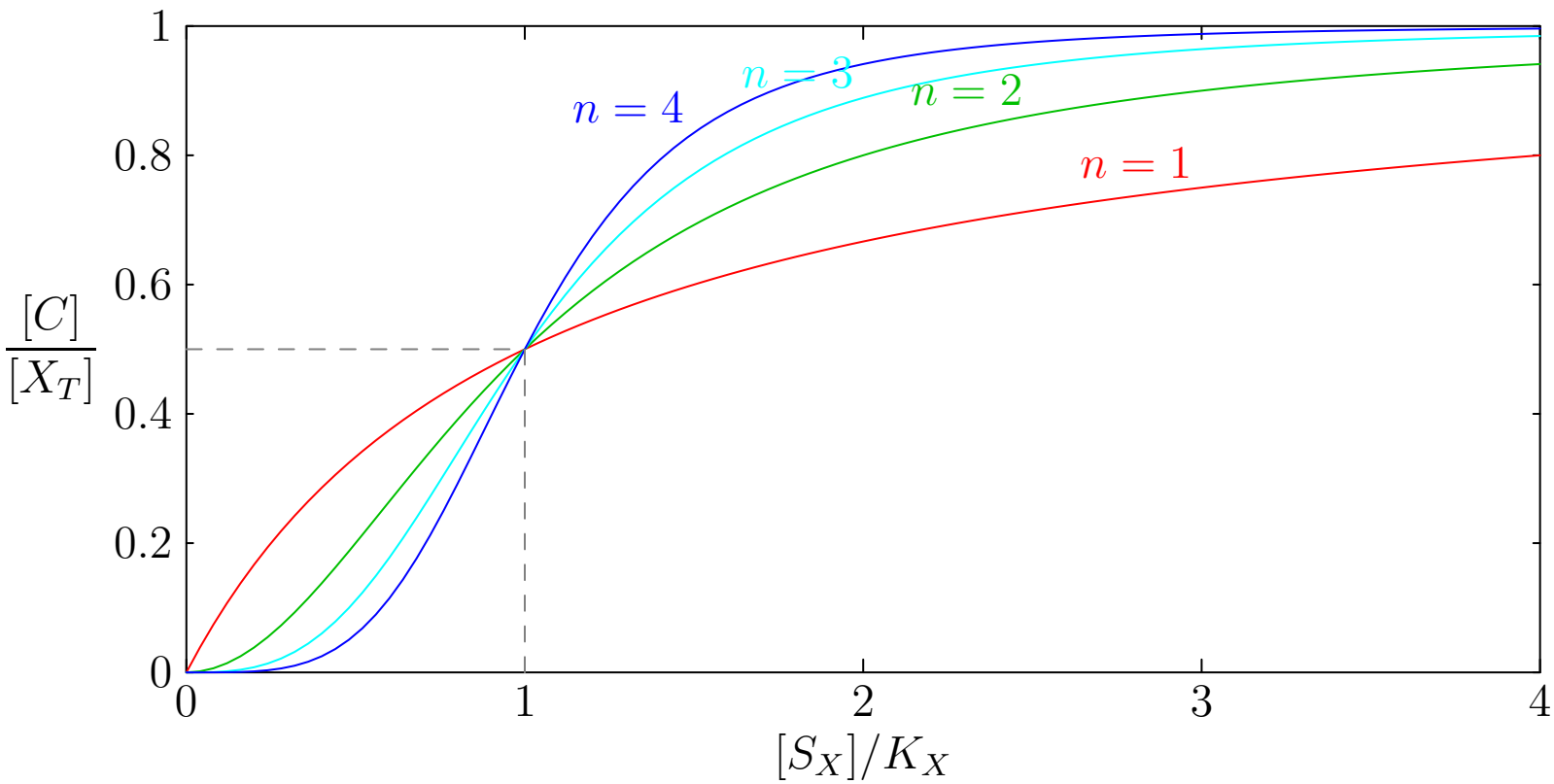

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- A protein that acts to increase the expression of a protein we call an **activator**

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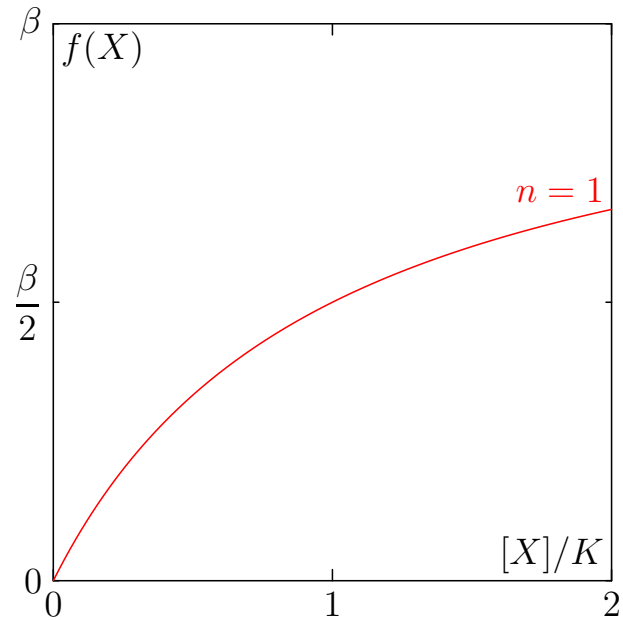


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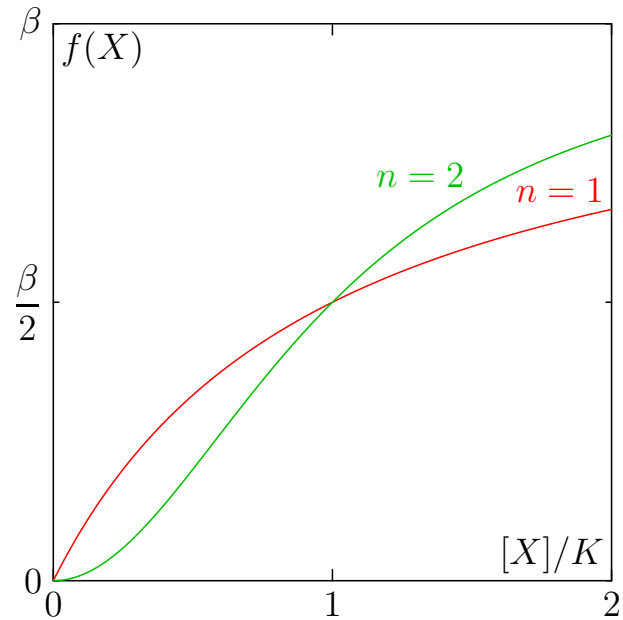


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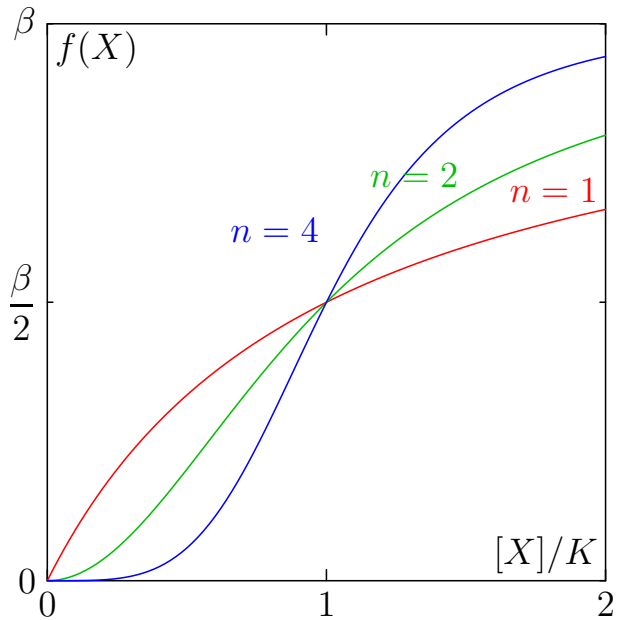


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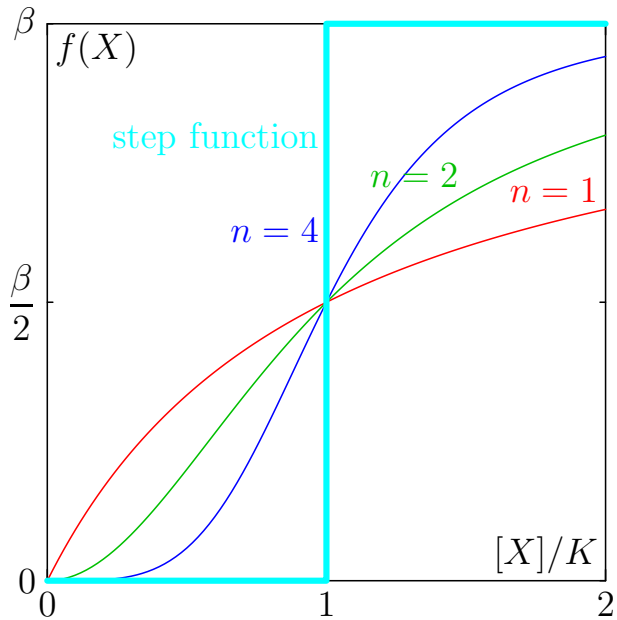


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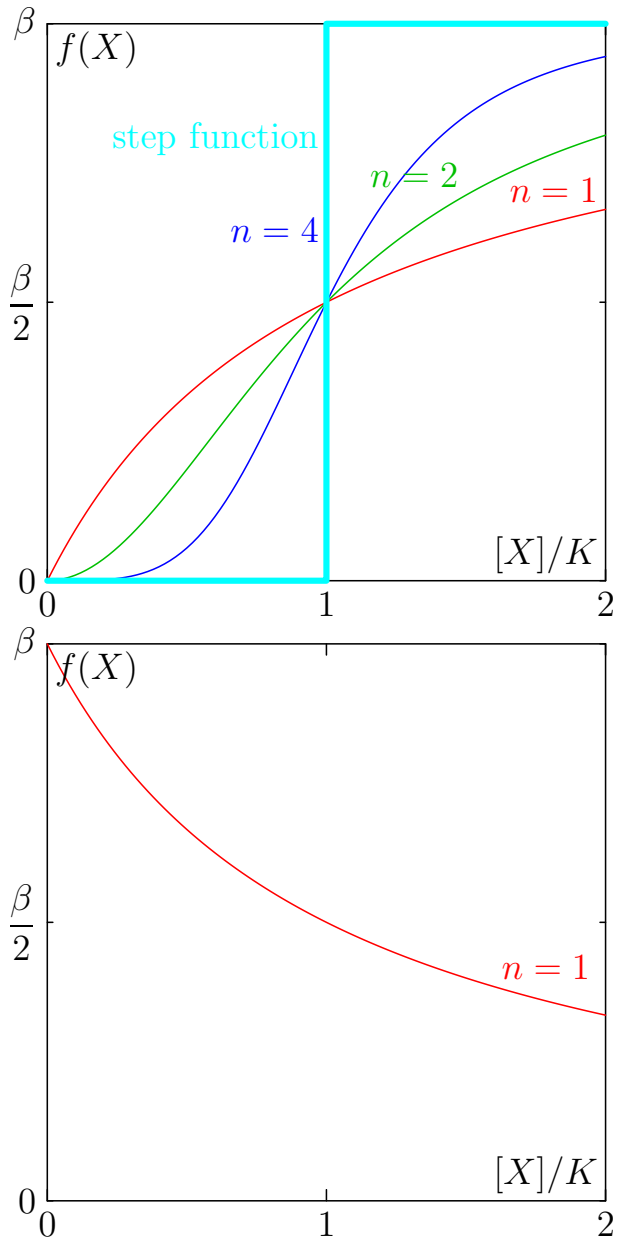

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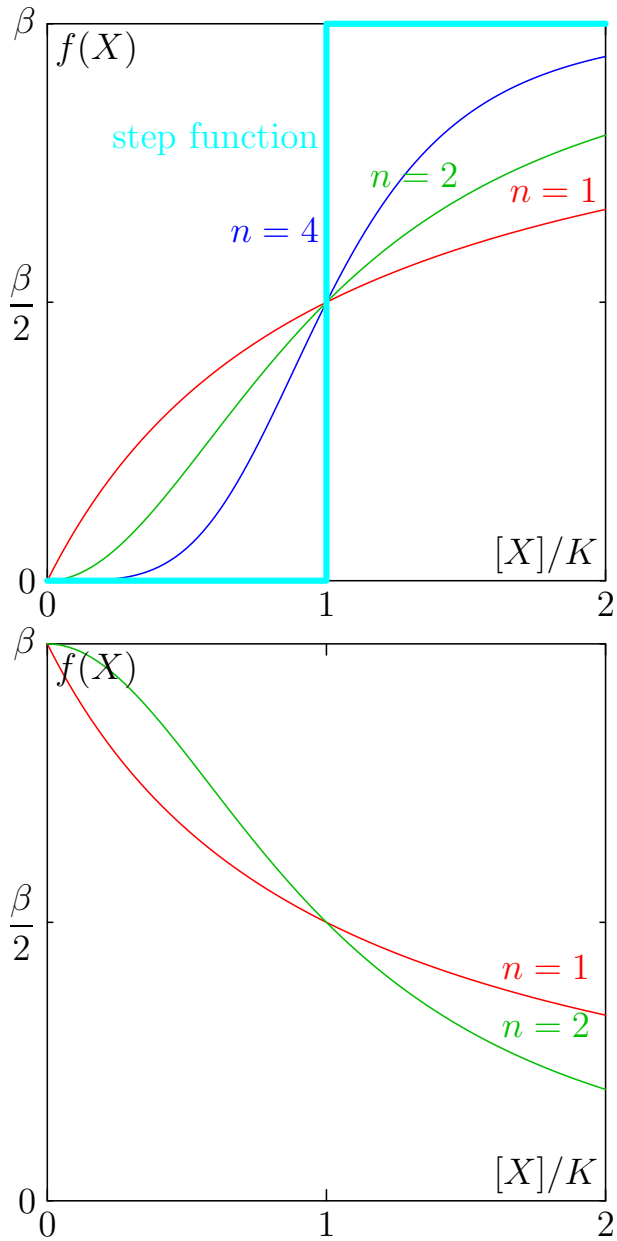

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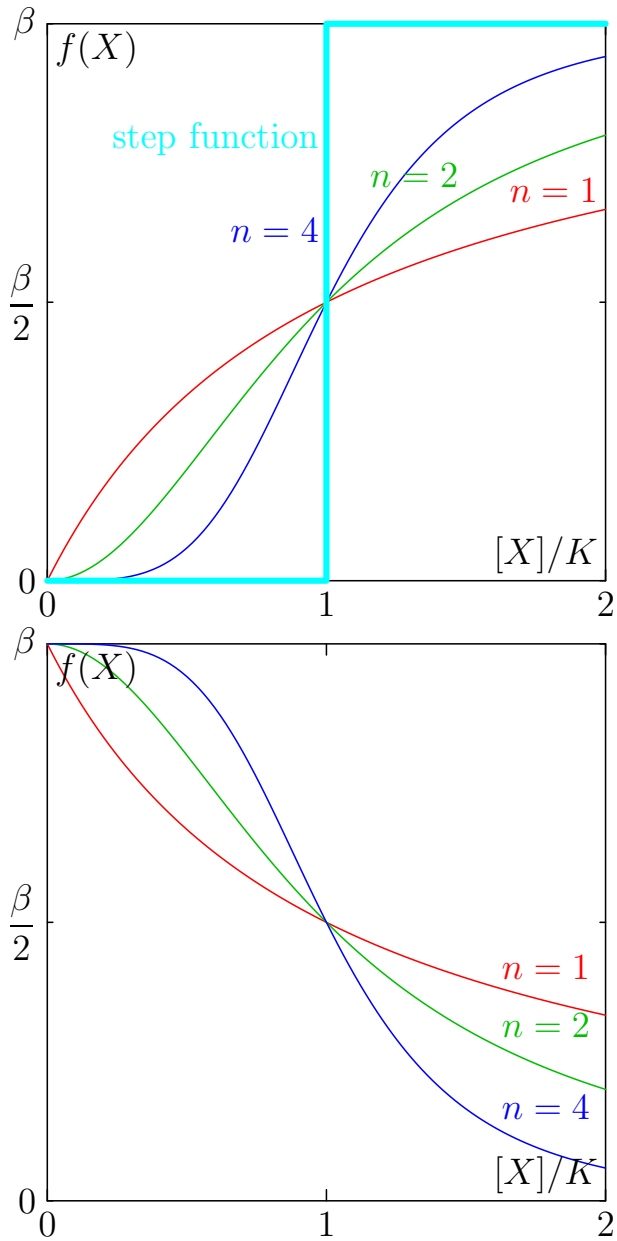

# Activator and Repressor

- A protein that acts to increase the expression of a protein we call an **activator**

- The activation coefficient is often given by the Hill-equation

$$f([X]) = \frac{\beta\,[X]^n}{K^n + [X]^n}$$

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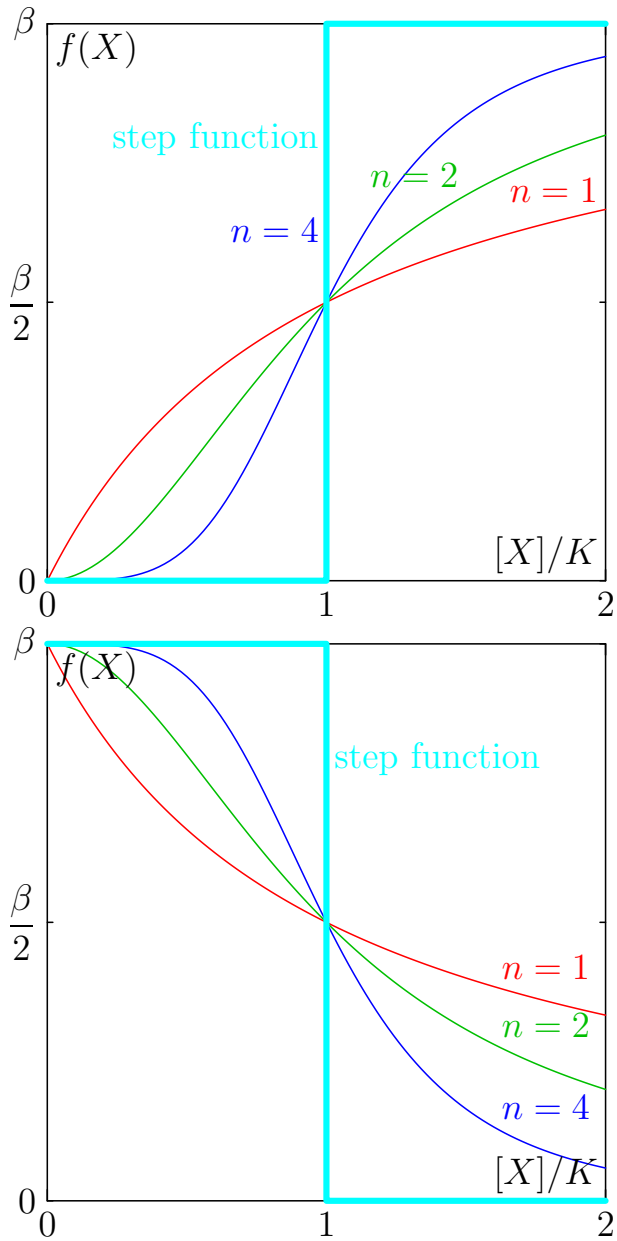
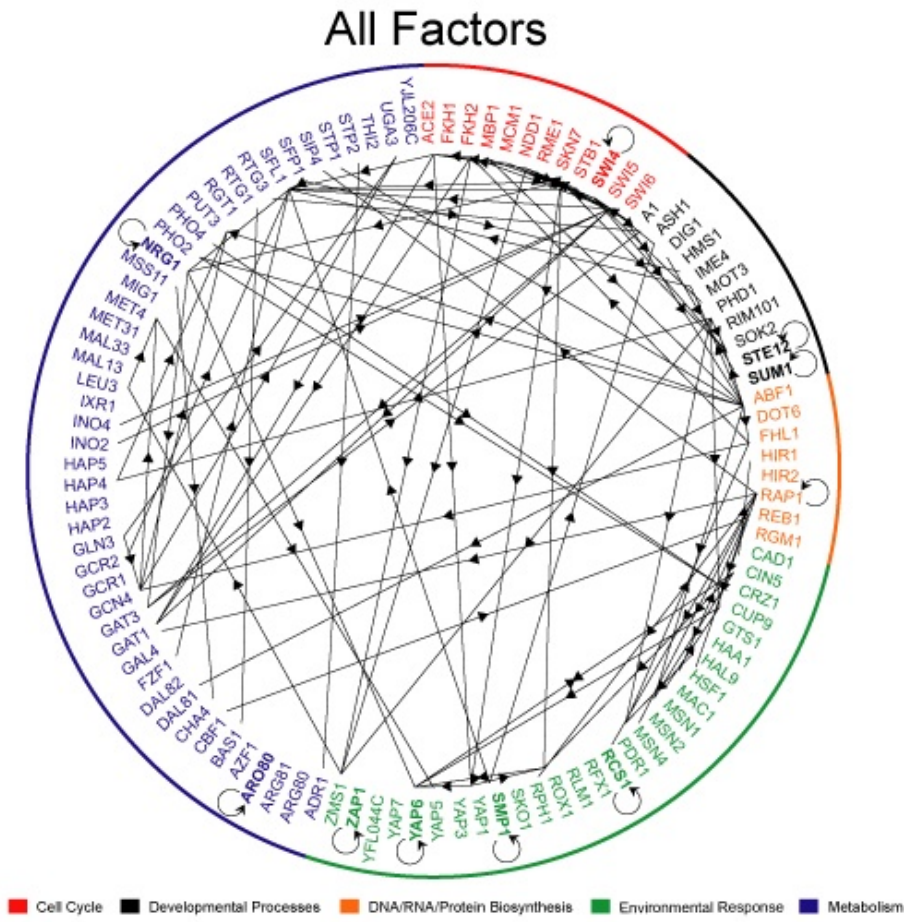


# Outline

1. Gene Regulation
2. Chemical Equations
3. **Motifs**



# Understanding Gene Regulation Networks

- We can infer interactions between proteins as we described in a previous lecture
- Gene regulation networks tend to be rather complicated
- To make sense of them we wish to look for patterns, "**motifs**", that occur very often
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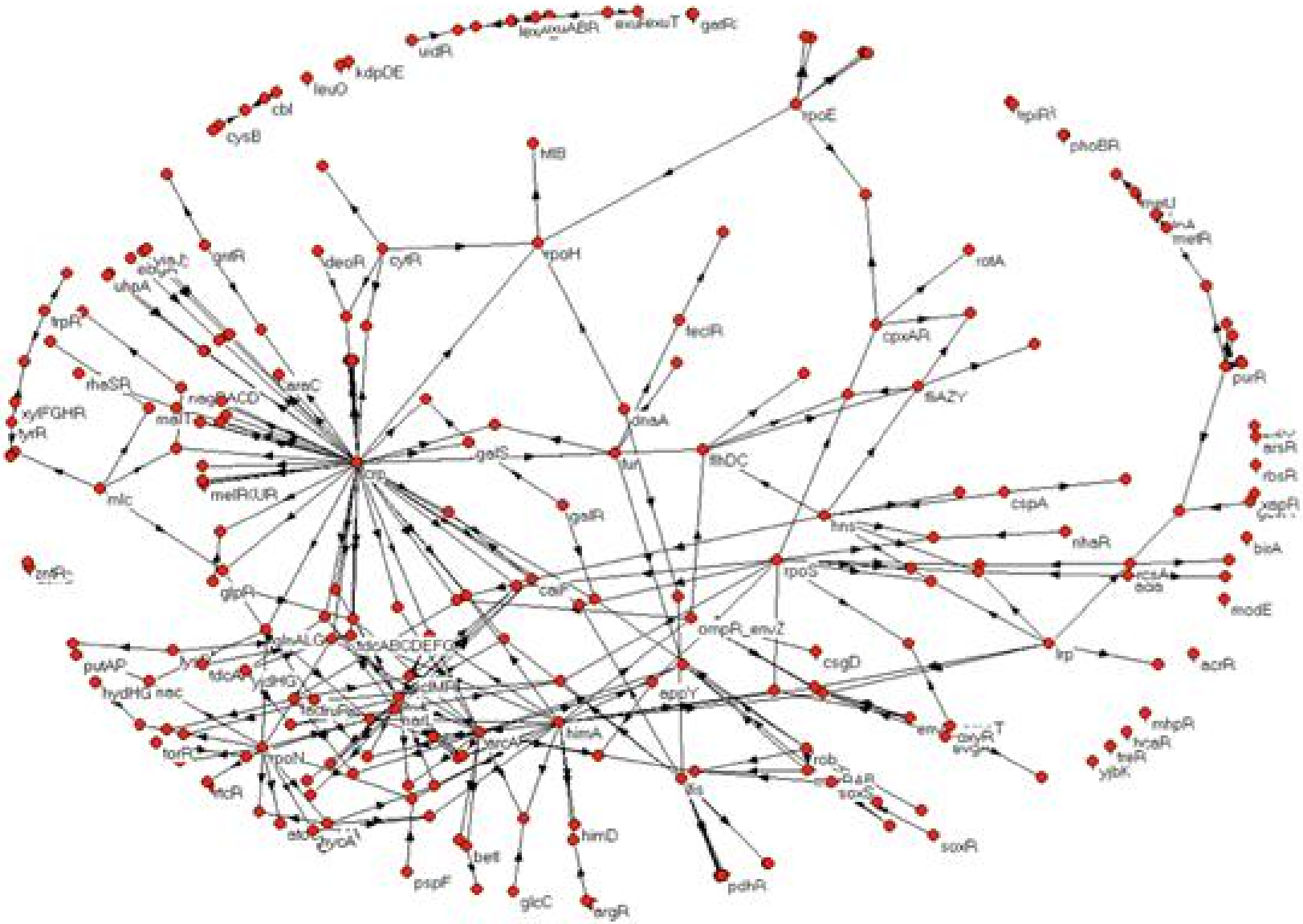
# E-Coli GRN


uidR
kdpDE
leuO
cbl
cysB
rpoE
phoBR
mlc
glpR
putAP
torR
rpoN
fnr
crp
rpoS
ompR_envZ
csgD
ompR
rpoH
cyaR
deoR
gntR
uhpA
malT
melR
cpxAR
fliAZY
fliDC
ctpA
nhaR
bioA
acrR
mhpR
soxR
pdhR
argR
glcC
pspF
hmD
hmA
ilv
rob
ansR
rbsR
purR
melR

# Auto-regulation

- In E-coli we have $N = 424$ nodes and $E = 519$ edges
- The total number of directed edges (including self-edges) is $N^2$
- The number of self edges is $N$ so the probability of a randomly chosen edge being a self-edge is $p = 1/N$
- The probability that if we choose $E$ edges $k$ will be self-edges is

$$\mathbb{P}\left(k|E,p\right) = \mathrm{Binom}(k|E,p) = \binom{E}{k} p^k \left(1-p\right)^{E-k}$$

- This has mean $\mathbb{E}\left[N_{\text{self}}\right] = E\,p \approx 1.2$ and standard deviation $\sigma = \sqrt{E\,p\left(1-p\right)} \approx 1.1$
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- We assume

$$\varnothing \xrightarrow{\hat{\beta}} \hat{X}$$

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- So on average

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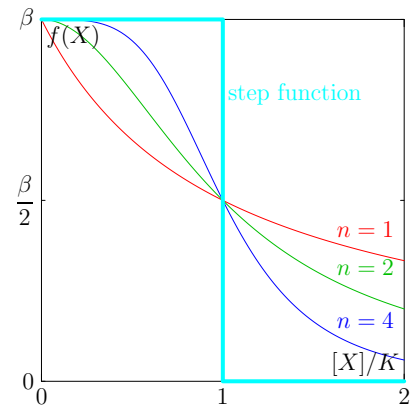


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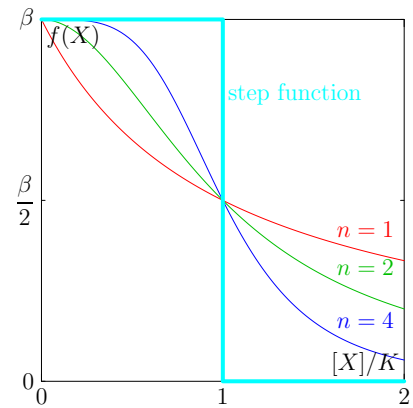


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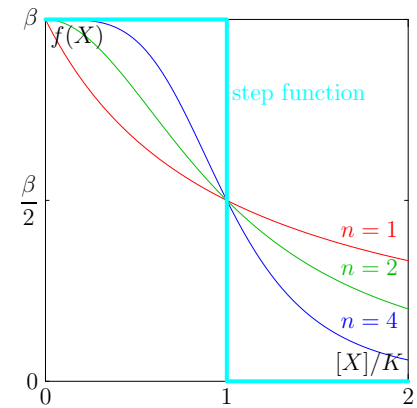


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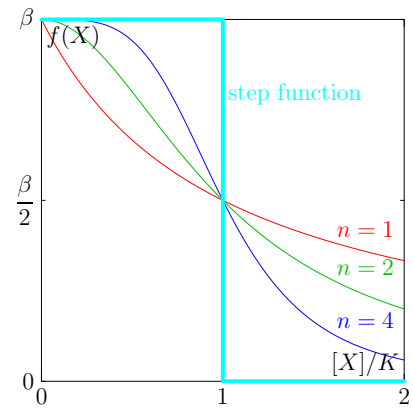


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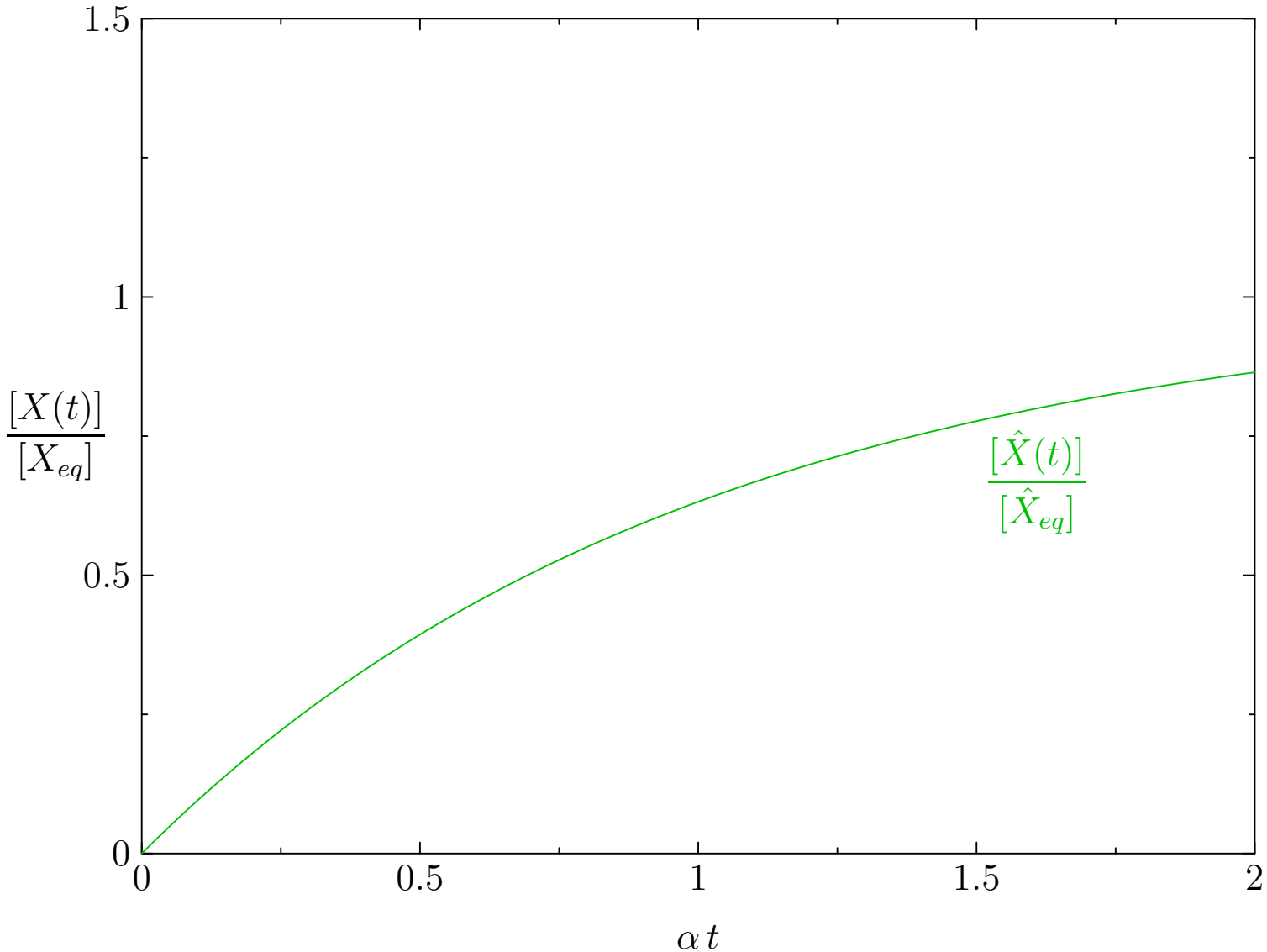

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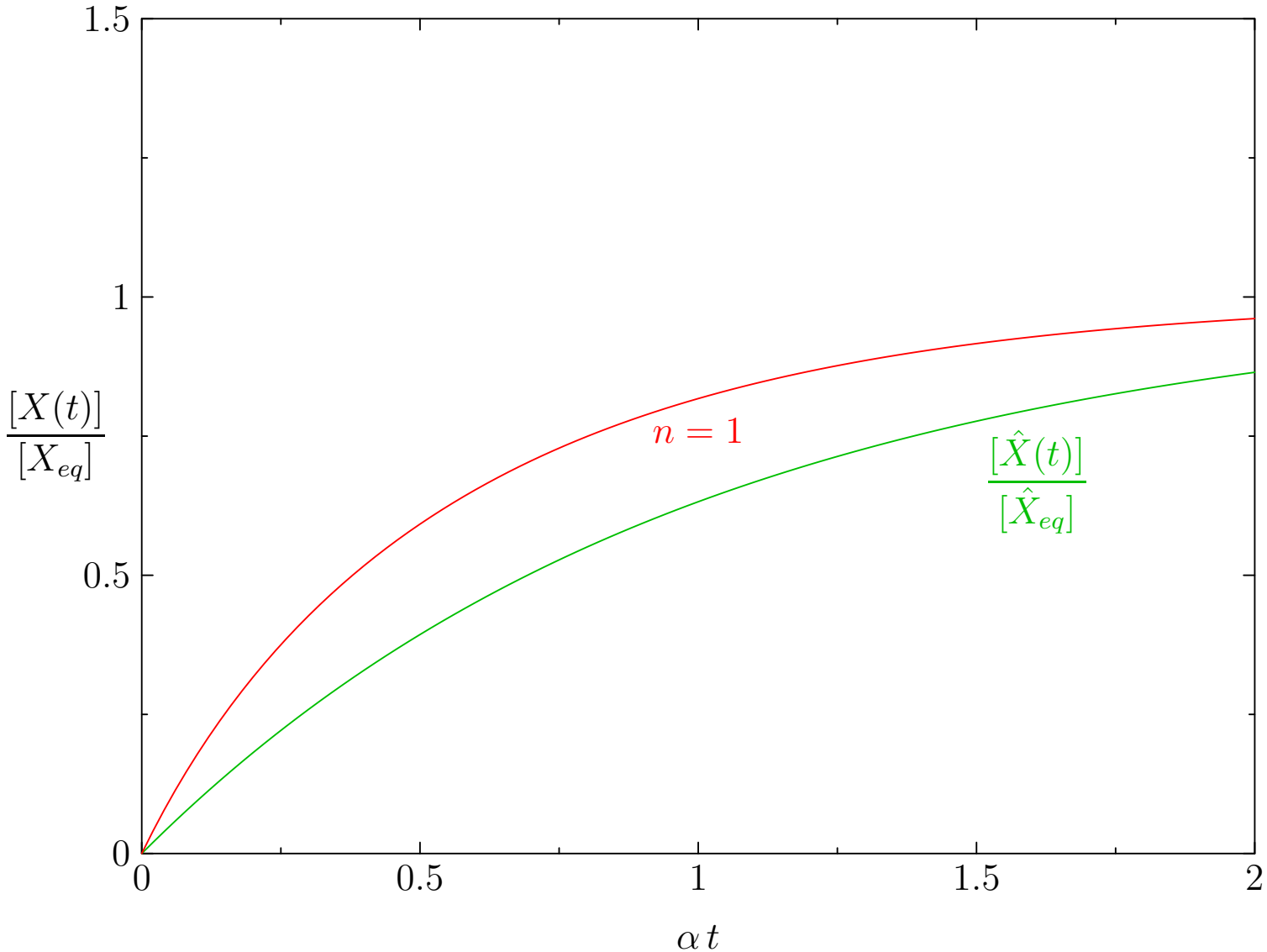

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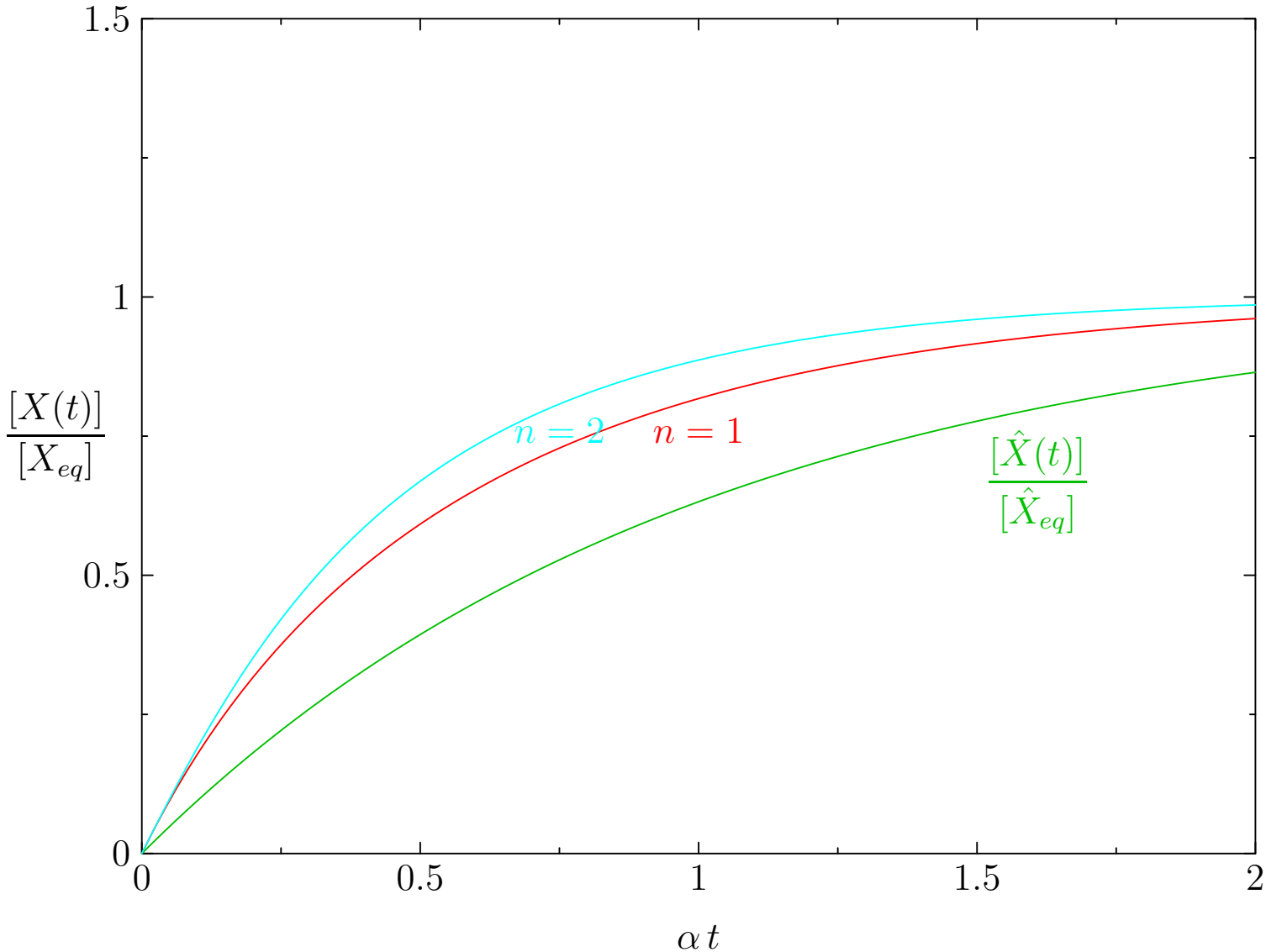

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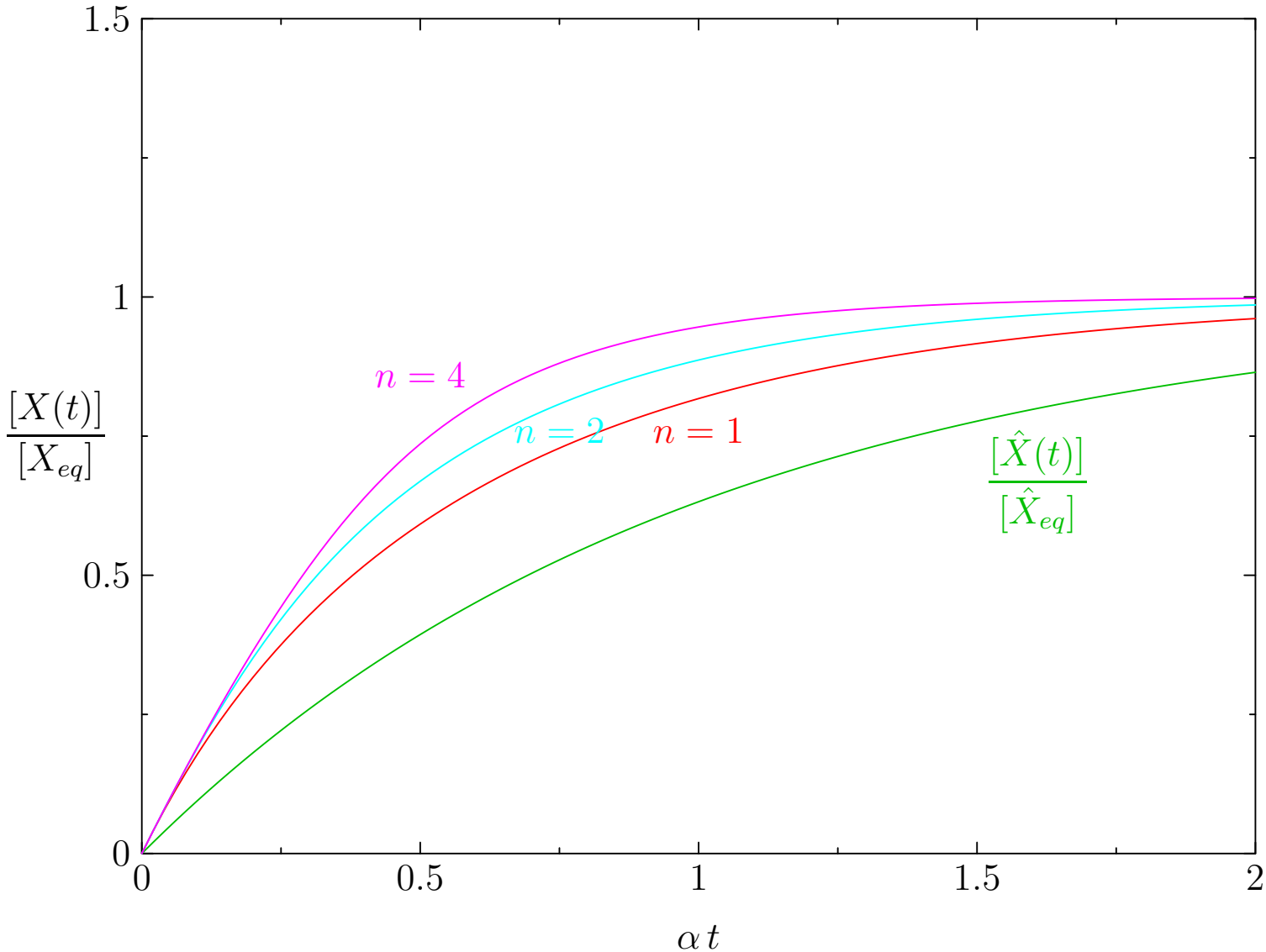

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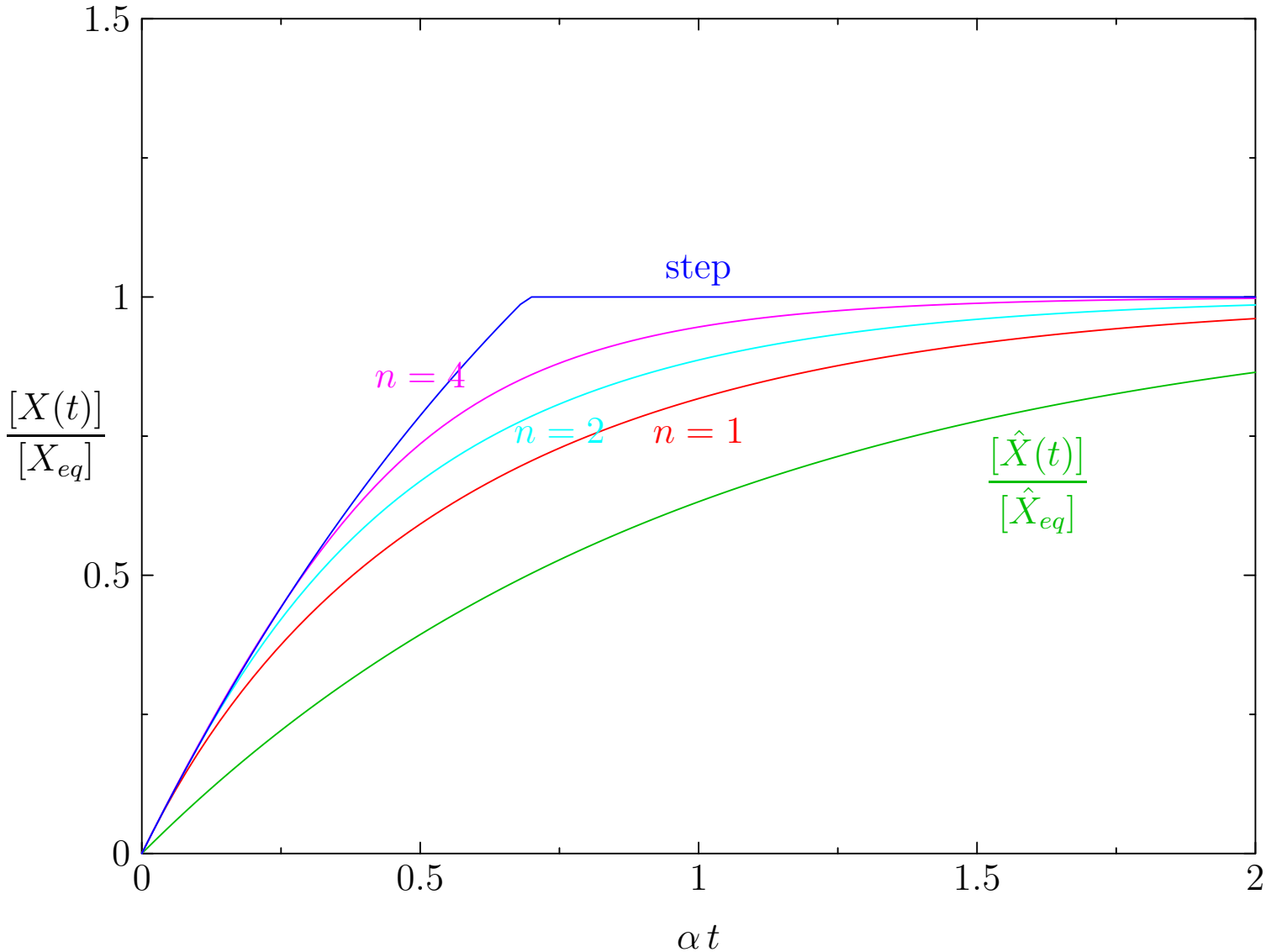
1.5
1
0.5
0
0
0.5
1
1.5
2
$\frac{[X(t)]}{[X_{eq}]}$
$\alpha\, t$
step
$n = 4$
$n = 2$
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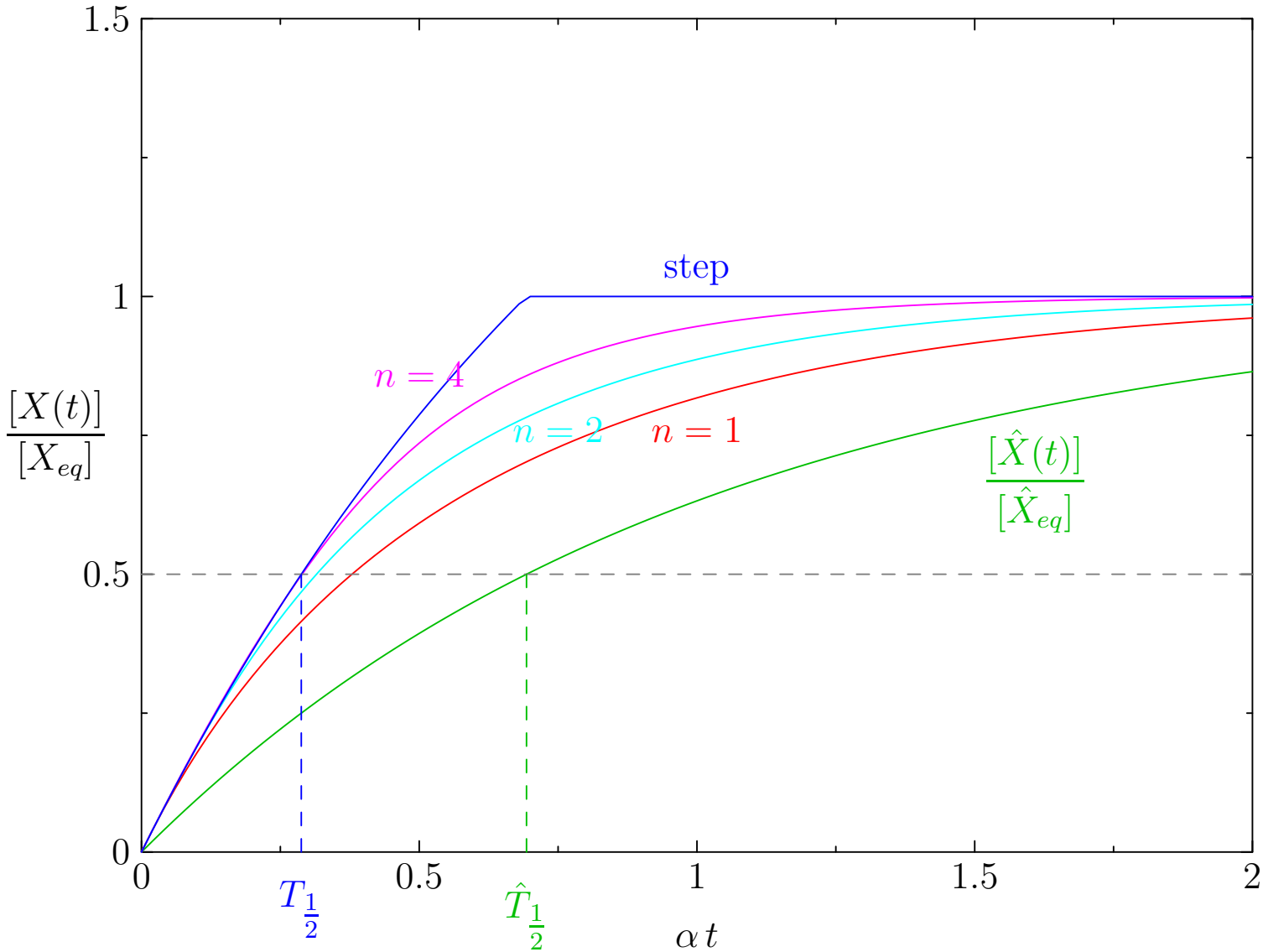
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# Advantages of Auto-regulation

- Auto-regulation allows us to use a large production rate $\beta$ for the same equilibrium, speeding up the equilibration time
- Note that we can independent evolve $K$ (by mutations to the binding site of $X$ in the promoter region) and $\beta$, by mutations in the binding site of RNA polymerase
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